

are used. High selectivity for the solute to be removed is desirable in order to reduce the amount of solid to be added. In any case, the adsorbent is applied in the form of a very finely ground powder, usually at least fine enough to pass entirely through a 200-mesh screen and frequently very much finer.

The highest convenient temperature should be used during the mixing, since the resulting decreased liquid viscosity increases both the rate of diffusion of solute and the ease with which the adsorbent particles can move through the liquid. Usually the equilibrium adsorption is decreased to a small extent at higher temperatures, but this is more than compensated for by the increased rate of approach to equilibrium. Operations are sometimes conducted at the boiling point of the liquid if this temperature will cause no injury to the substances involved. In the clay treatment of petroleum-lubricant fractions the adsorbent-oil mixture may be pumped through a tubular furnace to be heated to as much as 120 to 150°C, and for very heavy oils even to 300 to 380°C. If the adsorbed substance is volatile, however, the equilibrium extent of adsorption will be much more strongly affected by temperature and such material is best handled at ordinary temperatures.

Owing to the large quantity of solution usually treated relative to the amount of adsorption occurring, the temperature rise resulting from release of the heat of adsorption can usually be ignored.

The method of dealing with the spent adsorbent depends upon the particular system under consideration. The filter cake is usually washed to displace the solution held within the pores of the cake, but relatively little adsorbate will be removed in this manner. If the adsorbate is the desired product, it can be desorbed by contact of the solid with a solvent other than that which constitutes the original solution, one in which the adsorbate is more soluble. This can be done by washing the cake in the filter or by dispersing the solid into a quantity of the solvent. If the adsorbate is volatile, it can be desorbed by reduction of the partial pressure of the adsorbate over the solid by passage of steam or warm air through the solid. With activated carbon adsorbents, care must be taken to avoid too high temperatures in using air for this purpose, in order to avoid combustion of the carbon. In most decolorizing operations, the adsorbate is of no value and is desorbed with difficulty. The adsorbent can then be revived by burning off the adsorbate, followed by reactivation. Usually only a limited number of such revivifications is possible before the adsorbent ability is severely reduced, whereupon the solid is discarded.

Single-stage operation The schematic flowsheet for this type of operation in either batch or continuous fashion is shown in the upper part of Fig. 11.16. Here the circle represents all the equipment and procedures constituting one stage. The operation is essentially analogous to a single-stage gas absorption, where the solution to be treated corresponds to a gas and the solid adsorbent to a liquid. Since the amount of adsorbent used is ordinarily very small with respect to the amount of solution treated, and since the solute to be removed is adsorbed much more strongly than the other constituents present, the adsorption of the latter

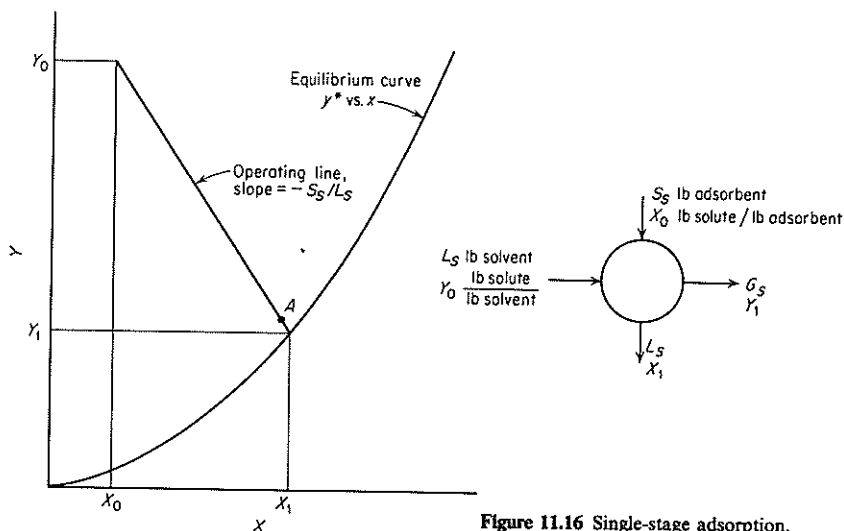


Figure 11.16 Single-stage adsorption.

can be ignored. Furthermore the adsorbent is insoluble in the solution. The solution to be treated contains L_S mass unadsorbed substance or solvent, and the adsorbable solute concentration is reduced from Y_0 to Y_1 mass solute/mass solvent. The adsorbent is added to the extent of S_S mass adsorbate-free solid, and the solute adsorbate content increases from X_0 to X_1 mass solute/mass adsorbent. If fresh adsorbent is used, $X_0 = 0$, and in cases of continuous operation L_S and S_S are measured in terms of mass/time.†

The solute removed from the liquid equals that picked up by the solid,

$$L_S(Y_0 - Y_1) = S_S(X_1 - X_0) \quad (11.4)$$

On X, Y coordinates this represents a straight operating line, through points of coordinates (X_0, Y_0) and (X_1, Y_1) of slope $-S_S/L_S$. If the stage is a theoretical or equilibrium stage, the effluent streams are in equilibrium, so that the point (X_1, Y_1) lies on the equilibrium adsorption isotherm. This is shown on the lower portion of Fig. 11.16. The equilibrium curve should be that obtaining at the final temperature of the operation. If insufficient time of contact is allowed, so that equilibrium is not reached, the final liquid and solid concentrations will correspond to some point such as A (Fig. 11.16), but ordinarily equilibrium is approached very closely.

† For the dilute solutions ordinarily used other consistent units can be applied to these terms. Thus Y can be expressed as kg solute/kg solution (or kg solute/m³ solution) and L_S as kilograms (or cubic meters, respectively) of solution. When the adsorbed solute is colored matter whose concentration is measured in arbitrary units, the latter can be considered as Y units of color per kilogram or cubic meter of solution and the adsorbate concentration on the solid X as units of color per kilogram of adsorbent.

The use of Eq. (11.4) assumes that the amount of liquid mechanically retained with the solid (but not adsorbed) after filtration or settling is negligible. This is quite satisfactory for most adsorption, since the quantity of solid employed is ordinarily very small with respect to that of the liquid treated. If the operation under consideration is *desorption*, and if again the quantity of liquid retained mechanically by the solid is negligible, Eq. (11.4) applies, but the operating line lies below the equilibrium curve on Fig. 11.16. In this case, however, it is much more likely that the quantity of liquid retained mechanically with the solid will be an appreciable portion of the total liquid used, and the methods of calculation described in Chap. 13 for leaching should be used.

Application of the Freundlich equation The Freundlich equation can frequently be applied to adsorption of this type, particularly since small adsorbable solute concentrations are usually involved. This can be written in the following form for the concentration units used here,

$$Y^* = mX^n \quad (11.5)$$

and, at the final equilibrium conditions,

$$X_1 = \left(\frac{Y_1}{m} \right)^{1/n} \quad (11.6)$$

Since the adsorbent used ordinarily contains no initial adsorbate and $X_0 = 0$, substitution in Eq. (11.4) yields

$$\frac{S_S}{L_S} = \frac{Y_0 - Y_1}{(Y_1/m)^{1/n}} \quad (11.7)$$

This permits analytical calculation of the adsorbent/solution ratio for a given change in solution concentration, Y_0 to Y_1 .

Refer to Fig. 11.17, where three typical Freundlich isotherms are shown. The isotherm is straight for $n = 1$, concave upward for $n > 1$, and concave

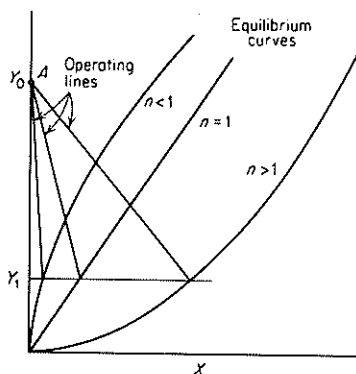


Figure 11.17 Single-stage adsorption, Freundlich equilibrium curves.

downward for $n < 1$. If in each case the solution concentration is to be reduced from Y_0 to Y_1 , the three operating lines radiating from point A apply. The slope of the operating line is in each case directly proportional to the adsorbent/solution ratio. It is generally stated [39] that values of n in the range 2 to 10 represent good, 1 to 2 moderately difficult, and less than 1 poor adsorption characteristics (although the value of m is important also). In the last case impractically large adsorbent dosages may be required for appreciable fractional removal of solute.

Multistage crosscurrent operation The removal of a given amount of solute can be accomplished with greater economy of adsorbent if the solution is treated with separate small batches of adsorbent rather than in a single batch, with filtration between each stage. This method of operation, sometimes called *split-feed treatment*, is usually done in batch fashion, although continuous operation is also possible. Economy is particularly important when activated carbon, a fairly expensive adsorbent, is used. The savings are greater the larger the number of batches used but are at the expense of greater filtration and other handling costs. It is therefore seldom economical to use more than two stages. In rare instances, the adsorption may be irreversible, so that separate adsorbent dosages can be applied without intermediate filtration at considerable savings in operating costs [38]. This is by far the exception rather than the rule, and when applied to ordinary reversible adsorption, it will provide the same end result as if all the adsorbent had been used in a single stage.

A schematic flowsheet and operating diagram for a typical operation of two equilibrium stages are shown in Fig. 11.18. The same quantity of solution is treated in each stage by amounts of adsorbent S_{S1} and S_{S2} in the two stages, respectively, to reduce the solute concentration of the solution from Y_0 to Y_2 . The material balances are, for stage 1,

$$L_S(Y_0 - Y_1) = S_{S1}(X_1 - X_0) \quad (11.8)$$

and for stage 2

$$L_S(Y_1 - Y_2) = S_{S2}(X_2 - X_0) \quad (11.9)$$

These provide the operating lines shown on the figure, each of a slope appropriate to the adsorbent quantity used in the corresponding stage. The extension to large numbers of stages is obvious. If the amounts of adsorbent used in each stage are equal, the operating lines on the diagram will be parallel. The least total amount of adsorbent will require unequal dosages in each stage except where the equilibrium isotherm is linear, and in the general case this can be established only by a trial-and-error computation.

Application of the Freundlich equation When the Freundlich expression [Eq. (11.5)] describes the adsorption isotherm satisfactorily and fresh adsorbent is used in each stage ($X_0 = 0$), the least total amount of adsorbent for a two-stage system can be computed directly [38, 100]. Thus, for stage 1,

$$\frac{S_{S1}}{L_S} = \frac{Y_0 - Y_1}{(Y_1/m)^{1/n}} \quad (11.10)$$

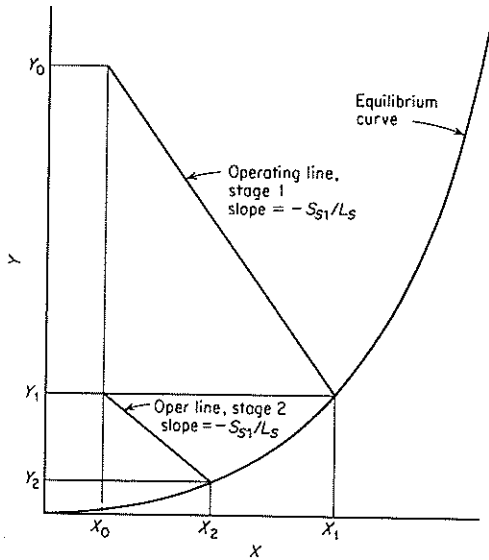
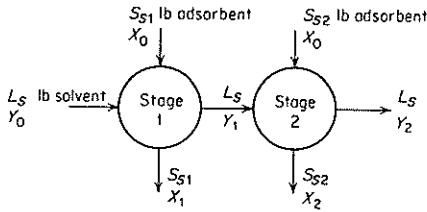


Figure 11.18 Two-stage crosscurrent adsorption.

and for stage 2

$$\frac{S_{S2}}{L_S} = \frac{Y_1 - Y_2}{(Y_2/m)^{1/n}} \quad (11.11)$$

The total amount of adsorbent used is

$$\frac{S_{S1} + S_{S2}}{L_S} = m^{1/n} \left(\frac{Y_0 - Y_1}{Y_1^{1/n}} + \frac{Y_1 - Y_2}{Y_2^{1/n}} \right) \quad (11.12)$$

For minimum total adsorbent, $d[(S_1 + S_2)/L_S]/dY_1$ is set equal to zero, and since for a given case m , n , Y_0 , and Y_2 are constants, this reduces to

$$\left(\frac{Y_1}{Y_2} \right)^{1/n} - \frac{1}{n} \frac{Y_0}{Y_1} = 1 - \frac{1}{n} \quad (11.13)$$

Equation (11.13) can be solved for the intermediate concentration Y_1 , and the adsorbed quantities calculated by Eqs. (11.10) and (11.11). Figure 11.19 permits solutions of Eq. (11.13) without trial and error.

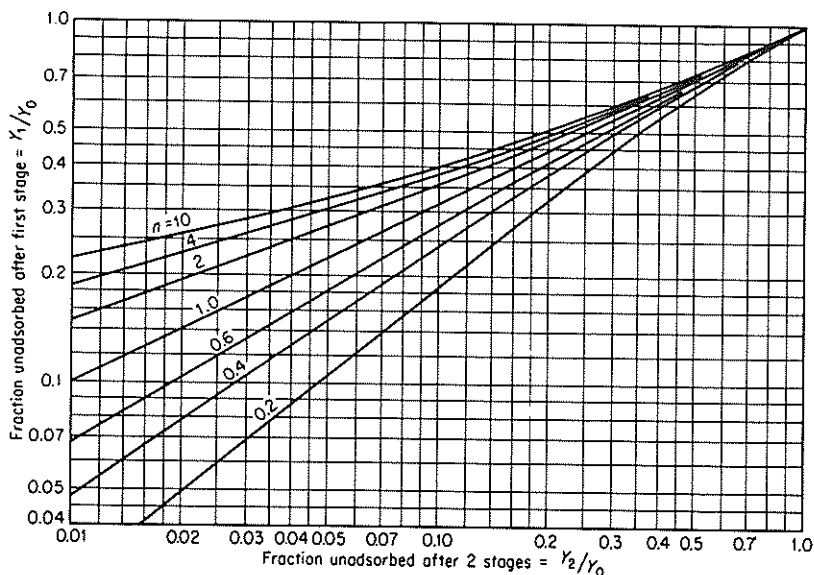


Figure 11.19 Solution to Eq. (11.13). Minimum total adsorbent, two-stage crosscurrent operation.

It has been established [60] that cross-flow operation with a split of the adsorbent, as in Fig. 11.18, is better than the alternative possibility, split of the treated solution, but that countercurrent operation is superior to both.

Multistage countercurrent operation Even greater economy of adsorbent can be obtained by countercurrent operation. When batch methods of treating liquids are used, this can only be simulated, for which the general scheme of batch simulation of countercurrent operations shown in Fig. 10.38 for liquid extraction is actually followed. The flowsheet of Fig. 11.20 then becomes the ultimate, steady-state result, reached only after a number of cycles. However, truly continuous operation has also been used, as in the simultaneous dissolution of gold and silver from finely ground ore by cyanide solution and adsorption of the dissolved metal upon granular activated carbon in a three-stage operation. Coarse screens between agitated vessels separate the large carbon particles from the pulped-ore-liquid mixture [19]. The fluidized-bed tower of Fig. 11.28 for gases and a similar apparatus for liquids, especially for ion exchange, also represent truly continuous stagewise operation.

A solute balance about the N_p stages is

$$L_S(Y_0 - Y_{N_p}) = S_S(X_1 - X_{N_p+1}) \quad (11.14)$$

which provides the operating line on the figure, through the coordinates of the terminal conditions (X_{N_p+1}, Y_{N_p}) and (X_1, Y_0) and of slope S_S/L_S . The number

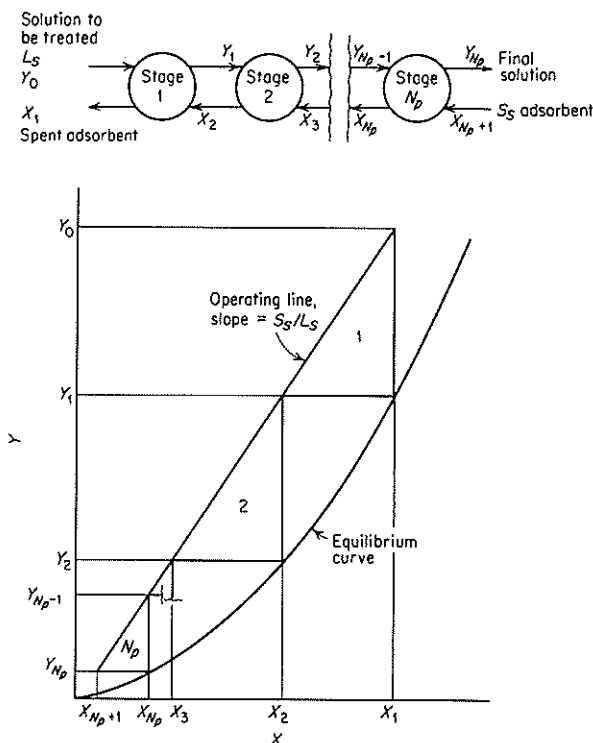


Figure 11.20 Countercurrent multistage adsorption.

of theoretical stages required is found by drawing the usual staircase construction between equilibrium curve and operating line in the manner shown. Alternatively the adsorbent/solution ratio for a predetermined number of stages can be found by trial-and-error location of the operating line. If the operation is a *desorption* (corresponding to stripping in gas-liquid contact), the operating line falls below the equilibrium curve.

The minimum adsorbent/solvent ratio will be the largest which results in an infinite number of stages for the desired change of concentration. This corresponds to the operating line of largest slope which touches the equilibrium curve within the specified range of concentrations. Where the equilibrium isotherm is straight or concave upward, as in Fig. 11.21a, this will cause a *pinch* at the concentrated end of the cascade, as at point A. If the isotherm is concave downward (Fig. 11.21b), the pinch may occur at a point of tangency, as at point B, if Y_0 is sufficiently large. The situations are entirely analogous to those found in gas absorption (Fig. 8.7).

As the number of stages in a cascade is increased, the amount of adsorbent required at first decreases rapidly but approaches the minimum value only

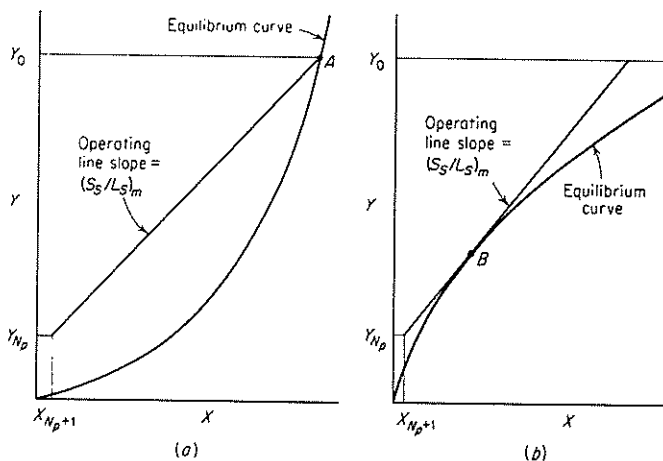


Figure 11.21 Operating line and minimum adsorbent/solvent ratio for infinite stages.

asymptotically. In practice, where intermediate filtration of solid from the liquid must be made between stages, it is rarely economical to use more than two stages in a countercurrent cascade. It will also be realized that, for treatment of gases, the symbol G_S can be substituted for L_S .

In small-scale processing of liquids, there may be appreciable variation in the amounts of solution to be treated from one batch to the next. Furthermore, long periods of time may pass between batches, so that partially spent adsorbent must be stored between stages. Activated carbon particularly may deteriorate during storage through oxidation, polymerization of the adsorbate, or other chemical change, and in such cases the crosscurrent flowsheet may be more practical.

Application of the Freundlich equation Trial-and-error calculation for the adsorbent/solvent ratio can be eliminated if the equilibrium curve can be conveniently described algebraically. If the equilibrium curve is linear, the Kremser equations (5.50) to (5.53) and Fig. 5.16 apply. More frequently the Freundlich expression (11.5) is useful, and fresh adsorbent ($X_{Np+1} = 0$) is used in the last stage [86]. For a typical two-stage cascade (Fig. 11.22) a solute material balance for the entire plant is

$$S_S(X_1 - 0) = L_S(Y_0 - Y_2) \quad (11.15)$$

Applying Eq. (11.5) to the effluents from the first equilibrium stage gives

$$X_1 = \left(\frac{Y_1}{m} \right)^{1/n} \quad (11.16)$$

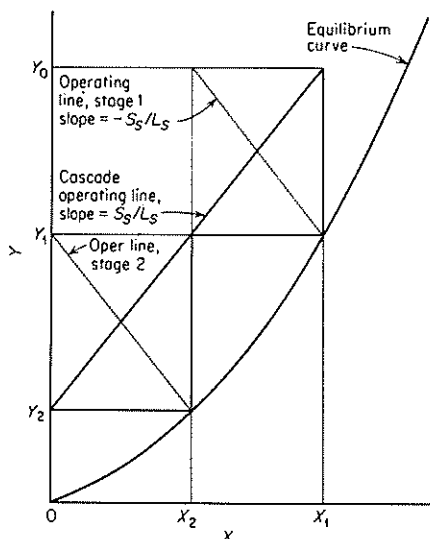
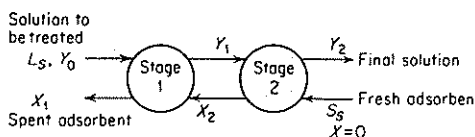


Figure 11.22 Two-stage countercurrent adsorption.

and combining them gives

$$\frac{S_S}{L_S} = \frac{Y_0 - Y_2}{(Y_1/m)^{1/n}} \quad (11.17)$$

The operating line for the second equilibrium stage is shown on the figure and is given by

$$L_S(Y_1 - Y_2) = S_S X_2 = S_S \left(\frac{Y_2}{m} \right)^{1/n} \quad (11.18)$$

Eliminating S_S/L_S between Eqs. (11.17) and (11.18) results in

$$\frac{Y_0}{Y_2} - 1 = \left(\frac{Y_1}{Y_2} \right)^{1/n} \left(\frac{Y_1}{Y_2} - 1 \right) \quad (11.19)$$

Equation (11.19) can be solved for the intermediate concentration Y_1 for specified terminal concentrations Y_0 and Y_2 , and S_S/L_S is then given by Eq. (11.17). Figure 11.23 will assist in the solution of Eq. (11.19). The greater the value of n the greater the savings in adsorbent by countercurrent operation over single stage.

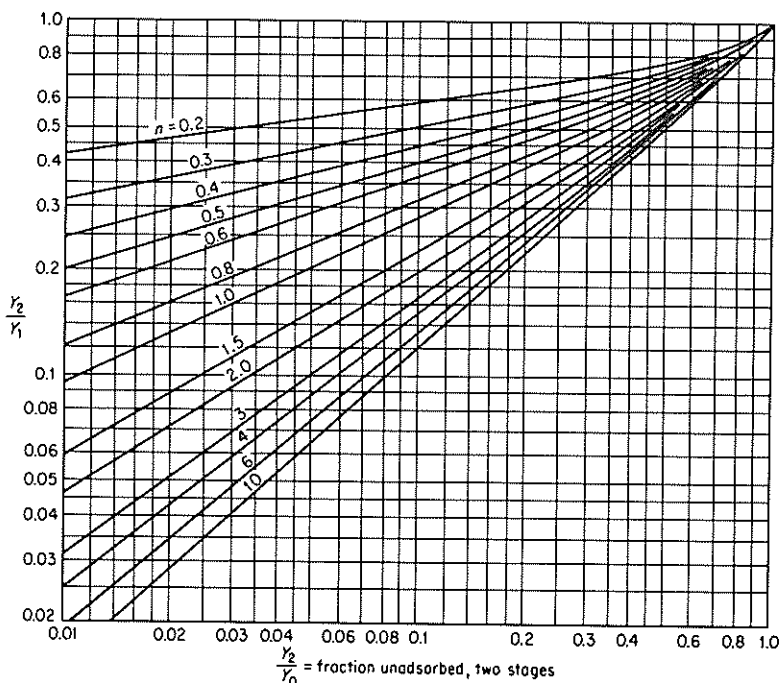


Figure 11.23 Solution to Eq. (11.19). Two-stage countercurrent adsorption.

Illustration 11.2 An aqueous solution containing a valuable solute is colored by small amounts of an impurity. Before crystallization, the impurity is to be removed by adsorption on a decolorizing carbon which adsorbs only insignificant amounts of the principal solute. A series of laboratory tests was made by stirring various amounts of the adsorbent into batches of the original solution until equilibrium was established, yielding the following data at constant temperature:

kg carbon/kg soln	0	0.001	0.004	0.008	0.02	0.04
Equilibrium color	9.6	8.1	6.3	4.3	1.7	0.7

The color intensity was measured on an arbitrary scale, proportional to the concentration of the colored substance. It is desired to reduce the color to 10% of its original value, 9.6. Determine the quantity of fresh carbon required per 1000 kg of solution for a single-stage operation, for a two-stage crosscurrent process using the minimum total amount of carbon, and for a two-stage countercurrent operation.

SOLUTION The experimental data must first be converted to a suitable form for plotting the equilibrium isotherm. For this purpose, define Y as units of color per kilogram of solution and X as units of color adsorbed per kilogram of carbon. The solutions may be considered as dilute in color, so that operating lines will be straight on X, Y coordinates expressed in this manner.

The calculations are made as indicated below.

$\frac{\text{kg carbon}}{\text{kg soln}}$	$Y^* = \text{equilibrium color, units/kg soln}$	$X = \text{adsorbate concentration, units/kg carbon}$
0	9.6	
0.001	8.6	$(9.6 - 8.6)/0.001 = 1000$
0.004	6.3	$(9.6 - 6.3)/0.004 = 825$
0.008	4.3	663
0.02	1.7	395
0.04	0.7	223

The equilibrium data, when plotted on logarithmic coordinates, provide a straight line, so that the Freundlich equation applies (see Fig. 11.24). The slope of the line is $1.66 = n$, and, at $X = 663$, $Y^* = 4.3$. Therefore [Eq. (11.5)]

$$m = \frac{4.3}{663^{1.66}} = 8.91 \times 10^{-5}$$

The Freundlich equation is therefore

$$Y^* = 8.91 \times 10^{-5} X^{1.66}$$

The equilibrium data can also be plotted on arithmetic coordinates (Fig. 11.25).

Single-stage operation $Y_0 = 9.6$ units of color/kg soln,

$$Y_1 = 0.10(9.6) = 0.96 \text{ unit/kg soln}$$

Let $L_S = 1000$ kg soln. Since fresh carbon is to be used, $X_0 = 0$. In Fig. 11.25, point A representing the initial solution and fresh adsorbent is located, and point B is located on the equilibrium curve at the color concentration of the final solution. At B , $X_1 = 260$. Therefore [Eq. (11.4)]

$$\frac{S_S}{L_S} = \frac{Y_0 - Y_1}{X_1 - X_0} = \frac{9.6 - 0.96}{270 - 0} = 0.032 \text{ kg carbon/kg soln}$$

and

$$S_S = 0.032(1000) = 32.0 \text{ kg carbon/1000 kg soln}$$

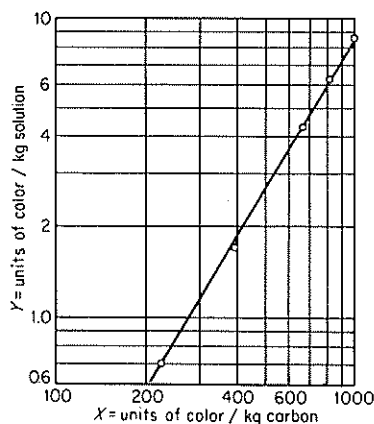


Figure 11.24 Equilibrium data, Illustration 11.2.

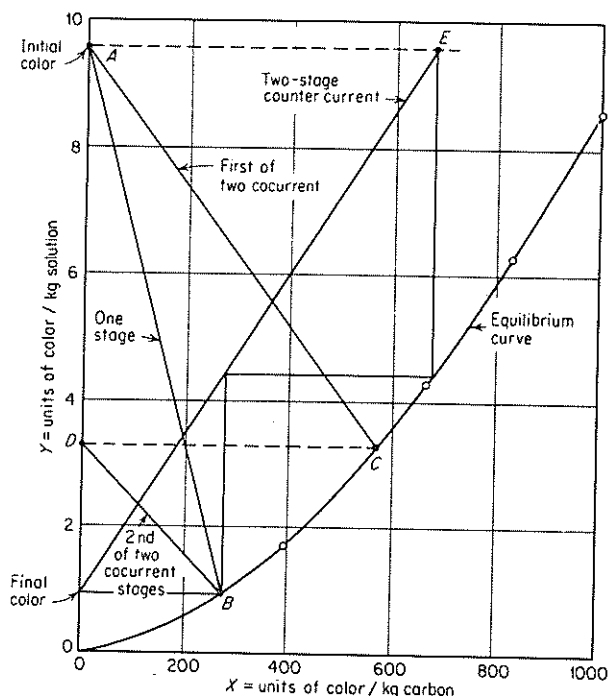


Figure 11.25 Solution to Illustration 11.2.

Alternatively, since the Freundlich equation applies, use Eq. (11.7):

$$\frac{S_S}{L_S} = \frac{Y_0 - Y_1}{(Y_1/m)^{1/n}} = \frac{9.6 - 0.96}{[0.96 / (8.91 \times 10^{-5})]^{1/1.66}} = 0.032 \text{ kg carbon/kg soln}$$

$$S_S = 0.032(1000) = 32.0 \text{ kg carbon/1000 kg soln}$$

Two-stage crosscurrent operation The minimum total amount of carbon can be found in Fig. 11.25 by a trial-and-error procedure. Thus, point C on the equilibrium curve is assumed, the operating lines AC and DB drawn, and the values of S_{S1} and S_{S2} computed by Eqs. (11.8) and (11.9). The position of point C is changed until the sum of S_{S1} and S_{S2} is a minimum. The position of C in Fig. 11.25 is the final value, and its coordinates are ($X_1 = 565$, $Y_1 = 3.30$). $X_2 = 270$ (at B). Eq. (11.8):

$$S_{S1} = \frac{L_S(Y_0 - Y_1)}{X_1 - X_0} = \frac{1000(9.6 - 3.30)}{565 - 0} = 11.14 \text{ kg}$$

Eq. (11.9):

$$S_{S2} = \frac{L_S(Y_1 - Y_2)}{X_2 - X_0} = \frac{1000(3.30 - 0.96)}{270 - 0} = 8.67 \text{ kg}$$

$$S_{S1} + S_{S2} = 11.14 + 8.67 = 19.81 \text{ kg carbon/1000 kg soln}$$

Alternatively, since the Freundlich equation applies, use Fig. 11.19: $Y_2/Y_0 = 0.96/9.6 = 0.10$,

$n = 1.66$. From the figure, $Y_1/Y_0 = 0.344$. $Y_1 = 0.344(9.6) = 3.30$. Eq. (11.10):

$$\frac{S_{S1}}{L_S} = \frac{Y_0 - Y_1}{(Y_1/m)^{1/n}} = \frac{9.6 - 3.30}{[3.30/(8.91 \times 10^{-5})]^{1/1.66}} = 0.01114 \text{ kg carbon/kg soln into 1st stage}$$

Eq. (11.11):

$$\frac{S_{S2}}{L_S} = \frac{Y_1 - Y_2}{(Y_2/m)^{1/n}} = \frac{3.30 - 0.96}{[0.96/(8.91 \times 10^{-5})]^{1/1.66}} = 0.00867 \text{ kg carbon/kg soln into 2d stage}$$

$$\text{Total carbon required} = (0.01114 + 0.00867)(1000) = 19.81 \text{ kg/1000 kg soln}$$

Two-stage countercurrent operation $Y_0 = 9.6$, $Y_2 = 0.96$, $X_{N_p+1} = 0$. The operating line is located by trial in Fig. 11.25 until two stages can be drawn between operating line and equilibrium curve, as shown. From the figure at E , $X_1 = 675$. Eq. (11.14):

$$S_S = \frac{L_S(Y_0 - Y_2)}{X_1 - X_{N_p+1}} = \frac{1000(9.6 - 0.96)}{675 - 0} = 12.80 \text{ kg carbon/1000 kg soln}$$

Alternatively, $Y_2/Y_0 = 0.96/9.6 = 0.10$; $n = 1.66$. From Fig. 11.24, $Y_2/Y_1 = 0.217$, and $Y_1 = Y_2/0.217 = 0.96/0.217 = 4.42$. Eq. (11.17):

$$\frac{S_S}{L_S} = \frac{Y_0 - Y_2}{(Y_1/m)^{1/n}} = \frac{9.6 - 0.96}{[4.42(8.91 \times 10^{-5})]^{1/1.66}} = 0.01280 \text{ kg carbon/kg soln}$$

and

$$L = 0.01280(1000) = 12.80 \text{ kg carbon/1000 kg soln}$$

Agitated Vessels for Liquid-Solid Contact

The vessels are commonly of the proportions described in Chap. 6, with liquid depths in the range 0.75 to 1.5 tank diameters. Four wall baffles, usually $T/12$ wide, are used to eliminate vortices in open vessels and to produce the vertical liquid motion needed to lift the solid particles. They are frequently arranged with a clearance, which may be half the baffle width, between baffle and tank wall to prevent accumulation of solids behind the baffles.

Flat-blade turbines, with or without disk, or pitched-blade turbines, as shown in Fig. 6.3, are most frequently used, arranged on an axially located shaft. Multiple turbines on the same shaft are sometimes used to obtain improved uniformity of the solid-liquid slurry throughout the vessel. The clearance C of the lowest impeller from the bottom of the tank may be $d_i/3$, but it is usually wise in any event to make C larger than the settled depth of the solids in a quiet liquid so that if the impeller stops, it will not become "sanded in." Large impellers operating at relatively low speeds require less power for a given liquid-pumping rate than small, high-speed impellers. Consequently if the solids tend to settle rapidly as a result of a large density or particle size, large impeller diameters are indicated. The recommendations of Lyons [65] for suspensions of solids in water, modified to permit application to liquids of other densities, are shown in Fig. 11.26. The figure shows two ratios T/d_i , one depending on densities and one on particle size; the average T/d_i is chosen. Should the solids consist of mixed sizes and densities, the densest and largest are used. In any event, T/d_i less than 1.6 is not desirable since then the free flow of mixture at

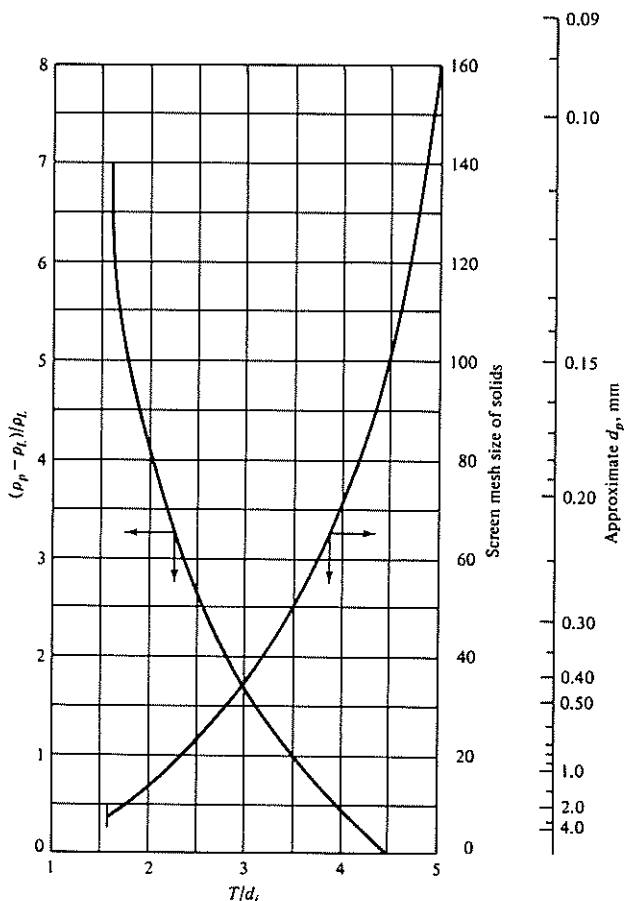


Figure 11.26 Recommended turbine diameters for suspension of solids in liquids of water-like viscosity. [Modified from E. J. Lyons, in V. W. Uhl and J. B. Gray (eds.), "Mixing," vol. 2, Academic Press, New York, with permission.]

the vessel wall is interfered with. The choice of impeller diameter can be modified by considerations of torque [79], since the larger impellers require greater torque to turn them and the speed-reducing gear is consequently expensive.

Solid Suspensions

Unless the solids are suspended in the liquid, the mass-transfer rates are seriously impaired. The agitator power $P_{\min}$ required just to lift all the particles from the vessel bottom, for particles of

density greater than that of the liquid, can be estimated from the expression [101]

$$\frac{P_{\min} g_c}{g v_T V_{iN} \Delta \rho} = 0.092 \left(\frac{\phi_{ST}}{1 - \phi_{ST}} \right)^{1/2} \frac{T}{d_i} e^{5.3C/T} \quad (11.20)$$

where $\Delta \rho = \rho_p - \rho_L$ and ρ_p = particle density. The latter is calculated from the mass and gross volume of the particle, including the volume of the internal pores. Volume v_T is that for a liquid depth equal to the vessel diameter, $\pi T^3/4$, and ϕ_{ST} is the volume fraction solids based on volume v_T . The limitations of Eq. (11.20) are substantial: particle size and density ranges studied were small; the largest vessel studied was about 0.3 m in diameter. The study was made with flat-blade turbines without disk, but the results agree with the more limited data for disk turbines [84]; it is probable that Eq. (11.20) is reasonably applicable to all types of turbines. Impeller Reynolds numbers Re must be at least 1000. The correlation requires that the terminal velocity V_{iN} of the particles be calculated for single spheres settling with the stagnant-liquid drag coefficient in the Newton's-law range ($d_p V_{iN} \rho_L / \mu_L = 1000$ to 200 000),

$$V_{iN} = 1.74 \left(\frac{g d_p^2 \Delta \rho}{\rho_L} \right)^{1/2} \quad (11.21)$$

even though in the agitated vessel it is known that the drag coefficients are substantially larger than in quiet liquids owing to the intense turbulence and the particle acceleration [87]. There are some additional data for very viscous liquids, for which Re will be small [43].

If larger impeller powers are used, the solids are lifted to heights above the bottom depending upon the power applied. The liquid above this height will be clear. It is particularly difficult to suspend large or heavy particles up to the liquid surface because the predominant liquid velocities in the upper regions are horizontal. The power P required to suspend particles of uniform size and density to a height Z' above the midplane of the uppermost impeller, with essentially the same limitations as those of Eq. (11.20), is given by [101]

$$\frac{P g_c}{g n \rho_m v_m V_{iS}} = \phi_{Sm}^{2/3} \left(\frac{T}{d_i} \right)^{1/2} \exp \left(4.35 \frac{Z'}{T} - 0.1 \right) \quad (11.22)$$

where ρ_m , v_m , and ϕ_{Sm} are the properties of the slurry below the height Z' and n is the number of impellers (1, 2, or 3) on the shaft. V_{iS} must be the terminal settling velocity of single spheres computed through Stokes' law,

$$V_{iS} = \frac{g d_p^2 \Delta \rho}{18 \mu_L} \quad (11.23)$$

For impeller Reynolds numbers Re less than 25 000, the power P must be multiplied by $4000 Re^{-0.8}$.

In batch operations, if the solids are of uniform size and density, the slurry produced below height Z of Eq. (11.22) will be of uniform solids concentration [101]. For mixed particles, the slurry can be quite nonuniform with respect to particle size and density [79]. However, for steady-state continuous-flow operation, the effluent slurry for nondissolving solids must be identical in solids size and density with the solids-feed mixture. The residence time of the different particles need not be uniform, however, and their relative concentration in the vessel need not necessarily be identical with that in the feed and effluent streams.

Impeller power For relatively dilute solid-liquid mixtures, except for fibrous solids, the power to agitate at a given speed is essentially the same as for the clear liquid [85]. Concentrated slurries and those containing fibrous solids are likely to be nonnewtonian in character; i.e., the velocity gradient in the moving mass is not directly proportional to shear stress. The viscosity of such fluids must be related to the agitator power, and the work of Metzner and his associates should be consulted [7, 68, 69]. Since such data are not usually available, a rough estimate can be made by using the clear-liquid

power-number-Reynolds-number curves of Fig. 6.5, with density and viscosity estimated from [52]

$$\rho_m = \phi_{sm}\rho_p + (1 - \phi_{sm})\rho_L \quad (11.24)$$

$$\mu_m = \frac{\mu_L}{(1 - \phi_{sm}/\phi_{ss})^{1.8}} \quad (11.25)$$

ϕ_{ss} is the volume fraction of solids in the bed of solids after final settling in quiet liquid. In the absence of measured data, this can be estimated roughly as 0.6. Results are poor for impeller Reynolds numbers less than 1000.

Illustration 11.3 A baffled vessel, 1 m diameter, filled to a depth of 1 m with water, is agitated with a 203-mm-diameter (8-in) six-bladed disk turbine, arranged axially 150 mm from the bottom of the vessel and turning at 8.33 r/s (500 r/min), then 50 kg (110 lb) sand, $\rho_s = 2300 \text{ kg/m}^3$ (143.5 lb/ft³), $d_p = 0.8 \text{ mm}$, is added. The temperature is 25°C. What agitator power is required?

SOLUTION From the definition of power number, $g_c P = \text{Po } \rho_m d_i^2 N^3$. For one impeller, $n = 1$. If the sand is lifted to a height Z' above the impeller,

$$v_m = \frac{\pi T^2 (Z' + C)}{4}$$

Since the solid volume is S/ρ_p ,

$$\phi_{sm} = \frac{S/\rho_p}{v_m} = \frac{4S}{\pi T^2 (Z' + C) \rho_p}$$

If these are substituted in Eq. (11.22), there results

$$(Z' + C)^{1/3} \exp \frac{4.35 Z'}{T - 0.1} = \frac{1.0839 \text{ Po } d^{11/2} N^3 \rho_p^{2/3}}{g V_{is} T^{7/6} S^{2/3}}$$

Tentatively, take $\text{Po} = 5$, $d_i = 0.203 \text{ m}$, $\rho_p = \rho_s = 2300 \text{ kg/m}^3$, $T = 1 \text{ m}$, $C = 0.150 \text{ m}$, $S = 50 \text{ kg}$, $g = 9.807 \text{ m/s}^2$, $d_p = 8 \times 10^{-4} \text{ m}$, $\mu_L = 8.94 \times 10^{-4} \text{ kg/m} \cdot \text{s}$, $\rho_L = 998 \text{ kg/m}^3$, $\Delta\rho = 1302 \text{ kg/m}^3$, $N = 8.33 \text{ s}^{-1}$. Eq. (11.23): $V_{is} = 0.508 \text{ m/s}$.

Substitution in the above equation yields

$$(Z' + 0.150)^{1/3} \exp(4.35 Z' - 0.1) = 1.257$$

By trial, $Z' = 0.104 \text{ m}$, and the depth to which the solid is lifted $= Z' + C = 0.314 \text{ m}$. $\phi_{sm} = 0.0882$, $\rho_m = 1112.8 \text{ kg/m}^3$. With $\phi_{ss} = 0.6$, Eq. (11.25) will yield $\mu_m = 1.19 \times 10^{-3} \text{ kg/m} \cdot \text{s}$. The impeller Reynolds number is then

$$\text{Re} = d_i^2 N \rho_m / \mu_m = 321\,000$$

Fig. 6.5: $\text{Po} = 5 = P g_c / \rho_m N^3 d_i^5$, whence $P = 1109 \text{ W}$ (1.5 hp) transmitted to the slurry. **Ans.**

Mass Transfer

Except for dissolving of crystals or crystallization, where the mass transfer is confined to the liquid phase, consideration must generally be given to transfer through the liquid surrounding the particles and to some kind of mass-transfer effect within the solid.

Liquid-phase mass transfer Although many studies have been made of mass-transfer rates in the liquid for solid-liquid slurries, the results are frequently conflicting and difficult to interpret. We do know that once the solids are fully suspended, further expenditure of agitator power does not produce commensurate improvement in the mass-transfer rates [51]. For small particles, the mass-transfer coefficient becomes smaller with increased particle size [36], whereas for large particles there is no effect of particle size [5]. The transition size is about 2 mm [87]. At least for moderate

density difference, the coefficients are independent of $\Delta\rho$, but for large differences, e.g., for metal particles in ordinary liquids, there may be an effect. There is no effect of solids concentration, at least up to 10 percent by volume [5], or of the presence or absence of baffles [77] and of agitator design at a given impeller power [9]. The effect of diffusivity is not well known, since large changes in diffusivity usually accompany corresponding changes in viscosity in such a fashion that the influence of the Schmidt number is difficult to establish [5].

It seems reasonable that the mass-transfer coefficients should depend upon some kind of Reynolds number. A great variety of such numbers has been suggested, many using the particle-liquid slip velocity (relative velocity), which is difficult to estimate, or the single-particle terminal settling velocity, which is not appropriate [87]. Refer to Eq. (3.30). The turbulent mass flux is

$$J_{A, \text{turb}} = E_D \frac{\Delta c}{l} = k_{L, \text{turb}} \Delta c = b_1 u' \Delta c$$

so that in an intensely turbulent liquid, as in an agitated vessel, k_L should be closely associated with the fluctuating velocity u' . For a fluctuating velocity over a distance corresponding to the particle size d_p [Eq. (3.26)] the power dissipation would be

$$\frac{Pg_c}{v_L} \propto \frac{\rho_L u'^3}{d_p} \quad (11.26)$$

and a turbulent-particle Reynolds number corresponding to this is

$$\text{Re}_p = \frac{d_p u' \rho_L}{\mu_L} \quad (11.27)$$

If we take u' from Eq. (11.26) and ignore proportionality constants in defining dimensionless groups, then

$$\text{Re}_p = d_p^{4/3} \left(\frac{Pg_c}{v_L} \right)^{1/3} \frac{\rho_L^{2/3}}{\mu_L} \quad (11.28)$$

where v_L is the volume of liquid in the vessel (not the rate of flow).

Small particles ($d_p < 2$ mm) fully suspended For moderate differences in $\Delta\rho$ [53, 62, 78],

$$\text{Sh}_L = 2 + 0.47 \text{Re}_p^{0.62} \left(\frac{d_p}{T} \right)^{0.17} \text{Sc}_L^{0.36} \quad (11.29)$$

Large particles ($d_p > 2$ mm) fully suspended The recommended correlation [11, 12, 71] is

$$\text{Sh}_L = 0.222 \text{Re}_p^{3/4} \text{Sc}_L^{1/3} \quad (11.30)$$

In this case the exponent on Re_p is chosen so as to make the mass-transfer coefficient independent of particle size, in accordance with observed data. It is important to note that the mass-transfer coefficients contained in the Sherwood numbers of the preceding equations are based on the outside, gross surface of the particles, ignoring the surface of any internal pore structure.

Solid-phase mass transfer The various mass-transfer processes that may occur within the solid can frequently be jointly described in terms of a mass-transfer coefficient k_s (based on external surface) or an effective diffusivity D_s . For a constant or nearly constant equilibrium distribution coefficient and spherical particles, the former can be related to the diffusivity through the approximation [30]

$$k_s a_p = \frac{60 D_s}{d_p^2} \quad (11.31)$$

or, since

$$a_p = \frac{6\phi_s}{d_p} \quad (11.32)$$

we get

$$k_S = \frac{10D_S}{d_p \phi_S} \quad (11.33)$$

The liquid- and solid-phase resistances can be added, in the case of constant distribution coefficient,

$$\frac{1}{K_L} = \frac{1}{k_L} + \frac{m}{k_S} \quad (11.34)$$

to produce an overall resistance.

Batch adsorption, negligible solid-phase mass-transfer resistance If the quantity S/mv_L , an adsorption factor, is sufficiently large (large solid doses and equilibrium favorable to adsorption), the solid-phase resistance can be neglected and only k_L need be considered. These are the conditions of what follows.

Let the volume of liquid to be treated be v_L , of initial concentration c_0 mol solute A/volume. The mass of adsorbate-free solid to be used is S_S , of external surface a_{pS} per unit mass of solid, initially of solute concentration X_0 mol solute/mass of solid. The final concentration in the liquid is to be c_1 .

At any instant, the adsorbed solute concentration within the solid is X mass solute/mass solute-free solid, and since the solid-phase resistance is negligible, this will be taken as uniform throughout the solid. The distribution coefficient m is then defined as

$$m = \frac{c^* M_A}{X} \quad (11.35)$$

where c^* is the equilibrium concentration in the liquid. Since the solutions involved are usually dilute, the coefficient k_L rather than F_L can be used. Then if c is the concentration in the bulk liquid at any time,

$$N_A = \frac{-dc}{a_{pL} d\theta} = k_L(c - c^*) \quad (11.36)$$

where $a_{pL} = a_{pS}S_S/v_L$. A solute balance is

$$\frac{S}{M_A}(X - X_0) = S_S\left(\frac{c^*}{m} - \frac{X_0}{M_A}\right) = v_L(c_0 - c) \quad (11.37)$$

Substitution of c^* from Eq. (11.37) into (11.36) and rearrangement produces

$$\int_{c_0}^{c_1} \frac{-dc}{c(1 + mv_L/S_S) - (mv_Lc_0/S_S + mX_0/M_A)} = k_L a_{pL} \int_0^\theta d\theta \quad (11.38)$$

or
$$\ln \frac{c_0 - mX_0/M_A}{c_1(1 + mv_L/S_S) - (mv_Lc_0/S_S + mX_0/M_A)} = \left(1 + \frac{mv_L}{S_S}\right) k_L a_{pL} \theta \quad (11.39)$$

If $X_0 = 0$, this can be rearranged to [28]

$$\frac{c_1}{c_0} = \frac{1}{1 + mv_L/S_S} \exp \left[- \left(1 + \frac{mv_L}{S_S}\right) k_L a_{pL} \theta \right] + \frac{mv_L/S_S}{1 + mv_L/S_S} \quad (11.40)$$

A method of computation for batch processes where both solid and liquid mass-transfer resistances are important is also available [99].

Illustration 11.4 A batch of kerosene containing dissolved water at a concentration 40 ppm = 0.0040 wt percent is to be dried to 5 ppm water by contacting with "dry" silica gel at 25°C. The kerosene density = 783 kg/m³, viscosity = 1.7×10^{-3} kg/m · s, and estimated av mol wt = 200. The silica gel, density ρ_p = 881 kg/m³, is in the form of 14-mesh spherical beads. The equilibrium distribution ratio for water is $m = c^*/X = 0.522$ (kg water/m³ kerosene)/(kg water/kg gel). (a) Calculate the minimum solid/liquid ratio which can be used. (b) For a liquid/solid ratio = 16 kg gel/m³ kerosene and batches of 1.7 m³ (60 ft³), specify the characteristics of a suitable baffled, agitated vessel and estimate the agitator power and contacting time.

SOLUTION Define c as kg water/m³ liquid.

(a) Water balance: $v_L(c_0 - c_1) = S_S(X_1 - X_0)$.

$$c_0 = 783(4 \times 10^{-5}) = 0.0313 \text{ kg H}_2\text{O/m}^3$$

$$c_1 = 783(5 \times 10^{-6}) = 0.00392 \text{ kg H}_2\text{O/m}^3$$

For $S_{S, \min}/v_L$, X_1 is in equilibrium with c_1

$$X_1 = \frac{c_1}{m} = \frac{0.00392}{0.522} = 7.51 \times 10^{-7} \text{ kg H}_2\text{O/kg gel}$$

therefore

$$\frac{S_{S, \min}}{v_L} = \frac{c_0 - c_1}{X_1 - X_0} = \frac{0.0313 - 0.00392}{7.51 \times 10^{-7} - 0} = 3.65 \text{ kg gel/m}^3 \text{ kerosene Ans.}$$

(b) Basis: 1 batch, 1.7 m³ kerosene.

$$S_S = 16(1.7) = 27.2 \text{ kg dry gel} \quad \text{vol solids} = \frac{27.2}{881} = 0.0309 \text{ m}^3$$

$$\text{Total batch volume} = 1.73 \text{ m}^3$$

Take $Z = T$. Then $\pi T^2 Z/4 = \pi T^3/4 = 1.73$, and $T = 1.3 \text{ m (4.25 ft)}$. To allow for adequate freeboard, make the vessel height = 1.75 m (5.75 ft).

Use a six-blade disk-turbine impeller. From Fig. 11.26, with d_p corresponding to 14 mesh = 1.4 mm, $T/d_i = 2$. With $(\rho_p - \rho_L)/\rho_L = (881 - 783)/783 = 0.125$, $T/d_i = 4.4$. Average $T/d_i = (4.4 + 2)/2 = 3.2$, $d_i = 1.3/3.2 = 0.406 \text{ m (16 in)}$.

Assuming that the settled volume fraction of solids = 0.6, the settled volume of solids = $0.309/0.6 = 0.0515 \text{ m}^3$. The depth of settled solids = $0.0515/[\pi(1.3)^2/4] = 0.0388 \text{ m}$, which is negligible. Locate the turbine 150 mm (6 in) from the bottom of the tank. $C = 0.150 \text{ m}$.

Power Use sufficient agitator power to lift the solids to 0.6 m above the bottom of the vessel. $Z' = 0.6 - 0.15 = 0.45 \text{ m}$. The properties of the slurry in the 0.6 m depth to which the solids are suspended are

$$v_m = \frac{0.6\pi(1.3)^2}{4} = 0.796 \text{ m}^3 \quad \phi_{sm} = \frac{0.0309}{0.796} = 0.0388 \text{ vol fraction solids}$$

Eq. (11.24): $\rho_m = 786.8 \text{ kg/m}^3$; Eq. (11.25) with $\phi_{ss} = 0.6$: $\mu_m = 1.917 \times 10^{-3} \text{ kg/m} \cdot \text{s}$. With $g_c = 1.0$, $g = 9.807 \text{ m/s}^2$, $\mu_L = 1.7 \times 10^{-3} \text{ kg/m} \cdot \text{s}$, $d_p = 0.0014 \text{ m}$, Eq. (11.23): $V_{LS} = 0.0616 \text{ m/s}$.

With $n = 1$ impeller, Eq. (11.22): $P = 315.8 \text{ W}$. Try $Po = 5$. Therefore $N = (g_c P / \rho_p d_p^5)^{1/3} = 1.94 \text{ r/s (116 r/min)}$. Use $2 \text{ r/s (120 r/min)}$. $Re = d_p^2 N \rho_m / \mu_m = 135 \text{ 300}$. Therefore $Po = 5$ (Fig. 6.5), and $P = 315.8(2/1.94)^3 = 346 \text{ W (0.46 hp)}$ applied to the slurry. Power to the motor will be larger, depending upon the efficiency of the motor and speed reducer (see Chap. 6).

Mass transfer Eq. (11.28), with $d_p = 0.0014 \text{ m}$, $v_L = 1.7 \text{ m}^3$: $Re_p = 44.8$.

D_L will be estimated through Eq. (2.44). $M_B = 200$, $T = 298 \text{ K}$, $\phi = 1$, $\mu = 1.7 \times 10^{-3}$, $v_A(\text{H}_2\text{O}) = 0.0756$ (Chap. 2 notation). Therefore $D_L = 1.369 \times 10^{-9} \text{ m}^2/\text{s}$. $Sc_L = \mu_L / \rho_L D_L = 1586$. Eq. (11.29): $Sh_L = k_L d_p / D_L = 59.8$. Therefore $k_L = 5.85 \times 10^{-5} \text{ kmol/m}^2 \cdot \text{s} \cdot (\text{kmol/m}^3)$.

$$\frac{\text{Mass}}{\text{particle}} = \frac{\pi}{6} (0.0014)^3 (881) = 1.266 \times 10^{-6} \text{ kg}$$

$$\frac{\text{Surface}}{\text{particle}} = \pi (0.0014)^2 = 6.158 \times 10^{-6} \text{ m}^2$$

Therefore

$$a_{pS} = \frac{6.158 \times 10^{-6}}{1.266 \times 10^{-6}} = 4.864 \text{ m}^2/\text{kg}$$

and

$$a_{pL} = 4.864(16) = 77.8 \text{ m}^2/\text{m}^3 \text{ liquid}$$

$$\frac{S_s}{v_L m} = \frac{16}{0.522} = 30.65$$

This is considered sufficiently large for only liquid-phase transfer to require consideration. Eq. (11.40):

$$\theta = \frac{\ln[(c_0/c_1)/(1 + mv_L/S_s - mv_L c_0/S_s c_1)]}{(1 + mv_L/S_s)k_L a_{pL}}$$

$$= 498 \text{ s or } 8.3 \text{ min contacting time required} \quad \text{Ans.}$$

Continuous cocurrent adsorption with liquid- and solid-phase mass-transfer resistances For continuous, cocurrent flow of solution and adsorbent through an agitated vessel, the individual particles experience an unsteady-state change of solute concentration during their time of contact. Nevertheless the process as a whole can be steady-state, the concentrations of effluents remaining unchanged with passage of time.

Gröber [33] computed the approach to equilibrium temperature of spherical particles immersed in an agitated fluid of constant temperature. Since the concentration of the solution in a well-stirred vessel in continuous flow is essentially uniform throughout at the effluent value, Gröber's result can be adapted to cocurrent adsorption through the heat-mass-transfer analogy, as in Fig. 11.27. Here the approach of the particles to equilibrium concentration with the effluent liquid is expressed as the Murphree stage efficiency,

$$E_{MS} = \frac{X_1 - X_0}{X_1^* - X_0} = \frac{X_1 - X_0}{M_A c_1 / m - X_0} \quad (11.41)$$

X is the average concentration of solute (adsorbate) in the solid. The figure can equally well be used for desorption (drying) and leaching of solute from the solid.

Illustration 11.5 Water containing copper ion Cu^{2+} at a concentration 100 ppm, flowing continuously at the rate $1.1 \times 10^{-4} \text{ m}^3/\text{s}$ (1.75 U.S. gal/min), is to be contacted with 0.0012 kg/s (0.15 lb/min) of an ion-exchange resin to adsorb the Cu^{2+} . The contacting will occur by cocurrent flow through a baffled, agitated vessel 1 m in diameter, 1 m tall, covered. The feed material will enter at the bottom, leave through an opening in the cover, and flow to a filter. The agitator, a six-bladed disk-type turbine, 0.3 m in diameter, will be centered. The temperature will be 25°C . Specify a suitable turbine speed, and estimate the Cu^{2+} concentration in the effluent water.

Data The ion-exchange particles, swollen when wet, are spherical, 0.8 mm diameter, density = 1120 kg/m^3 . The diffusivity of Cu^{2+} in the solid $D_s = 2 \times 10^{-11} \text{ m}^2/\text{s}$; that in the water = $7.3 \times 10^{-10} \text{ m}^2/\text{s}$. The equilibrium distribution coefficient is constant: $m = 0.2 \text{ (kg } \text{Cu}^{2+}/\text{m}^3 \text{ soln)}/(\text{kg } \text{Cu}^{2+}/\text{kg resin})$.

SOLUTION The particles will be lifted to the top of the vessel: $Z' = 0.5 \text{ m}$, $\rho_p = 1120 \text{ kg/m}^3$, $d_p = 8 \times 10^{-4} \text{ m}$, μ_L = viscosity of water, $25^\circ\text{C} = 8.94 \times 10^{-4} \text{ kg/m} \cdot \text{s}$, ρ_L = density of water = 998 kg/m^3 , $\Delta\rho = \rho_p - \rho_L = 122 \text{ kg/m}^3$, $g = 9.80 \text{ m/s}^2$. Eq. (11.23): $V_{IS} = 0.0476 \text{ m/s}$.

$v_m = \pi T^2 Z' / 4 = \pi (1)^3 / 4 = 0.785 \text{ m}^3$. $v_L = 1.1 \times 10^{-4} \text{ m}^3/\text{s}$. Volume rate of solid = $0.0012/1120 = 1.07 \times 10^{-6} \text{ m}^3/\text{s}$.

$\phi_{sm} = (1.07 \times 10^{-6}) / (1.1 \times 10^{-4} + 1.07 \times 10^{-6}) = 9.63 \times 10^{-3} \text{ vol fraction}$. Eq. (11.24): $\rho_m = 999 \text{ kg/m}^3$; $n = 1$ impeller, $d_i = 0.3 \text{ m}$, $g_c = 1.0$. Eq. (11.22): $P = 240.6 \text{ W}$ (0.32 hp).

To estimate the impeller speed, assume the power number $Po = 5$. Therefore $N = (g_c P / \rho_m d_i^3)^{1/3}$. With $d_i = 0.3 \text{ m}$, $N = 2.7 \text{ r/s}$ (162 r/min). For this speed, with $\mu_m = \mu_L$ since

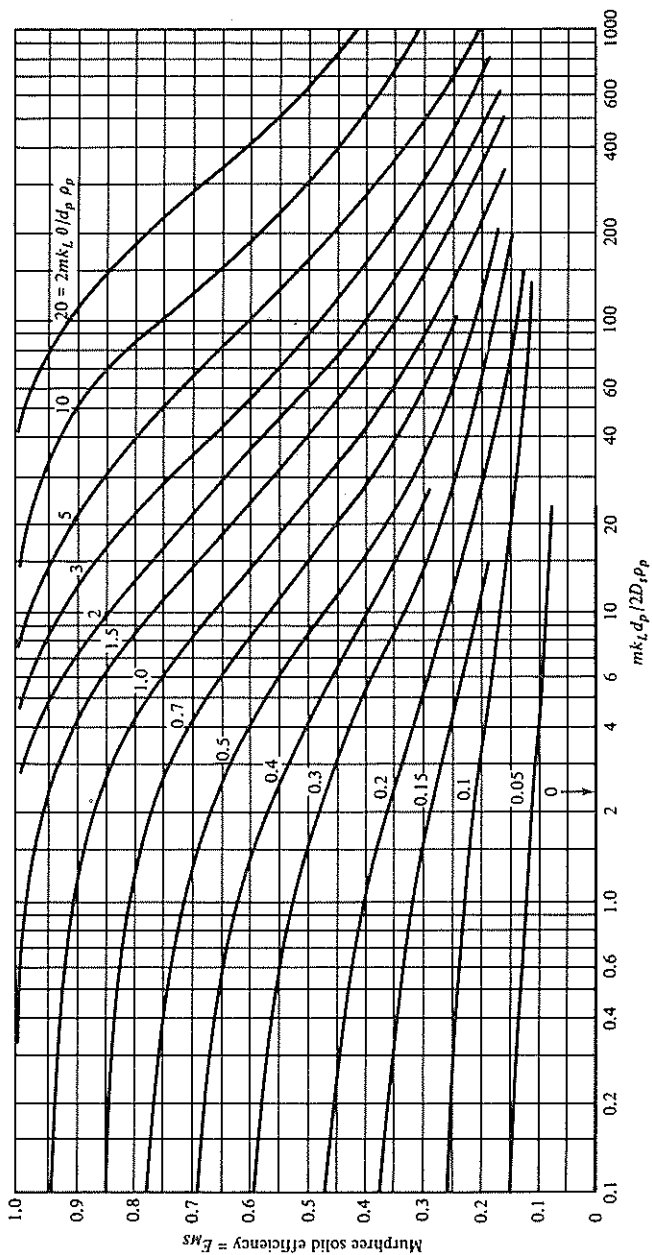


Figure 11.27 Stage efficiency for single-stage, continuous, cocurrent adsorption in well-stirred vessels, with spherical particles. [Adapted from Grober, *Ver. Dtsch. Ing.*, **69**, 705 (1925).]

the slurry is dilute, $Re = d_i^2 N \rho_m / \mu_L = 271\,500$, so that (Fig. 6.5) $Po = 5$ and the calculated speed is correct.

$$v_L = v_T(1 - \phi_{sm}) = \frac{\pi}{4}(1)^3(1 - 9.63 \times 10^{-3}) = 0.777 \text{ m}^3$$

Eq. (11.28):

$$Re_p = 56.3 \quad Sc_L = \frac{\mu_L}{\rho_L D_L} = 1226$$

Eq. (11.29):

$$Sh_L = 130.3 = \frac{k_L d_p}{D_L} \quad k_L = 1.19 \times 10^{-4} \text{ kmol/m}^2 \cdot \text{s} \cdot (\text{kmol/m}^3)$$

Since the dispersion is uniform throughout the vessel, the residence time for both liquid and solid = $0.777(1 - 9.63 \times 10^{-3}) \text{ m}^3 \text{ liquid} / (1.1 \times 10^{-4} \text{ m}^3/\text{s}) = 6996 \text{ s} = \theta$. Refer to Fig. 11.27.

$$\text{Abscissa} = \frac{0.2(1.19 \times 10^{-4})(0.0008)}{2(2 \times 10^{-11})(1120)} = 0.425$$

$$\text{Parameter} = \frac{2(0.2)(1.19 \times 10^{-4})(6996)}{(0.0008)(1120)} = 0.37$$

Therefore $E_{MS} = 0.63 = (X_1 - X_0)/(c_1/m - X_0) = (X_1 - 0)/(c_1/0.2 - 0)$ where c is defined as kg/m^3 . A solute balance gives

$$(X_1 - X_0)S_s = v_L(c_0 - c_1)$$

c_0 corresponding to 100 ppm = $100(998)/10^6 = 0.0998 \text{ kg/m}^3$.

$$(X_1 - 0)(0.0012) = (1.1 \times 10^{-4})(0.0998 - c_1)$$

The expression for stage efficiency and the solute balance are solved simultaneously, whence $c_1 = 2.82 \times 10^{-3} \text{ kg/m}^3$ effluent Cu^{2+} concentration. This corresponds to $(2.82 \times 10^{-3})(10^6)/998 = 2.8 \text{ ppm}$. Ans.

Fluidized and Teeter Beds

These have been used in increasing extents in recent years for recovery of vapors from vapor-gas mixtures [3, 22], extensively for desorption (drying, see Chap. 12), for fractionation of light hydrocarbon vapors with carbon [23], and several other purposes.

Consider a bed of granular solids up through which a gas flows, the gas being fairly uniformly distributed over the bottom cross section of the bed. At low gas rates, the gas suffers a pressure drop which can be estimated by Eq. (6.66). As the gas velocity is increased, the pressure drop eventually equals the sum of the weight of the solids per unit area of bed and the friction of the solids at the container walls. If the solids are free-flowing, an increment in the gas velocity causes the bed to expand and the gas-pressure drop will equal the weight/area of the bed. Further increase in gas velocity causes further enlargement of the bed, and voidage increases sufficiently for the solid particles to move about locally. This is the condition of a *quiescent fluidized bed*. Still further increase in gas velocity further expands the bed, solid particles move about freely (indeed, they are thoroughly circulated throughout the bed), and well-defined bubbles of gas may form and rise through the bed. The bed appears much

like a boiling liquid, a distinct interface between the top of the solids and the escaping gas remains evident, and the gas-pressure drop is not much different from that of the quiescent fluidized state. This is the condition used for adsorption. Further increase in gas velocity continues to expand the bed, and eventually the solids are carried away with the gas.

Squires [94] has distinguished between *fluidized* beds and *teeter* beds. Fluidized beds are produced by fine powders, usually smaller than 20-mesh and ranging down to 325-mesh, and will retain a great range of particle sizes within the bed, with little particle attrition. They usually operate with superficial gas velocities of the order of 0.6 m/s (2 ft/s) or less, which may be 10 times that for minimum fluidization. Teeter beds are produced with coarser particles, up to 10-mesh commonly. Gas velocities of from 1.5 to 3 m/s (5 to 10 ft/s) can be used, which may only be twice that for minimum fluidization. There is little gas bubbling, and the size range of particles retained is relatively small. In general, better fluid-solids contacting is obtained in teeter beds, and most adsorption applications are in this category. Solids can also be fluidized by liquids, and these will be ordinarily in the teeter-bed category.

Since the first introduction of these techniques for fluid-solid contact in 1948, with the principal application in the catalysis of gaseous chemical reactions, a very large technology for their design has developed. This is still not in a condition which permits summary for design purposes in the space available here, however, and more extended works should be consulted for such details [16, 54, 61, 103]. The discussion here is limited to the computation of stage requirements and does not include the mechanical design.

Slurry Adsorption of Gases and Vapors

A gas or vapor in a gas can be adsorbed on a solid which is slurried in a liquid. For example, sulfur dioxide can be adsorbed from a mixture with air on activated carbon slurried in water [8]. This procedure was suggested as early as 1910 [25], but interest in it has revived only recently. The reported details thus far are confined to laboratory studies of sparged vessels [73, 88] operated semibatch (gas flow continuous, slurry batch) or cocurrent flow or both [66]. A continuous countercurrent process has been described [4, 27]. The slurry is of course much easier to handle than dry solid, and it has been shown that the capacity of the slurried adsorbent is about the same as for dry solid [73], much larger than that of the liquid solvent alone [96].

Adsorption of a Vapor from a Gas Fluidized Beds

Figure 11.28 shows an arrangement typical of that required for adsorption of a vapor from a nonadsorbed gas, in this case drying of air with silica gel [22]. In the upper part of the tower, the gel is contacted countercurrently with the air to be dried on perforated trays in relatively shallow beds, the gel moving from tray to tray through downspouts. In the lower part of the tower, the gel is regenerated by similar contact with hot gas, which desorbs and carries off the moisture. The

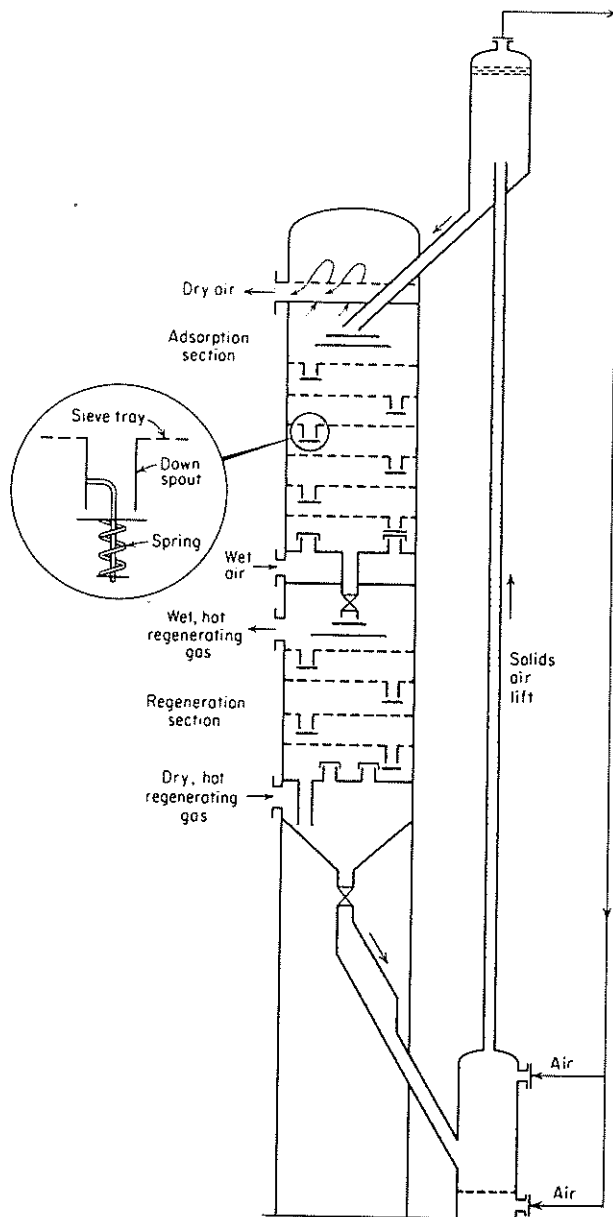


Figure 11.28 Fluidized bed, multistage countercurrent adsorber with regeneration. (After Ermenc [22]).

dried gel is then recirculated by air lift to the top of the tower. If the adsorbed vapor is to be recovered, regeneration might include steam stripping of a carbon adsorbent with distillation or decantation of the condensed water-organic-vapor mixture, followed by air drying of the adsorbent before reuse, and obvious changes in the flowsheet would be necessary.

If adsorption is isothermal, the calculations for theoretical stages can be made in the same manner as for contact filtration of liquids [Fig. 11.20, Eq. (11.14)], and the same is true for desorption-regeneration. Ordinarily, however, the adsorber will operate adiabatically, and there may be a considerable rise in temperature owing to adsorption (or fall, in the case of desorption). Since the equilibrium then changes with stage number, the calculations are done stage by stage, much in the manner of gas absorption. The same procedure will apply to any type of stage device and is not limited to fluidized beds. Thus, for the first n stages of the flowsheet of Fig. 11.20, a balance for adsorbable vapor is

$$G_S(Y_0 - Y_n) = S_S(X_1 - X_{n+1}) \quad (11.42)$$

and an enthalpy balance for adiabatic operation is

$$G_S(H_{G0} - H_{Gn}) = S_S(H_{S1} - H_{S_{n+1}}) \quad (11.43)$$

where H_G is the enthalpy of vapor-gas mixture, energy/mass, and H_S the enthalpy of solid plus adsorbate, energy/mass adsorbent. If enthalpies are referred to solid adsorbent, nonadsorbed gas (component C), and the adsorbate (component A) as a liquid, all at a base temperature t_0 , then

$$H_G = C_G(t_G - t_0) + Y[C_A(t_G - t_0) + \lambda_{A0}] \quad (11.44)$$

where C_G = gas heat capacity

C_A = vapor heat capacity

λ_{A0} = latent heat of vaporization of A at t_0

$$H_S = C_B(t_S - t_0) + XC_{A,L}(t_S - t_0) + \Delta H_A \quad (11.45)$$

where C_B = heat capacity of adsorbent (component B)

$C_{A,L}$ = heat capacity of liquid A

ΔH_A = integral heat of adsorption at X and t_0 , energy/mass adsorbent

Gas and solid leaving a theoretical stage are in thermal and concentration equilibrium. The use of these equations is precisely the same as for gas absorption (Illustration 8.4).

Fractionation of a vapor mixture For the separation of a vapor mixture consisting of, for example, two adsorbable components for which the adsorbent exhibits a relative adsorptivity, calculations are made in the manner for moving beds to locate operating lines and equilibrium curves [Eqs. (11.53) to (11.65) and Fig. 11.33]. Theoretical stages can then be determined by the usual step construction in the lower part of such a figure.

CONTINUOUS CONTACT

In these operations the fluid and adsorbent are in contact throughout the entire apparatus, without periodic separation of the phases. The operation can be carried out in strictly continuous, steady-state fashion, characterized by movement of the solid as well as the fluid. Alternatively, owing to the rigidity of the solid adsorbent particles, it is also possible to operate advantageously in semi-continuous fashion, characterized by a moving fluid but stationary solid. This results in unsteady-state conditions, where compositions in the system change with time.

Steady-State-Moving-Bed Adsorbers

Steady-state conditions require continuous movement of both fluid and adsorbent through the equipment at constant rate, with no change in composition at any point in the system with passage of time. If parallel flow of solid and fluid is used, the net result is at best a condition of equilibrium between the effluent streams, or the equivalent of one theoretical stage. It is the purpose of these applications to develop separations equivalent to many stages, however, and hence only countercurrent operations need be considered. There are applications especially in treating gases and liquids, for purposes of collecting solute and fractionation, through ordinary adsorption and ion exchange.

Equipment It is only in relatively recent years that satisfactory large-scale devices for the continuous contacting of a granular solid and a fluid have been developed. They have had to overcome the difficulties of obtaining uniform flow of solid particles and fluid without channeling or local irregularities, as well as those of introducing and removing the solid continuously into the vessel to be used.

A substantial effort has been expended in developing apparatus for counterflow of gas and solid, with plug flow of solid rather than fluidized beds. The hydraulics and mechanics of such a device were successfully solved, resulting in the *Hypersorber*, used for fractionating hydrocarbon gases with activated gas-adsorbent carbon [8, 48]. The device resembles a countercurrent distilling column with the solid phase, flowing down, analogous to the upward flow of heat in distillation. Several adsorbers on a very large scale were built, but the very brittle carbon was subject to serious attrition losses, and no new continuous-flow, countercurrent device for plug flow of solids and gas is believed to be in operation.

In liquid treating, far less elaborate devices have been used. Solids can be introduced to the top of the tower by a screw feeder or simply from a bin by gravity flow. Withdrawal of solids at the bottom can be accomplished by several types of devices, e.g., a revolving compartmented valve similar to revolving doors for buildings; it is not generally possible to remove solids entirely free of liquid.

The Higgins contactor [35, 42], developed initially for ion exchange but useful for solids-liquid contacting generally, is unique in the intermittent nature of its operation. Figure 11.29 shows it schematically as arranged for simple solute collection. In Fig. 11.29*a*, the temporarily stationary upper bed of solids is contacted with liquid flowing downward, so that fluidization does not occur. In the lower bed, the solid is regenerated by an eluting liquid. After several minutes, the liquid flow is stopped, valves are turned as shown in Fig. 11.29*b*, and the liquid-filled piston pump is moved as shown for a period of several seconds, whereupon solid is moved clockwise hydraulically. In Fig. 11.29*c*, with the valves readjusted to their original position, movement of solid is completed, and liquid flows are started to complete the cycle. Operation, although intermittent and cyclic, is nearly the same as for truly continuous countercurrent operation.

A truly continuous countercurrent scheme for ion-exchange treatment of liquids incorporates the resin in an endless, flexible belt which moves continuously over horizontal rollers and vertically with many passes, through which the solution to be treated must travel [46, 47]. The belt may pass successively from contact with solution to be treated, to regenerating liquid, and back to solution to be treated.

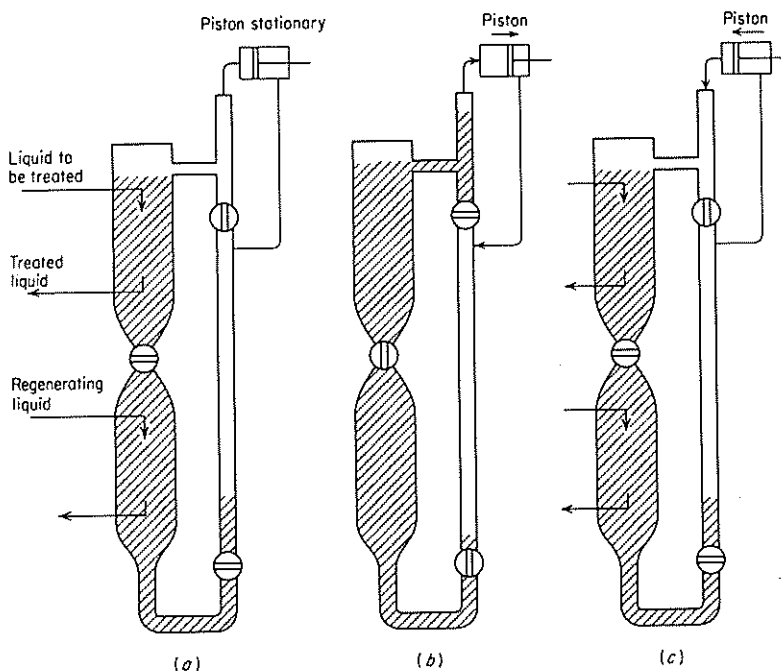


Figure 11.29 Higgins contactor, schematic (solids shown shaded).

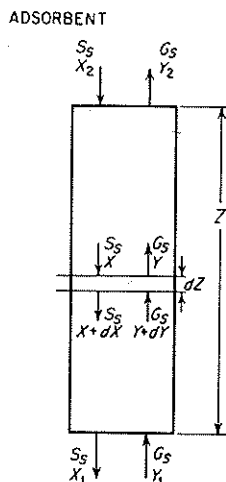


Figure 11.30 Continuous countercurrent adsorption of one component.

One component adsorbed (solute collection) For purposes of computation the operation can conveniently be considered analogous to gas absorption, with solid adsorbent replacing the liquid solvent. Refer to Fig. 11.30, where G_s and S_s are the solute-free fluid and solid mass velocities, respectively, mass/(cross-sectional area of column) (time). Solute concentrations are expressed as mass solute/mass solute-free substance. The development is shown for treating a gas, but can be applied equally well to treating a liquid by replacing G_s by L_s . A solute balance about the entire tower is

$$G_s(Y_1 - Y_2) = S_s(X_1 - X_2) \quad (11.46)$$

and about the upper part

$$G_s(Y - Y_2) = S_s(X - X_2) \quad (11.47)$$

These establish the operating line on X, Y coordinates, a straight line of slope S_s/G_s joining the terminal conditions (X_1, Y_1) and (X_2, Y_2) (Fig. 11.31). The solute concentrations X and Y at any level in the tower fall upon this line. An equilibrium-distribution curve appropriate to the system and to the prevailing temperature and pressure can also be plotted on the figure as shown. This will fall below the operating line for adsorption and above for desorption. In the same fashion as for absorbers, the minimum solid/fluid ratio is given by the operating line of maximum slope which anywhere touches the equilibrium curve.

The simplifying assumption found useful in gas absorption, that the temperature of the fluid remains substantially constant in adiabatic operations, will be satisfactory only in solute collection from dilute liquid solutions and is unsatisfactory for estimating temperatures in the case of gases. Calculation of the temperature effect when heats of adsorption are large is very complex [2, 89]. The present discussion is limited to isothermal operation.

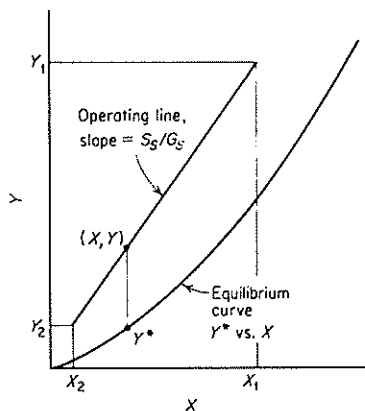


Figure 11.31 Continuous countercurrent adsorption of one component.

The resistance to mass transfer of solute from the fluid to the adsorbed state on the solid will include that residing in the gas surrounding the solid particles, that corresponding to the diffusion of solute through the gas within the pores of the solid, and possibly an additional resistance at the time of adsorption. During physical adsorption, the last of these will probably be negligible. If the remaining resistances can be characterized by an overall gas mass-transfer coefficient $K_Y a_p$ based on a_p , the outside surface of the solid particles, the rate of solute transfer over the differential height of adsorber dZ (Fig. 11.30) can be written in the usual manner as

$$S_G dX = G_S dY = K_Y a_p (Y - Y^*) dZ \quad (11.48)$$

where Y^* is the equilibrium composition in the gas corresponding to the adsorbate composition X . The driving force $Y - Y^*$ is then represented by the vertical distance between operating line and equilibrium curve (Fig. 11.31). Rearranging Eq. (11.48) and integrating define the number of transfer units N_{IOG} ,

$$N_{IOG} = \int_{Y_2}^{Y_1} \frac{dY}{Y - Y^*} = \frac{K_Y a_p}{G_S} \int_0^Z dZ = \frac{Z}{H_{IOG}} \quad (11.49)$$

where

$$H_{IOG} = \frac{G_S}{K_Y a_p} \quad (11.50)$$

The integral of Eq. (11.49) is ordinarily evaluated graphically and the active height Z determined through knowledge of the height of a transfer unit H_{IOG} , characteristic of the system.

The use of an overall coefficient or overall height of a transfer unit implies that the resistance to mass transfer within the pores of the solid particles can be characterized [21] by an individual mass-transfer coefficient $k_S a_p$ or height of a

transfer unit H_{IS} , thus,

$$\frac{G_S}{K_Y a_p} = \frac{G_S}{k_Y a_p} + \frac{m G_S}{S_S} \frac{S_S}{k_S a_p} \quad (11.51)$$

or

$$H_{IOG} = H_{IG} + \frac{m G_S}{S_S} H_{IS} \quad (11.52)$$

where $m = dY^*/dX$, the slope of the equilibrium curve.

The resistance within the fluid surrounding the particles H_{IG} or the corresponding coefficient $k_Y a_p$ can be estimated for moving beds through the correlations available for fixed beds (Table 3.3). Because of the rigidity of each solid particle and the unsteady-state diffusional conditions existing within each particle as it travels through the adsorber, the use of a mass-transfer coefficient k_S , with an implied linear concentration-difference driving force, is not strictly correct, and Eqs. (11.49) to (11.52) are sound only when the mass-transfer resistance of the fluid surrounding the particles is of controlling size. Ordinarily, however, the diffusional resistance within the particles is of major importance, and for these conditions, which space does not permit consideration of here, the work of Vermeulen [98, 99] should be consulted.

Illustration 11.6 Eagleton and Bliss [21] have measured the individual resistances to mass transfer residing in the fluid and within the solid during adsorption of water vapor from air by silica gel, using a fixed-bed semicontinuous technique. For low moisture contents of the air they found that

$$k_Y a_p = 31.6 G'^{0.55} \text{ kg H}_2\text{O/m}^3 \cdot \text{s} \cdot \Delta Y$$

and

$$k_S a_p = 0.965 \text{ kg H}_2\text{O/m}^3 \cdot \text{s} \cdot \Delta X$$

where G' is the mass velocity of the gas, $\text{kg/m}^2 \cdot \text{s}$. Their silica-gel bed had an apparent bed density 671.2 kg/m^3 and an average particle size 1.727 mm diameter, and the external surface of the particles was $2.167 \text{ m}^2/\text{kg}$.

We wish to estimate the height of a continuous countercurrent isothermal adsorber for drying air at 26.7°C , standard atmospheric pressure, from an initial humidity 0.005 to a final humidity $0.0001 \text{ kg H}_2\text{O/kg dry air}$. The entering gel will be dry. (Note: So-called "dry" silica gel must contain a minimum of about 5% water if it is to retain its adsorptive capacity. Moisture measurements as ordinarily reported do not include this.) A gel rate $0.680 \text{ kg/m}^2 \cdot \text{s}$ ($500 \text{ lb/ft}^2 \cdot \text{h}$) and an air rate $1.36 \text{ kg/m}^2 \cdot \text{s}$ ($1000 \text{ lb/ft}^2 \cdot \text{h}$) will be used. For the moisture contents expected here, the equilibrium adsorption isotherm at 26.7°C , 1 std atm (see Illustration 11.9), can be taken as substantially straight and described by the expression $Y^* = 0.0185X$.

SOLUTION $Y_1 = 0.005$, $Y_2 = 0.0001 \text{ kg H}_2\text{O/kg dry air}$. $S_S = 0.680 \text{ kg/m}^2 \cdot \text{s}$, $G_S = 1.36 \text{ kg/m}^2 \cdot \text{s}$, $X_2 = 0 \text{ kg H}_2\text{O/kg dry gel}$. Eq. (11.46):

$$X_1 = \frac{G_S(Y_1 - Y_2)}{S_S} + X_2 = \frac{1.36(0.005 - 0.0001)}{0.68} = 0.0098 \text{ kg H}_2\text{O/kg dry gel}$$

Y_2^* = humidity of air in equilibrium with entering gel = 0 . $Y_1^* = 0.0185X_1 = 0.0001814 \text{ kg H}_2\text{O/kg dry gel}$. Since operating line and equilibrium curve will both be straight on X , Y coordinates, the average driving force is the logarithmic average [Eqs. (8.47) and (8.48)]:

$$Y_1 - Y_1^* = 0.005 - 0.0001814 = 0.00482 \quad Y_2 - Y_2^* = 0.0001 - 0 = 0.0001$$

$$\Delta Y = \frac{0.00482 - 0.0001}{\ln(0.00482/0.0001)} = 0.001217$$

$$N_{IOG} = \frac{Y_1 - Y_2}{\Delta Y} = \frac{0.005 - 0.0001}{0.001217} = 4.03$$

If the fixed-bed data are to be used for estimating mass-transfer coefficients for a moving bed of solids, the relative mass velocity of air and solid is appropriate. The linear rate of flow of the solid downward is $0.680/671.2 = 1.013 \times 10^{-3}$ m/s, where 671.2 is the apparent density. The density of the substantially dry air at 26.7°C , 1 std atm, is 1.181 kg/m^3 , and its superficial linear velocity upward is $1.36/1.181 = 1.152$ m/s. The relative linear velocity of air and solid is $1.152 + 1.013 \times 10^{-3} = 1.153$ m/s, and the relative mass velocity of the air is $1.153(1.181) = 1.352 \text{ kg/m}^2 \cdot \text{s} = G'$.

$$H_{IG} = \frac{G_S}{31.6 G'^{0.55}} = \frac{1.36}{31.6(1.1532)^{0.55}} = 0.0365 \text{ m}$$

$$H_{IS} = \frac{S_S}{k_y a_p} = \frac{0.680}{0.965} = 0.7047 \text{ m} \quad \frac{mG_S}{S_S} = \frac{0.0185(1.36)}{0.680} = 0.037$$

Eq. (11.52):

$$H_{IOG} = 0.0365 + 0.037(0.7047) = 0.0625 \text{ m}$$

$$Z = N_{IOG} H_{IOG} = 4.03(0.0625) = 0.25 \text{ m} \quad \text{Ans.}$$

Note: The gas-phase mass-transfer coefficient in this case is smaller than that given by item 7, Table 3.3, perhaps because of the very small diameter (16 mm) of the bed in which it was measured. This could lead to an important *wall effect*; however, see Illustration 11.9.

Two components adsorbed; fractionation When several components of a gas mixture are appreciably adsorbed, fractionation is required for their separation. This discussion is confined to binary gas mixtures.

For purposes of computation, it is easiest to recall the similarity between the adsorption operation and continuous countercurrent extraction with reflux. Solid adsorbent as the added insoluble phase is analogous to extraction solvent, the adsorbate is analogous to the solvent-free extract, and the fluid stream is similar to the raffinate. Computations can then be made using the methods and equations of Chap. 10 [Eqs. (10.31) to (10.34) and (10.39) to (10.50) with Fig. 10.28]. Some simplification is possible, however, thanks to the complete insolubility of adsorbent in the mixture to be separated.

Refer to Fig. 11.32, a schematic representation of the adsorber. Feed enters at the rate of F mass/(area)(time), containing components A and C. The adsorbate-free adsorbent enters the top of the tower at the rate of B mass/(area)(time) and flows countercurrent to the gas entering at the bottom at the rate of R_1 mass/(area)(time). Compositions in the gas stream are expressed as x weight fraction C, the more strongly adsorbed substance. E represents the weight of adsorbent-free adsorbate A + C upon the solid, mass/(area)(time), and N the ratio, mass adsorbent/mass adsorbate. At any point in the adsorber, $B \approx NE$. The adsorbate composition is expressed as y weight fraction component C, on an adsorbent-free basis. Solid leaves the adsorber at the bottom and is stripped of its adsorbate in the desorption section, and the desorbed fluid is split into two streams, the reflux R_1 and the C-rich product P_E . At the top the fluid leaving is the A-rich product, R_2 mass/(area)(time). The arrangement is very similar to that shown in Fig. 10.27. Calculations are made on a phase diagram of the type shown in the upper part of Fig. 11.8a, with N plotted as ordinate, x and y as abscissa.

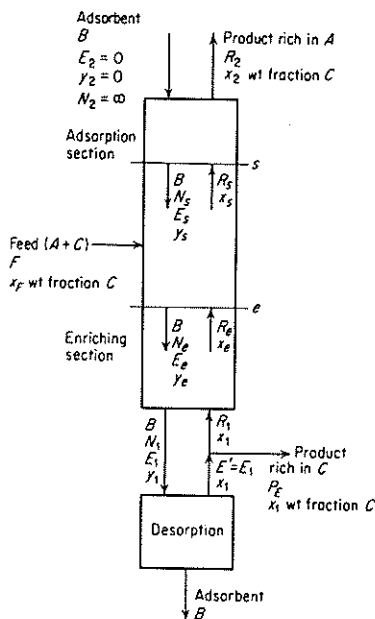


Figure 11.32 Continuous fractionation.

Let Δ_E represent the net adsorbent-free flow downward and out of the adsorber. Then $\Delta_E = P_E$, and the coordinates of point Δ_E on the phase diagram are $(N_{\Delta E} = B/P_E, y_{\Delta E} = x_1)$. At the bottom of the adsorber, $B = N_1 E_1$ and

$$E_1 = E' = P_E + R_1 = \Delta_E + R_1 \quad (11.53)$$

An A-C balance below section e of the enriching section is

$$E_e = P_E + R_e = \Delta_E + R_e \quad (11.54)$$

and, for substance C,

$$E_e y_e = P_E x_1 + R_e x_e = \Delta_E x_1 + R_e x_e \quad (11.55)$$

while for adsorbent it becomes

$$N_e E_e = N_{\Delta E} \Delta_E = B \quad (11.56)$$

Δ_E therefore represents the difference $E_e - R_e$ for any level in the enriching section. Equations (11.54) and (11.56) provide a measure of the internal reflux ratio,

$$\frac{R_e}{E_e} = \frac{N_{\Delta E} - N_e}{N_{\Delta E}} = 1 - \frac{N_e}{N_{\Delta E}} \quad (11.57)$$

and at the bottom of the tower the external reflux ratio is

$$\frac{R_1}{P_E} = \frac{R_1}{E_1 - R_1} = \frac{N_{\Delta E} - N_1}{N_1} = \frac{N_{\Delta E}}{N_1} - 1 \quad (11.58)$$

At the top of the adsorber let Δ_R represent the difference between flows of adsorbate and unadsorbed gas, and since $E_2 = 0$, $\Delta_R = -R_2$. The coordinates of Δ_R on the phase diagram are then ($N_{\Delta R} = -B/R_2$, $x_{\Delta R} = x_2$). Material balances above section s are then, for A and C,

$$R_s = E_s + R_2 = E_s - \Delta_R \quad (11.59)$$

for C,

$$R_s x_s = E_s y_s + R_2 x_2 = E_s y_s - \Delta_R x_2 \quad (11.60)$$

and for adsorbent,

$$B = N_s E_s = \Delta_R N_{\Delta R} \quad (11.61)$$

Equations (11.59) and (11.61) provide the internal reflux ratio.

$$\frac{R_s}{E_s} = \frac{N_s - N_{\Delta R}}{-N_{\Delta R}} = 1 - \frac{N_s}{N_{\Delta R}} \quad (11.62)$$

Overall balances about the entire plant are

$$F = R_2 + P_E \quad (11.63)$$

$$F x_F = R_2 x_2 + P_E x_1 \quad (11.64)$$

The definitions of Δ_R and Δ_E , together with Eq. (11.64), provide

$$\Delta_R + F = \Delta_E \quad (11.65)$$

The graphical interpretation of these relations on the phase diagram is the same as that for the corresponding situations in extraction and is shown in detail in Illustration 11.7 below.

If a stage-type device, e.g., fluidized beds, is used, the number of theoretical stages is readily obtained by the usual step construction on the lower part of a diagram such as Fig. 11.33. For continuous contact,

$$N_{IOG} = \frac{Z}{H_{IOG}} = \int_{\bar{p}_2}^{\bar{p}_1} \frac{d\bar{p}}{\bar{p} - \bar{p}^*} = \int_{x_2}^{x_1} \frac{dx}{x - x^*} - \ln \frac{1 + (r-1)x_1}{1 + (r-1)x_2} \quad (11.66)$$

where $r = M_A/M_C$ and H_{IOG} is defined by Eq. (11.50). This can be applied separately to the enriching and adsorption sections. It assumes equimolar counterdiffusion of A and C, which is not strictly the case, but refinement at this time is unwarranted in view of the sparsity of the data [45]. It is also subject to the same limitations considered for one component adsorbed (solute collection).

Illustration 11.7 Determine the number of transfer units and the adsorbent circulation rate required to separate a gas containing 60% ethylene C_2H_4 and 40% propane C_3H_8 by volume into products containing 5 and 95% C_2H_4 by volume, isothermally at $25^\circ C$ and 2.25 std atm, using activated carbon as the adsorbent and a reflux ratio of twice the minimum. *Note:* These adsorbents will customarily operate at reflux ratios closer to the minimum in order to reduce the adsorbent circulation rate. The present value is used in order to make the graphical solution clear.

SOLUTION Equilibrium data for this mixture at $25^\circ C$ and 2.25 std atm have been estimated from the data of Lewis et al., *Ind. Eng. Chem.*, 42, 1319, 1326 (1950), and are plotted in Fig.

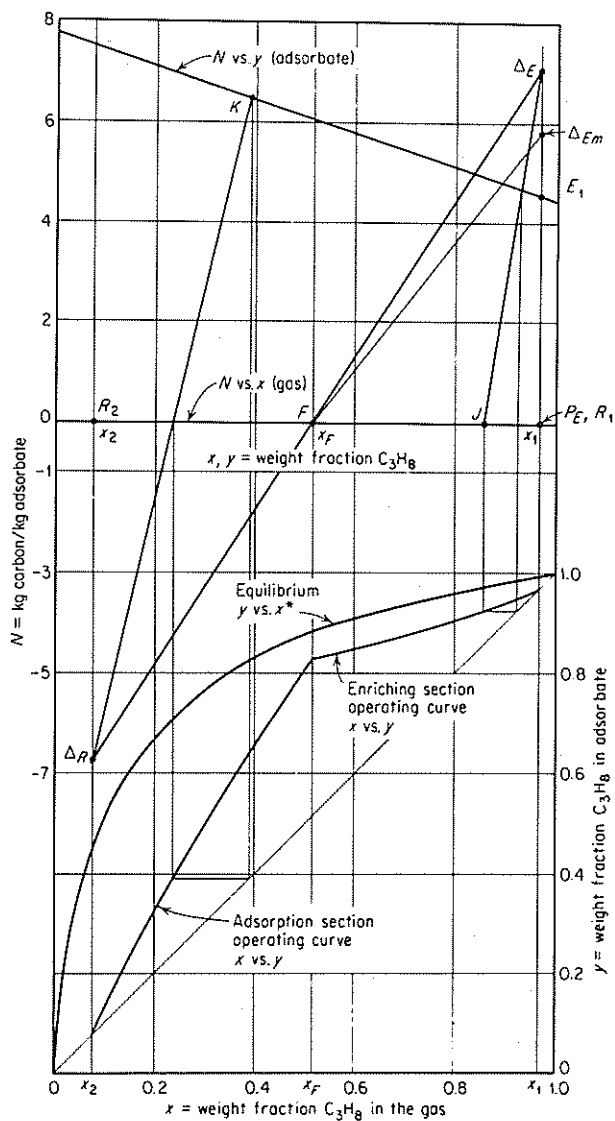


Figure 11.33 Solution to Illustration 11.7.

11.33. C_3H_8 is the more strongly adsorbed component, and compositions in the gas and adsorbate are expressed as weight fraction C_3H_8 . Tie lines have been omitted in the upper part of the plot, and the equilibrium compositions are shown instead in the lower part.

Molecular weights are 28.0 for C_2H_4 and 44.1 for C_3H_8 . The feed-gas composition is then $x_F = 0.4(44.1)/[0.4(44.1) + 0.6(28.0)] = 0.512$ wt fraction C_3H_8 . Similarly $x_1 = 0.967$, and $x_2 = 0.0763$ wt fraction C_3H_8 .

Basis: 100 kg feed gas. Eqs. (11.63) and (11.64):

$$100 = R_2 + P_E$$

$$100(0.512) = R_2(0.0763) + P_E(0.967)$$

Solving simultaneously gives $R_2 = 51.1$ kg, $P_E = 48.9$ kg.

Point F at x_F and point E_1 at x_1 are located on the diagram as shown. From the diagram, N_1 (at point E_1) = 4.57 kg carbon/kg adsorbate. The minimum reflux ratio is found as it is for extraction. In this case, the tie line through point F locates $(\Delta_E)_m$ and $(\Delta_E)_m = 5.80$.

$$\left(\frac{R_1}{P_E}\right)_m = \frac{5.80}{4.57} - 1 = 0.269 \text{ kg reflux gas/kg product}$$

$$(R_1)_m = 0.269 P_E = 0.269(48.9) = 13.15 \text{ kg}$$

$$(E_1)_m = (R_1)_m + P_E = 13.15 + 48.9 = 62.1 \text{ kg}$$

$$B_m = N_1(E_1)_m = 4.57(62.1) = 284 \text{ kg carbon/100 kg feed}$$

At twice the minimum reflux ratio, $R_1/P_E = 2(0.269) = 0.538$. Eq. (11.58):

$$0.538 = \frac{N_{\Delta E}}{4.57} - 1$$

$$N_{\Delta E} = 7.03 \text{ kg carbon/kg adsorbate}$$

and point Δ_E is located on the diagram.

$$R_1 = 0.538 P_E = 0.538(48.9) = 26.3 \text{ kg}$$

$$E_1 = R_1 + P_E = 26.3 + 48.9 = 75.2 \text{ kg}$$

$$B = N_1 E_1 = 4.57(75.2) = 344 \text{ kg carbon/100 kg feed}$$

Point Δ_R can be located by extending line $\Delta_E F$ to intersect the line $x = x_2$. Alternatively, $N_{\Delta R} = -B/R_2 = -344/51.1 = -6.74$ kg carbon/kg adsorbate. Random lines such as line $\Delta_R K$ are drawn from Δ_R , and the intersections with the equilibrium curves are projected downward in the manner shown to provide the adsorption-section operating curve. Similarly random lines such as line $\Delta_E J$ are drawn from Δ_E , and intersections are projected downward to provide the enriching-section operating curve. The horizontal distance between operating and equilibrium curves on the lower diagram is the driving force $x - x^*$ of Eq. (11.66). The following were determined from the diagram:

x	x^*	$\frac{1}{x - x^*}$	x	x^*	$\frac{1}{x - x^*}$
$x_1 = 0.967$	0.825	7.05	$x_F = 0.512$	0.39	8.20
0.90	0.710	5.26	0.40	0.193	4.83
0.80	0.60	5.00	0.30	0.090	4.76
0.70	0.50	5.00	0.20	0.041	6.29
0.60	0.43	5.89	$x_2 = 0.0763$	0.003	13.65

The third column was plotted as ordinate against the first as abscissa, the area under the curve between x_1 and x_F was 2.65, and that between x_F and x_2 was 2.67. Further, $r = 28.0/44.1$

= 0.635. Applying Eq. (11.66) to the enriching section, gives

$$N_{\text{LOG}} = 2.65 - \ln \frac{1 + (0.635 - 1)(0.967)}{1 + (0.635 - 1)(0.512)} = 2.52$$

and to the adsorption section,

$$N_{\text{LOG}} = 2.67 - \ln \frac{1 + (0.635 - 1)(0.512)}{1 + (0.635 - 1)(0.0763)} = 2.53$$

The total $N_{\text{LOG}} = 2.52 + 2.53 = 5.1$.

Simulation of moving beds The arrangement shown in Fig. 11.34 for fractionating liquids avoids the problems of moving beds of solid downward through a tower and transporting the solid up to the top [10]. In the adsorber, divided into sections by sieve plates, the adsorbent is in the form of fixed, i.e., unmoving, beds. The rotary valve, turning counterclockwise periodically, moves the position of introducing the feed successively from positions *a* through *c* and back to *a*. Similarly the desorbent and the withdrawn streams are moved periodically, the topmost stream (*A* + *D*) moving from position *a* at the top to *b* and *c* at the bottom, and back to *a*. The recirculating-liquid pump is programmed to change its rate of pumping as needed. The degree of approach to a true countercurrent system increases as the number of compartments increases, at the expense, of course, of complexity of the piping. The adsorber vessel may

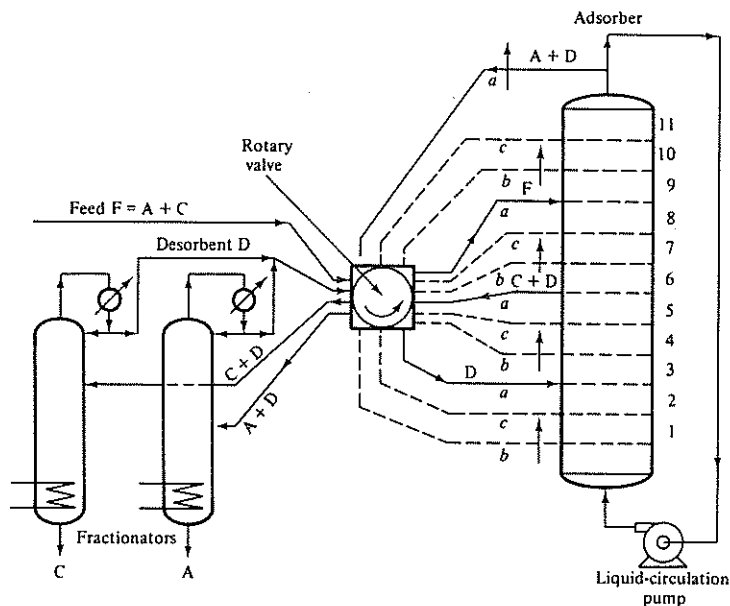


Figure 11.34 Simulation of moving beds. [Broughton, *Chem. Eng. Prog.*, 64(8), 60 (1968), with permission.]

be horizontal [80]. Many plants have been built, using molecular sieves for a variety of hydrocarbon separations.

Unsteady State—Fixed-Bed Adsorbers

The inconvenience and relatively high cost of continuously transporting solid particles as required in steady-state operations frequently make it more economical to pass the fluid mixture to be treated through a stationary bed of adsorbent. As increasing amounts of fluid are passed through such a bed, the solid adsorbs increasing amounts of solute, and an unsteady state prevails. This technique is very widely used and finds application in such diverse fields as the recovery of valuable solvent vapors from gases, purifying air as with gas masks, dehydration of gases and liquids, decolorizing mineral and vegetable oils, the concentration of valuable solutes from liquid solutions, and many others.

The adsorption wave Consider a binary solution, either gas or liquid, containing a strongly adsorbed solute at concentration c_0 . The fluid is to be passed continuously down through a relatively deep bed of adsorbent initially free of adsorbate. The uppermost layer of solid, in contact with the strong solution entering, at first adsorbs solute rapidly and effectively, and what little solute is left in the solution is substantially all removed by the layers of solid in the lower part of the bed. The effluent from the bottom of the bed is practically solute-free, as at c_a in the lower part of Fig. 11.35. The distribution of adsorbate in the solid bed is indicated in the sketch in the upper part of this figure at a , where the relative density of the horizontal lines in the bed is meant to indicate the relative concentration of adsorbate. The uppermost layer of the bed is practically saturated, and the bulk of the adsorption takes place over a relatively narrow adsorption zone in which the concentration changes rapidly, as shown. As solution continues to flow, the adsorption zone moves downward as a wave, at a rate ordinarily very much slower than the linear velocity of the fluid through the bed. At a later time, as at b in the figure, roughly half the bed is saturated with solute, but the effluent concentration c_b is still substantially zero. At c in the figure the lower portion of the adsorption zone has just reached the bottom of the bed, and the concentration of solute in the effluent has suddenly risen to an appreciable value c_c for the first time. The system is said to have reached the *breakpoint*. The solute concentration in the effluent now rises rapidly as the adsorption zone passes through the bottom of the bed and at d has substantially reached the initial value c_0 . The portion of the effluent concentration curve between positions c and d is termed the *breakthrough curve*. If solution continues to flow, little additional adsorption takes place since the bed is for all practical purposes entirely in equilibrium with the feed solution.

If a vapor is being adsorbed adiabatically from a gas mixture in this manner, the evolution of the heat of adsorption causes a temperature wave to flow through the adsorbent bed in a manner somewhat similar to the adsorption wave [56] and the rise in temperature of the bed at the fluid outlet can

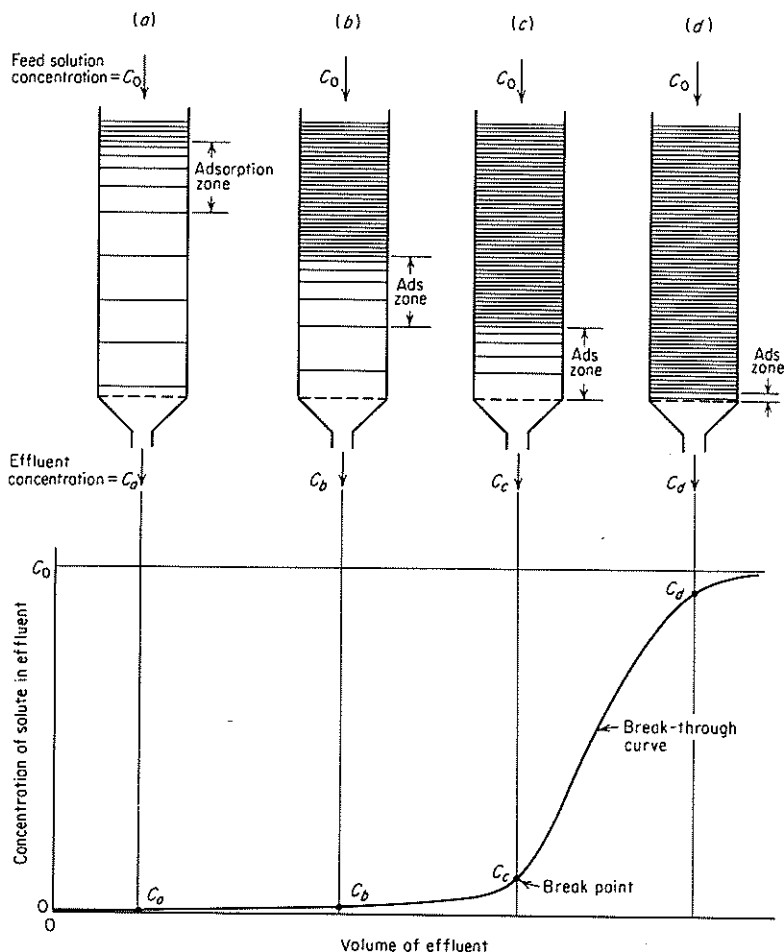


Figure 11.35 The adsorption wave.

sometimes be used as a rough indication of the breakpoint. In adsorption from liquids the temperature rise is usually relatively small.

The shape and time of appearance of the breakthrough curve greatly influence the method of operating a fixed-bed adsorber. The curves generally have an S shape, but they may be steep or relatively flat and in some cases considerably distorted. If the adsorption process were infinitely rapid, the breakthrough curve would be a straight vertical line in the lower part of Fig. 11.35. The actual rate and mechanism of the adsorption process, the nature of the adsorption equilibrium, the fluid velocity, the concentration of solute in the feed, and the length of the adsorber bed (particularly if the solute concentration

in the feed is high) all contribute to the shape of the curve produced for any system. The breakpoint is very sharply defined in some cases and in others poorly defined. Generally the breakpoint time decreases with decreased bed height, increased particle size of adsorbent, increased rate of flow of fluid through the bed, and increased initial solute content of the feed. There is a critical minimum bed height below which the solute concentration in the effluent will rise rapidly from the first appearance of effluent. In planning new processes it is best to determine the breakpoint and breakthrough curve for a particular system experimentally under conditions resembling as much as possible those expected in the process.

Adsorption of vapors One of the most important applications of fixed-bed adsorbers is in the recovery of valuable solvent vapors. Solids saturated with solvents such as alcohol, acetone, carbon disulfide, benzene, and others can be dried by evaporation of the solvent into an airstream, and the solvent vapor can be recovered by passing the resulting vapor-gas mixture through a bed of activated carbon. The very favorable adsorption equilibrium provided by a good grade of carbon for vapors of this sort permits substantially complete vapor recovery, 99 to 99.8 percent, from gas mixtures containing as little as 0.5 to 0.05 percent of the vapor by volume [15]. Air-vapor mixtures of concentration well below the explosive limits can thus be handled. In most such adsorption plants it is necessary to operate with a small drop in pressure through the bed of adsorbent in order to keep power costs low. Therefore granular rather than powdered adsorbents are used, and bed depths are relatively shallow (0.3 to 1 m) and large in cross section. The superficial gas velocity may be in the range 0.25 to 0.6 m/s. A typical arrangement of the adsorption vessel is shown in Fig. 11.36.

In a typical operation the air-vapor mixture, if necessary cooled to 30 to 40°C and filtered free of dust particles which might clog the pores of the adsorbent, is admitted to the adsorbent bed. If the breakthrough curve is steep, the effluent air, substantially free of vapor, may be discharged to the atmosphere until the breakpoint is reached, whereupon the influent stream must be diverted to a second adsorber while the first is regenerated. On the other hand, if the breakthrough curve is flat, so that at the breakpoint a substantial portion of the adsorbent remains unsaturated with adsorbate, the gas may be permitted to flow through a second adsorber in series with the first until the carbon in the first is substantially all saturated. The influent mixture is then passed through the second and a third adsorber in series while the first is regenerated.

After gas flow has been diverted from an adsorber, the carbon is usually desorbed by admission of low-pressure steam. This lowers the partial pressure of the vapor in contact with the solid and provides by condensation the necessary heat of desorption. The steam-vapor effluent from the carbon is condensed and the condensed solvent recovered by decantation from the water if it is water-insoluble or by rectification if an aqueous solution results. When desorption is complete, the carbon is saturated with adsorbed water. This moisture is readily

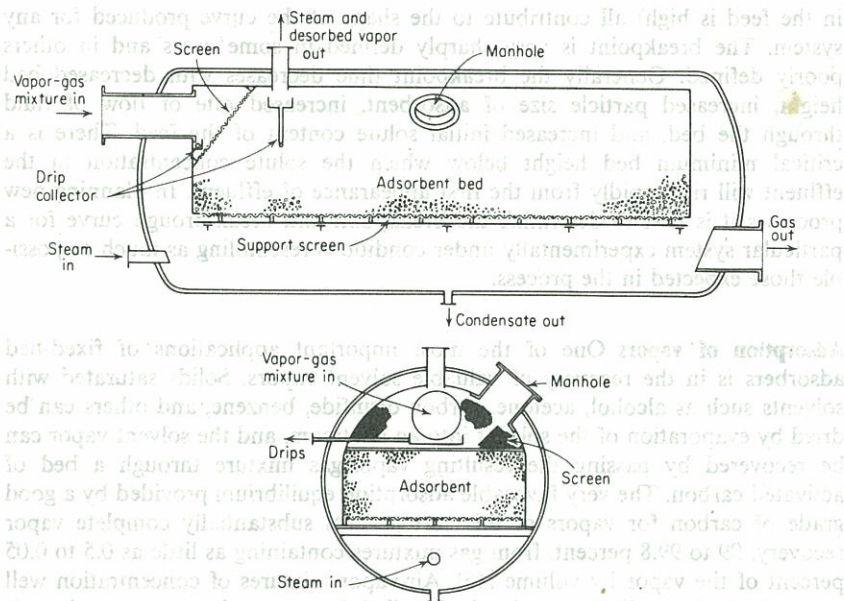


Figure 11.36 Adsorber for solvent vapors at low pressures, schematic. [Logan, U.S. Patent 2,180,712 (1939).]

displaced by many vapors and evaporated into the air when the air-vapor mixture is readmitted to the carbon, and indeed much of the heat evolved during adsorption of the vapor can be used in desorbing the water, thus maintaining moderate bed temperatures. If the moisture interferes with vapor adsorption, the bed can first be dried by admitting heated air and then cooled by unheated air, before reuse for vapor recovery. Figure 11.37 shows a typical plant layout arranged for this method. An alternative procedure for regenerating the bed is to heat it electrically or with embedded steam pipes and pump off the adsorbed vapor with a vacuum pump. An arrangement which avoids the intermittent operation usually associated with fixed beds is shown in Fig. 11.38. Operation is reminiscent of continuous filters. As the inner drum rotates, the adsorbent is successively subject to adsorption, regeneration, and drying with cooling. All these, which use heat as a regenerative means, are sometimes referred to as *thermal-swing procedures*.

Moist gases can be dried of their water content by passing them through beds of activated silica gel, alumina, bauxite, or molecular sieves. Especially if the gases are under appreciable pressures, moderately deep beds are used since the pressure drop will still be only a small fraction of the total pressure. Towers containing the adsorbent may be as much as 10 m tall or more, but in such instances the solid is best supported on trays at intervals of 1 to 2 m in order to

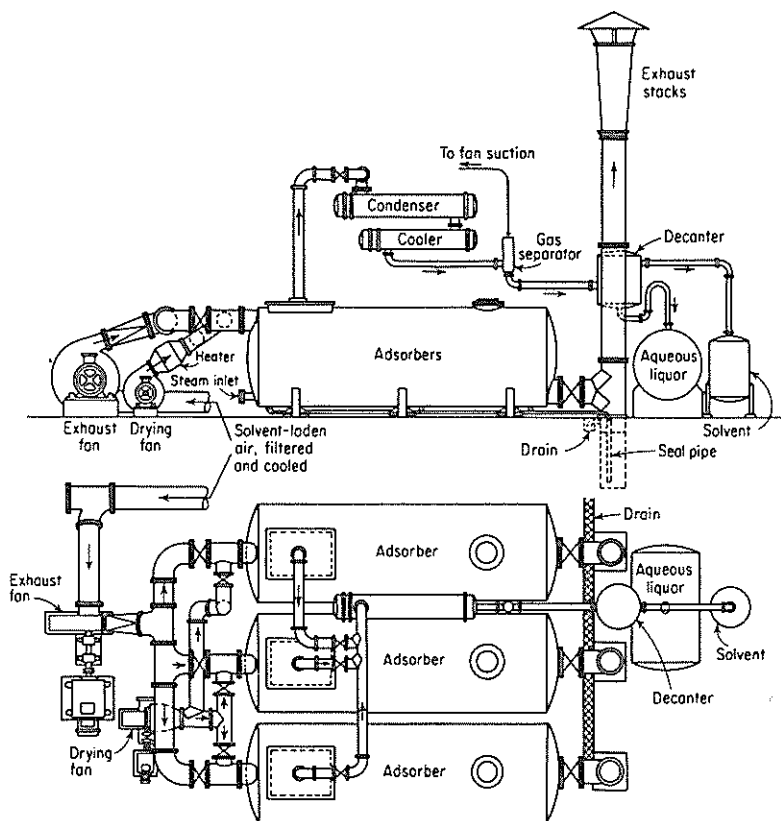


Figure 11.37 Solvent-recovery adsorption plant. [After Benson and Courouleau, *Chem. Eng. Prog.*, 44, 466 (1948), with permission.]

minimize the compression of the bed resulting from pressure drop. A vessel for treating gases under pressure can be designed as in Fig. 11.39. After the bed has reached the maximum practical moisture content, the adsorbent can be regenerated, either by application of heat by steam coils embedded in the adsorbent or by admitting heated air or gas. Liquids such as gasoline, kerosene, and transformer oil can also be dehydrated by passage through beds of activated alumina. High-temperature (230°C) steam followed by application of vacuum provided by a steam-jet ejector can be used for regenerating the adsorbent.

Illustration 11.8 A solvent-recovery plant is to recover 0.1 kg/s (800 lb/h) of ethyl acetate vapor from a mixture with air at a concentration of $3 \text{ kg vapor}/100 \text{ m}^3$ ($1.9 \text{ lb}/1000 \text{ ft}^3$), 1 std atm pressure. The adsorbent will be activated carbon, 6- to 8-mesh (av particle diam = 2.8 mm), apparent density of individual particles 720 kg/m^3 (45 lb/ft^3). Apparent density of the packed bed = 480 kg/m^3 (30 lb/ft^3). The carbon is capable of adsorbing 0.45 kg vapor/kg

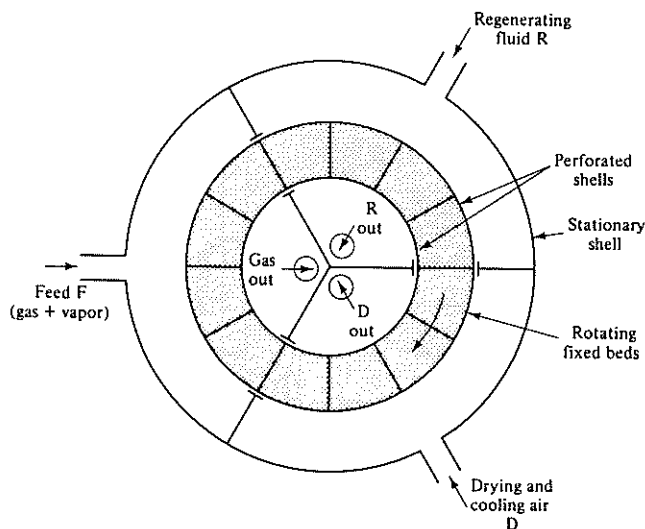


Figure 11.38 Rotating fixed-bed adsorber (schematic).

carbon up to the breakpoint. The adsorption cycle will be set at 3 h to allow sufficient time for regeneration. Determine the amount of carbon required, choose suitable dimensions for the carbon beds, and estimate the pressure drop.

SOLUTION In 3 h, the vapor to be adsorbed is $3(3600)(0.1) = 1080$ kg. The carbon required per bed = $1080/0.45 = 2400$ kg. Two beds will be necessary, one adsorbing while the other is being regenerated. Total carbon required = $2(2400) = 4800$ kg.

The volume of each bed will be $2400/480 = 5$ m³. If the bed depth is 0.5 m, the cross-sectional area is $5/0.5 = 10$ m², say 2 by 5 m, and it can be installed horizontally in a suitable vessel, such as that shown in Fig. 11.36.

The pressure drop can be estimated by Eq. (6.66). At 35°C, the viscosity of air = 0.0182 cP = 1.82×10^{-5} kg/m · s, and the density = $(29/22.41)(273/308) = 1.148$ kg/m³. Since 1 m³ of the bed, weighing 480 kg, contains $480/720 = 0.667$ m³ of solid particles, the fractional void volume of the bed (space between particles, not including pore volume) = $1 - 0.667 = 0.333 = \epsilon$.

The volumetric rate of airflow = $0.1(100/3) = 3.33$ m³/s, and the superficial mass velocity = $3.33(1.148)/10 = 0.383$ kg/m² · s (linear velocity = 0.333 m/s). In Chap. 6 notation, the particle Reynolds number $Re = d_p G/\mu = 0.0028(0.383)/(1.82 \times 10^{-5}) = 58.9$, $Z = 0.5$ m, $g_c = 1.0$, and, by Eq. (6.66), $\Delta p = 1421$ N/m² (5.7 in H₂O). **Ans.**

The heatless adsorber (pressure swing) The method of operation shown in Fig. 11.40 for removing a vapor from a gas under pressure uses mechanical work of compression instead of heat to regenerate the adsorbent [92]. Wet gas under pressure passes through four-way valve 3 into adsorber bed 1, and the dry gas leaves at the top. A portion of the product gas is passed through valve 4 and at a lower pressure into bed 2 to evaporate the adsorbate accumulated there. It

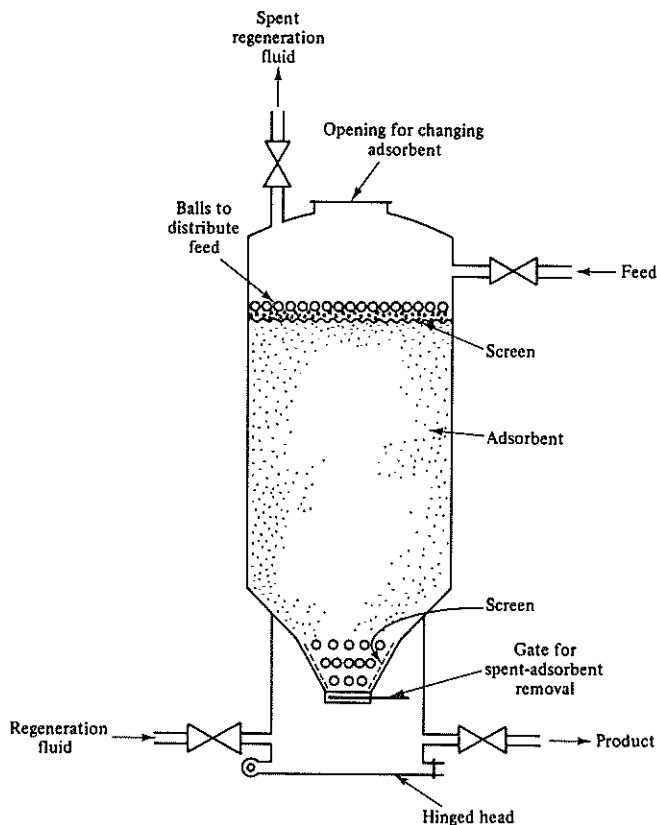


Figure 11.39 Fixed-bed adsorber for vapors at high pressures, and for liquid percolation. [After Johnston, *Chem. Eng.*, 79, 87 (Nov. 27, 1972).]

leaves through valve 3. Valves 5 and 6 are check valves allowing flow to the right and left, respectively. After a short interval (30 to 300 s), valve 3 is turned, beds 1 and 2 switch roles, and flow through valve 4 is reversed. The cycle time is short enough to ensure that the adsorption zone always remains in the central part of the beds, the lower portion of the adsorbent being saturated with adsorbate and the upper part being adsorbate-free. The heat of adsorption is confined to a small volume of the bed and is used in purging the bed. The saturated humidity of a gas is given by $Y = [p/(p_i - p)](M_C/M_A)$, where p is the saturation vapor pressure of the more strongly adsorbed solute C. For a given temperature, with p fixed, Y is larger the lower the total pressure p_i . It then follows that to ensure that as much vapor is desorbed as is adsorbed, thus maintaining a steady flow of dry product, the ratio mole of purge gas/moles of feed must exceed the ratio low purge pressure/high feed pressure.

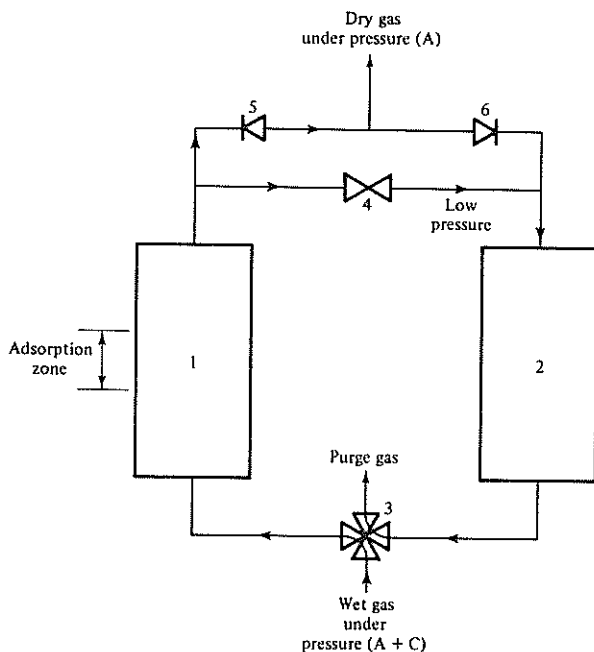


Figure 11.40 The heatless adsorber.

The original application, drying of gases, has been extended to a variety of fractionations, including separation of oxygen from air [17, 57] and the separation of hydrogen from refinery gas. Such processes usually require a more elaborate flowsheet than that in Fig. 11.40.

Adsorption of liquids; percolation The dehydration of liquids by stationary beds of adsorbent has already been mentioned. In addition, the colors of petroleum products, such as lubricating oils and transformer oils, and of vegetable oils are commonly reduced by percolation through beds of decolorizing clays; sugar solutions are deashed and decolorized by percolation through bone char; and many other liquid-treating operations use these semibatch methods.

In such service, the bed of adsorbent is commonly called a *filter*. It may be installed in a vertical cylindrical vessel fitted with a dished or conical bottom, of diameter ranging up to as much as 4.5 m and height up to 10 m. In the case of decolorizing clays, the beds may contain as much as 50 000 kg of adsorbent. The granular adsorbent is supported on a screen or blanket, in turn supported by a perforated plate.

The liquid flow is ordinarily downward, either under the force of gravity alone or under pressure from above. At the start of the operation, the adsorbent bed is frequently allowed to soak for a time in the first portions of the feed

liquid, in order to displace air from the adsorbent particles before the percolation is begun. In decolorizing operations, the concentration of impurity in the initial effluent liquid is usually much smaller than the specifications of the product require, and the breakthrough curve is frequently rather flat. Consequently it is common practice to allow the effluent liquid to accumulate in a receiving tank below the filter and to blend or composite it until the blended liquid reaches the largest acceptable concentration of impurity. In this way, the largest possible adsorbate concentration can be accumulated upon the solid.

When the solid requires revivification, the flow of liquid is stopped and the liquid in the filter drained. The solid can then be washed in place with an appropriate solvent, e.g., water for sugar-refining filters and naphtha for petroleum products. If necessary the solvent is then removed by admission of steam, following which the solid can be dumped from the filter and reactivated by burning or other suitable procedure, depending upon the nature of the adsorbent.

Elution Desorption of the adsorbed solute by a solvent is called *elution*. The desorption solvent is the *elutant*, and the effluent stream containing the desorbed solute and eluting solvent is the *eluate*. The *elution curve* is a plot of the solute concentration in the eluate against quantity of eluate, as in Fig. 11.41. The initial rise in solute concentration of the eluate, *OA* in the figure, is found when the void spaces between the adsorbent particles are initially filled with fluid remaining from the adsorption. For liquids, if the bed is drained before elution, the elution curve starts at *A*. If elution is stopped after eluate corresponding to *C* has been withdrawn, the area under the curve *OABC* represents the quantity of solute desorbed. For a successful process, this must equal the solute adsorbed during an adsorption cycle; otherwise solute will build up in the bed from one cycle to the next.

Chromatography Imagine a solution containing two solutes, *A* and *B*, which are differently adsorbed at equilibrium, *A* more strongly. A small quantity of this solution, insufficient to saturate all but a small quantity of the adsorbent, is passed through an adsorbent bed, whereupon both solutes are retained in the

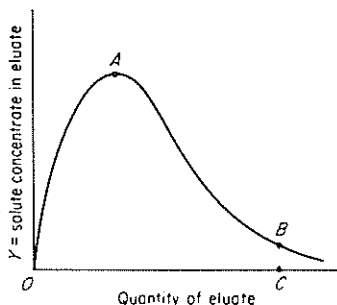


Figure 11.41 Elution of a fixed bed.

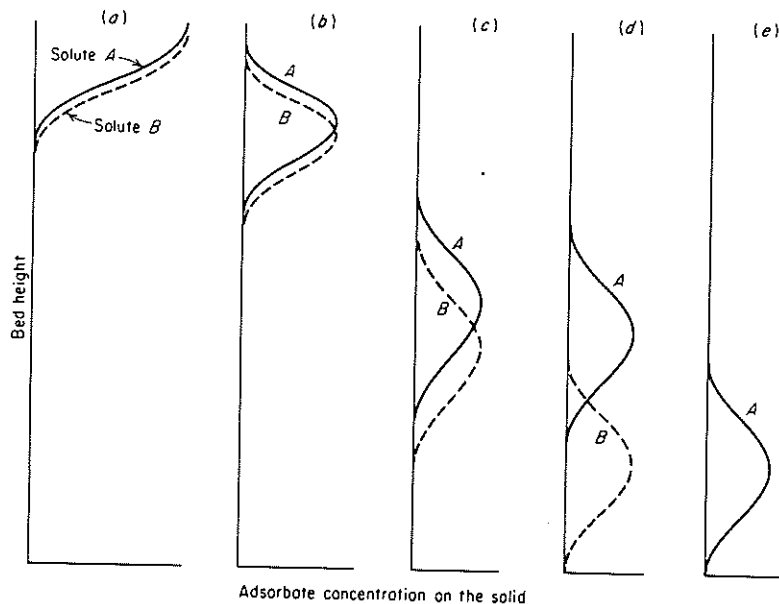


Figure 11.42 Chromatographic separation of two solutes.

upper portion of the bed to give rise to adsorbate concentrations as indicated in Fig. 11.42a. A suitable elutant is now passed through the bed, whereupon solute B is more readily desorbed than A. At *b*, *c*, and *d* in the figure, both solutes are seen to be desorbed, only to be readsorbed and redesorbed at lower positions in the bed, but the wave of B concentration moves more rapidly downward than that of A.† At *e*, solute B has been washed out of the solid, leaving essentially all the A behind. The curves of Fig. 11.42 are idealized, since in reality their shape may change as the concentration waves pass down the column. If the adsorption (or desorption) zone heights are short relative to the height of the bed, and if the selectivity of the adsorbent is sufficiently great, essentially complete separation of the solutes is possible. This technique is the basis of very powerful methods of analysis of mixtures, both gas and liquid, and variants are used for industrial purposes [91]. *Partition chromatography* accomplishes a similar result, using a liquid contained in the pores of a solid as an immobilized solvent for solute separation, rather than adsorption itself.

Rate of adsorption in fixed beds The design of a fixed-bed adsorber and prediction of the length of the adsorption cycle between revivifications require knowledge of the percentage approach to saturation at the breakpoint, as shown

† The adsorption bands in some cases have different colors, depending upon the chemical nature of the solutes, and from this arose the terms *chromatographic adsorption* and *chromatography*.

in Illustration 11.8. This in turn requires the designer to predict the time of the breakpoint and the shape of the breakthrough curve. The unsteady-state circumstances of fixed-bed adsorption and the many factors which influence the adsorption make such computations for the general case very difficult [99]. The following simplified treatment due to Michaels [70] is readily used but is limited to isothermal adsorption from dilute feed mixtures and to cases where the equilibrium adsorption isotherm is concave to the solution-concentration axis, where the adsorption zone is constant in height as it travels through the adsorption column, and where the height of the adsorbent bed is large relative to the height of the adsorption zone. Many industrial applications fall within these restrictions. The development here is in terms of adsorption from a gas, but it is equally applicable to treatment of liquids. Nonisothermal cases have also been treated [56].

Consider the idealized breakthrough curve of Fig. 11.43. This results from the flow of a solvent gas through an adsorbent bed at the rate of G_s mass/(area)(time), entering with an initial solute concentration Y_0 mass solute/mass solvent gas. The total solute-free effluent after any time is w mass/area of bed cross section. The breakthrough curve is steep, and the solute concentration in the effluent rises rapidly from essentially zero to that in the incoming gas. Some low value Y_B is arbitrarily chosen as the breakpoint concentration, and the adsorbent is considered as essentially exhausted when the effluent concentration has risen to some arbitrarily chosen value Y_E , close to Y_0 . We are concerned principally with the quantity of effluent w_B at the breakpoint and the shape of the curve between w_B and w_E . The total effluent accumulated during the appearance of the breakthrough curve is $w_a = w_E - w_B$. The adsorption zone, of

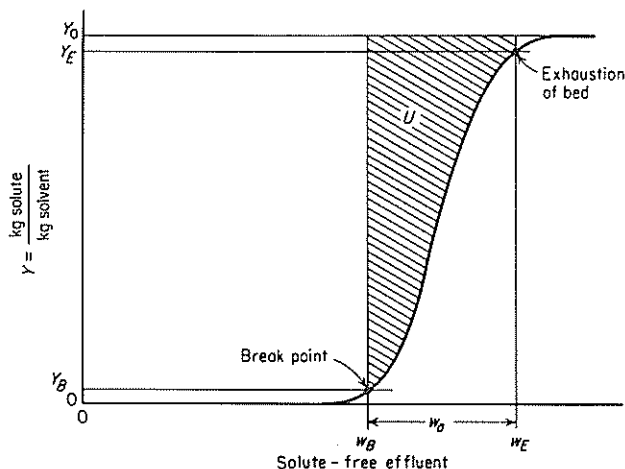


Figure 11.43 Idealized breakthrough curve.

constant height Z_a ft, is that part of the bed in which the concentration change from Y_B to Y_E is occurring at any time.

Let θ_a be the time required for the adsorption zone to move its own height down the column, after the zone has been established. Then

$$\theta_a = \frac{w_a}{G_S} \quad (11.67)$$

Let θ_E be the time required for the adsorption zone to establish itself and move out of the bed. Then

$$\theta_E = \frac{w_E}{G_S} \quad (11.68)$$

If the height of the adsorbent bed is Z , and if θ_F is the time required for the formation of the adsorption zone,

$$Z_a = Z \frac{\theta_a}{\theta_E - \theta_F} \quad (11.69)$$

The quantity of solute removed from the gas in the adsorption zone from the breakpoint to exhaustion is U mass solute/area of bed cross section. This is given by the shaded area of Fig. 11.43, which is

$$U = \int_{w_B}^{w_E} (Y_0 - Y) dw \quad (11.70)$$

If, however, all the adsorbent in the zone were saturated with solute, it would contain $Y_0 w_a$ mass solute/area. Consequently at the breakpoint, when the zone is still within the column, the fractional ability of the adsorbent in the zone still to adsorb solute is

$$f = \frac{U}{Y_0 w_a} = \frac{\int_{w_B}^{w_E} (Y_0 - Y) dw}{Y_0 w_a} \quad (11.71)$$

If $f = 0$, so that the adsorbent in the zone is essentially saturated, the time of formation of the zone at the top of the bed θ_F should be substantially the same as the time θ_a required for the zone to travel a distance equal to its own height. On the other hand, if $f = 1.0$, so that the solid in the zone contains essentially no adsorbate, the zone-formation time should be very short, essentially zero. These limiting conditions, at least, are described by

$$\theta_F = (1 - f)\theta_a \quad (11.72)$$

Equations (11.69) and (11.72) provide

$$Z_a = Z \frac{\theta_a}{\theta_E - (1 - f)\theta_a} = Z \frac{w_a}{w_E - (1 - f)w_a} \quad (11.73)$$

The adsorption column, Z tall and of unit cross-sectional area, contains a mass $Z\rho_S$ of adsorbent, where ρ_S is the apparent packed density of the solid in the bed. If this were all in equilibrium with the entering gas and therefore completely saturated at an adsorbate concentration X_T mass adsorbate/mass

solid, the adsorbate mass would be $Z\rho_S X_T$. At the breakpoint, the adsorption zone of height Z_a is still in the column at the bottom, but the rest of the column, $Z - Z_a$, is substantially saturated. At the breakpoint, therefore, the adsorbed solute is $(Z - Z_a)\rho_S X_T + Z_a\rho_S(1 - f)X_T$. The fractional approach to saturation of the column at the breakpoint is therefore [34]

$$\text{Degree of saturation} = \frac{(Z - Z_a)\rho_S X_T + Z_a\rho_S(1 - f)X_T}{Z\rho_S X_T} = \frac{Z - f Z_a}{Z} \quad (11.74)$$

In the fixed bed of adsorbent, the adsorption zone in reality moves downward through the solid, as we have seen. Imagine instead, however, that the solid moves upward through the column countercurrent to the fluid fast enough for the adsorption zone to remain stationary within the column, as in Fig. 11.44a. Here the solid leaving at the top of the column is shown in equilibrium with the entering gas, and all solute is shown as having been removed from the effluent gas. This would, of course, require an infinitely tall column, but our concern will be primarily with the concentrations at the levels corresponding to the extremities of the adsorption zone. The operating line over the entire tower is

$$G_S(Y_0 - 0) = S_S(X_T - 0) \quad (11.75)$$

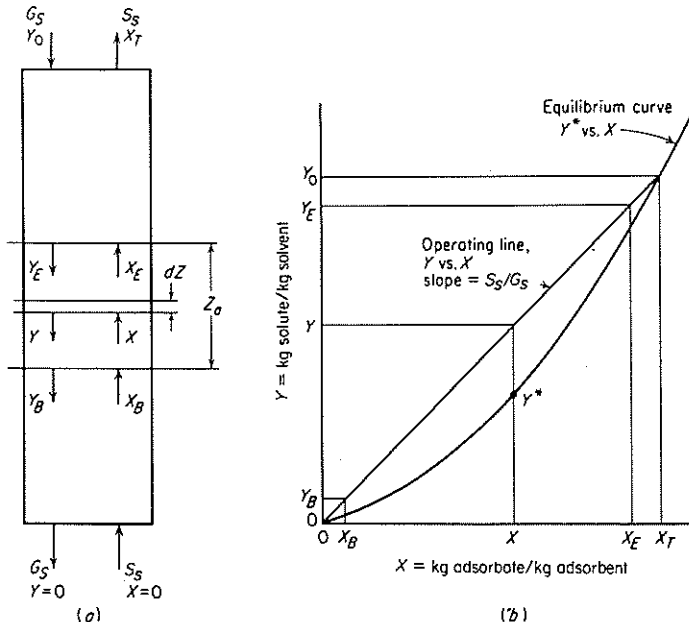


Figure 11.44 The adsorption zone.

$$\text{or} \quad \frac{S_S^*}{G_S} = \frac{Y_0}{X_T} \quad (11.76)$$

Since the operating line passes through the origin of Fig. 11.44*b*, at any level in the column the concentrations of solute in the gas Y and of adsorbate upon the solid X are then related by

$$G_S Y = S_S X \quad (11.77)$$

Over the differential height dZ , the ratio of adsorption is

$$G_S dY = K_Y a_p (Y - Y^*) dZ \quad (11.78)$$

For the adsorption zone, therefore,

$$N_{IOG} = \int_{Y_B}^{Y_E} \frac{dY}{Y - Y^*} = \frac{Z_a}{H_{IOG}} = \frac{Z_a}{G_S / K_Y a_p} \quad (11.79)$$

where N_{IOG} is the overall number of gas transfer units in the adsorption zone. For any value of Z less than Z_a , assuming H_{IOG} remains constant with changing concentrations,

$$\frac{Z \text{ at } Y}{Z_a} = \frac{w - w_B}{w_a} = \frac{\int_{Y_B}^Y \frac{dY}{Y - Y^*}}{\int_{Y_B}^{Y_E} \frac{dY}{Y - Y^*}} \quad (11.80)$$

Equation (11.80) should permit plotting the breakthrough curve by graphical evaluation of the integrals.

In addition to the restrictions outlined at the beginning, the success of this analysis largely hinges upon the constancy of $K_Y a_p$ or H_{IOG} for the concentrations within the adsorption zone. This will, of course, depend upon the relative constancy of the resistances to mass transfer in the fluid and within the pores of the solid. Illustration 11.9 demonstrates the method of using the equations in a typical case.

Illustration 11.9 Air at 26.7°C, 1 std atm, with a humidity of 0.00267 kg water/kg dry air is to be dehumidified by passage through a fixed bed of the silica gel used in Illustration 11.6. The depth of the adsorbent bed is to be 0.61 m. The air will be passed through the bed at a superficial mass velocity of 0.1295 kg/m²·s, and the adsorption will be assumed to be isothermal. The breakpoint will be considered as that time when the effluent air has a humidity of 0.0001 kg water/kg dry air, and the bed will be considered exhausted when the effluent humidity is 0.0024 kg water/kg dry air. Mass-transfer coefficients are given in Illustration 11.6 for this gel. Estimate the time required to reach the breakpoint.

SOLUTION The equilibrium data are plotted in Fig. 11.45. The gel is initially 'dry,' and the effluent air initially of so low a humidity as to be substantially dry, so that the operating line passes through the origin of the figure. The operating line is then drawn to intersect the equilibrium curve at $Y_0 = 0.00267$ kg water/kg dry air. $Y_B = 0.0001$, $Y_E = 0.0024$ kg water/kg dry air.

In the accompanying table, column 1 lists values of Y on the operating line between Y_B and Y_E and column 2 the corresponding values of Y^* taken from the equilibrium curve at the same value of X . From these the data of column 3 were computed. A curve (not shown) of column 1 as abscissa, column 3 as ordinate was prepared and integrated graphically between each value of Y in the table and Y_B , to give in column 4 the numbers of transfer units

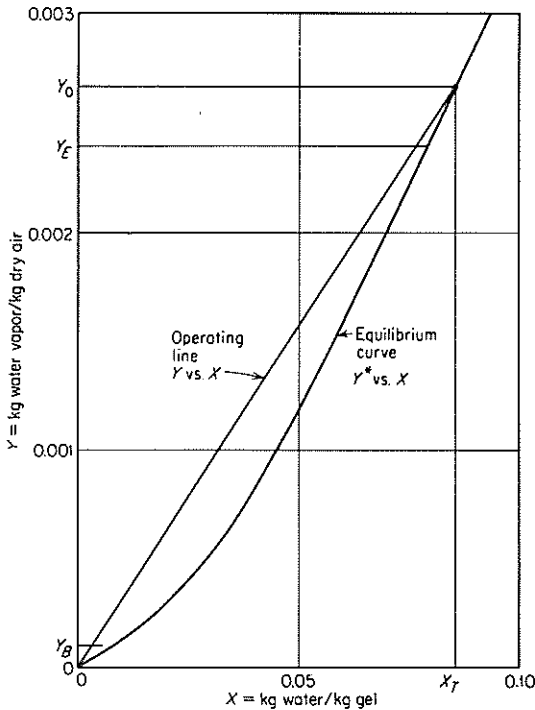


Figure 11.45 Solution to Illustration 11.9.

corresponding to each value of Y (thus, for example, the area under the curve from $Y = 0.0012$ to $Y = 0.0001$ equals 4.438). The total number of transfer units corresponding to the adsorption zone is $N_{tOG} = 9.304$, in accordance with Eq. (11.79).

Y , kg H ₂ O kg dry air (1)	Y^* , kg H ₂ O kg dry air (2)	$\frac{1}{Y - Y^*}$ (3)	$\int_{Y_B}^Y \frac{dY}{Y - Y^*}$ (4)	$\frac{w - w_B}{w_a}$ (5)	$\frac{Y}{Y_0}$ (6)
$Y_B = 0.0001$	0.00003	14 300	0	0	0.0374
0.0002	0.00007	7 700	1.100	0.1183	0.0749
0.0004	0.00016	4 160	2.219	0.2365	0.1498
0.0006	0.00027	3 030	2.930	0.314	0.225
0.0008	0.00041	2 560	3.487	0.375	0.300
0.0010	0.00057	2 325	3.976	0.427	0.374
0.0012	0.000765	2 300	4.438	0.477	0.450
0.0014	0.000995	2 470	4.915	0.529	0.525
0.0016	0.00123	2 700	5.432	0.584	0.599
0.0018	0.00148	3 130	6.015	0.646	0.674
0.0020	0.00175	4 000	6.728	0.723	0.750
0.0022	0.00203	5 880	7.716	0.830	0.825
$Y_E = 0.0024$	0.00230	10 000	9.304	1.000	0.899

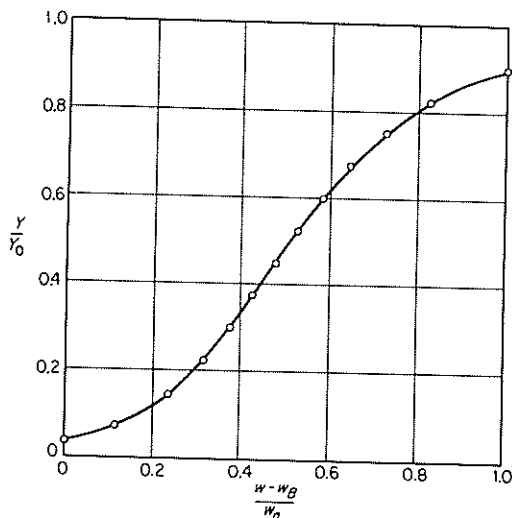


Figure 11.46 Calculated breakthrough curve for Illustration 11.9.

By dividing each entry in column 4 by 9.304, the values in column 5 were determined in accordance with Eq. (11.80). Column 6 was obtained by dividing each entry in column 1 by $Y_0 = 0.00267$, and column 6 plotted against column 5 provides a dimensionless form of the breakthrough curve between w_B and w_E (Fig. 11.46). Equation (11.74) can be written as

$$f = \frac{\int_{w_B}^{w_E} (Y_0 - Y) dw}{Y_0 w_a} = \int_0^{1.0} \left(1 - \frac{Y}{Y_0}\right) d\frac{w - w_B}{w_a}$$

from which it is seen that f equals the entire area above the curve of Fig. 11.46 up to $Y/Y_0 = 1.0$. By graphical integration, $f = 0.530$.

The mass-transfer rates are given in equation form in Illustration 11.6. For a mass velocity of air $G' = 0.1295 \text{ kg/m}^2 \cdot \text{s}$, $k_Y a_p = 31.6(0.1295)^{0.55} = 10.27 \text{ kg H}_2\text{O/m}^3 \cdot \text{s} \cdot \Delta Y$, and $k_S a_p = 0.965 \text{ kg H}_2\text{O/m}^3 \cdot \text{s} \cdot \Delta X$.

From Fig. 11.45, $X_T = 0.0858 \text{ kg H}_2\text{O/kg gel}$. Eq. (11.76):

$$S_S = \frac{Y_0 G_S}{X_T} = \frac{0.0026(0.1295)}{0.0858} = 3.92 \times 10^{-3}$$

The average slope of the equilibrium curve is $\Delta Y/\Delta X = 0.0185$, whence $mG_S/S_S = 0.0185(0.1295)/(3.92 \times 10^{-3}) = 0.61$. Eqs. (11.51) and (11.52):

$$H_{IG} = \frac{G_S}{k_Y a_p} = \frac{0.1295}{10.27} = 0.0126 \text{ m} \quad H_{IS} = \frac{S_S}{k_S a_p} = \frac{3.92 \times 10^{-3}}{0.965} = 4.062 \times 10^{-3} \text{ m}$$

$$H_{IOG} = H_{IG} + \frac{mG_S}{S_S} H_{IS} = 0.0126 + 0.61(4.062 \times 10^{-3}) = 0.0151 \text{ m}$$

Eq. (11.79): $Z_a = N_{IOG} H_{IOG} = 9.304(0.0151) = 0.14 \text{ m}$. The height of the bed = $Z = 0.61 \text{ m}$; therefore [Eq. (11.74)]:

$$\text{Degree of saturation at breakpoint} = \frac{0.61 - 0.53(0.14)}{0.61} = 0.878 = 87.8\%$$

The bed contains $0.61 \text{ m}^3 \text{ gel/m}^2$ of cross section, and since the bed density is 671.2 kg/m^3 (Illustration 11.6), the mass of the gel = $0.61(671.2) = 409.4 \text{ kg/m}^2$. At 87.8% of equilibrium

with the incoming air, the gel contains $409.4(0.878)(0.0858) = 30.8 \text{ kg H}_2\text{O/m}^2$ cross section. The air introduces $0.1295(0.00267) = 3.46 \times 10^{-4} \text{ kg H}_2\text{O/m}^2 \cdot \text{s}$, and hence the breakpoint occurs at $30.8/(3.46 \times 10^{-4}) = 89\,140 \text{ s}$ or 24.8 h after air is admitted initially, and $w_B = 89\,140(0.1295) = 11\,540 \text{ kg air/m}^2$ cross section. If the entire bed were in equilibrium with the entering gas, the adsorbed water would be $409.4(0.0858) = 35.1 \text{ kg/m}^2$, and hence $U = 35.1 - 30.8 = 4.3 \text{ kg/m}^2 = \text{ability of the bed still to adsorb}$. Therefore $w_a = U/f Y_0 = 4.3/0.53(0.0026) = 3038 \text{ kg air/m}^2$.

The circumstances of the calculation correspond to run S2 of Eagleton and Bliss [21], whose observed breakthrough curve (their fig. 5) agrees excellently with Fig. 11.46. However, they observed that 879 kg air/m^2 flowed through while the effluent humidity rose from $Y/Y_0 = 0.1$ to 0.8 , whereas the curve of Fig. 11.46 predicts this amount to be $(0.79 - 0.17)(3038) = 1883 \text{ kg/m}^2$. It is noteworthy that better agreement is obtained if the correlation of Table 3.3 is used for the gas-phase mass-transfer coefficient in the above calculation.

The preceding method is useful if both fluid- and solid-phase mass-transfer coefficients are available. The solid-phase coefficient particularly is not usually available for most industrial developments. In such cases the following extension is a useful approximation, also limited to cases of favorable adsorption equilibrium [14].

Consider Fig. 11.47. If the mass-transfer rate were infinitely rapid, the breakthrough curve would be the vertical line at θ_s , which can be located so that the shaded areas are equal. The adsorption zone of Fig. 11.35 can then be idealized as reduced to a plane, with the length of bed Z_s upstream of the plane at concentration X_T and the length $Z - Z_s$ downstream equal to the *length of unused bed* (LUB). At breakthrough, the length of the bed is taken to be the sum of LUB and a length saturated with solute in equilibrium with the feed stream. If $V = \text{velocity of advancement of the "adsorption plane,"}$ then at any time, $Z_s = V\theta$, at time θ_s , $Z = V\theta_s$, and at breakthrough, $Z_s = V\theta_B$; therefore

$$\text{LUB} = Z - Z_s = V(\theta_s - \theta_B) = \frac{Z}{\theta_s}(\theta_s - \theta_B) \quad (11.81)$$

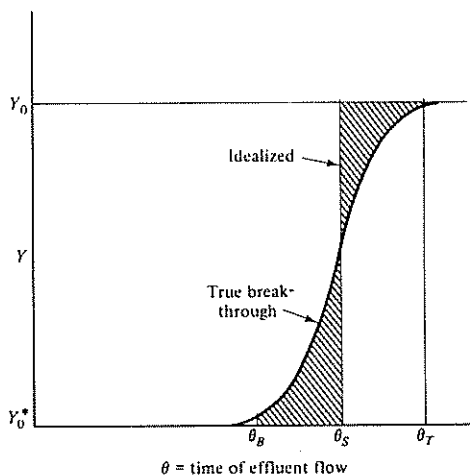


Figure 11.47 Idealized breakthrough curve for infinitely rapid mass transfer.

A solute balance is

$$G_S(Y_0 - Y_0^*)\theta = Z_s \rho_s (X_T - X_0) \quad (11.82)$$

where Y_0^* is the fluid concentration in equilibrium with X_0 .

In a laboratory experiment at the same G_S to be used on the large scale, the breakthrough curve can be measured, θ_B and θ_s (by equal areas) determined, and LUB calculated from Eq. (11.81). This should be the same on the large scale, provided radial mixing or bed channeling are not important.

If measurement is made on a production adsorber which cannot be operated beyond the breakpoint, the estimate can be made as follows. At θ_s , $Z = Z_s$, and Eq. (11.82) becomes

$$X_T - X_0 = \frac{G_S \theta_s}{Z_s \rho_s} (Y_0 - Y_0^*) \quad (11.83)$$

If we let $Z_s = V\theta$, Eq. (11.82) becomes

$$V = \frac{G_S}{\rho_s} \frac{Y_0 - Y_0^*}{X_T - X_0} \quad (11.84)$$

Elimination of θ_s from Eqs. (11.81) and (11.83) yields

$$\text{LUB} = Z - \frac{G_S \theta_B (Y_0 - Y_0^*)}{\rho_s (X_T - X_0)} \quad (11.85)$$

If the bed depth on the large scale is to be calculated for a given breakpoint time, Z_s can be calculated through Eq. (11.82), letting $\theta = \theta_B$. Then Eq. (11.85) provides LUB.

Illustration 11.10 The following laboratory test was made using type 4A molecular sieves, to remove water vapor from nitrogen (Collins' run B [14]):

Bed depth = 0.268 m (0.88 ft)
 Initial water concn in solid = 0.01 kg H₂O/kg solid = X_0
 Bulk density of bed = 712.8 kg/m³ (44.5 lb/ft³)
 Temperature = 28.3°C
 Gas pressure = 593 kN/m² (86 lb_f/in²)
 Nitrogen rate = 4052 kg/m² · h (830 lb/ft² · h)
 Initial H₂O concn in gas = 1440×10^{-6} mole fraction

Time, h	H ₂ O concn of effluent, mole fraction $\times 10^6$	Time, h	H ₂ O concn of effluent, mole fraction $\times 10^6$	Time, h	H ₂ O concn of effluent, mole fraction $\times 10^6$
0	< 1	10.2	238	11.75	1235
9 = θ_B	1	10.4	365	12.0	1330
9.2	4	10.6	498	12.5	1410
9.4	9	10.8	650	12.8 = θ_T	1440
9.6	33	11.0	808	13.0	1440
9.8	80	11.25	980	15.0	1440
10.0	142	11.5	1115		

Y_0^* in equilibrium with $X_0 \approx 0$

X_0^* in equilibrium with $Y_0 \approx 0.22$ kg H₂O/kg dry solid

$$Y = \frac{\text{mole fraction H}_2\text{O}}{1 - \text{mole fraction H}_2\text{O}} \frac{M_{\text{H}_2\text{O}}}{M_{\text{N}_2}} \approx \text{mole fraction H}_2\text{O} \frac{M_{\text{H}_2\text{O}}}{M_{\text{N}_2}}$$

The breakthrough data are plotted in the manner of Fig. 11.47, and θ_b is determined (equal areas) to be 10.9 h. Eq. (11.81):

$$\text{LUB} = \frac{0.268}{10.9} (10.9 - 9) = 0.0467 \text{ m}$$

It is desired to estimate the depth of bed required for the same molecular sieves operated at the same gas mass velocity, the same Y_0 and X_0 , and the same breakpoint concentration, 1×10^{-6} mole fraction water, as in the laboratory test, but the breakpoint time θ_b is to be 15 h.

$$Y_0 = \frac{1440 \times 10^{-6}}{1 - 1440 \times 10^{-6}} \frac{18}{29} = 9.0 \times 10^{-4} \text{ kg H}_2\text{O/kg N}_2 \quad Y_0^* \approx 0$$

Eq. (11.82):

$$Z_s = \frac{G_s(Y_0 - Y_0^*)\theta_b}{\rho_s(X_T - X_0)} = \frac{4052(9 \times 10^{-4})}{712.8(0.21 - 0.01)} = 0.365 \text{ m}$$

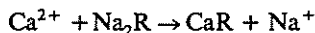
$$Z = \text{LUB} + Z_s = 0.0467 + 0.365 = 0.41 \text{ m (1.35 ft)} \quad \text{Ans.}$$

Note: Collins' run A [14], bed depth = 0.44 m, 26.1°C, $Y_0 = 1490 \times 10^{-6}$, $G_s = 4002$ (conditions not quite the same as above), records a breakpoint at 15 h.

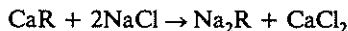
Ion Exchange

Ion-exchange operations are essentially metathetical chemical reactions between an electrolyte in solution and an insoluble electrolyte with which the solution is contacted. The mechanisms of these reactions and the techniques used to bring them about resemble those of adsorption so closely that for most engineering purposes ion exchange can simply be considered as a special case of adsorption.

Principles of ion exchange [20, 40, 55] The ion-exchange solids first used were porous, natural or synthetic minerals containing silica, the zeolites, such as the mineral $\text{Na}_2\text{O} \cdot \text{Al}_2\text{O}_3 \cdot 4\text{SiO}_2 \cdot 2\text{H}_2\text{O}$, for example. Positively charged ions (cations) of a solution which are capable of diffusing through the pores will exchange with the Na^+ ions of such a mineral, which is therefore called a *cation exchanger*. For example,



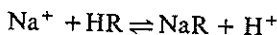
where R represents the residual material of the zeolite. In this manner "hard" water containing Ca^{2+} can be softened by contact with the zeolite, the less objectionable Na^+ replacing the Ca^{2+} in solution and the latter becoming immobilized in the solid. The reaction is reversible, and after saturation with Ca^{2+} the zeolite can be regenerated by contact with a solution of salt,



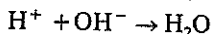
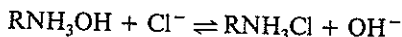
Later certain carbonaceous cation exchangers were manufactured by treating substances such as coal with reagents such as fuming sulfuric acid, and the like.

The resulting exchangers can be regenerated to a hydrogen form, HR, by treatment with acid rather than salt. Thus, hard water containing $\text{Ca}(\text{HCO}_3)_2$ would contain H_2CO_3 after removal of the Ca^{2+} by exchange, and since the carbonic acid is readily removed by degasification procedures, the total solids content of the water can be reduced in this manner. Early applications of ion exchangers using these principles were largely limited to water-softening problems.

In 1935, synthetic resinous ion exchangers were introduced. For example, certain synthetic, insoluble polymeric resins containing sulfonic, carboxylic, or phenolic groups can be considered as consisting of an exceedingly large anion and a replaceable or exchangeable cation. These make exchanges of the following type possible,



and different cations will exchange with the resin with different relative ease. The Na^+ immobilized in the resin can be exchanged with other cations or with H^+ , for example, much as one solute can replace another adsorbed upon a conventional adsorbent. Similarly synthetic, insoluble polymeric resins containing amine groups and anions can be used to exchange anions in solution. The mechanism of this action is evidently not so simple as in the case of the cation exchangers, but for present purposes it can be considered simply as an ion exchange. For example,



where RNH_3 represents the immobile cationic portion of the resin. Such resins can be regenerated by contact with solutions of sodium carbonate or hydroxide. The synthetic ion-exchange resins are available in a variety of formulations of different exchange abilities, usually in the form of fine, granular solids or beads, 16- to 325-mesh. The individual beads are frequently nearly perfect spheres.

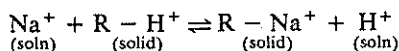
Techniques and applications All the operational techniques ordinarily used for adsorption are used also for ion exchange. Thus we have batch or stagewise treatment of solutions, fluidized- and fixed-bed operations, and continuous countercurrent operations. Fixed-bed percolations are most common. Chromatographic methods have been used for fractionation of multicomponent ionic mixtures. Applications have been made in the treatment of ore slurries ("resin-pulp") for collection of metal values.

In addition to the water-softening applications mentioned above, the complete deionization of water can be accomplished by percolation first through a cation exchanger and then through an anion exchanger. By using a bed formed of an intimate mixture of equivalent amounts of a strong cationic and a strong anionic exchange resin, simultaneous removal of all ions at neutrality is possible. For purposes of regeneration, such mixed-bed resins are separated by hydraulic classification through particle-size and density differences for the two resin types, and these are regenerated separately. The ion exchangers have also been

used for treatment and concentration of dilute waste solutions. Perhaps the most remarkable application of exchange resins has been to the separation of the rare-earth metals, using chromatographic techniques.

In *ion exclusion*, a resin is presaturated with the same ions as in a solution. It can then reject ions in such a solution but at the same time absorb nonionic organic substances such as glycerin, and the like, which may also be in the solution. The organic matter can later be washed from the resin in an ion-free state.

Equilibria The equilibrium distribution of an ion between an exchange solid and a solution can be described graphically by plotting isotherms in much the same manner used for ordinary adsorption. Various empirical equations for these isotherms, such as the Freundlich equation (11.3), have sometimes been applied to them. It is also possible to apply equations of the mass-action type to the exchange reaction. For example, for the cationic exchange



the mass-action-law constant is

$$\alpha = \frac{[\text{R} - \text{Na}^+]_{\text{solid}} [\text{H}^+]_{\text{soln}}}{[\text{R} - \text{H}^+]_{\text{solid}} [\text{Na}^+]_{\text{soln}}} = \left[\frac{\text{Na}^+}{\text{H}^+} \right]_{\text{solid}} \left[\frac{\text{H}^+}{\text{Na}^+} \right]_{\text{soln}} \quad (11.86)$$

where the square brackets indicate the use of some suitable equilibrium-concentration unit. The quantity α is thus seen to be an expression of relative adsorptivity, in this case of relative adsorptivity of Na^+ to H^+ . Since the solution and the solid remain electrically neutral during the exchange process, we can write

$$\alpha = \frac{X}{X_0 - X} \frac{c_0 - c^*}{c^*} = \frac{X/X_0}{1 - X/X_0} \frac{1 - c^*/c_0}{c^*/c_0} \quad (11.87)$$

where c_0 = initial concentration of $\text{Na}^+ + \text{H}^+$ in solution and consequently their total at any time

c^* = equilibrium Na^+ concentration after exchange

X = equilibrium concentration of Na^+ in solid

X_0 = concentration if all H^+ were replaced by Na^+

All these are expressed as equivalents per unit volume or mass. In the general case for any system, the relative adsorptivity α at a given temperature varies with total cationic concentration c_0 in the solution and also with c . In some cases, α has been found to be essentially constant with varying c at fixed c_0 .

Rate of ion exchange The rate of ion exchange depends, as in ordinary adsorption, upon rates of the following individual processes: (1) diffusion of ions from the bulk of the liquid to the external surface of an exchanger particle, (2) inward diffusion of ions through the solid to the site of exchange, (3) exchange of the ions, (4) outward diffusion of the released ions to the surface of the solid, and (5) diffusion of the released ions from the surface of the solid to the bulk of the

liquid. In some instances, the kinetics of the exchange reaction (3) may be controlling, but in others the rate of reaction is apparently very rapid in comparison with the rate of diffusion. The diffusion rates can be described by appropriate mass-transfer coefficients for equi-equivalent counterdiffusion through the solid and through the liquid, and in some instances at least it appears that the resistance to diffusion in the liquid phase may be controlling.

When the exchange reactions are rapid in comparison with the rates of mass transfer, the methods of design developed for conventional adsorbers can be applied to ion-exchange operations directly. Some modification of the units of the terms in the various equations may be desirable, owing to the customary use of concentrations expressed as equivalents per unit volume in the cgs system. The following example illustrates this.

Illustration 11.11 A synthetic ion-exchange resin in bead form is to be used for collecting and concentrating the copper in a dilute waste solution. The feed contains CuSO_4 at a concentration of 20 milligram equivalents (meq) Cu^{2+} /l and is to be treated at the rate of 37 850 l/h. A continuous system is planned: the solution to be treated and regenerated resin will flow countercurrently through a vertical tower, where 99% of the Cu^{2+} of the feed will be exchanged; the resin will be regenerated in a second tower by countercurrent contact with 2 *N* sulfuric acid. The necessary data are provided by Selke and Bliss, *Chem. Eng. Prog.*, 47, 529 (1951).

For collection of Cu^{2+} A superficial liquid velocity of $2.2 \text{ l/cm}^2 \cdot \text{h}$ will be used, for which the mass-transfer rate is $2.0 \text{ meq Cu}^{2+}/(\text{g resin}) \cdot \text{h} \cdot (\text{meq Cu}^{2+}/\text{l})$. The regenerated resin will contain $0.30 \text{ meq Cu}^{2+}/\text{g}$, and 1.2 times the minimum resin/solution ratio will be used.

For regeneration of the resin The superficial liquid velocity will be $0.17 \text{ l/cm}^2 \cdot \text{h}$, for which the mass-transfer rate is $0.018 \text{ meq Cu}^{2+}/(\text{g resin}) \cdot \text{h} \cdot (\text{meq Cu}^{2+}/\text{l})$. The acid will be utilized to the extent of 70%.

Compute the necessary rates of flow of resin and the amount of resin holdup in each tower.

SOLUTION Equilibria for the Cu^{2+} - H^+ exchange are provided by Selke and Bliss at two concentration levels, 20 and 2000 meq cation/l, shown in Fig. 11.48*a* and *b*, respectively.

Collection of Cu^{2+} Feed soln = 37 850 l/h. $c_1 = 20 \text{ meq Cu}^{2+}/\text{l}$, $c_2 = 0.01(20) = 0.20 \text{ meq Cu}^{2+}/\text{l}$. Cu^{2+} exchanged = $37\,850(20 - 0.20) = 750\,000 \text{ meq/h}$.

$X_2 = 0.30 \text{ meq Cu}^{2+}/\text{g}$. The point (c_2, X_2) is plotted in Fig. 11.48*a*. For the minimum resin/solution ratio and an infinitely tall tower, the operating line passes through point *P* at $X = 4.9$ on this figure, corresponding to equilibrium with c_1 . The minimum resin rate is then $750\,000/(4.9 - 0.30) = 163\,000 \text{ g/h}$. For 1.2 times the minimum the resin rate is $1.2(163\,000) = 196\,000 \text{ g/h}$. A copper balance is

$$750\,000 = 196\,000(X_1 - 0.30)$$

$$X_1 = 4.12 \text{ meq Cu}^{2+}/\text{g resin}$$

The point (c_1, X_1) is plotted on Fig. 11.48*a*, and the operating line can be drawn as a straight line at these low concentrations.

The quantity of resin in the tower can be obtained by application of the rate equation written in a form appropriate to the units of the quantities involved. Adapting Eq. (11.48) to this case, we have

$$V \, dc = \frac{K'_L a_p}{\rho_s} (c - c^*) \, d(SZ\rho_s)$$

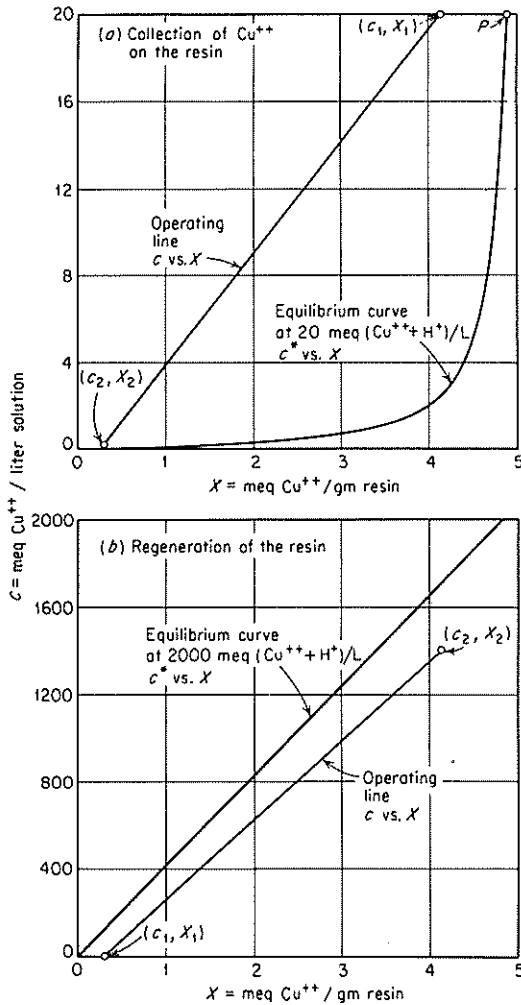


Figure 11.48 Solution to Illustration 11.11.

where $V = 1$ liquid/h

c = concn Cu^{2+} , meq/l, in soln

c^* = concn Cu^{2+} in soln at equilibrium with resin

$K'_L a_p / \rho_s$ = overall liquid mass-transfer coefficient, meq/(g resin) $\cdot$ h $\cdot$ (meq/l)

K'_L = overall liquid mass-transfer coefficient, meq/ $\text{cm}^2 \cdot \text{h} \cdot$ (meq/l)

a_p = surface of resin particles, cm^2/cm^3

ρ_s = packed density of resin, g/ cm^3

$SZ\rho_s$ = resin in tower, g

S = cross section of tower, cm^2

Z = height of tower, cm

Rearranging this equation and integrating gives

$$SZ\rho_S = \frac{V}{K'_L a_p / \rho_S} \int_{c_2}^{c_1} \frac{dc}{c - c^*}$$

For values of c on the operating line between c_1 and c_2 , the corresponding values of c^* from the equilibrium curve at the same value of X are obtained as follows:

c	20	16	12	8	4	2	1	0.2
c^*	2.4	1.9	0.5	0.25	0.10	0.05	0.02	0
$\frac{1}{c - c^*}$	0.0568	0.0710	0.0870	0.129	0.256	0.513	1.02	5.0

A curve (not shown) of $1/(c - c^*)$ as ordinate, c as abscissa, is plotted and integrated graphically between the limits c_1 and c_2 . The area under the curve is 5.72. (Note: This is the number of transfer units N_{IOL} .) Substituting in the integrated equation, we get

$$\text{Resin holdup} = SZ\rho_S = \frac{37\,850(5.72)}{2.0} = 108\,300 \text{ g}$$

Regeneration of resin Cu^{2+} to be exchanged = 750 000 meq/h, requiring as many meq H^+ /h. For a 70% utilization of acid, the acid feed must contain $750\,000/0.70 = 1\,071\,000$ meq H^+ /h, or $1\,071\,000/2000 = 536$ l/h of 2 N acid.

$c_1 = 0$, $c_2 = 750\,000/536 = 1400$ meq Cu^{2+} /l. $X_1 = 0.30$, $X_2 = 4.12$ meq Cu^{2+} /g resin. The points (c_1, X_1) and (c_2, X_2) are plotted on Fig. 11.48b and the operating line drawn. Integration of the rate equation for this case, where both operating and equilibrium lines are straight, provides

$$V(c_2 - c_1) = \frac{K'_L a_p}{\rho_S} (SZ\rho_S)(c^* - c)_m$$

where $(c^* - c)_m$ is the logarithmic average of the driving forces at the extremities of the tower and the other symbols have the same meaning as before.

$$c_1^* - c_1 = 120 - 0 = 120 \quad c_2^* - c_2 = 1700 - 1400 = 300 \text{ meq Cu}^{2+}/\text{l}$$

$$(c^* - c)_m = \frac{300 - 120}{\ln(300/120)} = 196.5 \text{ meq Cu}^{2+}/\text{l}$$

Substituting in the rate equation, we get

$$750\,000 = (0.018SZ\rho_S)(196.5)$$

$$SZ\rho_S = 212\,000 \text{ g resin holdup in regeneration tower}$$

The resin should be water-rinsed before reintroducing it into the adsorption tower. The Cu^{2+} in the effluent solution has been concentrated $1400/20 = 70$ times, equivalent to the evaporation of 37 300 l/h of water from the original solution.

NOTATION FOR CHAPTER 11

Any consistent set of units may be used, except as noted.

a_p	surface of solids/volume solid-fluid mixture, L^2/L^3
a_{pL}	surface of solids/volume liquid, L^2/L^3
a_{pS}	surface of solids/mass solids, L^2/M
A	mass velocity of adsorbed solute, $\text{M}/\text{L}^2\Theta$
B	mass velocity of adsorbate-free adsorbent, $\text{M}/\text{L}^2\Theta$

c	solute concentration in solution, M/L^3 or mole/ L^3 ; for ion exchange, equivalent/ L^3
Δc	liquid-phase concentration-difference driving force, mole/ L^3
C	distance, impeller to bottom of tank, L heat capacity, FL/MT
	mass velocity of more strongly adsorbed solute, $M/L^2\Theta$
d_i	impeller diameter, L
d_p	particle diameter, L
D	diffusivity, L^2/Θ
E	mass velocity of adsorbent-free adsorbate, $M/L^2\Theta$
E_D	eddy diffusivity of mass, $L^2\Theta$
E_{MS}	Murphree solid-phase stage efficiency, fractional
f	fractional ability of adsorption zone to adsorb solute, dimensionless
F	mass velocity of feed, $M/L^2\Theta$
F_L	liquid-phase mass-transfer coefficient, mole/ $L^2\Theta$
g	acceleration of gravity, L^2/Θ
g_c	conversion factor, $ML/F\Theta^2$
G	total gas mass velocity, $M/L^2\Theta$
G_S	mass velocity of unadsorbed gas based on container cross section, $M/L^2\Theta$
$\bar{H}$	differential heat of adsorption, FL/M
H_G	enthalpy of a gas, solute-free basis, FL/M
H_S	enthalpy of solid and adsorbed solute, per unit mass of adsorbate-free solid, FL/M
H_{IG}	height of a gas transfer unit, L
H_{IOG}	height of an overall gas transfer unit, L
H_{IS}	height of a solid transfer unit, L
ΔH_A	integral heat of adsorption referred to liquid adsorbate, per unit mass of adsorbent, FL/M
$\Delta H'_A$	integral heat of adsorption referred to vapor adsorbate, per unit mass of adsorbent, FL/M
J	mass-transfer flux, mole/ $L^2\Theta$
k	a constant
k_L	liquid-phase mass-transfer coefficient, mole/ $L^2\Theta$ (mole/ L^3)
k_S	solid-phase mass-transfer coefficient, mole/ $L^2\Theta$ (mole/ L^3)
k_Y	gas-phase mass-transfer coefficient, $M/L^2\Theta(M/M)$
K_Y	overall gas-phase mass-transfer coefficient, mole/ $L^2\Theta$ (mole fraction)
K_Y	overall gas-phase mass-transfer coefficient, $M/L^2\Theta(M/M)$
l	a distance, L
L_S	solvent liquid: mass in a batch process, M ; mass velocity in a continuous process, $M/L^2\Theta$
LUB	length of unused portion of fixed bed, L
m	slope of equilibrium adsorption isotherm [Eqs. (11.51) and (11.52)], dY^*/dX , dimensionless; also (mole/ L^3)/(mole/ M) or (M/L^3)/(M/M)
	const
M	molecular weight, M/mole
n	const
	number of impellers on a shaft, dimensionless
N	rotational speed, Θ^{-1} flux of mass transfer based on external solid surface, mole/ $L^2\Theta$
	mass adsorbent/mass adsorbate (fractionation), M/M
N_{IOG}	number of overall gas transfer units, dimensionless
p	equilibrium vapor pressure, F/L^2
$\bar{p}$	partial pressure, F/L^2
P	agitator power transmitted to fluid or slurry, FL/Θ

P_E	mass velocity of C-rich product, $M/L^2\Theta$
Po	power number, $P_{Gc}/d^3N^3\rho_m$, dimensionless
r	molecular weight poorly adsorbed gas/molecular weight strongly adsorbed gas, dimensionless
R	mass velocity of reflux to fractionator, $M/L^2\Theta$
	universal gas constant, $FL/\text{mole } T$
Re	impeller Reynolds number, $d^2N\rho_L/\mu_L$, dimensionless
Re_p	particle Reynolds number [Eq. (11.28)], dimensionless
S_S	adsorbate-free solid: mass in a batch process, M ; mass velocity in a continuous process, $M/L^2\Theta$
Sc_L	liquid-phase Schmidt number, $\mu_L/\rho_L D_L$, dimensionless
Sh_L	liquid-phase Sherwood number, $k_L d_p/D_L$, dimensionless
T	absolute temperature, T
	tank diameter, L
u'	turbulent fluctuating velocity, L/Θ
U	solute adsorbed in adsorption zone, $M/L^2\Theta$
v	volume of solute/mass adsorbed, L^3/M
v_L	liquid volume in a batch process, L^3 ; volume rate of liquid in a continuous process, L^3/Θ
v_m	slurry volume, L^3
v_T	volume of liquid in a depth equal to the tank diameter, L^3
V	velocity of advance of adsorption plane, L/Θ
V_{IN}	terminal settling velocity corresponding to Newton's law, L/Θ
V_{IS}	terminal settling velocity corresponding to Stokes' law, L/Θ
w	quantity of effluent from a fixed-bed adsorber, M/L^2
w_a	$w_E - w_B$, M/L^2
w_B	quantity of effluent from a fixed-bed adsorber at breakpoint, M/L^2
w_E	quantity of effluent from a fixed-bed adsorber at bed exhaustion, M/L^2
x	weight fraction of component C in fluid stream, dimensionless
X	adsorbate concentration, mass solute/mass adsorbent, M/M ; for ion exchange, equivalent/ M or equivalent/ L^3
X_T	adsorbate concentration in equilibrium with entering fluid, mass solute/mass adsorbent, M/M ; for ion exchange, equivalent/ M or equivalent/ L^3
y	weight fraction component C in adsorbate, dimensionless
Y	concentration of solute in fluid, mass solute/mass solvent, M/M
Y_B	concentration of solute in effluent at breakpoint, mass solute/mass solvent, M/M
Y_E	concentration of solute in effluent at bed exhaustion, mass solute/mass solid, M/M
Z	height of adsorbent column, L
Z'	height above the midplane of uppermost impeller, L
Z_a	height of adsorption zone, L
Z_s	length of adsorber bed in equilibrium with feed, L
α	mass-action-law constant or relative adsorptivity, dimensionless
Δ	difference
θ	time, Θ
θ_a	time required for the adsorption zone to move a distance Z_a , Θ
θ_E	time required to reach bed exhaustion, Θ
θ_F	time of formation of adsorption zone, Θ
θ_i	time to idealized breakthrough (Fig. 11.47), Θ
θ_T	time to bed saturation, Θ
λ	latent heat of vaporization, FL/M
μ	viscosity, $M/L\Theta$
μ_m	viscosity of slurry below Z' , $M/L\Theta$
ρ	fluid density, M/L^3
ρ_m	density of slurry below Z' , M/L^3

ρ_s	apparent density of adsorbent bed, mass solid/packed volume, M/L^3
$\Delta\rho$	absolute value of density difference, solid and liquid, M/L^3
φ_{sm}	volume fraction solids in slurry below Z' , dimensionless
φ_{ss}	volume fraction solids in fully settled bed, dimensionless
φ_{sT}	volume fraction solids based on v_T , dimensionless

Subscripts

A	weakly adsorbed solute; adsorbate
C	strongly adsorbed solute
e	within the enriching section of a continuous fractionator
L	liquid
min	minimum
n	stage n
p	particle
r	reference substance
s	within the adsorption section of a continuous fractionator
S	solid; adsorbent
0	initial
1	stage 1; bottom of a continuous-contact adsorber
2	stage 2; top of a continuous-contact adsorber

Superscript

*	at equilibrium
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PROBLEMS

11.1 The equilibrium adsorption of acetone vapor on an activated carbon at 30°C is given by the following data:

g adsorbed/g carbon	0	0.1	0.2	0.3	0.35
Partial pressure acetone, mmHg	0	2	12	42	92

The vapor pressure of acetone at 30°C is 283 mmHg.

A 1-l flask contains air and acetone vapor at 1 std atm and 30°C, with a relative saturation of the vapor of 35%. After 2 g of fresh activated carbon has been introduced into the flask, the flask is sealed. Compute the final vapor concentration at 30°C and the final pressure. Neglect the adsorption of air.

11.2 A solution of washed, raw cane sugar, 48% sucrose by weight, is colored by the presence of small quantities of impurities. It is to be decolorized at 80°C by treatment with an adsorptive carbon in a contact filtration plant. The data for an equilibrium adsorption isotherm were obtained by adding various amounts of the carbon to separate batches of the original solution and observing the equilibrium color reached in each case. The data, with the quantity of carbon expressed on the basis of the sugar content of the solution, are as follows:

kg carbon/kg dry sugar	0	0.005	0.01	0.015	0.02	0.03
Color removed, %	0	47	70	83	90	95

The original solution has a color concentration of 20, measured on an arbitrary scale, and it is desired to reduce the color to 2.5% of its original value.

(a) Convert the equilibrium data to Y^* = color units/kg sugar, X = color units/kg carbon. Do they follow the Freundlich equation? If so, what are the equation constants?

(b) Calculate the necessary dosage of fresh carbon, per 1000 kg of solution, for a single-stage process. **Ans.:** 20.4 kg.

(c) Calculate the necessary carbon dosages per 1000 kg of solution for a two-stage crosscurrent treatment, using the minimum total amount of fresh carbon. **Ans.:** 10.54 kg.

(d) Calculate the necessary carbon dosage per 1000 kg of solution for a two-stage countercurrent treatment. **Ans.:** 6.24 kg.

11.3 The sugar refinery of Prob. 11.2 must treat also a raw cane sugar solution, 48 wt % sucrose, of original color 50, based on the same color scale used in Prob. 11.2. The color scale is such that colors are additive, i.e., equal weights of solution of color 20 and color 50 will give a solution of color $(20 + 50)/2 = 35$. The same adsorption isotherm describes the color removal of the darker solution as that of Prob. 11.2. Equal quantities of the dark solution and that of Prob. 11.2 must both be decolorized to a color 0.5.

(a) In a single-stage process, will it be more economical of carbon first to blend the original solutions and to treat the blend, or to treat each separately to color 0.5 and to blend the finished products?

(b) Repeat for a two-stage crosscurrent treatment, fresh carbon in each stage, arranged for the minimum carbon in each case.

(c) Repeat for a countercurrent two-stage treatment.

(d) The following treating scheme was suggested. The light-colored solution is to be treated in a two-stage countercurrent plant to the final desired color. The spent carbon from this operation is to be used to treat an equal weight of the dark solution, and the carbon is then revived. The

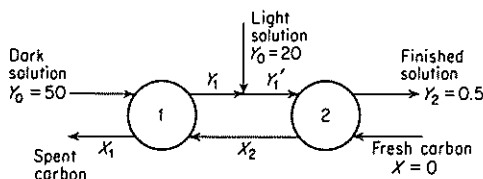


Figure 11.49 Flowsheet for Prob. 11.3e.

residual dark solution is then finished to the desired final color with the necessary amount of fresh carbon. Sketch a flowsheet and an operating diagram (freehand) for the entire process. Determine whether there is any saving of carbon over that for the arrangement of part (c).

(e) Determine whether the scheme of Fig. 11.49 offers any economies of carbon.

11.4 Prove that for crosscurrent two-stage treatment of liquid solutions by contact filtration, when the adsorption isotherm is linear, the least total adsorbent results if the amounts used in each stage are equal.

11.5 For adsorption from dilute liquid solutions in stagewise countercurrent operations, where the Freundlich equation describes the adsorption equilibrium, derive analytical expressions in terms of n , m , Y_0 , and Y_N for the minimum adsorbent/solvent ratio when fresh adsorbent is used.

11.6 A batch of water containing residual chlorine from a treating process, at a concentration 12 ppm, is to be treated with activated carbon at 25°C to reduce the chlorine concentration to 0.5 ppm. The carbon consists of 30-mesh granules, density = 561 kg/m³ (35 lb/ft³) = mass of particle/gross volume of particle. Adsorbate diffusional resistance is expected to be small relative to that in the liquid. The equilibrium distribution coefficient = $c^*/X = 0.80$ (kg Cl₂/m³ liquid)/(kg Cl₂/kg C) = 0.05 (lb Cl₂/ft³ liquid)/(lb Cl₂/lb C).

(a) Calculate the minimum mass of carbon/unit vol water which can be used. Ans.: 18.4 kg/m³.

(b) A batch of 2 m³ (528 U.S. gal) is to be contacted with 40 kg (88 lb) of the carbon in an agitated vessel. Specify the dimensions of the vessel, choose a suitable impeller and agitator speed, and estimate the required contact time for the concentration change specified above.

11.7 Refer to Illustration 11.5. (a) For the same concentration of Cu²⁺ in the effluent, what is the minimum solids rate that could be used? Ans.: 7.56×10^{-4} kg/s.

(b) If 0.0025 kg/s (0.33 lb/min) resin were used in the vessel with the same agitator speed and the same effluent concentration of Cu²⁺ in the water as in the illustration, what is the maximum volume rate of water that can be treated? Ans. 3.2×10^{-4} m³/s.

11.8 Nitrogen dioxide, NO₂, produced by a thermal process for fixation of nitrogen is to be removed from a dilute mixture with air by adsorption on silica gel in a continuous countercurrent adsorber. The gas entering the adsorber at the rate of 0.126 kg/s contains 1.5% NO₂ by volume, and 90% of the NO₂ is to be removed. Operation is to be isothermal at 25°C, 1 std atm. The entering gel will be free of NO₂. The equilibrium adsorption isotherm at this temperature is given by the following data [Foster and Daniels, *Ind. Eng. Chem.*, 43, 986 (1951)]:

Partial pressure NO ₂ , mmHg	0	2	4	6	8	10	12
kg NO ₂ /100 kg gel	0	0.4	0.9	1.65	2.60	3.65	4.85

(a) Calculate the minimum weight of gel required per hour.

(b) For twice the minimum gel rate, calculate the number of transfer units required.

(c) A superficial air rate of 0.407 kg/m² · s is to be used. Assume that the characteristics of the gel are the same as those described in Illustration 11.6. Modify the gas mass-transfer coefficient of Illustration 11.6 so that it will apply to the transfer of NO₂ rather than water. Modify the solid-phase mass-transfer coefficient to apply for NO₂ on the assumption that the transfer in the pores of the solid is by molecular diffusion through the gas filling the pores. The diffusivity of NO₂ in air is estimated to be 1.36×10^{-5} m²/s at 25°C, 1 std atm.

Estimate the value of H_{iOG} and calculate the corresponding height of the adsorber.

Ans.: 3 m.

11.9 Lewis et al., *J. Am. Chem. Soc.*, 72, 1157 (1950), report the following for the simultaneous adsorption of acetylene and ethylene from mixtures of the two on silica gel at 1 std atm, 25°C

(reprinted with permission of the American Chemical Society):

Mole fraction ethylene in adsorbate	Mole fraction ethylene in gas, at equilibrium	Gram moles mixture adsorbed/kg adsorbent
0.0686	0.2422	1.622
0.292	0.562	1.397
0.458	0.714	1.298
0.592	0.814	1.193
0.630	0.838	1.170
0.864	0.932	1.078

A gas containing equimolar amounts of acetylene and ethylene is to be fractionated in a continuous countercurrent adsorber, to yield products containing 98 and 2% acetylene by volume. Assume the temperature to remain constant at 25°C and the pressure to be 1 std atm. Calculate the number of transfer units and the gel circulation rate per 1000 m³ feed gas, using 1.2 times the minimum gel circulation rate. Ans.: $N_{T,OG} = 15.3$.

11.10 The sulfur content of an oil is to be reduced by percolation through a bed of adsorbent clay. Laboratory tests with the clay and oil in a representative percolation filter show the following instantaneous sulfur contents of the effluent oil as a function of the total oil passing through the filter [adapted from Kaufman, *Chem. Met. Eng.*, 30, 153 (1924)]:

bbl oil/ton clay	0	10	20	50	100	200	300	400
10 ³ m ³ oil/kg clay	0	1.752	3.504	8.760	17.52	35.04	52.56	70.08
Sulfur	0.011	0.020	0.041	0.067	0.0935	0.118	0.126	0.129

Assume that the specific gravity of the oil is unchanged during the percolation. The untreated oil has a sulfur content of 0.134%, and a product containing 0.090% sulfur is desired.

(a) If the effluent from the filter is composited, what yield of satisfactory product can be obtained per ton of clay? Ans.: 240 bbl/ton = 0.042 m³/kg.

(b) If the effluent from the filter is continually and immediately withdrawn and blended with just sufficient untreated oil to give the desired sulfur content in the blend, what quantity of product can be obtained per ton of clay? Ans.: 159.4 bbl/ton = 0.0279 m³/kg.

11.11 A laboratory fixed-bed adsorption column filled with a synthetic sulfonic acid cation-exchange resin in the acid form is to be used to remove Na⁺ ions from an aqueous solution of sodium chloride. The bed depth is 33.5 cm, and the solution to be percolated through the bed contains 0.120 meq Na⁺/cm³. At saturation, the resin contains 2.02 meq Na⁺/cm³ resin. The solution will be passed through the bed at a superficial linear velocity of 0.31 cm/s. For this resin, Michaels [70] reports that the overall liquid mass-transfer rate $K_L a_p = 0.86 v_L^{0.5}$, where v_L is the superficial liquid velocity, cm/s, and $K_L a_p$ is expressed as meq Na⁺/cm³ · s · (meq/cm³). The relative adsorptivity of Na⁺ with respect to H⁺ for this resin is $\alpha = 1.20$, and this is constant for the prevailing concentration level. Define the breakpoint concentration as 5% of the initial solution concentration, and assume that practical bed exhaustion occurs when the effluent concentration is 95% of the initial. Estimate the volume of effluent at the breakpoint, per unit bed cross section.

Ans.: 360 cm³/cm².

Note: For these circumstances, Michaels [70] observed that the adsorption-zone height was 23.8 cm, that the breakpoint occurred after 382 ± 10 cm³ effluent/cm² bed cross section was collected, and that the holdup of liquid in the bed was 14.5 ± 2.5 cm³ solution/cm² bed cross section. Compare the calculated results with these.

DRYING

The term *drying* refers generally to the removal of moisture from a substance. It is so loosely and inconsistently applied that some restriction in its meaning is necessary in the treatment to be given the subject here. For example, a wet solid such as wood, cloth, or paper can be dried by evaporation of the moisture either into a gas stream or without the benefit of the gas to carry away the vapor, but the mechanical removal of such moisture by expression or centrifuging is not ordinarily considered drying. A solution can be "dried" by spraying it in fine droplets into a hot, dry gas, which results in evaporation of the liquid, but evaporation of the solution by boiling in the absence of a gas to carry away the moisture is not ordinarily considered a drying operation. A liquid such as benzene can be "dried" of any small water content by an operation which is really distillation, but the removal of a small amount of acetone by the same process would not usually be called drying. Gases and liquids containing small amounts of water can be dried by adsorption operations, as discussed in Chap. 11. This discussion will be largely limited to the removal of moisture from solids and liquids by evaporation into a gas stream. In practice, the moisture is so frequently water and the gas so frequently air that this combination will provide the basis for most of the discussion. It is important to emphasize, however, that the equipment, techniques, and relationships are equally applicable to other systems.

EQUILIBRIUM

The moisture contained in a wet solid or liquid solution exerts a vapor pressure to an extent depending upon the nature of the moisture, the nature of the solid,

and the temperature. If then a wet solid is exposed to a continuous supply of fresh gas containing a fixed partial pressure of the vapor $\bar{p}$, the solid will either lose moisture by evaporation or gain moisture from the gas until the vapor pressure of the moisture of the solid equals $\bar{p}$. The solid and the gas are then in equilibrium, and the moisture content of the solid is termed its *equilibrium-moisture content* at the prevailing conditions.

Insoluble Solids

A few typical equilibrium-moisture relationships are shown in Fig. 12.1, where the moisture in each case is water. Here the equilibrium partial pressure $\bar{p}$ of the water vapor in the gas stream has been divided by the vapor pressure of pure water p to give the relative saturation, or relative humidity (see Chap. 7), of the gas, since the curves are then applicable over a modest range of temperatures instead of being useful for one temperature only. Consider the curve for wood. If the wood contained initially a very high moisture content, say 0.35 kg water/kg dry solid, and were exposed to a continuous supply of air of 0.6 relative humidity, the wood would lose moisture by evaporation until its equilibrium concentration corresponding to point *A* on the curve was eventually reached. Further exposure to this air, for even indefinitely long periods, would not bring about additional loss of moisture from the solid. The moisture content could be reduced further, however, by exposure of the solid to air of lower

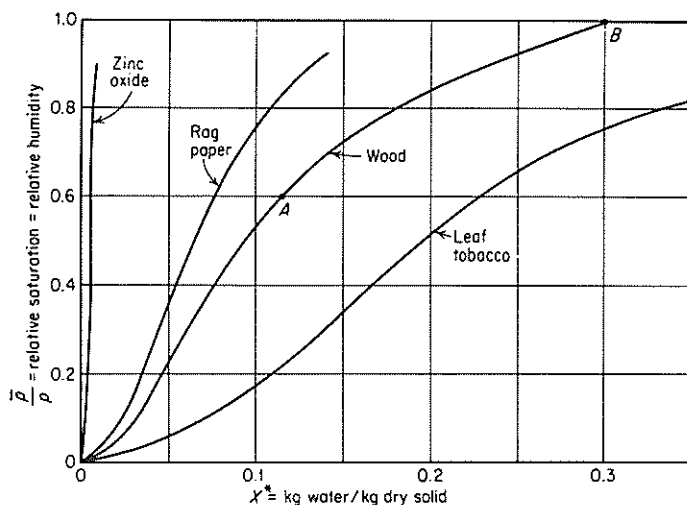


Figure 12.1 Equilibrium water content of some common solids at about 25°C. (From "International Critical Tables," vol. II, pp. 322–325, with permission.)

relative humidity, but to remove all the moisture would require exposure to perfectly dry air, corresponding to the origin of the curve. The moisture contained in the wood up to a concentration corresponding to point *B* in the figure, which exerts a vapor pressure less than that of pure water, may be moisture contained inside the cell walls of the plant structure, moisture in loose chemical combination with the cellulosic material, moisture present as a liquid solution of soluble portions of the solid and as a solid solution, or moisture held in small capillaries and crevasses throughout the solid or otherwise adsorbed upon the surface. Such moisture is called *bound water*. If exposed to saturated air, the wood may have any moisture content greater than 0.3 kg/kg dry solid (point *B*), and moisture in excess of that at *B*, *unbound water*, exerts the vapor pressure of pure water at the prevailing temperature.

The equilibrium moisture for a given species of solid may depend upon the particle size or specific surface, if the moisture is largely physically adsorbed. Different solids have different equilibrium-moisture curves, as shown in the figure. Generally, inorganic solids which are insoluble in the liquid and which show no special adsorptive properties, such as the zinc oxide in the figure, show relatively low equilibrium-moisture content, while spongy, cellular materials, especially those of vegetable origin such as the tobacco in the figure, generally show large equilibrium-moisture content. The equilibrium partial pressure for a solid is independent of the nature of the dry gas provided the latter is inert to the solid and is the same in the absence of noncondensable gas also. The same solids, if wet with liquids other than water, will show different equilibrium curves. The effect of large changes in temperature can frequently be shown in the form of a reference-substance plot, as in Fig. 11.5 [57]. It is seen that the equilibrium moisture is similar in many respects to the adsorption equilibria discussed in Chap. 11.

Hysteresis

Many solids exhibit different equilibrium-moisture characteristics depending upon whether the equilibrium is reached by condensation (adsorption) or evaporation (desorption) of the moisture. A typical example is shown in Fig. 12.2, which somewhat resembles the curve of Fig. 11.4. In drying operations, it is the desorption equilibrium which is of particular interest, and this will always show the larger of the two equilibrium-moisture contents for a given partial pressure of vapor. The moisture picked up by a dry solid when exposed to moist air, i.e., the adsorption equilibrium, is sometimes called *regain*, and knowledge of this has practical value in the consideration of drying operations. For example, in the case of Fig. 12.2, it will be of little use to dry the solid to a water content below that corresponding to point *A* if it is expected to expose the dried material to air of 0.6 relative humidity later. If it is important that the solid be kept at a lower moisture content, it would have to be packaged or stored immediately out of contact with the air in a moisture-impervious container.

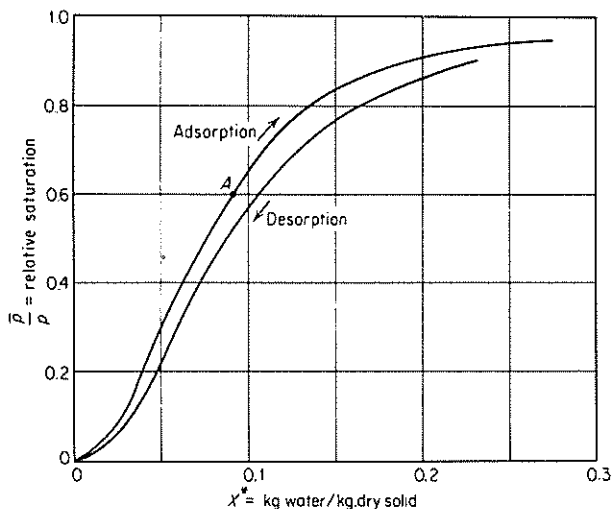


Figure 12.2 Equilibrium water content of a sulfite pulp, showing hysteresis. [Seborg, *Ind. Eng. Chem.*, 29, 160 (1937).]

Soluble Solids

Solids which are soluble in the liquid in question ordinarily show insignificant equilibrium-moisture content when exposed to gases whose partial pressure of vapor is less than that of the saturated solution of the solid. Refer to Fig. 12.3, where the characteristics of sodium nitrate–water are shown. A saturated solution of sodium nitrate in water at 25°C exerts a partial pressure of water (B) equal to 17.7 mmHg, and more dilute solutions exert higher partial pressures, as shown by curve BC. When exposed to air containing a partial pressure of water less than 17.7 mmHg, a solution will evaporate and the residual solid will retain only a negligible amount of adsorbed moisture, as shown by the curve from the origin to point A, and will appear dry. If the solid is exposed to air containing a higher water-vapor content, say 20 mmHg, moisture will be adsorbed to such an extent that the solid will completely dissolve, or *deliquesce*, to produce the corresponding solution at C. Solids of very low solubility, when exposed to ordinary atmospheric air, will not deliquesce since the equilibrium partial pressure of their saturated solutions is greater than that ordinarily found in the air.

Hydrated crystals may show more complicated relationships, such as those of Fig. 12.4 for the system copper sulfate–water at 25°C. Three hydrates are formed in this system, as the figure indicates. The anhydrous salt shows a negligible equilibrium-moisture content, which would in any case consist merely of adsorbed water upon the surface of the crystals. If exposed to air containing a partial pressure of water less than 7.8 and more than 5.6 mmHg, the salt will

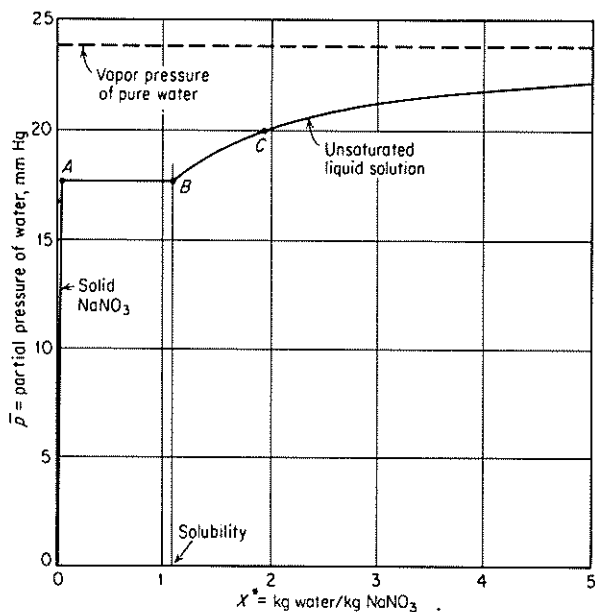


Figure 12.3 Equilibrium moisture content of sodium nitrate at 25°C.

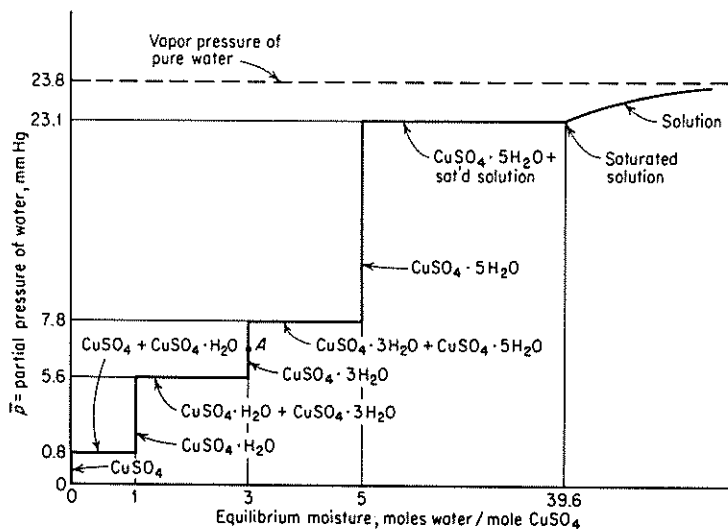


Figure 12.4 Equilibrium moisture of copper sulfate at 25°C (not to scale).

take on sufficient water to form the trihydrate and the crystals will have negligible adsorbed water other than the water of crystallization. The conditions will correspond to a point such as point *A* on the figure. If the moisture content of the air is then reduced to slightly less than 5.6 mmHg, the trihydrate will lose moisture (*effloresce*) to form the monohydrate, while at 5.6 mmHg any proportion of mono- and trihydrate can coexist. Similarly, if the moisture content of the air is increased to slightly more than 7.8 mmHg, additional moisture will be adsorbed until the pentahydrate is formed. If the moisture content of the air exceeds 23.1 mmHg, the salt will deliquesce.

Definitions

For convenient reference, certain terms used to describe the moisture content of substances are summarized below:

Moisture content, wet basis. The moisture content of a solid or solution is usually described in terms of weight percent moisture, and unless otherwise qualified this is ordinarily understood to be expressed on the wet basis, i.e., as $(\text{kg moisture/kg wet solid})100 = [\text{kg moisture}/(\text{kg dry solid} + \text{kg moisture})]100 = 100X/(1 + X)$.

Moisture content, dry basis. This is expressed as $\text{kg moisture/kg dry solid} = X$. Percentage moisture, dry basis = $100X$.

Equilibrium moisture X^ .* This is the moisture content of a substance when at equilibrium with a given partial pressure of the vapor.

Bound moisture. This refers to the moisture contained by a substance which exerts an equilibrium vapor pressure less than that of the pure liquid at the same temperature.

Unbound moisture. This refers to the moisture contained by a substance which exerts an equilibrium vapor pressure equal to that of the pure liquid at the same temperature.

Free moisture. Free moisture is that moisture contained by a substance in excess of the equilibrium moisture: $X - X^*$. Only free moisture can be evaporated, and the free-moisture content of a solid depends upon the vapor concentration in the gas.

These relations are shown graphically in Fig. 12.5 for a solid of moisture content X exposed to a gas of relative humidity A .

Illustration 12.1 A wet solid is to be dried from 80 to 5% moisture, wet basis. Compute the moisture to be evaporated, per 1000 kg of dried product.

SOLUTION

$$\text{Initial moisture content} = \frac{0.80}{1 - 0.80} = 4.00 \text{ kg water/kg dry solid}$$

$$\text{Final moisture content} = \frac{0.05}{1 - 0.05} = 0.0527 \text{ kg water/kg dry solid}$$

$$\text{Dry solid in product} = 1000(0.95) = 950 \text{ kg}$$

$$\text{Moisture to be evaporated} = 950(4 - 0.0527) = 3750 \text{ kg}$$

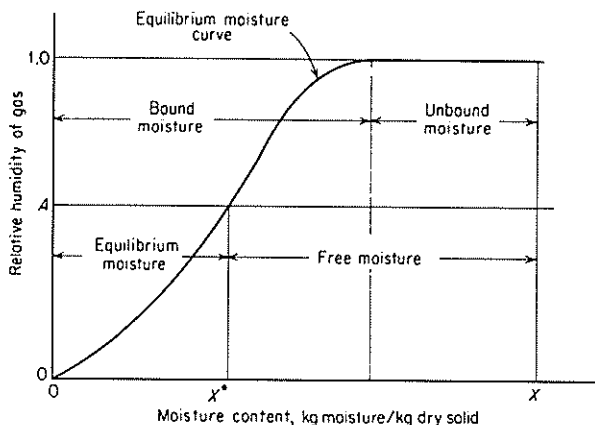


Figure 12.5 Types of moisture.

DRYING OPERATIONS

Drying operations can be broadly classified according to whether they are batch or continuous. These terms are applied specifically from the point of view of the substance being dried. Thus the operation termed *batch drying* is usually in fact a semibatch process wherein a quantity of the substance to be dried is exposed to a continuously flowing stream of air into which the moisture evaporates. In continuous operations, the substance to be dried as well as the gas passes continually through the equipment. No typically stagewise methods are ordinarily used, and all operations involve continuous contact of the gas and the drying substance.

Equipment used for drying can be classified according to equipment type [34] and the nature of the drying process [10]. The following classification is useful for purposes of outlining theories of drying and methods of design. Excellent discussions of equipment and theories of drying are also provided in Refs. 22, 40, and 50.

1. *Method of operation, i.e., batch or continuous.* Batch, or semibatch, equipment is operated intermittently or cyclically under unsteady-state conditions: the drier is charged with the substance, which remains in the equipment until dry, whereupon the drier is emptied and recharged with a fresh batch. Continuous driers are usually operated in steady-state fashion.
2. *Method of supplying the heat necessary for evaporation of the moisture.* In *direct* driers, the heat is supplied entirely by direct contact of the substance with the hot gas into which evaporation takes place. In *indirect* driers, the heat is supplied quite independently of the gas used to carry away the vaporized moisture. For example, heat may be supplied by conduction through a metal

wall in contact with the substance, or less frequently by exposure of the substance to infrared radiation or by dielectric heating. With the last, the heat is generated inside the solid by a high-frequency electric field.

3. *Nature of the substance to be dried.* The substance may be a rigid solid such as wood or fiberboard, a flexible material such as cloth or paper, a granular solid such as a mass of crystals, a thick paste or a thin slurry, or a solution. If it is a solid, it may be fragile or sturdy. The physical form of the substance and the diverse methods of handling necessarily have perhaps the greatest influence on the type of drier used.

BATCH DRYING

Drying in batches is a relatively expensive operation and is consequently limited to small-scale operations, to pilot-plant and development work, and to drying valuable materials whose total cost will be little influenced by added expense in the drying operation.

Direct Driers

The construction of such driers depends greatly upon the nature of the substance being dried. *Tray driers*, also called cabinet, compartment, or shelf driers, are used for drying solids which must be supported on trays, e.g., pasty materials such as wet filter cakes from filter presses, lumpy solids which must be spread upon trays, and similar materials. A typical device, shown schematically in Fig. 12.6, consists of a cabinet containing removable trays on which the solid to be

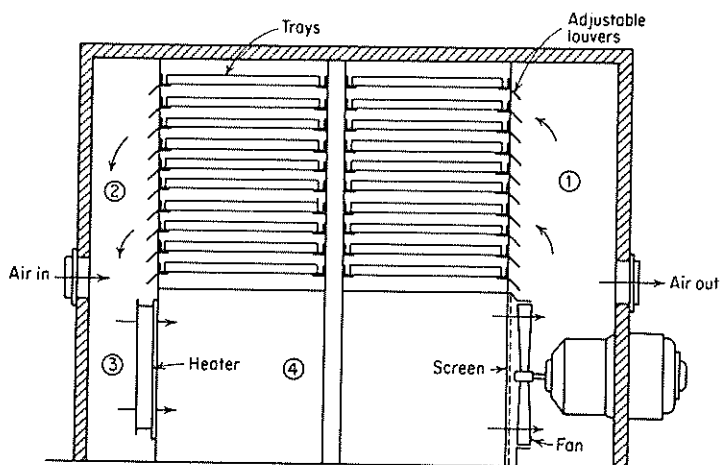


Figure 12.6 Typical tray drier. (Proctor and Schwartz, Inc.)

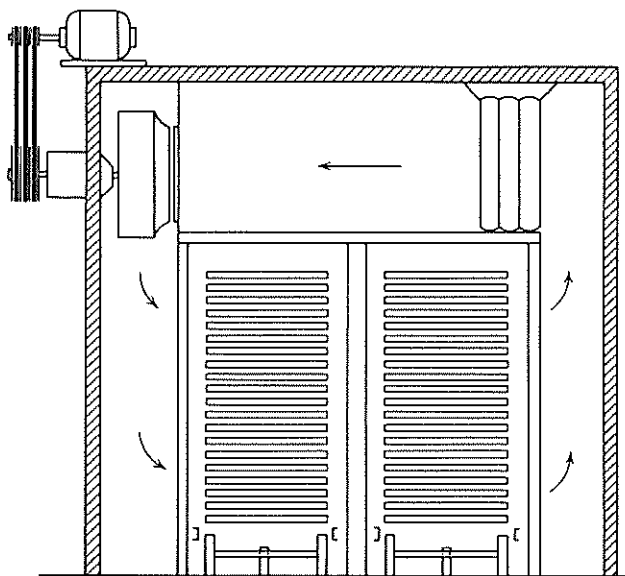


Figure 12.7 Two-truck drier. (Proctor and Schwartz, Inc.)

dried is spread. After loading, the cabinet is closed, and steam-heated air is blown across and between the trays to evaporate the moisture (cross-circulation drying). Inert gas, even superheated steam [7, 24] (which has the advantage of a large heat capacity) rather than air can be used if the liquid to be evaporated is combustible or if oxygen is damaging to the solid. When the solid has reached the desired degree of dryness, the cabinet is opened and the trays replaced with a new batch. Figure 12.7 shows a simple modification, a *truck drier*, where the trays are racked upon trucks which can be rolled into and out of the cabinet. Since the trucks can be loaded and unloaded outside the drier, considerable time can be saved between drying cycles. Other obvious modifications of the design are also used, depending upon the nature of the drying substance. Thus, skeins of fibers such as rayon can be hung from poles, and wood or boardlike materials can be stacked in piles, the layers separated from each other by spacer blocks.

With granular materials, the solid can be arranged in thin beds supported on screens so that air or other gas can be passed through the beds. This results in very much more rapid drying. A typical device for this purpose, a batch *through-circulation drier*, is shown schematically in Fig. 12.8. Crystalline solids and materials which are naturally granular such as silica gel can be dried in this manner directly. With others, some sort of preliminary treatment to put them into satisfactory form, *preforming*, is necessary. Pastes, e.g., those resulting from precipitation of pigments or other solids, can be preformed by (1) extrusion into short, spaghetti-like rods, (2) granulation, i.e., forcing through screens, or (3) briquetting [31].

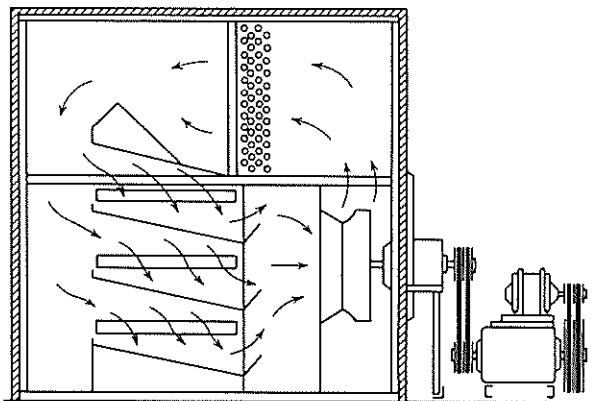


Figure 12.8 Through-circulation drier. (Proctor and Schwartz, Inc.)

One of the most important difficulties in the use of driers of the type described is the nonuniformity of moisture content found in the finished product taken from various parts of the drier. This is largely the result of inadequate and nonuniform air movement inside the drier. It is important to eliminate stagnant air pockets and to maintain reasonably uniform air humidity and temperature throughout the drier. In order to do this, large volumes of air must be blown over the trays, if possible at velocities ranging up to 3 or 4 m/s (10 or 20 ft/s) if the solid will not blow from the trays at these air rates. This can be accomplished by blowing large quantities of heated fresh air only once through the drier, but the loss of heat in the discharged air will then usually be prohibitive in cost. Instead, it is the practice to admit only relatively small quantities of fresh air and to recirculate the bulk of it, sometimes as much as 80 to 95 percent [21, 54]. This can be done inside the drier, as shown, for example, in Fig. 12.6, with dampers in the inlet and outlet pipes to regulate the extent of recirculation. The louvers at each tray level can then be adjusted to ensure as nearly uniform air velocity over each tray as possible. Alternatively, the heaters and fans can be installed outside the drier, with ductwork and dampers to permit more careful control of the relative amounts of fresh and recirculated air admitted to the drier itself. It is important also that the trays in such driers be filled level to the brim but not overloaded, so that uniform free space for air movement is available between trays.

The recirculation of large quantities of air necessarily raises the humidity of the air in the drier considerably above that of the fresh air. Low percentage humidity and consequently reasonably rapid drying rates are then obtained by using as high a temperature as practicable. The drier must then be thoroughly insulated, not only to conserve heat but also to maintain the inside walls at temperatures above the dew point of the air to prevent condensation of moisture upon the walls. Specially conditioned, low-humidity air is not used except where low-temperature drying is necessary to avoid damage to the product.

Illustration 12.2 The drier of Fig. 12.6 contains trays arranged in a tier of 10, each on racks 100 mm apart, each tray 38 mm deep and 1 m wide, with 14 m² of drying surface. The air entering the trays (position 1 in the figure) is to have a dry-bulb temperature of 95°C and humidity 0.05 kg water/kg dry air. Atmospheric air enters at 25°C, humidity 0.01. The air velocity over the trays at the entrance to them is to be 3 m/s. When the solid being dried is losing water at a constant rate of 7.5×10^{-3} kg/s, determine the percentage recirculation of air and the conditions of the air in the various parts of the drier.

SOLUTION At position 1, $Y_1 = 0.05$ kg water/kg dry air, $t_{G1} = 95^\circ\text{C}$, and the humid volume (Table 7.1) is $[0.00283 + 0.00456(0.05)](95 + 273) = 1.125$ m³/kg dry air.

Free area for flow between trays = $1[(100 - 38)/1000]11 = 0.682$ m².

Rate of air flow to trays = $3(0.682) = 2.046$ m³/s, or $2.046/1.125 = 1.819$ kg dry air/s (at position 1).

The rate of evaporation is 0.0075 kg water/s, and the humidity at 2 (Fig. 12.6) is therefore $0.05 + 0.0075/1.819 = 0.0541$ kg water/kg dry air. Assuming adiabatic drying, the temperature at 2 can be found on the adiabatic-saturation line (Fig. 7.5) drawn through the conditions at 1, and at $Y_2 = 0.0541$, $t_{G2} = 86^\circ\text{C}$.

The condition of the air at 4, and the discharged air, must be the same as at 1. An overall water balance about the drier therefore is

$$G(0.05 - 0.01) = 0.0075 \text{ kg water evaporated/s}$$

$$G = 0.1875 \text{ kg dry air/s enters and leaves}$$

The rate of airflow at 3 and 4 (Fig. 12.6) is therefore $1.819 + 0.1875 = 2.01$ kg dry air/s, and at position 4 the humid volume must be 1.125 m³/kg dry air. The volumetric rate through the fan is therefore $2.01/1.125 = 1.787$ m³/s. The percentage of air recycled is $(1.819/2.01)100 = 90.5\%$.

The enthalpy of air at 2 (and at 1) is 233 kJ/kg dry air (Fig. 7.5, saturated enthalpy at the adiabatic-saturation temperature, 46.5°C), and that of the fresh air is 50 kJ/kg dry air. Assuming complete mixing, the enthalpy of the air at 3 is, by an enthalpy balance, $[233(1.819) + 50(0.1875)]/2.01 = 215.5$ kJ/kg dry air. Since its humidity is 0.05, its dry-bulb temperature (Table 7.1) is $[215.500 - 2.502.300(0.05)]/[1005 + 1884(0.05)] = 82.2^\circ\text{C}$. The heater must apply $2.01(233 - 215.5) = 35.2$ kW, neglecting heat losses.

The dew-point temperature of the air (Fig. 7.5) at 1, 3, and 4 is 40.4°C, and at 2 it is 41.8°C. The drier should be well enough insulated to ensure that the inside surface will not fall to a temperature of 41.8°C.

The general humidity level in the drier can be altered during the drying cycle. This may be especially important in the case of certain solids which warp, shrink, develop surface cracks, or case harden when dried too rapidly. A cake of soap of high moisture content, for example, if exposed to very dry, hot air, will lose moisture by evaporation from the surface so fast that water will not move rapidly enough from the center of the cake to the surface to keep pace with the evaporation. The surface then becomes hard and impervious to moisture (case hardened), and drying stops even though the average water content of the cake is still very high. With other solids, e.g., wood, shrinkage at the surface may cause cracks or warping. Such substances should be dried slowly at first with air of high humidity, and drier air should be used only after the bulk of the water has been removed.

Driers of the type described are relatively cheap to build and have low maintenance costs. They are expensive to operate; however, owing to low heat economy and high labor costs. Each time the drier is opened for unloading and

loading, the temperature of the interior falls and all the metal parts of the drier must be heated again to the operating temperature when operation is resumed. Steam consumption for heating the air will generally not be less than 2.5 kg steam/kg water evaporated and may be as high as 10, especially when the moisture content of the product is reduced to very low levels [25]. The labor requirement for loading, unloading, and supervision of the drying cycle is high.

Indirect Driers

Vacuum shelf driers are tray driers whose cabinets, made of cast-iron or steel plates, are fitted with tightly closing doors so that they can be operated at subatmospheric pressure. No air is blown or recirculated through such driers. The trays containing the solid to be dried rest upon hollow shelves through which warm water or steam is passed to provide the necessary heat for vaporization of moisture. The heat is conducted to the solid through the metal of the shelves and trays. After loading and sealing, the air in the drier is evacuated by a mechanical vacuum pump or steam-jet ejector, and distillation of the moisture proceeds. The vapors usually pass to a condenser, where they are liquefied and collected, and only noncondensable gas is removed by the pump. *Agitated pan driers* [52], which can be used to dry pastes or slurries in small batches, are shallow, circular pans, 1 to 2 m in diameter and 0.3 to 0.6 m deep, with flat bottoms and vertical sides. The pans are jacketed for admission of steam or hot water for heating. The paste, or slurry, in the pan is stirred and scraped by a set of rotating plows, in order to expose new material to the heated surface. Moisture is evaporated into the atmosphere in atmosphere pan driers, or the pan may be covered and operated under vacuum. *Vacuum rotary driers* are steam-jacketed cylindrical shells, arranged horizontally, in which a slurry, or paste, can be dried in vacuum. The slurry is stirred by a set of rotating agitator blades attached to a central horizontal shaft which passes through the ends of the cylindrical shell. Vaporized moisture passes through an opening in the top to a condenser, and noncondensable gas is removed by a vacuum pump. The dried solid is discharged through a door in the bottom of the drier.

Driers of this category are expensive to build and to operate. Consequently they are used only for valuable materials which must be dried at low temperatures or in the absence of air to prevent damage, such as certain pharmaceutical products, or where the moisture to be removed is an expensive or poisonous organic solvent which must be recovered more or less completely.

Freeze drying (sublimation drying) Substances which cannot be heated even to moderate temperatures, such as foodstuffs and certain pharmaceuticals, can be dried by this method [27]. The substance to be dried is customarily frozen by exposure to very cold air and placed in a vacuum chamber, where the moisture sublimates and is pumped off by steam-jet ejectors or mechanical vacuum pumps. An alternative method of freezing is by flash vaporization of part of the moisture under vacuum, although foodstuffs which are not rigid in the unfrozen state may be damaged by this procedure. Some foods, e.g., beef, evidently

contain capillary channels, and the water vapor diffuses from the receding ice surface through these channels as drying proceeds [18]. In other cases diffusion through cell walls must occur. In any event, one of the major problems is to supply the heat necessary for sublimation: as the plane of sublimation recedes, heat must be driven through larger thicknesses of dried matter of poor thermal conductivity, requiring increasing temperature differences, which may damage the product. Alternatively, the heat can be introduced through the "back face," i.e., through the surface such as a shelf or tray on which the solid rests, opposite the surface through which the evaporated moisture leaves. This very substantially improves the rate of drying [11]. Radiant heat is sometimes used. Dielectric heat is a possibility although an expensive one: because of the high dielectric constant of water, the heat is liberated directly into the water. Still an additional method, useful for granular products, is through-circulation drying with air instead of pumping off the water by vacuum pump.

The Rate of Batch Drying

In order to set up drying schedules and to determine the size of equipment, it is necessary to know the time required to dry a substance from one moisture content to another under specified conditions. We shall also wish to estimate the influence that different drying conditions will have upon the time for drying. Our knowledge of the mechanism of drying is so incomplete that it is necessary with few exceptions to rely upon at least some experimental measurements for these purposes. Measurements of the rate of batch drying are relatively simple to make and provide much information not only for batch but also for continuous operation.

Drying tests The rate of drying can be determined for a sample of a substance by suspending it in a cabinet or duct, in a stream of air, from a balance. The weight of the drying sample can then be measured as a function of time. Certain precautions must be observed if the data are to be of maximum utility. The sample should not be too small. Further, the following conditions should resemble as closely as possible those expected to prevail in the contemplated large-scale operation: (1) the sample should be similarly supported in a tray or frame; (2) it should have the same ratio of drying to nondrying surface; (3) it should be subjected to similar conditions of radiant-heat transfer; and (4) the air should have the same temperature, humidity, and velocity (both speed and direction with respect to the sample). If possible, several tests should be made on samples of different thicknesses. The dry weight of the sample should also be obtained.

The exposure of the sample to air of constant temperature, humidity, and velocity constitutes drying under *constant drying conditions*.

Rate-of-drying curve From the data obtained during such a test, a curve of moisture content as a function of time (Fig. 12.9) can be plotted. This will be useful directly in determining the time required for drying larger batches under

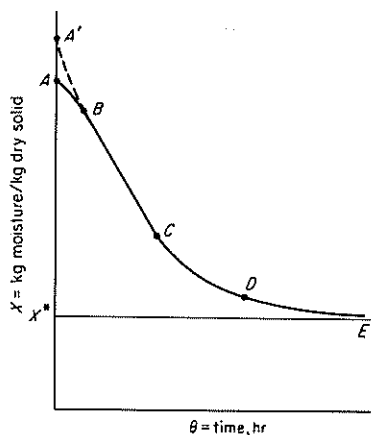


Figure 12.9 Batch drying, constant drying conditions.

the same drying conditions. Much information can be obtained if the data are converted into rates (or fluxes) of drying, expressed as N mass/(area)(time), and plotted against moisture content, as in Fig. 12.10. This can be done by measuring the slopes of tangents drawn to the curve of Fig. 12.9 or by determining from the curve small changes in moisture content ΔX for corresponding small changes in time $\Delta \theta$ and calculating the rate as $N = -S_g \Delta X / A \Delta \theta$. Here S_g is the mass

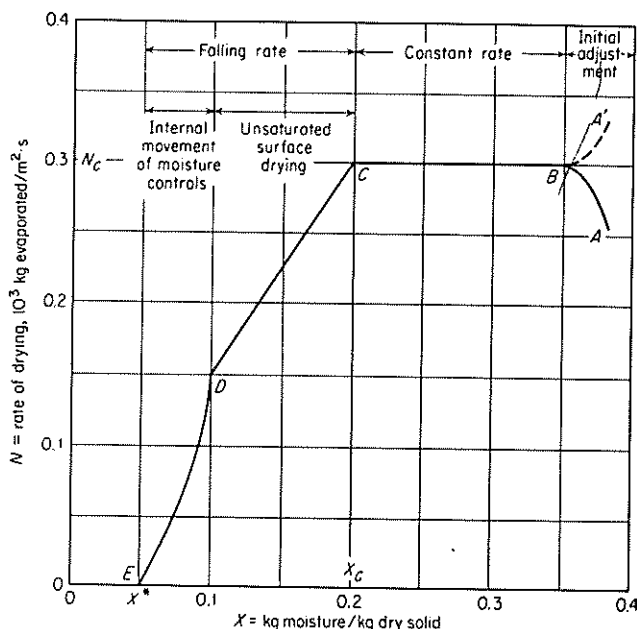


Figure 12.10 Typical rate-of-drying curve, constant drying conditions.

of dry solid, and A is the wet surface over which the gas blows and through which evaporation takes place in the case of cross-air circulation drying. In the case of through-circulation drying, A is the cross section of the bed measured at right angles to the direction of gas flow.

The rate-of-drying curve is sometimes plotted with the ordinate expressed as mass moisture evaporated/(mass dry solid)(time), which in the present notation is $-dX/d\theta$.

There are usually two major parts to the rate curve of Fig. 12.10, a period of constant rate and one of falling rate, as marked on the figure. While different solids and different conditions of drying often give rise to curves of very different shape in the falling-rate period, the curve shown occurs frequently. Some of the differences which may arise will be considered later, but for the present let us briefly review the reasons generally advanced for the various parts of the curve shown [16, 34, 48, 49].

If a solid is initially very wet, the surface will be covered with a thin film of liquid, which we shall assume is entirely unbound moisture. When it is exposed to relatively dry air, evaporation will take place from the surface. The rate at which moisture evaporates can be described in terms of a gas mass-transfer coefficient k_Y and the difference in humidity of the gas at the liquid surface Y_s and in the main stream Y . Thus, for cross-circulation drying

$$N_c = k_Y(Y_s - Y) \quad (12.1)$$

The coefficient k_Y can be expected to remain constant as long as the speed and direction of gas flow past the surface do not change. The humidity Y_s is the saturated humidity at the liquid-surface temperature t_s and will therefore depend upon this temperature. Since evaporation of moisture absorbs latent heat, the liquid surface will come to, and remain at, an equilibrium temperature such that the rate of heat flow from the surroundings to the surface exactly equals the rate of heat absorption. Y_s therefore remains constant. The capillaries and interstices of the solid, filled with liquid, can deliver liquid to the surface as rapidly as it evaporates there. Since in addition Y remains unchanged under constant drying conditions, the rate of evaporation must remain constant at the value N_c , as shown in Figs. 12.9 and 12.10 between points B and C . In the beginning, the solid and the liquid surface are usually colder than the ultimate surface temperature t_s , and the evaporation rate will increase while the surface temperature rises to its ultimate value during the period AB on these curves. Alternatively the equilibrium temperature t_s may be lower than the initial value, which will give rise to a curve $A'B$ while the initial adjustment occurs. The initial period is usually so short that it is ordinarily ignored in subsequent analysis of the drying times.

When the average moisture content of the solid has reached a value X_c , the *critical moisture content* (Fig. 12.10), the surface film of moisture has been so reduced by evaporation that further drying causes dry spots to appear upon the surface; these spots occupy increasingly larger proportions of the exposed surface as drying proceeds. Since, however, the rate N is computed by means of the constant gross surface A , the value of N must fall even though the rate per

unit of wet surface remains constant. This gives rise to the first part of the falling-rate period, the period of *unsaturated surface drying*, from points *C* to *D* (Figs. 12.9 and 12.10). Ultimately the original surface film of liquid will have entirely evaporated at an average moisture content for the solid corresponding to point *D*. This part of the curve may be missing entirely, or it may constitute the whole of the falling-rate period. With some textiles, other explanations for the linear falling-rate period have been necessary [39].

On further drying, the rate at which moisture can move through the solid, as a result of concentration gradients existing between the deeper parts and the surface, is the controlling step. As the moisture concentration generally is lowered by the drying, the rate of internal movement of moisture decreases. In some cases, evaporation may take place beneath the surface of the solid in a plane or zone which retreats deeper into the solid as drying proceeds. In any event, the rate of drying falls even more rapidly than before, as from *D* to *E* (Fig. 12.10). At point *E*, the moisture content of the solid has fallen to the equilibrium value X^* for the prevailing air humidity, and drying stops. The moisture distribution within the solid during the falling-rate period has been calculated and displayed graphically [17].

Time of drying If one wishes to determine the time of drying a solid under the same conditions for which a drying curve such as Fig. 12.9 has been completely determined, one need merely read the difference in the times corresponding to the initial and final moisture contents from the curve.

Within limits, it is sometimes possible to estimate the appearance of a rate-of-drying curve such as Fig. 12.10 for conditions different from those used in the experiments. In order to determine the time for drying for such a curve, we proceed as follows. The rate of drying is, by definition,

$$N = \frac{-S_s dX}{A d\theta} \quad (12.2)$$

Rearranging and integrating over the time interval while the moisture content changes from its initial value X_1 to its final value X_2 gives

$$\theta = \int_0^\theta d\theta = \frac{S_s}{A} \int_{X_2}^{X_1} \frac{dX}{N} \quad (12.3)$$

1. *The constant-rate period.* If the drying takes place entirely within the constant-rate period, so that X_1 and $X_2 > X_c$ and $N = N_c$, Eq. (12.3) becomes

$$\theta = \frac{S_s(X_1 - X_2)}{AN_c} \quad (12.4)$$

2. *The falling-rate period.* If X_1 and X_2 are both less than X_c , so that drying occurs under conditions of changing N , we proceed as follows:

- a. *General case.* For any shape of falling-rate curve whatsoever, Eq. (12.3) can be integrated graphically by determining the area under a curve of

$1/N$ as ordinate, X as abscissa, the data for which can be obtained from the rate-of-drying curve.

- b. *Special case.* N is linear in X , as in the region BC of Fig. 12.10. In this case,

$$N = mX + b \quad (12.5)$$

where m is the slope of the linear portion of the curve and b is a constant. Substitution in Eq. (12.3) provides

$$\theta = \frac{S_s}{A} \int_{X_2}^{X_1} \frac{dX}{mX + b} = \frac{S_s}{mA} \ln \frac{mX_1 + b}{mX_2 + b} \quad (12.6)$$

But since $N_1 = mX_1 + b$, $N_2 = mX_2 + b$, and $m = (N_1 - N_2)/(X_1 - X_2)$, Eq. (12.6) becomes

$$\theta = \frac{S_s(X_1 - X_2)}{A(N_1 - N_2)} \ln \frac{N_1}{N_2} = \frac{S_s(X_1 - X_2)}{AN_m} \quad (12.7)$$

where N_m is the logarithmic average of the rate N_1 , at moisture content X_1 , and N_2 at X_2 .

Frequently the entire falling-rate curve can be taken as a straight line between points C and E (Fig. 12.10). It is often assumed to be so for lack of more detailed data. In this case

$$N = m(X - X^*) = \frac{N_c(X - X^*)}{X_c - X^*} \quad (12.8)$$

and Eq. (12.7) becomes

$$\theta = \frac{S_s(X_c - X^*)}{N_c A} \ln \frac{X_1 - X^*}{X_2 - X^*} \quad (12.9)$$

In any particular drying problem, either or both constant- and falling-rate periods may be involved, depending upon the relative values of X_1 , X_2 , and X_c . The appropriate equations and limits must then be chosen.

Illustration 12.3 A batch of the solid for which Fig. 12.10 is the drying curve is to be dried from 25 to 6% moisture under conditions identical to those for which the figure applies. The initial weight of the wet solid is 160 kg, and the drying surface is $1 \text{ m}^2/40 \text{ kg dry weight}$. Determine the time for drying.

SOLUTION The total weight of the batch is unimportant. $S_s/A = 40$. At 25% moisture, $X_1 = 0.25/(1 - 0.25) = 0.333 \text{ kg moisture/kg dry solid}$. At 6% moisture, $X_2 = 0.06/(1 - 0.06) = 0.064 \text{ kg moisture/kg dry solid}$. Inspection of Fig. 12.10 shows that both constant- and falling-rate periods are involved. The limits of moisture content in the equations for the different periods will be chosen accordingly.

Constant-rate period This is from $X_1 = 0.333$ to $X_c = 0.200$. $N_c = 0.30 \times 10^{-3}$. Eq. (12.4):

$$\theta = \frac{S_s(X_1 - X_c)}{AN_c} = \frac{40(0.333 - 0.200)}{1(0.30 \times 10^{-3})} = 17\,730 \text{ s}$$

Falling-rate period This is from $X_c = 0.200$ to $X_2 = 0.064$. Use Eq. (12.3). The following table is prepared from data of Fig. 12.10:

x	0.20	0.18	0.16	0.14	0.12	0.10	0.09	0.08	0.07	0.064
$10^3 N$	0.300	0.266	0.239	0.208	0.180	0.150	0.097	0.070	0.043	0.025
$\frac{1}{N} \times 10^{-3}$	3.33	3.76	4.18	4.80	5.55	6.67	10.3	14.3	23.3	40.0

A curve, not shown, is prepared of $1/N$ as ordinate, X as abscissa, and the area under the curve between $X = 0.20$ and 0.064 is 1060. Eq. (12.3):

$$\theta = \frac{40}{1} (1060) = 42\,400 \text{ s}$$

The total drying time is therefore $17\,730 + 42\,400 = 60\,130 \text{ s} = 16.7 \text{ h}$.

Alternatively, since the drying curve is straight from $X = 0.20$ to 0.10 , Eq. (12.7) can be used in this range of moisture content,

$$\theta = \frac{S_s(X_c - X_D)}{A(N_c - N_D)} \ln \frac{N_c}{N_D} = \frac{40(0.20 - 0.10)}{1(0.30 - 0.15 \times 10^{-3})} \ln \frac{0.30 \times 10^{-3}}{0.15 \times 10^{-3}} = 18\,500 \text{ s}$$

Graphical integration in the range $X = 0.1$ to 0.064 , through Eq. (12.3), provides an additional $23\,940 \text{ s}$, so that the falling-rate time is $18\,500 + 23\,940 = 42\,440 \text{ s}$.

As an approximation, the falling rate can be represented by a straight line from C to E (Fig. 12.10). The corresponding falling-rate time is, by Eq. (12.9),

$$\theta = \frac{S_s(X_c - X^*)}{N_c A} \ln \frac{X_c - X^*}{X_2 - X^*} = \frac{40(0.20 - 0.05)}{(0.30 \times 10^{-3})(1)} \ln \frac{0.20 - 0.05}{0.064 - 0.05} = 47\,430 \text{ s}$$

THE MECHANISMS OF BATCH DRYING

We now consider the various portions of the rate-of-drying curve in more detail. Our present knowledge permits us to describe the drying process in the constant-rate period reasonably well, but our understanding of the falling-rate periods is very limited.

Cross-Circulation Drying

The constant-rate period In this period, where surface evaporation of unbound moisture occurs, it has been shown that the rate of drying is established by a balance of the heat requirements for evaporation and the rate at which heat reaches the surface. Consider the section of a material drying in a stream of gas as shown in Fig. 12.11. The solid of thickness z_s is placed on a tray of thickness z_M . The whole is immersed in a stream of drying gas at temperature T_G and humidity Y mass moisture/mass dry gas, flowing at a mass velocity G mass/(time) (area). The evaporation of moisture takes place from the upper surface, area A , which is at a temperature T_s . The drying surface receives heat from several sources: (1) q_c by convection from the gas stream; (2) q_k by conduction through the solid; (3) q_R by direct radiation from a hot surface at temperature T_R , as shown, all expressed as a flux, energy/(area of solid for heat transfer)

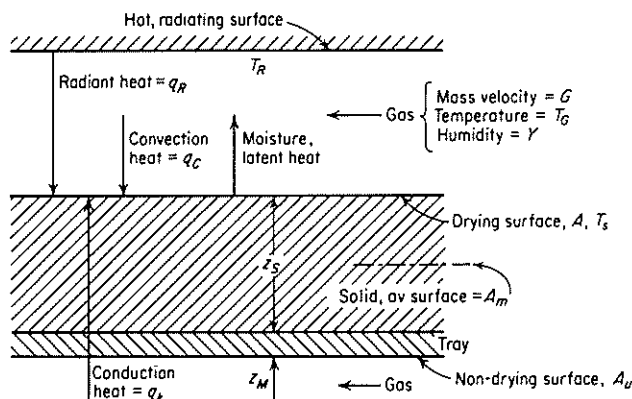


Figure 12.11 Constant-rate drying.

(time). In accordance with the mechanism discussed earlier, the heat arriving at the surface by these methods is removed by the evaporating moisture, so that the surface temperature remains constant at T_s . The entire mechanism resembles the wet-bulb-thermometer process, complicated by the additional source of heat.

The rate of evaporation and the surface temperature can then be obtained by a heat balance [46]. If q represents the total heat arriving at the surface, then

$$q = q_c + q_R + q_k \quad (12.10)$$

If we neglect the heat required to superheat the evaporated moisture to the gas temperature and consider only the latent heat of vaporization λ_s , the flux of evaporation N_c and the flux of heat flow are related,

$$N_c \lambda_s = q \quad (12.11)$$

The heat received at the surface by convection is controlled by the appropriate convection heat-transfer coefficient h_c ,

$$q_c = h_c (T_G - T_s) \quad (12.12)$$

The heat received by radiation can be estimated by the usual means [50, 55] and can also be expressed as a heat-transfer coefficient h_R ,†

$$q_R = \epsilon (5.729 \times 10^{-8}) (T_R^4 - T_s^4) = h_R (T_R - T_s) \quad (12.13)$$

$$h_R = \frac{\epsilon (5.729 \times 10^{-8}) (T_R^4 - T_s^4)}{T_R - T_s} \quad (12.14)$$

† Equations (12.13) and (12.14) are written for SI units: q_R in W/m^2 , T in kelvins, h_R in $\text{W/m}^2 \cdot \text{K}$. For q_R in $\text{Btu/ft}^2 \cdot \text{h}$, T in degrees Rankine, and h_R in $\text{Btu/ft}^2 \cdot \text{h} \cdot ^\circ\text{F}$ the constant is 1.730×10^{-9} . Note: The theoretical value of the Stefan-Boltzmann constant is $5.6693 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4$, but the accepted experimental value [50] is 5.729×10^{-8} .

where ϵ is the emissivity of the drying surface and T_R and T_s are the absolute temperatures of the radiating and drying surfaces. The heat received by convection and conduction through the solid can be computed by the usual methods for heat transfer through a series of resistances,

$$q_k = U_k(T_G - T_s) \quad (12.15)$$

$$U_k = \frac{1}{(1/h_c)(A/A_u) + (z_M/k_M)(A/A_u) + (z_S/k_S)(A/A_m)} \quad (12.16)$$

where h_c = convection coefficient for tray; ordinarily can be taken as same as that for drying surface

k_M, k_s = thermal conductivities of tray material and drying solid, respectively
 A_u, A_m = nondrying surface and average area of the drying solid, respectively
 A thermal resistance at the junction of the drying solid and the tray material and an effect of radiation to the tray can be added to the terms of Eq. (12.16) if desired.

Combining Eqs. (12.1) and (12.10) to (12.15) permits calculation of the rate of drying,

$$N_c = \frac{q}{\lambda_s} = \frac{(h_c + U_k)(T_G - T_s) + h_R(T_R - T_s)}{\lambda_s} = k_Y(Y_s - Y) \quad (12.17)$$

The surface temperature must be known in order to use the relationship. It can be obtained by consideration of the left-hand portions of Eq. (12.17), which can be rearranged to read

$$\frac{(Y_s - Y)\lambda_s}{h_c/k_Y} = \left(1 + \frac{U_k}{h_c}\right)(T_G - T_s) + \frac{h_R}{h_c}(T_R - T_s) \quad (12.18)$$

The ratio h_c/k_Y , applicable to flow of gases past wet-bulb thermometers [Eqs. (7.27) and (7.28)] can be used for present purposes, and for the system air-water vapor this ratio was shown to be substantially the same as the humid heat of the gas C_s . Since Y_s is the saturated humidity of the gas stream corresponding to T_s when unbound moisture is being evaporated, both these quantities can be found by solving Eq. (12.18) simultaneously with the saturated-humidity curve on a psychrometric chart.

If conduction through the solid and radiation effects are absent, Eq. (12.18) reduces to that for the wet-bulb thermometer [Eq. (7.26)] and the surface temperature is the wet-bulb temperature of the gas. The drying surfaces will also be at the wet-bulb temperature if the solid is dried from all surfaces in the absence of radiation. When pans or trays of drying material are placed one above another, as in Figs. 12.6 and 12.7, most of the solid will receive radiation only from the bottom of the pan immediately above it, and unless gas temperatures are very high, this is not likely to be very important. It is essential therefore not to overemphasize the heat received by radiation in conducting drying tests on single pairs of trays.

For flow of gas parallel to a surface and confined between parallel plates, as between the trays of a tray drier, the transfer coefficients h_c and k_Y are described

generally by item 3, Table 3.3. Application of the heat-mass-transfer analogy, for $Re_e = 2600$ to 22 000, results in

$$j_H = \frac{h_c}{C_p G} Pr^{2/3} = j_D = \frac{k_Y}{G_s} Sc^{2/3} = 0.11 Re_e^{-0.29} \quad (12.19)$$

where $Re_e = d_e G / \mu$ and d_e is the equivalent diameter of the airflow space. With the properties of air at 95°C, this becomes, in the SI†

$$h_c = 5.90 \frac{G^{0.71}}{d_e^{0.29}} \quad (12.20)$$

whereas in a detailed study of drying of sand in trays [46], h_c was given (SI) by

$$h_c = 14.3 G^{0.8} \quad (12.20a)$$

Such discrepancies can be assigned to a number of causes, e.g., different "calming" length ahead of the drying surface [41] or the shape of the leading edge of the drying surface [7]. In the absence of more specific information, Eqs. (12.19) and (12.20) are recommended.

Airflow perpendicular to the surface [37], for $G = 1.08$ to $5.04 \text{ kg/m}^2 \cdot \text{s}$ (0.9 to 4.5 m/s), in the SI,‡

$$h_c = 24.2 G^{0.37} \quad (12.21)$$

The relationships developed in Eqs. (12.17) to (12.21) permit direct estimates of the rate of drying during the constant-rate period, but they should not be considered as complete substitutes for experimental measurements. Perhaps their greatest value is in conjunction with limited experimental data in order to predict the effect of changing the drying conditions.

Effect of gas velocity. If radiation and conduction through the solid are negligible, N_c is proportional to $G^{0.71}$ for parallel flow of gas and to $G^{0.37}$ for perpendicular flow. If radiation and conduction are present, the effect of gas rate will be less important.

Effect of gas temperature. Increased air temperature increases the quantity $T_G - T_s$ and hence increases N_c . In the absence of radiation effects, if the variation of λ over moderate temperature ranges is neglected, N_c is directly proportional to $T_G - T_s$.

Effect of gas humidity. N_c varies directly as $Y_s - Y$, and consequently increasing the humidity lowers the rate of drying. Usually, changes in Y and T_G involve simultaneous changes in T_s and Y_s , and the effects are best estimated by direct application of Eq. (12.17).

Effect of thickness of drying solid. If heat conduction through the solid occurs, Eqs. (12.15) and (12.16) indicate lowered values of N_c with increased solid

† For G in $\text{lb/ft}^2 \cdot \text{h}$, d_e in feet, and h_c in $\text{Btu/ft}^2 \cdot \text{h} \cdot ^\circ\text{F}$ the coefficient of Eq. (12.20) is 0.0135.

‡ For $G = 800$ to $4000 \text{ lb/ft}^2 \cdot \text{h}$ (3 to 15 ft/s) and h_c in $\text{Btu/ft}^2 \cdot \text{h} \cdot ^\circ\text{F}$ the coefficient of Eq. (12.21) becomes 0.37.

thickness. However, conduction of heat through edge surfaces of pans and trays may be an important source of heat which can result in increased rate of drying if the edge surface is large. If nondrying surfaces are heat-insulated, or if drying occurs from all surfaces of the solid, N_c is independent of thickness. The time for drying between fixed moisture contents within the constant-rate period will then be directly proportional to thickness.

Illustration 12.4 An insoluble crystalline solid wet with water is placed in a rectangular pan 0.7 m (2.3 ft) by 0.7 m and 25 mm (1 in) deep, made of 0.8-mm-thick ($\frac{1}{32}$ -in) galvanized iron. The pan is placed in an airstream at 65°C, humidity 0.01 kg water/kg dry air, flowing parallel to the upper and lower surface at a velocity of 3 m/s (10 ft/s). The top surface of the solid is in direct sight of steam-heated pipes whose surface temperature is 120°C (248°F), at a distance from the top surface of 100 mm (4 in).

(a) Estimate the rate of drying at constant rate.

(b) Reestimate the rate if the pan is thoroughly heat-insulated and there is no radiation from the steam pipes.

SOLUTION (a) $Y = 0.01$ kg water/kg dry air, $t_G = 65^\circ\text{C}$. The humid volume of the air (Table 7.1) is $[0.00283 + 0.00456(0.01)](65 + 273) = 0.972 \text{ m}^3/\text{kg dry air}$.

$$\rho_G = \text{density of gas} = \frac{1.01}{0.972} = 1.04 \text{ kg/m}^3 \quad G = 3(1.04) = 3.12 \text{ kg/m}^2 \cdot \text{s}$$

$$\text{Estimated } d_c = \frac{4(\text{cross section for flow})}{\text{perimeter}} = \frac{4(0.7)(0.1)}{(0.7 + 0.1)2} = 0.175 \text{ m}$$

Eq. (12.20):

$$h_c = \frac{5.90(3.12)^{0.71}}{(0.175)^{0.29}} = 21.9 \text{ W/m}^2 \cdot \text{K}$$

Take the emissivity of the solid as $\epsilon = 0.94$. $T_R = 120 + 273 = 393 \text{ K}$. Tentatively estimate t_s as 38°C , $T_s = 38 + 273 = 311 \text{ K}$. Eq. (12.14):

$$h_R = \frac{0.94(5.73 \times 10^{-8})(393^4 - 311^4)}{393 - 311} = 9.51 \text{ W/m}^2 \cdot \text{K}$$

Take $A_m = A = (0.7)^2 = 0.49 \text{ m}^2$. The area of the sides of the pan $= 4(0.7)(0.025) = 0.07 \text{ m}^2$ and $A_u = (0.7)^2 + 0.07 = 0.56 \text{ m}^2$ for bottom and sides (this method of including heat transfer through the sides is admittedly an oversimplification but adequate for present purposes). Thermal conductivities are $k_M = 45$ for the metal of the pan and $k_S = 3.5$ for the wet solid [46], both as $\text{W/m} \cdot \text{K}$. The latter value must be carefully chosen, and it may bear no simple relation to the conductivity of either the dry solid or its moisture. $z_S = 0.025 \text{ m}$, $z_M = 0.0008 \text{ m}$. Eq. (12.16):

$$\frac{1}{U_k} = \frac{0.49}{21.9(0.56)} + \frac{0.0008}{45} \frac{0.49}{0.56} + \frac{0.025}{3.5} \frac{0.49}{0.49}$$

$$U_k = 21.2 \text{ W/m}^2 \cdot \text{K}$$

The humid heat of the air (Table 7.1) is $C_s = 1005 + 1884(0.01) = 1023.8$, and λ_s at the estimated 38°C is 2411.4 kJ/kg . Eq. (12.18):

$$\frac{(Y_s - 0.01)(2411.4 \times 10^3)}{1023.8} = \left(1 + \frac{21.2}{21.9}\right)(65 - t_s) + \frac{9.51}{21.9}(120 - t_s)$$

This reduces to $Y_s = 0.0864 - 10.194 \times 10^{-4}t_s$, which must be solved simultaneously with the saturated-humidity curve of the psychrometric chart for air–water vapor. The line marked *a* on

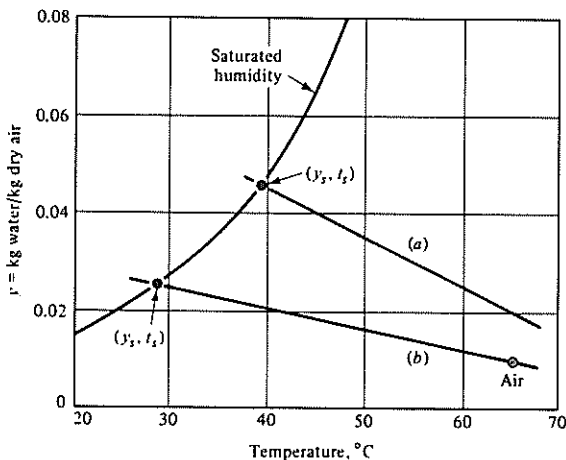


Figure 12.12 Solution to Illustration 12.4.

Fig. 12.12 is the above expression, which intersects the saturated-humidity curve at $Y_s = 0.0460$, $t_s = 39^\circ\text{C}$, the surface temperature, which is sufficiently close to the 38°C estimated previously to make iteration unnecessary. At 39°C , $\lambda_s = 2409.7 \text{ kJ/kg}$. Eq. (12.17):

$$N_c = \frac{(21.9 + 21.2)(65 - 39) + 9.51(120 - 39)}{2409.7 \times 10^3} = 7.85 \times 10^{-4} \text{ kg water evapd/m}^2 \cdot \text{s}$$

and the evaporation rate is $(7.85 \times 10^{-4})(0.49) = 3.85 \times 10^{-4} \text{ kg/s}$ or 3.06 lb/h .

(b) When no radiation or conduction of heat through the solid occurs, the drying surface assumes the wet-bulb temperature of the air. For the system air-water at this humidity, the adiabatic-saturation lines of the psychrometric chart serve as wet-bulb lines, and on Fig. 12.12, line b is the adiabatic-saturation line through the point representing the air ($t_G = 65^\circ\text{C}$, $Y = 0.01$). The line intersects the saturation-humidity curve at the wet-bulb condition, $t_s = 28.5^\circ\text{C}$, $Y_s = 0.025$. At this temperature, $\lambda_s = 2435.0 \text{ kJ/kg}$. Eq. (12.17):

$$N_c = \frac{h_c(t_G - t_s)}{\lambda_s} = \frac{21.9(65 - 28.5)}{2435 \times 10^3} = 3.28 \times 10^{-4} \text{ kg/m}^2 \cdot \text{s}$$

and the evaporation rate is $(3.28 \times 10^{-4})(0.49) = 1.609 \times 10^{-4} \text{ kg/s}$ or 1.277 lb/h .

When the air suffers a considerable change in temperature and humidity in its passage over the solid, as indicated in Illustration 12.2 for example, the rate of drying at the leading and trailing edge of the solid will differ and this accounts in part for the nonuniform drying frequently obtained in tray driers. This can be counteracted in part by periodic reversal of the airflow.

Movement of moisture within the solid When surface evaporation occurs, there must be a movement of moisture from the depths of the solid to the surface. The nature of the movement influences the drying during the falling-rate periods. In order to appreciate the diverse nature of the falling-rate portions of the drying curve which have been observed, let us review very briefly some of the theories

advanced to explain moisture movement and their relation to the falling-rate curves.

Liquid diffusion Diffusion of liquid moisture may result because of concentration gradients between the depths of the solid, where the concentration is high, and the surface, where it is low. These gradients are set up during drying from the surface. This method of moisture transport is probably limited to cases where single-phase solid solutions are formed with the moisture, as in the case of soap, glue, gelatin, and the like, and to certain cases where bound moisture is being dried, as in the drying of the last portions of water from clays, flour, textiles, paper, and wood [19]. The general mechanism of this process is described in Chap. 4. It has been found that the moisture diffusivity usually decreases rapidly with decreased moisture content.

During the constant-rate period of drying such solids, the surface-moisture concentration is reduced, but the concentration in the depths of the solid remains high. The resulting high diffusivities permit movement of the moisture to the surface as fast as it can be evaporated, and the rate remains constant. When dry spots appear because portions of the solid project into the gas film, a period of unsaturated surface evaporation results. The surface eventually dries to the equilibrium-moisture content for the prevailing gas. Further drying occurs at rates which are entirely controlled by the diffusion rates within the solid, since these are slow at low moisture contents. If the initial constant-rate drying is very rapid, the period of unsaturated surface evaporation may not appear, and the diffusion-controlled falling-rate period begins immediately after the constant-rate period is completed [47], as in Fig. 12.13.

For many cases of drying where the diffusion mechanism has satisfactorily explained the rate of drying as a function of average moisture content, the distribution of moisture within the solid at the various stages of drying has not conformed to this mechanism [19, 38]. The superficial applicability of the diffusion mechanism is then apparently accidental.

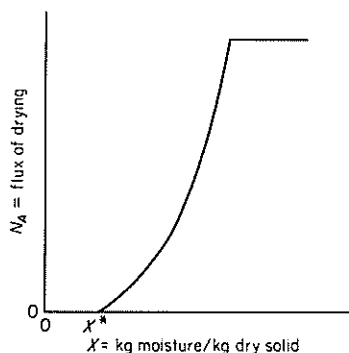


Figure 12.13 Diffusion-controlled falling rate.

Capillary movement [6, 19, 38, 42, 56] Unbound moisture in granular and porous solids such as clays, sand, paint pigments, and the like, moves through the capillaries and interstices of the solids by a mechanism involving surface tension, the way oil moves through a lamp wick. The capillaries extend from small reservoirs of moisture in the solid to the drying surface. As drying proceeds, at first moisture moves by capillarity to the surface rapidly enough to maintain a uniformly wetted surface and the rate of drying is constant. The water is replaced by air entering the solid through relatively few openings and cracks. The surface moisture is eventually drawn to spaces between the granules of the surface, the wetted area at the surface decreases, and the unsaturated-surface drying period follows. The subsurface reservoirs eventually dry up, the liquid surface recedes into the capillaries, evaporation occurs below the surface in a zone or plane which gradually recedes deeper into the solid, and a second falling-rate period results. During this period, diffusion of vapor within the solid will occur from the place of vaporization to the surface.

With certain pastes dried in pans, the adhesion of the wet cake to the bottom of the pan may not permit ventilation of the subsurface passageways by gas. This can give rise to curves of the sort shown in Fig. 12.14. In this case, the usual constant-rate period prevailed during *a*. When the surface moisture was first depleted, liquid could not be brought to the surface by the tension in the capillaries since no air could enter to replace the liquid, the surface of moisture receded into the capillaries, and the rate fell during *b*. The solid eventually crumpled, admitting air to replace the liquid, whereupon capillary action brought this to the surface and the rate rose again, as at *c*.

Vapor diffusion [42] Especially if heat is supplied to one surface of a solid while drying proceeds from another, the moisture may evaporate beneath the surface and diffuse outward as a vapor. Moisture particles in granular solids which have been isolated from the main portion of the moisture flowing through capillaries may also be evaporated below the surface.

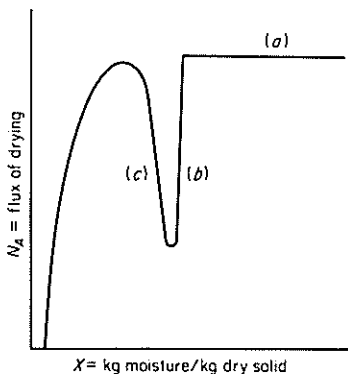


Figure 12.14 Effect of adhesion of drying paste to the pan. [After Ernst, et al., *Ind. Eng. Chem.*, 30 1119 (1938).]

Pressure Shrinkage of the outside layers of a solid on drying may squeeze moisture to the surface.

Usually we can only speculate about which mechanism is appropriate to a particular solid and must rely on more or less empirical treatment of the experimental rates of drying.

Unsaturated-surface drying During such a period the rate of drying N will usually vary linearly with moisture content X . Since the mechanism of evaporation during this period is the same as that in the constant-rate period, the effects of such variables as temperature, humidity, and velocity of the gas and thickness of the solid are the same as for constant-rate drying.

In some cases this period may constitute the whole of the falling-rate drying, giving rise to a curve of the type shown in Fig. 12.15. Equations (12.8) and (12.9) then apply. Combining Eqs. (12.2), (12.8), and (12.17) gives

$$-\frac{dX}{d\theta} = \frac{k_Y A (X - X^*) (Y_s - Y)}{S_S (X_c - X^*)} \quad (12.22)$$

Noting that $S_S = z_S \rho_S$, and letting $k_Y = f(G)$, we get

$$-\frac{dX}{d\theta} = \frac{f(G)(X - X^*)(Y_s - Y)}{z_S \rho_S (X_c - X^*)} = \frac{\alpha f(G)(X - X^*)(Y_s - Y)}{z_S} \quad (12.23)$$

where $\alpha = \text{const.}$ This expression is sometimes used as an empirical description of the rates of drying for such cases. Alternatively, when the time θ is defined as that when the moisture content is X , Eq. (12.9) is readily transformed into

$$\ln \frac{X - X^*}{X_1 - X^*} = \frac{-N_c \theta}{\rho_S z_S (X_c - X^*)} \quad (12.24)$$

which suggests that falling-rate data of this nature will plot as a straight line, line a , on the semilogarithmic coordinates of Fig. 12.16. If drying tests are made

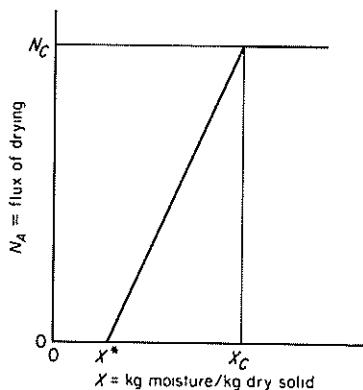


Figure 12.15 Linear falling rate.

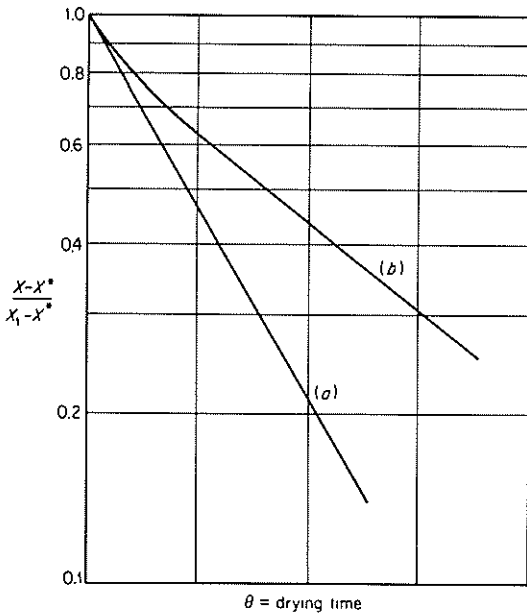


Figure 12.16 Empirical treatment of falling-rate data: (a) N varies linearly with $\log(X - X^*)$; (b) diffusion-controlled falling rate.

under the same conditions for samples of different thickness, and so that heat is applied to the solid only through the drying surface, the slopes of such lines on this chart should be proportional to $-1/z_s$. The drying time between fixed moisture contents is then directly proportional to the solid thickness.

Internal-diffusion controlling If a period of drying is developed where internal diffusion of moisture controls the rate, it can be expected that variables which influence the gas coefficients of heat or mass transfer will not influence the rate of drying. The drying rates should be independent of gas velocity, and humidity will be of importance only insofar as it controls the equilibrium-moisture concentration. When a semilogarithmic plot (Fig. 12.16) is prepared from drying data of this type, the curve b which results resembles those of Fig. 4.2. For a slab, the curves should be substantially straight at values of the ordinate below 0.6, with slopes proportional to $-1/z_s^2$. The drying time between fixed moisture contents should be proportional to the square of the thickness. Discrepancies may result because the initial moisture distribution is not uniform throughout the solid if a drying period precedes that for which diffusion controls and because the diffusivity varies with moisture content.

Attempts have been made to describe rates of drying in the falling-rate period by means of overall coefficients of heat and mass transfer [35] but these have not been too successful owing to the change in the individual resistances to transfer in the solid during the course of the drying. Various expressions for the

time of drying in the falling-rate period for different kinds of solids are available [30].

Critical moisture content The available data indicate that the average critical moisture content for a given type of solid depends upon the surface-moisture concentration. If drying during the constant-rate period is very rapid, and if the solid is thick, steep concentration gradients are developed within the solid and the falling rate begins at high average moisture contents. Generally, the critical moisture content will increase with increased drying rate and thickness of solid. It must usually be measured experimentally. McCormick [34] lists approximate values for many industrial solids.

Through-Circulation Drying

When a gas passes through a bed of wet, granular solids, both a constant-rate and a falling-rate period of drying may result and the rate-of-drying curves may appear very much like that shown in Fig. 12.10 [31]. Consider the case where the bed of solids has an appreciable thickness with respect to the size of the particles, as in Fig. 12.17 [1]. The evaporation of unbound moisture into the gas occurs in a relatively narrow zone which moves slowly through the bed, and unless the bed is internally heated, the gas leaving this zone is for all practical purposes saturated at the adiabatic-saturation temperature of the entering gas. This is also the surface temperature of the wet particles. The rate of drying is constant as long as the zone is entirely within the bed. When the zone first reaches the end of the bed, the rate of drying begins to fall because the gas no longer leaves in a saturated condition. In other words, a *desorption* wave passes through the bed, and the situation is much like that described for elution in fixed beds (Chap. 11). However, the point of view of interest is the moisture content of the solid rather than the concentration changes occurring in the exit gas. In the case of shallow beds composed of large particles, the gas leaves the bed unsaturated from the beginning [31], but as long as each particle surface remains

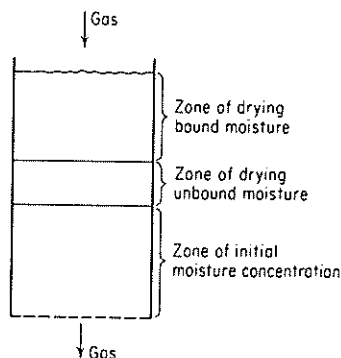


Figure 12.17 Through-circulation drying of thick beds of solids.

fully wet, there will still be a constant-rate period. The falling rate then begins when the surface moisture is depleted.

The rate of drying of unbound moisture [1]. Consider a bed of uniform cross section as in Fig. 12.17, fed with a gas of humidity Y_1 at the rate of G_S mass dry gas/(area bed cross section) (time). The maximum rate of drying $N_{\max}$ will occur if the gas leaving the bed is saturated at the adiabatic-saturation temperature, with humidity Y_{as} ,

$$N_{\max} = G_S(Y_{as} - Y_1) \quad (12.25)$$

where N is expressed as mass moisture evaporated/(area bed cross section) (time). In general, the gas will leave the bed at humidity Y_2 , and the instantaneous rate of drying is

$$N = G_S(Y_2 - Y_1) \quad (12.26)$$

For a differential section of the bed where the gas undergoes a change in humidity dY and leaves at a humidity Y , the rate of drying is

$$dN = G_S dY = k_Y dS(Y_{as} - Y) \quad (12.27)$$

where S is the interfacial surface per unit area of bed cross section. Letting a represent the interfacial surface per unit volume of bed whose thickness is z_S , we get

$$dS = a dz_S \quad (12.28)$$

and Eq. (12.27) becomes

$$\int_{Y_1}^{Y_2} \frac{dY}{Y_{as} - Y} = \int_0^{z_S} \frac{k_Y a dz_S}{G_S} \quad (12.29)$$

$$\ln \frac{Y_{as} - Y_1}{Y_{as} - Y_2} = N_{IG} = \frac{k_Y a z_S}{G_S} \quad (12.30)$$

where N_{IG} is the number of gas transfer units in the bed. This equation is the same as Eq. (7.57), developed for a somewhat similar situation. The mean driving force for evaporation is then the logarithmic mean of $Y_{as} - Y_1$ and $Y_{as} - Y_2$, in accordance with Eq. (7.61). Combining Eqs. (12.25), (12.26), and (12.30) gives

$$\frac{N}{N_{\max}} = \frac{Y_2 - Y_1}{Y_{as} - Y_1} = 1 - \frac{Y_{as} - Y_2}{Y_{as} - Y_1} = 1 - e^{-N_{IG}} = 1 - e^{-k_Y a z_S / G_S} \quad (12.31)$$

Equation (12.31) provides the rate of drying N if values of $k_Y a$ or N_{IG} can be determined. These have been established for certain special cases as follows:

1. *Particles small (10- to 200-mesh, or 2.03 to 0.074 mm diameter) with respect to bed depth (greater than 11.4 mm); drying of unbound water from the surface of*

nonporous particles [1]. For this case,† the constant rate is given by $N_{\max}$ [Eq. (12.25)]. Equation (12.31) can be used for both constant and falling rates, since at high moisture contents the exponential term becomes negligible. The interfacial surface a varies with moisture content, and it is most convenient to express N_{rG} empirically as‡

$$N_{rG} = \frac{0.273}{d_p^{0.35}} \left(\frac{d_p G}{\mu} \right)^{0.215} (X \rho_s z_s)^{0.64} \quad (12.32)$$

where d_p is the particle diameter and ρ_s the apparent density of the bed, mass dry solid/volume. Through-drying of such beds in the ordinary equipment may involve a pressure drop for gas flow which is too high for practical purposes, especially if the particles are very small. Drying of such beds is done on continuous rotary filters (crystal-filter driers), however.

2. *Particles large (3.2 to 20 mm diameter) in shallow beds (10 to 64 mm thick); drying of unbound moisture from porous or nonporous particles.* During the constant-rate period the gas leaves the bed unsaturated, and the constant rate of drying is given by Eq. (12.31). For this purpose, k_Y is given by

$$k_Y = \frac{j_D G_S}{Sc^{2/3}} \quad (12.33)$$

and j_D in turn by the first two entries of item 7, Table 3.3. The interfacial surface may be taken as the surface of the particles. For air drying of water from solids, $Sc = 0.6$. Additional experimental data on a large number of preformed materials are also available [31]. During the falling-rate period, internal resistance to moisture movement may be important, and no general treatment is available. In many cases [31] it is found that semilogarithmic plots of the form of Fig. 12.16 are useful. If the line on this plot is straight, Eqs. (12.8) and (12.9) are applicable.

Drying of bound moisture Presumably some adaptation of the methods used for adsorption in fixed beds (Chap. 11) could be made to describe this. No experimental confirmation is available, however.

Illustration 12.5 A cake of a crystalline precipitate is to be dried by drawing air through the cake. The particles of the cake are nonporous, of average diameter 0.20 mm, and since they are insoluble in water have a negligible equilibrium-moisture content. The cake is 18 mm thick, and the apparent density is 1350 kg dry solid/m³ (85 lb/ft³). It is to be dried from 2.5 to 0.1% moisture. The air will enter the cake at 0.24 kg dry air/(m² bed cross section) · s (= 177 lb/ft² · h), at a dry-bulb temperature 32°C and 50% humidity. Estimate the time for drying.

†In beds of finely packed solids containing a large percentage of liquid, the liquid is largely forced from between the solid particles by a mechanical process when gas is forced through the bed. See particularly Brownell and Katz, *Chem. Eng. Prog.*, 43, 537, 601, 703 (1947). Only the last traces of moisture are removed by the drying process considered here.

‡ Equation (12.32) is empirical, to be used with SI units. For units of feet, pounds mass, and hours the coefficient is 1.14.

SOLUTION $X_1 = 0.025/(1 - 0.025) = 0.0256$ kg water/kg dry solid, $X_2 = 0.001/(1 - 0.001) = 0.001001$. From Fig. 7.5, $Y_1 = 0.0153$ kg water/kg dry air, $t_{as} = 24^\circ\text{C}$ (the adiabatic-saturation temperature), and $Y_{as} = 0.0190$ kg water/kg dry air. $G_S = 0.24$ kg dry air/m² · s.

$$\text{Approx av } G = 0.24 + \frac{0.24(0.0153 + 0.0190)}{2} = 0.244 \text{ kg dry air/m}^2 \cdot \text{s}$$

Eq. (12.26): $N_{\max} = 0.24(0.0190 - 0.0153) = 8.88 \times 10^{-4}$ kg evapd/m² · s, $z_s = 0.018$ m, $d_p = 2 \times 10^{-4}$ m, $\rho_s = 1350$ kg/m³. Viscosity of air at $(32 + 24)/2 = 28^\circ\text{C}$ is 1.8×10^{-5} kg/m · s.

$$\frac{d_p G}{\mu} = \frac{(2 \times 10^{-4})(0.244)}{1.8 \times 10^{-5}} = 2.71$$

Eq. (12.32):

$$N_{IG} = \frac{0.273}{(2 \times 10^{-4})^{0.35}} (2.71)^{0.215} [X(1350)(0.018)]^{0.64} = 51.37 X^{0.64}$$

Eq. (12.31):

$$N = (8.88 \times 10^{-4}) [1 - \exp(-51.37 X^{0.64})]$$

This provides the following values of N , kg water evapd/m² · s, for values of X within the range of interest. They are the coordinates of the rate-of-drying curve.

X	0.0256	0.02	0.015	0.010	0.008	0.006	0.004	0.002	0.001
$N \times 10^4$	8.815	8.747	8.610	8.281	8.022	7.609	6.898	5.488	4.092

Calculation of the time of drying requires a value of the integral $\int dX/N$, in accordance with Eq. (12.3), between X_1 and X_2 . This can be evaluated numerically, or graphically from a plot of $1/N$ vs. X . The integral = 31.24. Since $S_S/A = \rho_s z_s = 1350(0.018) = 24.3$ kg dry solid/m², the time for drying is, by Eq. (12.3),

$$\theta = 24.3(31.24) = 759 \text{ s} = 12.7 \text{ min}$$

Illustration 12.6 Wet, porous catalyst pellets in the form of small cylinders, 13.5 mm diameter, 13.0 mm long, are to be dried of their water content in a through-circulation drier. The pellets are to be arranged in beds 50 mm deep on screens and dried by air flowing at the rate 1.1 kg dry air/(m² bed cross section) · s, entering at 82°C dry-bulb temperature, humidity 0.01 kg water/kg dry air. The apparent density of the bed is 600 kg dry solids/m³, and the particle surface is 280 m²/m³ of bed. Estimate the rate of drying and the humidity and temperature of the air leaving the bed during the constant-rate period.

SOLUTION $Y_1 = 0.01$ kg water/kg dry gas, and from Fig. 7.5, $Y_{as} = 0.031$ at the corresponding adiabatic-saturation temperature, 32°C. $G_S = 1.1$ kg dry air/m² · s and the approximate average $G = 1.1 + 1.1(0.031 + 0.01)/2 = 1.123$ kg/m² · s. The approximate average air viscosity is 1.9×10^{-5} kg/m · s.

The surface of each particle = $\pi(0.0135)^2(2)/4 + \pi(0.0135)(0.0130) = 8.376 \times 10^{-4}$ m². The diameter of a sphere of equal area = $d_p = [(8.376 \times 10^{-4})/\pi]^{0.5} = 0.01633$ mm. $a = 280$ m²/m³; $z_s = 0.050$ m.

The Reynolds number for the particles (Re'' of item 7, Table 3.3) = $d_p G/\mu = 0.01633(1.123)/(1.9 \times 10^{-5}) = 965$. ϵ = fraction voids (see Table 3.3) = $1 - d_p a/6 = 1 - 0.01633(280)/6 = 0.237$. Hence $j_D = (2.06/\epsilon) Re''^{-0.575} = 0.1671$. Sc for air-water vapor = 0.6. Eq. (12.33):

$$k_Y = \frac{0.1671(1.1)}{(0.6)^{2/3}} = 0.258 \text{ kg H}_2\text{O/m}^2 \cdot \text{s} \cdot \Delta Y$$

$$N_{IG} = \frac{k_Y a z_s}{G_S} = \frac{0.258(280)(0.050)}{1.1} = 3.28$$

Eq. (12.25):

$$N_{\max} = G_S(Y_{\text{air}} - Y_1) = 1.1(0.031 - 0.01) = 0.0231 \text{ kg/m}^2 \cdot \text{s}$$

Eq. (12.31):

$$\frac{N}{0.0231} = \frac{Y_2 - 0.01}{0.031 - 0.01} = 1 - \exp(-3.28)$$

Therefore $N = 0.0222 \text{ kg water evaporated/m}^2 \cdot \text{s}$ in the constant-rate period, and $Y_2 = 0.0302 \text{ kg water/kg dry air}$ for the outlet air. The corresponding outlet air temperature, from the adiabatic-saturation curve of Fig. 7.5 for the entering air, is 33.0°C . Since

$$\frac{S_S}{A} = \rho_S z_S = 600(0.050) = 30 \text{ kg dry solid/m}^2$$

from Eq. (12.2)

$$\frac{-dX}{d\theta} = \frac{NA}{S_S} = \frac{0.0222}{30} = 7.4 \times 10^{-4} \text{ kg H}_2\text{O}/(\text{kg dry solid}) \cdot \text{s}$$

This must be considered as an estimate only, subject to check by drying-rate tests.

CONTINUOUS DRYING

Continuous drying offers the advantages that usually the equipment necessary is small relative to the quantity of product, the operation is readily integrated with continuous chemical manufacture without intermediate storage, the product has a more uniform moisture content, and the cost of drying per unit of product is relatively small. As in batch drying, the nature of the equipment used is greatly dependent upon the type of material to be dried. Either direct or indirect heating, and sometimes both, may be used.

In many of the direct driers to be described, the solid is moved through a drier while in contact with a moving gas stream. The gas and solid may flow in parallel or in countercurrent, or the gas may flow across the path of the solid. If heat is neither supplied within the drier nor lost to the surroundings, operation is adiabatic and the gas will lose sensible heat and cool down as the evaporated moisture absorbs latent heat of vaporization. By supplying heat within the drier, the gas can be maintained at constant temperature.

In *countercurrent* adiabatic operation, the hottest gas is in contact with the driest solid, and the discharged solid is therefore heated to a temperature which may approach that of the entering gas. This provides the most rapid drying, since especially in the case of bound moisture the last traces are the most difficult to remove, and this is done more rapidly at high temperatures. On the other hand, the dry solid may be damaged by being heated to high temperatures in this manner. In addition, the hot discharged solid will carry away considerable sensible heat, thus lowering the thermal efficiency of the drying operation.

In *parallel* adiabatic operation, the wet solid is contacted with the hottest gas. As long as unbound surface moisture is present, the solid will be heated only to the wet-bulb temperature of the gas, and for this reason even heat-sensitive solids can frequently be dried by fairly hot gas in parallel flow. For example, a typical flue gas resulting from combustion of a fuel, which may have

a humidity of 0.03 kg water vapor/kg dry gas at 430°C, has a wet-bulb temperature of only about 65°C. In any event, the wet-bulb temperature can never exceed the boiling point of the liquid at the prevailing pressure. At the outlet of the drier, the gas will have been considerably cooled, and no damage will result to the dry solid. Parallel flow also permits greater control of the moisture content of the discharged solid when the solid must not be completely dried, through control of the quantity of gas passing through the drier and consequently its exit temperature and humidity. For this reason also it is used to avoid case hardening and other problems associated with batch drying.

Tunnel Driers

These direct driers are essentially adaptations of the truck drier to continuous operation. They consist of relatively long tunnels through which trucks, loaded with trays filled with the drying solid, are moved in contact with a current of gas to evaporate the moisture. The trucks may be pulled continuously through the drier by a moving chain, to which they are attached. In a simpler arrangement, the loaded trucks are introduced periodically at one end of the drier, each displacing a truck at the other end. The time of residence in the drier must be long enough to reduce the moisture content of the solid to the desired value. For relatively low-temperature operation the gas is usually steam-heated air, while for higher temperatures and especially for products which need not be kept scrupulously clean, flue gas from the combustion of a fuel can be used. Parallel or countercurrent flow of gas and solid can be used, or in some cases fans placed along the sides of the tunnel blow the gas through the trucks in crossflow. Operation may be essentially adiabatic, or the gas may be heated by steam coils along its path through the drier, and operation may then be substantially at constant temperature. Part of the gas may be recycled, much as in the case of batch driers, for heat economy. Truck-type tunnel driers can be used for any material which can be dried on trays: crystals, filter cakes, pastes, pottery, and the like.

There are many modifications of the tunnel drier which are essentially the same in principle but different in detailed design owing to the nature of the material being dried. For example, skeins of wet yarn may be suspended from poles or racks which move through the tunnel drier. Hides may be stretched on frames which hang from conveyor chains passing through the drier. Material in continuous sheets, such as cloth, may move through the drier under tension, as in a continuous belt over a series of rollers, or be hung in festoons from moving racks if it is to be dried in the absence of tension.

Turbo-Type (Rotating Shelf) Driers

Solids which ordinarily can be dried on trays, such as powdery and granular materials, heavy sludges and pastes, beads and crystalline solids, can be continuously dried in a turbo-type drier, a form of direct drier. The simplest of these is shown in Fig. 12.18. The drier is fitted with a series of annular trays arranged in

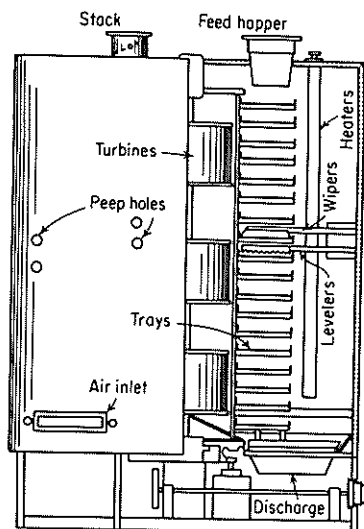


Figure 12.18 Turbo-type drier. (Wyssmont Co., Inc.)

a vertical stack. These rotate slowly (from a few to 60 r/s) about a vertical shaft. Each tray is provided with a slot cut into the tray, as well as a leveling rake for spreading the solid. Solid fed in at the top is spread upon the top tray to a uniform thickness, and as the tray revolves, the solid is pushed through the slot by a separate wiper rake, to fall upon the tray beneath. In this way, with overturning and respreading on each tray, the solid progresses to the discharge chute at the bottom of the drier. The drying gas flows upward through the drier, is circulated over the trays by slowly revolving turbine fans, and is reheated by internal heating pipes as shown. The rate of drying will be faster than that experienced in a tray-equipped tunnel drier, thanks to the frequent reloading of the solid on each tray [26]. Alternative arrangements are possible: external heating of the gas, recirculation of the gas, and arrangements for recovery of evaporated solvents may be provided. In some installations the solid is carried upon a moving, endless conveyor, which is wound about the vertical axis of the drier in a close-pitched, screw-type spiral. These driers are regularly built in sizes ranging from 2 to 6 m (6 to 20 ft) in diameter and 2 to 7.5 m (6 to 25 ft) high. A few installations as tall as 18 m (60 ft) have been made.

Similar devices, without the central turbines and built like multiple-hearth furnaces, are known as *plate driers*.

Through-Circulation Driers

Granular solids can be arranged in thin beds for through circulation of the gas, and, if necessary, pastes and filter cakes can be preformed into granules, pellets, or noodles, as described in the case of batch driers. In the continuous through-circulation drier of Fig. 12.19 [20], the solid is spread to a depth of 38 to 50 mm upon a moving endless conveyor which passes through the drier. The conveyor

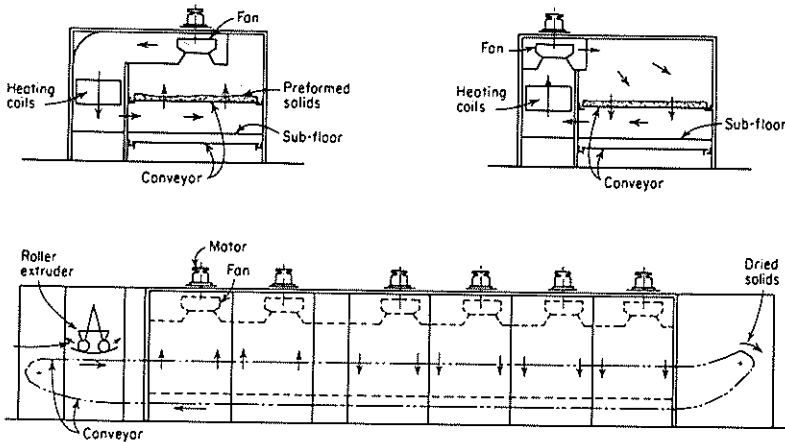


Figure 12.19 Continuous through-circulation (single-conveyor) drier with roller extruder. (*Proctor and Schwartz, Inc.*)

is made of perforated plates or woven wire screens in hinged sections in order to avoid failure from repeated flexing of the screen. Fans blow the heated air through the solid, usually upward through the wet solid and downward after initial drying has occurred. In this way, a more uniform moisture concentration throughout the bed is attained. Much of the gas is usually recycled, and a portion is discarded continuously at each fan position in the drier. For materials which permit the flow of gas through the bed in the manner shown, drying is much more rapid than for tray-type tunnel driers.

Rotary Driers

This is a most important group of driers, suitable for handling free-flowing granular materials which can be tumbled about without concern over breakage. Figure 12.20 shows one form of such a drier, a direct countercurrent hot-air drier. The solid to be dried is continuously introduced into one end of a rotating cylinder, as shown, while heated air flows into the other. The cylinder is installed at a small angle to the horizontal, and the solid consequently moves slowly through the device. Inside the drier, lifting flights extending from the cylinder wall for the full length of the drier lift the solid and shower it down in a moving curtain through the air, thus exposing it thoroughly to the drying action of the gas. This lifting action also assists in the forward motion of the solid. At the feed end of the drier, a few short spiral flights assist in imparting the initial forward motion to the solid before the principal flights are reached. The solid must clearly be one which is neither sticky nor gummy, which might stick to the sides of the drier or tend to ball up. In such cases, recycling of a portion of the dried product may nevertheless permit use of a rotary drier.

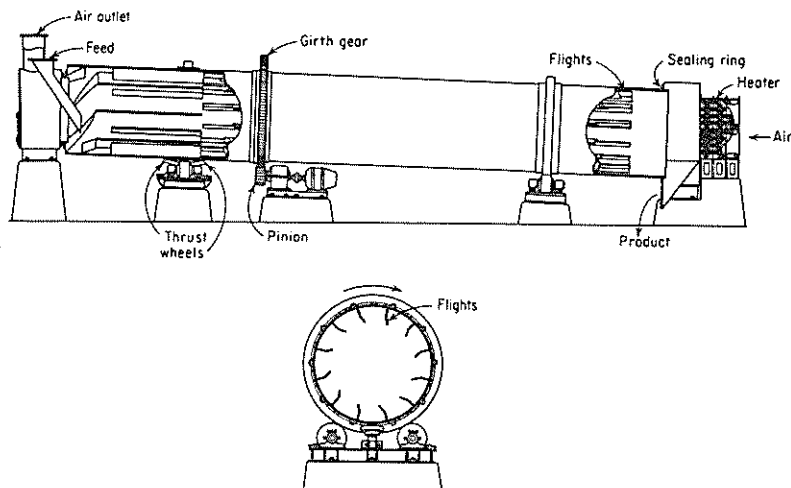


Figure 12.20 Ruggles-Coles XW hot-air drier. (Hardinge Co., Inc.)

The drier may be fed with hot flue gas rather than air, and if the gas leaves the drier at a high enough temperature, discharging it through a stack may provide adequate natural draft to provide sufficient gas for drying. Ordinarily, however, an exhaust fan is used to pull the gas through the drier, since this provides more complete control of the gas flow. A dust collector, of the cyclone, filter, or washing type, may be interposed between the fan and the gas exit. A blower may also be provided at the gas entrance, thus maintaining a pressure close to atmospheric in the drier; this prevents leakage of cool air in at the end housings of the drier, and if the pressure is well balanced, outward leakage will also be minimized.

Rotary driers are made for a variety of operations. The following classification includes the major types.

1. *Direct heat, countercurrent flow.* For materials which can be heated to high temperatures, such as minerals, sand, limestone, clays, etc., hot flue gas can be used as the drying gas. For substances which should not be heated excessively, such as certain crystalline chemical products like ammonium sulfate and cane sugar, heated air can be used. The general arrangement is that shown in Fig. 12.20, and if flue gas is used, the heating coils are replaced by a furnace burning gas, oil, or coal.
2. *Direct heat, cocurrent flow.* Solids which can be dried with flue gas without fear of contamination but which must not be heated to high temperatures for fear of damage, such as gypsum, iron pyrites, and organic material such as peat and alfalfa, should be dried in a cocurrent-flow drier. The general construction is much like that of Fig. 12.20, except that the gas and solid both enter at the same end of the drier.

3. *Indirect heat, countercurrent flow.* For solids such as white pigments, and the like, which can be heated to high temperatures but which must remain out of contact with flue gas, the indirect drier indicated schematically in Fig. 12.21*a* can be used. As an alternative construction, the drier may be enclosed in a brick structure and completely surrounded by the hot flue gases. The airflow in such a drier can be kept to a minimum since the heat is supplied by conduction through the shell or central tube, and finely pulverized solids which dust severely can then be handled. For solids which must not be heated to high temperatures and for which indirect heat is desirable such as cattle feed, brewers' grains, feathers, and the like, the steam-tube drier, shown in Fig. 12.21*b*, can be used. This drier may or may not have lifting flights and may be built with one, two, or more concentric rows of steam-heated tubes.

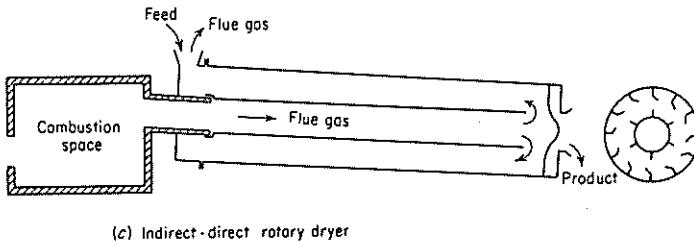
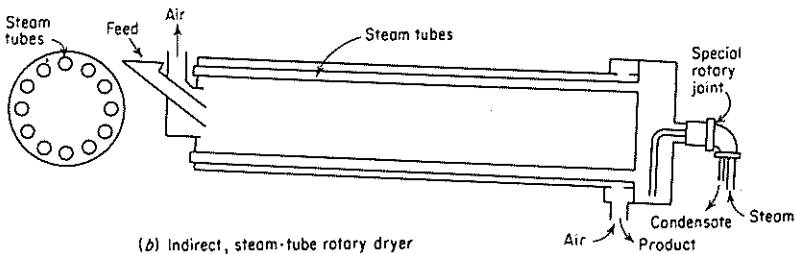
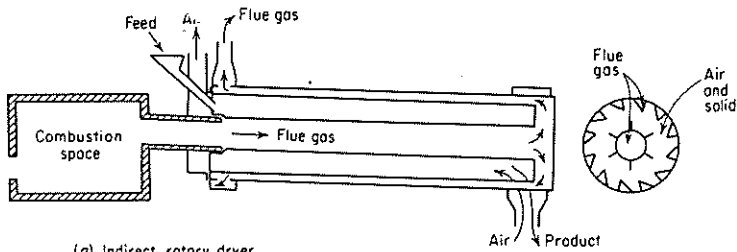


Figure 12.21 Some rotary driers (schematic).

The tubes revolve with the drier, necessitating a special rotary joint where the steam is introduced and the condensate removed. This type of drier is frequently used when recovery of the evaporated liquid is necessary.

4. *Direct-indirect.* These driers, more economical to operate than the direct driers, can be used for solids which can be dried at high temperatures by flue gas, especially when fuel costs are high and when large percentages of moisture must be removed from the solid. A typical schematic arrangement is shown in Fig. 12.21c. In such a drier, the hot gas may enter the center tube at 650 to 980°C (1200 to 1800°F), cool to 200 to 480°C (400 to 900°F) in its first passage through the drier, and on returning through the annular drying space cool further to 60 to 70°C (140 to 170°F) at discharge. Lignite, coal, and coke can be dried in the inert atmosphere of such a drier at relatively high temperatures without danger of burning or dust explosion.

The *Solidaire* drier contains a large number of paddles attached to an axial shaft, the paddles extending nearly to the inside of the shell. These rotate at relatively high speed, 10 to 20 m/s (2000 to 4000 ft/min), and the centrifugal force thus imparted to the solids keeps them in contact with the heated shell. Hot gas flowing cocurrently with the solids provides their forward motion.

High-frequency insonation has been shown to increase the rate of drying in rotary driers substantially [13].

All these driers are available from various manufacturers in standard sizes, ranging from approximately 1 m in diameter by 4 m long to 3 m in diameter by 30 m long.

Holdup in Rotary Driers

The average time of passage, or retention time, of the solid in a drier must equal the required drying time if the solid is to emerge at the desired moisture content. It must be recognized of course that the retention time of individual particles may differ appreciably from the average [36, 45], and this can lead to nonuniformity of product quality. Several agencies bring about the movement of the solid particles through a rotary drier. Flight action is the lifting and dropping of particles by the flights on the drier shell: in the absence of air flow, each time the solid is lifted and dropped, it advances a distance equal to the product of the length of the drop and the slope of the drier. Kiln action is the forward rolling of the particles on top of each other in the bottom of the drier, as in a kiln without flights. The particles also bounce in a forward direction after being dropped from the flight. In addition, the forward motion of the solid is hindered by a counterflowing gas or assisted by parallel flow.

The holdup ϕ_D of solid is defined as the fraction of the drier volume occupied by the solid at any instant, and the average time of retention θ can be computed by dividing the holdup by the volumetric feed rate,

$$\theta = \frac{\phi_D Z \pi T_D^2 / 4}{(S_S / \rho_S)(\pi T_D^2 / 4)} = \frac{Z \phi_D \rho_S}{S_S} \quad (12.34)$$

where S_S / ρ_S = volumetric feed rate/cross-sectional area of drier

S_S = mass velocity of dry solids, mass/(area)(time)

ρ_S = apparent solid density, mass dry solid/volume

Z = drier length

T_D = drier diameter

Although the influence of the character of the solids can be considerable [51], Friedman and Marshall [14] found that the holdup of a large number of solids under a variety of typical operating conditions could be expressed simply as

$$\phi_D = \phi_{D0} \pm KG \quad (12.35)$$

where ϕ_{D0} is the holdup with no gas flow and $\pm KG$ is the correction for influence of gas rate, G mass/(area)(time). The plus sign is used for countercurrent flow of gas and solid and the minus sign for cocurrent flow. Holdup for conditions of no gas flow depends to some extent upon flight design and the nature of the solid but under typical conditions and for ϕ_{D0} not exceeding 0.08 their data can be described by†

$$\phi_{D0} = \frac{0.3344S_s}{\rho_s s N^{0.9} T_D} \quad (12.36)$$

where s = slope of drier, m/m

N = rotational speed, r/s

T_D = drier diameter, m

The constant K is dependent upon the properties of the solid, and for rough estimates it may be taken, for SI units, as‡

$$K = \frac{0.6085}{\rho_s d_p^{1/2}} \quad (12.37)$$

where d_p is the average particle diameter. Holdups in the range 0.05 to 0.15 appear to be best. Higher holdup results in increased kiln action, with consequent poor exposure of the solid to the gas and an increase in power required for operating the drier.

These empirical relationships are applicable only under conditions of reasonable gas rates which do not cause excessive dusting or blowing of the solid particles from the drier. It is ordinarily desirable to keep dust down to 2 to 5 percent of the feed material as a maximum, and the corresponding permissible gas rates depend greatly upon the nature of the solid. From 0.27 to 13.6 kg gas/m² · s (200 to 10 000 lb/ft² · h) is used, depending upon the solid; for most 35-mesh solids (d_p approximately 0.4 mm) 1.36 kg/m² · s (1000 lb/ft² · h) is amply safe [14]. Dusting is less severe for countercurrent than for cocurrent flow, since then the damp feed acts to some extent as a dust collector, and it is also influenced by design of feed chutes and end breechings. In any case, it is best to depend upon actual tests for final design and to use the equations for initial estimates only.

Driers are most readily built with length/diameter ratios Z/T_D = 4 to 10. For most purposes, the flights extend from the wall of the drier a distance of 8 to 12 percent of the diameter, and their number range from $6T_D$ to $10T_D$ (T_D in meters). They should be able to lift the entire solids holdup, thus minimizing kiln action, which leads to low retention time. Rates of rotation are such as to provide peripheral speeds of 0.2 to 0.5 m/s (40 to 100 ft/min) and the slopes are usually in the range 0 to 0.08 m/m [14]. Negative slopes are sometimes necessary in cocurrent-flow driers.

Through-Circulation Rotary Driers

Driers of the type indicated in Fig. 12.22 combine the features of the through-circulation and rotary driers. The direct drier shown, the Roto-Louvre [12], consists of a slowly revolving tapered drum fitted with louvers to support the drying solid and to permit entrance of the hot gas beneath the solid. The hot gas is admitted only to those louvers which are underneath the bed of solid. There is

† Equation (12.36) is empirical. For units of feet, pounds mass, and hours (N in min⁻¹), the coefficient is 0.0037.

‡ For units of feet, pounds mass, and hours the coefficient is 9.33×10^{-5} .

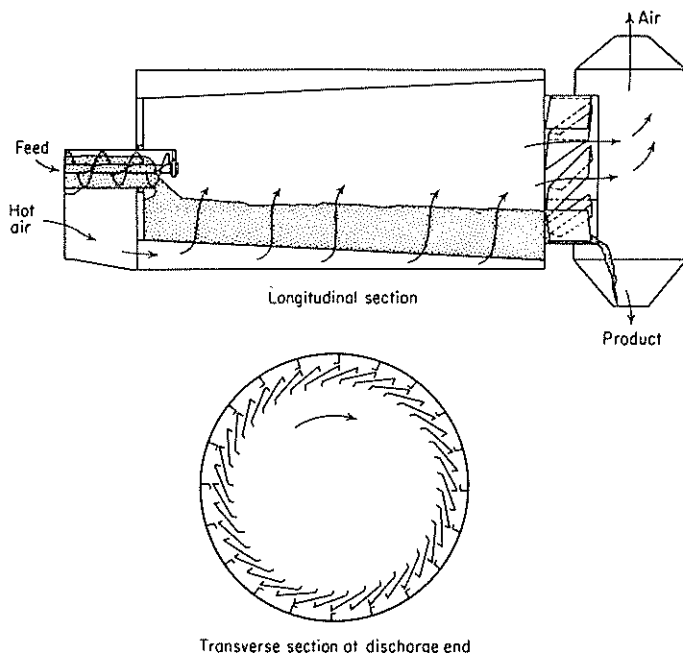


Figure 12.22 Continuous through-circulation rotary drier (Roto-Louvre). (*Link-Belt Co.*)

substantially no showering of the solid through the gas stream, and consequently a minimum of dusting results. The device is satisfactory for both low- and high-temperature drying of the same materials ordinarily treated in a rotary drier.

Drum Driers

Fluid and semifluid materials such as solutions, slurries, pastes, and sludges can be dried on an indirect drier, of which Fig. 12.23, a dip-feed drum drier, gives an example. A slowly revolving internally steam-heated metal drum continuously dips into a trough containing the substance to be dried, and a thin film of the substance is retained on the drum surface. The thickness of the film is regulated by a spreader knife, as shown, and as the drum revolves, moisture is evaporated into the surrounding air by heat transferred through the metal of the drum. The dried material is then continuously scraped from the drum surface by a knife. For such a drier, heat transfer rather than diffusion is the controlling factor. The liquid or solution is first heated to its boiling point; moisture is then evolved by boiling at constant temperature if a solute precipitates from a solution at constant concentration, or at increasing temperatures if the concentration change is gradual; and finally the dried solid is heated to approach the

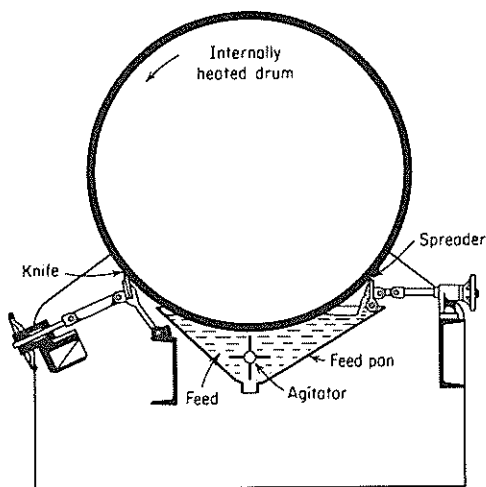


Figure 12.23 Dip-feed single-drum drier.
(Blaw-Knox Co.)

temperature of the drum surface. With slurries or pastes of insoluble solids, the temperature remains essentially constant at the solvent boiling point as long as the solid is completely wet and increases only during the last stages of drying. The vapors are frequently collected by a ventilated hood built directly over the drier.

The ability of various slurries, solutions, and pastes to adhere to a heated drum varies considerably, and diverse methods of feeding the drum are accordingly resorted to. Slurries of solids dispersed in liquids are frequently fed to the bottom of the drum on an inclined pan, and the excess nonadherent material is recycled to the feed reservoir. Vegetable glues and the like can be pumped against the bottom surface of the drum. The dip-feed arrangement shown in Fig. 12.23 is useful for heavy sludges while for materials which stick to the drum only with difficulty, the feed can be spattered on by a rapidly revolving roll. Double-drum driers, consisting of two drums placed close together and revolving in opposite directions, can be fed from above by admitting the feed into the depression between the drums. The entire drum drier may, on occasion, be placed inside a large evacuated chamber for low-temperature evaporation of the moisture.

Cylinder driers are drum driers used for material in continuous sheet form, such as paper and cloth. The wet solid is fed continuously over the revolving drum, or a series of such drums, each internally heated by steam or other heating fluid.

Spray Driers

Solutions, slurries, and pastes can be dried by spraying them as fine droplets into a stream of hot gas in a spray drier [29, 32]. One such device is shown in

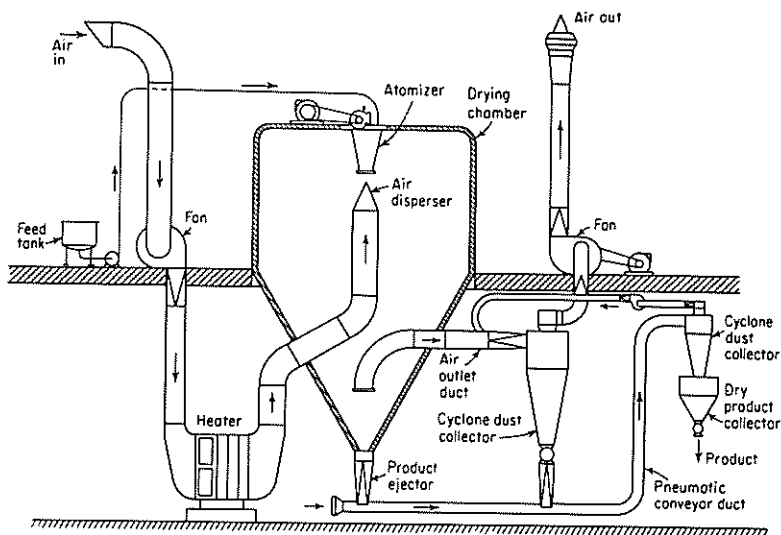


Figure 12.24 Spray drier. (Nichols Engineering and Research Corp.)

Fig. 12.24. The liquid to be dried is atomized and introduced into the large drying chamber, where the droplets are dispersed into a stream of heated air. The particles of liquid evaporate rapidly and dry before they can be carried to the sides of the chamber, and the bulk of the dried powder which results falls to the conical bottom of the chamber to be removed by a stream of air to the dust collector. The principal portion of the exit gas is also led to a dust collector, as shown, before being discharged. Many other arrangements are possible, involving both parallel and counterflow of gas and spray [28]. Installations may be very large, as much as 12 m in diameter and 30 m high (40 by 100 ft). Arrangements and detailed designs vary considerably, depending upon the manufacturer. Spray driers are used for a wide variety of products, including such diverse materials as organic and inorganic chemicals, pharmaceuticals, food products such as milk, eggs, and soluble coffee, as well as soap and detergent products.

In order to obtain rapid drying, atomization of the feed must provide small particles of high surface/weight ratio, whose diameter is usually in the range 10 to 60 μm . For this purpose, spray nozzles or rapidly rotating disks can be used. Spray nozzles are of two major types: pressure nozzles, in which the liquid is pumped at high pressure and with a rapid circular motion through a small orifice, and two fluid nozzles, in which a gas such as air or steam at relatively low pressures is used to tear the liquid into droplets. Nozzles are relatively inflexible in their operating characteristics and do not permit even moderate variation in liquid-flow rates without large changes in droplet size. They are also subject to rapid erosion and wear. Rotating disks are therefore favored in the chemical industry. They may be plane, vaned, or cup-shaped, up to about 0.3 m

(12 in) in diameter, and rotate at speeds in the range 50 to 200 r/s. The liquid or slurry is fed onto the disk near the center and is centrifugally accelerated to the periphery, from which it is thrown in an umbrella-shaped spray. Appreciable variation in liquid properties and feed rates can be handled satisfactorily, and even thick slurries or pastes can be atomized without clogging the device provided they can be pumped to the disk.

The drying gas, either flue gas or air, can enter at the highest practical temperature, 80 to 760°C (175 to 1400°F), limited only by the heat sensitivity of the product. Since the contact time for product and gas is so short, relatively high temperatures are feasible. The short time of drying requires effective gas-spray mixing, and attempts to improve upon this account in part for the large number of designs of spray chambers. Cool air is sometimes admitted at the drying-chamber walls in order to prevent sticking of the product to the sides. The effluent gas may convey all the dried product out of the drier or only the fines, but in either case the gas must be passed through some type of dust collector such as cyclones or bag filters, and these are sometimes followed by wet scrubbers for the last traces of dust. Recirculation of hot gas to the drier for purposes of heat economy is not practical, since the dust-recovery operation cannot usually be accomplished without appreciable heat loss.

The drops of liquid reach their terminal velocity in the gas stream quickly, within inches of the atomizing device. Evaporation takes place from the surface of the drops, and with many products solid material may accumulate as an impervious shell at the surface. Since heat is nevertheless rapidly being transmitted to the particles from the hot gas, the entrapped liquid portion of the drop vaporizes and expands the still-plastic wall of the drop to 3 to 10 times the original size, eventually exploding a small blowhole in the wall and escaping, to leave a hollow, dried shell of solid as the product. In other cases, the central liquid core diffuses through the shell to the outside, and the reduced internal pressure causes an implosion. In any event, the dried product is frequently in the form of small hollow beads of low bulk density [8]. Some control over the bulk density is usually possible through control of the particle size during atomization or through the temperature of the drying gas (increased gas temperature causes decreased product bulk density by more extensive expansion of the drop contents). For high-density products, the dried beads can be crushed.

Spray drying offers the advantage of extremely rapid drying for heat-sensitive products, a product particle size and density which are controllable within limits, and relatively low operating costs, especially in large-capacity driers. Extensive data have now been accumulated on drop sizes, drop trajectories, relative velocities of gas and drop, and rates of drying [32]. Logical procedures for design of spray driers with a minimum of experimental work have been outlined in detail in the work of Gauvin and coworkers [3, 15].

Fluidized and Spouted Beds

Granular solids, fluidized (see Chap. 11) by a drying medium such as hot air, can be dried and cooled in a similar fluidized bed [53], shown schematically in

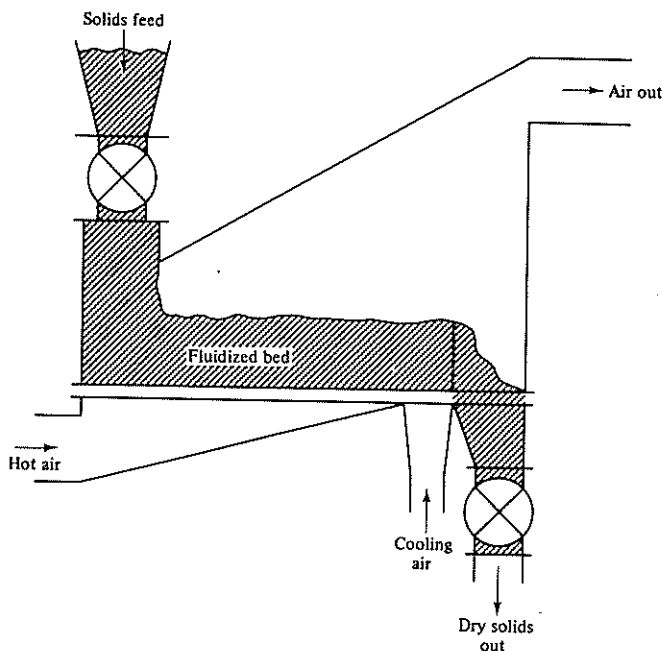


Figure 12.25 Fluidized bed drier (schematic). (Strong Scott Mfg. Co.)

Fig. 12.25. The principal characteristics of such beds include cross flow of solid and drying gas, a solids residence time controllable from seconds to hours, and suitability for any gas temperature. It is necessary that the solids be free-flowing, of a size range 0.1 to 36 mm [59]. Since the mass flow rate of gas for thermal requirements is substantially less than that required for fluidization, the bed is most economically operated at the minimum velocity for fluidization. Multi-stage, cross-flow operation (fresh air for each stage) is a possibility [2], as is a two-stage countercurrent arrangement, as in Fig. 11.28 [58]. A tentative design procedure has been proposed [40].

Coarse solids too large for ready fluidization can be handled in a *spouted bed* [33]. Here the fluid is introduced into the cone-shaped bottom of the container for the solids instead of uniformly over the cross section. It flows upward through the center of the bed in a column, causing a fountainlike spout of solids at the top. The solids circulate downward around the fluid column. Such a bed has found particular use in drying wheat, peas, flax, and the like [43].

Pneumatic (Flash) Driers

If the gas velocity in a fluidized bed is increased to the terminal velocity of the individual solid particles, they are lifted from the bed and are carried along by the fluidizing gas. Such gas-solid-particle mixtures are characteristic of *flash*, or

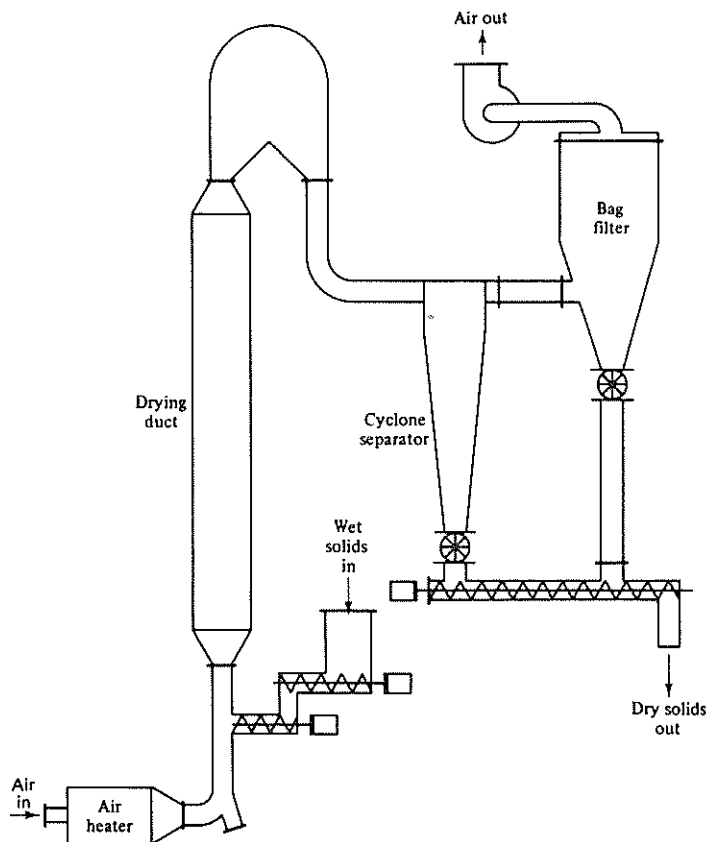


Figure 12.26 Pneumatic drier. (Strong-Scott Mfg. Co.)

pneumatic driers. The granular, free-flowing solids are dispersed in a rapidly flowing (~ 25 m/s) hot gas stream, as in Fig. 12.26, with an exposure time of the order of seconds [34]. Such short drying times limit the method to cases of surface moisture only, where internal diffusion of moisture within the solid is unimportant. Although a beginning has been made to systematize the drying-rate parameters [9] and a computer-design procedure has been prepared [5], pilot tests are nevertheless necessary.

Material and Enthalpy Balances

A general flow diagram for a continuous drier, arranged for countercurrent flow, is shown in Fig. 12.27. Solid enters at the rate S_S mass dry solid/(area)(time),[†] is

[†] For purposes of material and enthalpy balances alone, rates of flow of gas and solid can equally well be expressed as mass/time.

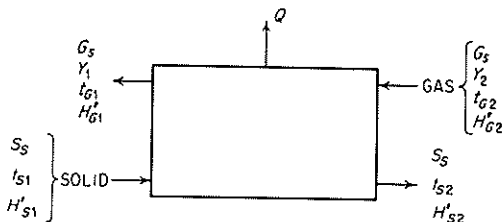


Figure 12.27 Material and enthalpy balances, continuous drier.

dried from X_1 to X_2 mass moisture/mass dry solid, and undergoes a temperature change t_{S1} to t_{S2} . The gas flows at the rate G_S mass dry gas/(area)(time) and undergoes a humidity change Y_2 to Y_1 mass moisture/mass dry gas and a temperature change t_{G2} to t_{G1} . A moisture balance is then

$$S_S X_1 + G_S Y_2 = S_S X_2 + G_S Y_1 \quad (12.38)$$

or

$$S_S (X_1 - X_2) = G_S (Y_1 - Y_2) \quad (12.39)$$

The enthalpy of the wet solid is given by Eq. (11.45),

$$H'_S = C_S(t_S - t_0) + X C_A(t_S - t_0) + \Delta H_A \quad (11.45)$$

where H'_S = enthalpy of wet solid at t_S , referred to solid and liquid at reference temperature t_0 , energy/mass dry solid

C_S = heat capacity of dry solid, energy/(mass)(temperature)

C_A = heat capacity of moisture, as a liquid, energy/(mass)(temperature)

ΔH_A = integral heat of wetting (or of adsorption, hydration, or solution) referred to pure liquid and solid, at t_0 , energy/mass dry solid

Bound moisture will generally exhibit a heat of wetting (see Chap. 11), although data are largely lacking. The enthalpy of the gas, H'_G energy/mass dry gas, is given by Eq. (7.13). If the net heat lost from the drier is Q energy/time, the enthalpy balance becomes

$$S_S H'_{S1} + G_S H'_{G2} = S_S H'_{S2} + G_S H'_{G1} + Q \quad (12.40)$$

For adiabatic operation, $Q = 0$, and if heat is added within the drier to an extent greater than the heat losses, Q is negative. If the solid is carried on trucks or other support, the sensible heat of the support should also be included in the balance. Obvious changes in the equations can be made for parallel-flow driers.

Illustration 12.7 An uninsulated, hot-air countercurrent rotary drier of the type shown in Fig. 12.20 is to be used to dry ammonium sulfate from 3.5 to 0.2% moisture. The drier is 1.2 m (4 ft) in diameter, 6.7 m (22 ft) long. Atmospheric air at 25°C, 50% humidity, will be heated by passage over steam coils to 90°C before it enters the drier and is expected to be discharged at 32°C. The solid will enter at 25°C and is expected to be discharged at 60°C. Product will be delivered at a rate of 900 kg/h. Estimate the air and heat requirements for the drier.

SOLUTION Define the rates of flow in terms of kg/h. $X_2 = 0.2/(100 - 0.2) = 0.0020$; $X_1 = 3.5/(100 - 3.5) = 0.0363$ kg water/kg dry solid. $S_S = 900(1 - 0.0020) = 898.2$ kg dry solid/h. The rate of drying = $898.2(0.0363 - 0.0020) = 30.81$ kg water evaporated/h.

At 25°C, 50% humidity, the absolute humidity of the available air = 0.010 kg water/kg dry air = Y_2 . Since $t_{G2} = 90^\circ\text{C}$, and with $t_0 = 0^\circ\text{C}$, the enthalpy of the air entering the drier (Table 7.1) is

$$H'_{G2} = [1005 + 1884(0.01)]90 + 2\,502\,300(0.01) = 117\,200 \text{ J/kg dry air}$$

For the outlet air, $t_{G1} = 32^\circ\text{C}$,

$$H_{G1} = (1005 + 1884Y_2)(32) + 2\,502\,300Y_1 = 32\,160 + 2\,504\,200Y_1$$

The heat capacity of dry ammonium sulfate is $C_S = 1507$ and that of water 4187 J/kg · K. ΔH_A will be assumed to be negligible for lack of better information. Taking $t_0 = 0^\circ\text{C}$, so that enthalpies of gas and solid are consistent, and since $t_{S1} = 25^\circ\text{C}$, $t_{S2} = 60^\circ\text{C}$, the solid enthalpies, J/kg dry solid, are [Eq. (11.45)]:

$$H'_{S2} = 1507(60 - 0) + 0.002(4187)(60 - 0) = 90\,922$$

$$H'_{S1} = 1507(25 - 0) + 0.0363(4187)(25 - 0) = 41\,475$$

The estimated combined natural convection and radiation heat-transfer coefficient from the drier to the surroundings [23] is $12 \text{ W/m}^2 \cdot \text{K}$. The mean Δt between drier and surroundings is taken as $[(90 - 25) + (32 - 25)]/2 = 36^\circ\text{C}$, and the exposed area as $\pi(1.2)(6.7) = 25.3 \text{ m}^2$, whence the estimated heat loss is

$$Q = 12(3600)(25.3)(36) = 39\,350 \text{ kJ/h}$$

Moisture balance [Eq. (12.39)]:

$$898.2(0.0363 - 0.002) = G_S(Y_1 - 0.01)$$

Enthalpy balance [Eq. (12.40)]:

$$898.2(41\,475) + G_S(117\,200) = 898.2(90\,922) + G_S(32\,160 + 2\,504\,200Y_1) + 39\,350\,000$$

When solved simultaneously, these yield

$$G_S = 2682 \text{ kg dry air/h} \quad Y_1 = 0.0215 \text{ kg water/kg dry air}$$

The enthalpy of the fresh air (Fig. 7.5) is 56 kJ/kg dry air, and hence the heat load for the heater is $2682(117\,200 - 56\,000) = 164\,140 \text{ kJ/h}$. If steam at pressure 70 kN/m² (10.2 lb_f/in²) gauge and latent heat 2215 kJ/kg is used, the steam required is $164\,140/2215 = 74.1 \text{ kg steam/h}$, or $74.1/30.81 = 2.4 \text{ kg steam/kg water evaporated}$.

Rate of Drying for Continuous Direct-Heat Driers

Direct-heat driers are best placed in two categories, according to whether high or low temperatures prevail. For operation at temperatures above the boiling point of the moisture to be evaporated, the humidity of the gas has only a minor influence on the rate of drying, and it is easiest to work directly with the rate of heat transfer. At temperatures below the boiling point, mass-transfer driving forces are conveniently established. In any case, it must be emphasized that our imperfect knowledge of the complex drying mechanisms makes experimental testing of the drying necessary. Calculations are useful only for the roughest estimate.

Drying at high temperatures In a typical situation, three separate zones are distinguished in such driers, recognizable by the variation in temperatures of the gas and solid in the various parts of the drier [14]. Refer to Fig. 12.28, where typical temperatures are shown schematically by the solid lines for a countercurrent drier. In zone I, the preheat zone, the solid is heated by the gas until the

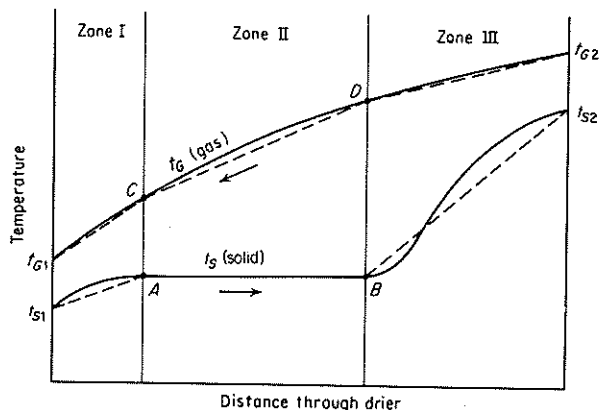


Figure 12.28 Temperature gradients in a continuous countercurrent drier.

rate of heat transfer to the solid is balanced by the heat requirements for evaporation of moisture. Little actual drying will usually occur here. In zone II, the equilibrium temperature of the solid remains substantially constant while surface and unbound moisture are evaporated. At point *B*, the critical moisture of the solid is reached, and, in zone III, unsaturated surface drying and evaporation of bound moisture occur. Assuming that the heat-transfer coefficients remain essentially constant, the decreased rate of evaporation in zone III results in increased solid temperature, and the discharge temperature of the solid approaches the inlet temperature of the gas.

Zone II represents the major portion for many driers, and it is of interest to consider the temperature-humidity relationship of the gas as it passes through this section. On the psychrometric chart (Fig. 12.29), point *D* represents the gas conditions at the corresponding point *D* of Fig. 12.28. If drying is adiabatic, i.e., without addition to, or loss of heat from, the drier, the adiabatic-saturation line DC_1 will represent the variation of humidity and temperature of the gas as it passes through this section of the drier, and the conditions of the gas leaving this zone (point *C*, Fig. 12.28) are shown at C_1 on Fig. 12.29. The surface temperature of the solid, which can be estimated by the methods described earlier in the case of batch drying, will vary from that at S_1 (corresponding to point *B* of Fig. 12.28) to S'_1 (corresponding to point *A*). If radiation and conduction through the solid can be neglected, these are the wet-bulb temperatures corresponding to *D* and C_1 , respectively. For the system air-water, whose wet-bulb and adiabatic-saturation temperatures are the same, these will be both given by an extension of the adiabatic-saturation line DC_1 to the saturation-humidity curve. Heat losses may cause the gas to follow some such path as DC_2 . On the other hand, if heat is added to the gas in this section, the path will be represented by a line such as DC_3 and, if the gas is kept at constant temperature, by line DC_4 . In the case of the last, the surface temperature of the solid will vary from that at S_1 to that at

and if the heat-transfer coefficient is constant,

$$N_{IOG} = \frac{\Delta t'_G}{\Delta t_m} = \frac{Z}{H_{IOG}} \quad (12.45)$$

$$H_{IOG} = \frac{G_s C_s}{Ua} \quad (12.46)$$

where N_{IOG} = number of heat-transfer units

H_{IOG} = length of heat-transfer unit

$\Delta t'_G$ = change in gas temperature owing to *heat transfer to solid only*

Δt_m = appropriate average temperature difference between gas and solid

If the temperature profiles in the drier can be idealized as straight lines, such as the broken lines of Fig. 12.28, then *for each zone taken separately* Δt_m is the logarithmic average of the terminal temperature differences and N_{IOG} the corresponding number of transfer units for each zone. In the case of zone III, this simplification will be satisfactory for the evaporation of unsaturated surface moisture but not for bound moisture or where internal diffusion of moisture controls the rate of drying.

Tunnel driers In drying solids by cross circulation of air over the surface, as in the case of materials on trays or solids in sheet form, the surface temperature in zone II can be estimated through Eq. (12.18). Unless all surfaces are exposed to heat transfer by radiation, this feature of the heat transfer is better ignored, and U in Eq. (12.46) can be taken as $h_c + U_r$. This value will also serve in zone I, and in zone III only for cases of unsaturated surface drying. The quantity a can be computed from the method of loading the drier.

Rotary driers The surface of the solid exposed to drying gas cannot be conveniently measured, so the group Ua must be considered together. In counter-current driers, Ua is influenced by changes in holdup due to changes in gas flow and solid feed rate, but effects of changes in slope or rate of rotation of the drier are small [14, 44]. The effect of gas mass velocity is somewhat uncertain. In the absence of experimental data, the value of Ua in Eqs. (12.43) to (12.46) for commercial driers manufactured in the United States is recommended [34], in SI units, as†

$$Ua = \frac{237 G^{0.67}}{T_D} \quad (12.47)$$

Here, especially in zone II, the “appropriate” average temperature difference is the average wet-bulb depression of the gas since the surface of the wet solid is at the wet-bulb temperature. Use of temperature differences based on the bulk solid temperature will lead to conservative results. Obviously the character of the solids must also have an influence, not recognized in Eq. (12.47).

† In units of Btu, feet, pounds mass, hours, and degrees Fahrenheit the coefficient of Eq. (12.47) is 0.5.

Illustration 12.8 A preliminary estimate of the size of a countercurrent, direct-heat, rotary drier for drying an ore-flotation concentrate is to be made. The solid is to be delivered from a continuous filter and introduced into the drier at 8% moisture, 27°C, and is to be discharged from the drier at 150°C, 0.5% moisture. There will be 0.63 kg/s (5000 lb/h) of dried product. The drying gas is a flue gas analyzing 2.5% CO₂, 14.7% O₂, 76.0% N₂, and 6.8% H₂O by volume. It will enter the drier at 480°C. Heat losses will be estimated at 15% of the heat in the entering gas. The ore concentrate is ground to 200-μm average particle diameter and has a bulk density of 1300 kg dry solid/m³ (81.1 lb/ft³) and a heat capacity of 837 J/kg · K (0.2 Btu/lb · °F), dry. The gas rate ought not to exceed 0.70 kg/m² · s to avoid excessive dusting.

SOLUTION $X_1 = 8/(100 - 8) = 0.0870$, $X_2 = 0.5/(100 - 0.5) = 0.00503$ kg water/kg dry solid. Temporarily define S_S and G_S as kg dry substance/s. $S_S = 0.63(1 - 0.00503) = 0.627$ kg dry solid/s. Water to be evaporated = $0.627(0.0870 - 0.00503) = 0.0514$ kg/s.

Basis: 1 kmol gas in. Dry gas = $1 - 0.068 = 0.932$ kmol.

	kmol	kg	Av heat capacity, kJ/kmol · K 480 - 0°C
CO ₂	0.025	1.10	45.6
O ₂	0.147	4.72	29.9
N ₂	0.760	21.3	29.9
Total dry wt		27.1	

Av mol wt dry gas = $27.1/0.932 = 29.1$ kg/kmol, nearly the same as that of air.

$$Y_2 = \frac{0.068(18.02)}{0.932(29.1)} = 0.0452 \text{ kg water/kg dry gas} \quad t_{G2} = 480^\circ\text{C}$$

$$\begin{aligned} \text{Av ht capacity of dry gas} &= \frac{0.025(45.6) + (0.147 + 0.760)(29.9)}{0.932(29.1)} \\ &= 1.042 \text{ kJ/kmol} \cdot \text{K} \end{aligned}$$

The exit gas temperature will be tentatively taken as 120°C. This is subject to revision after the number of transfer units has been computed. In a manner similar to that above, the average heat capacity of the dry gas, 120 to 0°C, is 1.005 kJ/kmol · K. Eq. (7.13):

$$H'_{G2} = [1.042 + 1.97(0.0452)](480 - 0) + 2502.3(0.0452) = 656.0 \text{ kJ/kg dry gas}$$

$$H'_{G1} = (1.005 + 1.884 Y_1)(120 - 0) + 2502.3 Y_1 = 120.5 + 2728 Y_1$$

Take $\Delta H_A = 0$. Eq. (11.45):

$$H'_{S1} = 0.837(27 - 0) + 0.087(4.187)(27 - 0) = 32.43 \text{ kJ/kg dry solid}$$

$$H'_{S2} = 0.837(150 - 0) + 0.00503(4.187)(150 - 0) = 128.7 \text{ kJ/kg dry solid}$$

$$Q = \text{heat loss} = 0.15(656)G_S = 98.4G_S \text{ kJ/s}$$

Eq. (12.39):

$$0.0514 = G_S(Y_1 - 0.0452)$$

Eq. (12.40):

$$0.627(32.43) + G_S(656) = 0.627(128.7) + G_S(120.5 + 2728 Y_1) + 98.4G_S$$

Simultaneous solution:

$$G_S = 0.639 \text{ kg dry gas/s} \quad Y_1 = 0.1256 \text{ kg water/kg dry gas}$$

$$H'_{G1} = 463 \text{ kJ/kg dry gas} \quad Q = 62.9 \text{ kJ/s}$$

Assuming that the psychrometric ratio of the gas is the same as that of air, its wet-bulb temperature for zone II [Eq. (7.26)] is estimated to be about 65°C. The surface of the drying

solid particles is subject to radiation from the hot walls of the drier, and the surface of the solid in zone II is then estimated to be about 68°C. *Note:* This should be recalculated after the temperature at point D, Fig. 12.28, is known.

For lack of information on the critical moisture content of the solid, which is probably quite low for the conditions found in such a drier, it will be assumed that all moisture is evaporated in zone II at 68°C. Zone I will be taken as a preheat zone for warming the solid to 68°C, without drying. Enthalpy of the solid at 68°C, $X = 0.0870$ (point A, Fig. 12.28) = $0.837(68 - 0) + 0.0870(4.187)(68 - 0) = 81.7$ kJ/kg dry solid. Similarly, enthalpy of the solid at 68°C, $X = 0.00503$ (point B, Fig. 12.28) = 58.3 kJ/kg dry solid.

Assuming that heat losses in the three zones are proportional to the number of transfer units in each zone and to the average temperature difference between the gas and the surrounding air (27°C), the losses are apportioned (by a trial-and-error calculation) as 14% in zone I, 65% in zone II, 21% in zone III.

Calculation for zone III Humid heat of entering gas = $1.042 + 1.97(0.0452) = 1.131$ kJ/(kg dry gas) · K. A heat balance:

$$0.639(1.131)(480 - t_{GD}) = 0.627(128.7 - 58.3) + 0.21(62.9)$$

$$t_{GD} = \text{gas temp at D (Fig. 12.28)} = 401^\circ\text{C}$$

The change in gas temperature, exclusive of that resulting from heat losses, is

$$\Delta t'_G = \frac{0.627(128.7 - 58.3)}{0.639(1.131)} = 61.1^\circ\text{C}$$

Average temp difference between gas and solid = average of $480 - 150$ and $401 - 68 = 332^\circ\text{C}$
 $= \Delta t_m$

$$N_{IOG} = \frac{\Delta t'_G}{\Delta t_m} = \frac{61.1}{332} = 0.18$$

Calculation for zone I Humid heat of outlet gas = $1.005 + 1.884(0.1256) = 1.242$ kJ/(kg dry gas) · K. A heat balance:

$$0.639(1.242)(t_{GC} - 120) = 0.627(81.7 - 32.43) + 0.14(62.9)$$

$$t_{GC} = \text{gas temp at C (Fig. 12.28)} = 170^\circ\text{C}$$

$$\Delta t'_G = \frac{0.627(81.7 - 32.43)}{0.639(1.242)} = 38.9^\circ\text{C}$$

$$\Delta t_m = \text{average of } 170 - 68 \text{ and } 120 - 27 = 98^\circ\text{C}$$

$$N_{IOG} = \frac{\Delta t'_G}{\Delta t_m} = \frac{38.9}{98} = 0.39$$

Calculation for zone II

$$\text{Average humid heat of gas} = \frac{1.042 + 1.242}{2} = 1.142 \text{ kJ/kg} \cdot \text{K}$$

$$\text{True change in gas temp} = 401 - 170 = 231^\circ\text{C}$$

$$\text{Change in temp resulting from heat loss} = \frac{0.65(62.9)}{0.639(1.142)} = 56^\circ\text{C}$$

$$\Delta t'_G \text{ resulting from heat transfer to solid} = 231 - 56 = 175^\circ\text{C}$$

$$\Delta t_m = \frac{(401 - 68) - (170 - 68)}{\ln[(401 - 68)/(170 - 68)]} = 195^\circ\text{C}$$

$$N_{IOG} = \frac{\Delta t'_G}{\Delta t_m} = \frac{175}{195} = 0.90$$

$$\text{Total } N_{IOG} = 0.39 + 0.90 + 0.18 = 1.47$$

Size of drier Standard diameters available are 1, 1.2, and 1.4 m, and larger. The 1-m diameter is too small. For $T_D = 1.2$ m, the cross-sectional area = $\pi(1.2)^2/4 = 1.13$ m². Expressing the rates of flow as kg dry substance/m² · s gives $G_S = 0.639/1.13 = 0.565$, $S_S = 0.627/1.13 = 0.555$ kg/m² · s.

$$AvG = G_S(1 + Y_{av}) = 0.565 \left(1 + \frac{0.1256 + 0.0452}{2} \right) = 0.613 \text{ kg/m}^2 \cdot \text{s}$$

For lack of more specific information, use Eq. (12.47):

$$Ua = \frac{237(0.613)^{0.67}}{1.2} = 142.3 \text{ W/m}^2 \cdot \text{K}$$

$$H_{IOG} = \frac{G_S C_s}{Ua} = \frac{0.565(1142)}{142.3} = 4.5 \text{ m} \quad Z = N_{IOG} H_{IOG} = 1.47(4.5) = 6.6 \text{ m (21.7 ft)}$$

The nearest standard length is 7.5 m (24.6 ft).

Take the peripheral speed as 0.35 m/s (70 ft/min), whence the rate of revolution = $N = 0.35/\pi T_D = 0.093 \text{ s}^{-1}$ (5.6 r/min).

$$d_p = 200 \times 10^{-6} \text{ m} \quad \rho_s = 1300 \text{ kg/m}^3$$

Eq. (12.37):

$$K = \frac{0.6085}{1300(200 \times 10^{-6})^{0.5}} = 0.0331$$

Take the holdup $\phi_D = 0.05$. Eq. (12.35): $\phi_{D0} = \phi_D - KG = 0.05 - 0.0331(0.613) = 0.0297$. Eq. (12.36):

$$s = \frac{0.3344 S_S}{\phi_{D0} \rho_s N^{0.9} T_D} = \frac{0.3344(0.555)}{0.05(1300)(0.093)^{0.9} 1.2} = 0.02 \text{ m/m, drier slope}$$

Drying at low temperatures Continuous driers operating at low temperatures can be divided into zones the same as high-temperature driers. Since the surface moisture will evaporate at a comparatively low temperature in zone II, the preheat zone can generally be ignored and only zones II and III need be considered. Refer to Fig. 12.30, which shows an arrangement for countercurrent flow. In zone II, unbound and surface moisture is evaporated as discussed previously, and the moisture content of the solid falls to the critical value X_c . The rate of drying in this zone would be constant if it were not for the varying conditions of the gas. In zone III, unsaturated-surface drying and evaporation of bound moisture occur, and the gas humidity rises from its initial value Y_2 to Y_c . The latter can be calculated by applying the material-balance relation (12.39) to either zone separately. The retention time can be calculated by integration of Eq. (12.3),

$$\theta = \theta_{II} + \theta_{III} = \frac{S_S}{A} \left(\int_{X_c}^{X_1} \frac{dX}{N} + \int_{X_2}^{X_c} \frac{dX}{N} \right) \quad (12.48)$$

where A/S_S is the specific exposed drying surface, area/mass dry solid.

Zone II, $X > X_c$ The rate N is given by Eq. (12.17), which, when substituted in the first part of Eq. (12.48), provides

$$\theta_{II} = \frac{S_S}{A} \frac{1}{k_Y} \int_{X_c}^{X_1} \frac{dX}{Y_s - Y} \quad (12.49)$$

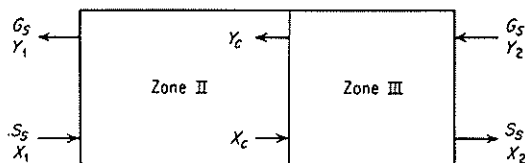


Figure 12.30 Continuous low-temperature countercurrent drier.

Since $G_S dY = S_S dX$, Eq. (12.49) becomes

$$\theta_{II} = \frac{G_S}{S_S} \frac{S_S}{A} \frac{1}{k_Y} \int_{Y_c}^{Y_1} \frac{dY}{Y_s - Y} \quad (12.50)$$

Integration of Eq. (12.50) must take into account the variation of Y_s , the humidity of the gas at the solid surface, with Y . If the gas temperature, for example, is held constant in this zone by application of heat, the path of the gas resembles line DC_4 in Fig. 12.29. If, furthermore, radiation and conduction effects can be neglected, Y_s for any value of Y on line DC_4 is the saturated humidity at the corresponding wet-bulb temperature. Equation (12.50) can then be integrated graphically.

For the case where Y_s is constant, as for adiabatic drying of water into air, Eq. (12.50) becomes

$$\theta_{II} = \frac{G_S}{S_S} \frac{S_S}{A} \frac{1}{k_Y} \ln \frac{Y_s - Y_c}{Y_s - Y_1} \quad (12.51)$$

Zone III, $X < X_c$ Some simplification is necessary for mathematical treatment. For the case where unsaturated-surface drying occurs and the drying rate is dependent strictly upon the conditions prevailing at any instant, independent of the immediate past history of the drying, Eqs. (12.8) and (12.17) apply. These provide

$$N = \frac{N_c(X - X^*)}{X_c - X^*} = \frac{k_Y(Y_s - Y)(X - X^*)}{X_c - X} \quad (12.52)$$

When this is substituted in the second part of Eq. (12.48), the result is

$$\theta_{III} = \frac{S_S}{A} \frac{X_c - X^*}{k_Y} \int_{X_2}^{X_c} \frac{dX}{(Y_s - Y)(X - X^*)} \quad (12.53)$$

This can be evaluated graphically after determining the relationship between X , X^* , Y_s , and Y . For this purpose, the material balance can be written as

$$Y = Y_2 + (X - X_2) \frac{S_S}{G_S} \quad (12.54)$$

The surface humidity Y_s is found in the manner previously described, and X^* is given by the equilibrium-moisture curve for the appropriate Y .

For the special case where the bound moisture is negligible ($X^* = 0$) and Y_s is constant (adiabatic drying), substitution of Eq. (12.54) and its differential

$G_S dY = L_S dX$ in Eq. (12.53) provides

$$\theta_{III} = \frac{G_S}{S_S} \frac{S_S}{A} \frac{X_c}{k_Y} \int_{Y_2}^{Y_1} \frac{dY}{(Y_s - Y)(Y - Y_2)G_S/S_S + X_2} \quad (12.55)$$

$$\theta_{III} = \frac{G_S}{S_S} \frac{S_S}{A} \frac{X_c}{k_Y} \frac{1}{(Y_s - Y_2)G_S/S_S + X_2} \ln \frac{X_c(Y_s - Y_2)}{X_2(Y_s - Y_c)} \quad (12.56)$$

These methods must not be applied to solids whose internal resistance to movement of moisture is large, where internal diffusion controls the rate of drying, or where case hardening occurs. In these circumstances the instantaneous rate of drying under variable conditions is not merely a function of the prevailing conditions but depends upon the immediate past drying history as well. For such solids, the time for drying is best determined experimentally in a carefully planned test which simulates the countercurrent action of the continuous drier [4].

In applying Eqs. (12.48) to (12.56), it is clear that, should drying take place only above or only below the critical moisture content, appropriate changes in the limits of moisture content and gas humidities must be made. In parallel-flow driers, where gas enters at humidity Y_1 and leaves at Y_2 while solid enters at moisture content X_1 and leaves at X_2 , Eqs. (12.51) and (12.56) become

$$\theta_{II} = \frac{G_S}{S_S} \frac{S_S}{A} \frac{1}{k_Y} \int_{Y_1}^{Y_2} \frac{dY}{Y_s - Y} = \frac{G_S}{S_S} \frac{S_S}{A} \frac{1}{k_Y} \ln \frac{Y_s - Y_1}{Y_s - Y_2} \quad (12.57)$$

$$\begin{aligned} \theta_{III} &= \frac{S_S}{A} \frac{X_c - X^*}{k_Y} \int_{X_2}^{X_1} \frac{dX}{(Y_s - Y)(X - X^*)} \\ &= \frac{G_S}{S_S} \frac{S_S}{A} \frac{X_c}{k_Y} \frac{1}{(Y_s - Y_c)G_S/S_S - X_2} \ln \frac{X_c(Y_s - Y_2)}{X_2(Y_s - Y_c)} \end{aligned} \quad (12.58)$$

Illustration 12.9 Wet rayon skeins, after centrifuging, are to be air-dried from 46 to 8.5% water content in a continuous countercurrent tunnel drier. The skeins are hung on poles which travel through the drier. The air is to enter at 82°C, humidity 0.03 kg water/kg dry air, and is to be discharged at a humidity 0.08. The air temperature is to be kept constant at 82°C by heating coils within the drier. The air rate is to be 1.36 kg/m² · s (1000 lb/ft² · h).

The critical moisture content of rayon skeins is 50%, and its percent equilibrium moisture at 82°C can be taken as one-fourth of the percent relative humidity of the air. The rate of drying is then [Simons, Koffolt, and Withrow, *Trans. AIChE*, 39, 133 (1943)]

$$\frac{-dX}{d\theta} = 0.0137 G^{1.47} (X - X^*)(Y_w - Y)$$

where Y_w is the saturation humidity of the air at the wet-bulb temperature corresponding to Y . *Note:* Here G is expressed as kg/m² · s and θ as seconds; for feet, pounds mass, and hours, the original paper gives the coefficient as 0.003.

Determine the time the rayon should remain in the drier.

SOLUTION $X_1 = 0.46/(1 - 0.46) = 0.852$; $X_2 = 0.085/(1 - 0.085) = 0.093$ kg water/kg dry solid. $Y_1 = 0.08$; $Y_2 = 0.03$ kg water/kg dry air. A water balance [Eq. (12.39)] is

$$\frac{S_S}{G_S} = \frac{0.08 - 0.03}{0.852 - 0.093} = 0.0660 \text{ kg dry solid/kg air}$$

Since the initial moisture content of the rayon is less than the critical, drying takes place entirely within zone III. The form of the rate equation is the same as that of Eq. (12.22), where $k_Y A / S_S (X_c - X^*) = 0.0137 G^{1.47}$. Rearranging the rate equation gives

$$\theta_{III} = \int_0^\theta d\theta = \frac{1}{0.0137 G^{1.47}} \int_{X_2}^{X_1} \frac{dX}{(X - X^*)(Y_W - Y)}$$

which is in the form of Eq. (12.53). On substituting $G = 1.36$, this becomes

$$\theta_{III} = 46.4 \int_{0.093}^{0.852} \frac{dX}{(X - X^*)(Y_W - Y)} \quad (12.59)$$

Consider that part of the drier where the moisture content of the rayon is $X = 0.4$. Eq. (12.54):

$$Y = 0.03 + (0.4 - 0.093)0.066 = 0.0503 \text{ kg water/kg dry gas}$$

At 82°C , $Y = 0.0503$, the wet-bulb temperature is 45°C , and the corresponding saturation humidity $Y_W = 0.068$ (Fig. 7.5). Eq. (7.8):

$$0.0503 = \frac{\bar{p}}{101\,330 - \bar{p}} \frac{18}{29}$$

$$\bar{p} = \text{partial pressure of water} = 7584 \text{ N/m}^2$$

The vapor pressure of water at 82°C $p = 51\,780 \text{ N/m}^2$, and the relative humidity of the air $= (7584/51\,780)100 = 14.63\%$. The equilibrium moisture is $14.63/4 = 3.66\%$, and $X^* = 3.66/(100 - 3.66) = 0.038 \text{ kg water/kg dry solid}$. Therefore

$$\frac{1}{(X - X^*)(Y_W - Y)} = \frac{1}{(0.4 - 0.038)(0.068 - 0.0503)} = 156$$

In similar fashion, other values of this quantity are calculated for other values of X , as follows:

X	Y	Y_W	Relative humidity, %	X^*	$\frac{1}{(X - X^*)(Y_W - Y)}$
0.852	0.080	0.0950	22.4	0.0594	84
0.80	0.0767	0.0920	21.5	0.0568	88
0.60	0.0635	0.0790	18.17	0.0488	117
0.40	0.0503	0.0680	14.63	0.0380	156
0.20	0.0371	0.0550	11.05	0.0284	325
0.093	0.030	0.0490	9.04	0.0231	755

The integral of Eq. (12.59) is evaluated either graphically or numerically; the area is 151.6, whence, by Eq. (12.59),

$$\theta_{III} = 46.4(151.6) = 7034 \text{ s} = 1.95 \approx 2 \text{ h} \quad \text{Ans.}$$

NOTATION FOR CHAPTER 12

Any consistent set of units can be used, except as noted.

a	specific interfacial surface of the solid, L^2/L^3
A	cross-sectional area perpendicular to the direction of flow for through-circulation drying, L^2
	drying surface for cross-circulation drying, L^2
A_m	average cross-sectional area of drying solid, L^2
b	const

C_A	heat capacity of moisture as a liquid, FL/MT
C_p	heat capacity at constant pressure, FL/MT
C_g	humid heat, heat capacity of wet gas per unit mass dry gas content, FL/MT
C_S	heat capacity of dry solid, FL/MT
d_e	equivalent diameter, (cross-sectional area)/perimeter, L
d_p	particle diameter, L
D	diffusivity, L^2/Θ
G	mass velocity of gas, $M/L^2\Theta$
G_S	mass velocity of dry gas, $M/L^2\Theta$
h_c	heat-transfer coefficient for convection, $FL/L^2\Theta$
h_R	heat-transfer coefficient for radiation, $FL/L^2\Theta$
ΔH_A	integral heat of wetting (or of adsorption, hydration, or solution) referred to pure liquid and solid, per unit mass of dry solid, FL/M
H'_G	enthalpy of moist gas per unit mass of dry gas, FL/M
H'_S	enthalpy of wet solid per unit mass of dry solid, FL/M
H_{IOG}	length of an overall gas transfer unit, L
j_D	$k_Y Sc^{2/3}/G_S$, dimensionless
j_H	$h_c Pr^{2/3}/C_p G$, dimensionless
k_M	thermal conductivity of tray, $FL^2/L^2\Theta$
k_S	thermal conductivity of drying solid, $FL^2/L^2\Theta$
k_Y	gas-phase mass-transfer coefficient, mass evaporated/(area)(time) (humidity difference), $M/L^2\Theta(M/M)$
K	const
m	const
N	flux of drying, mass of moisture evaporated/(area)(time), $M/L^2\Theta$
N_c	rate of revolution, Θ^{-1}
N_c	constant flux of drying, $M/L^2\Theta$
N_{IG}	number of gas transfer units, dimensionless
N_{IOG}	number of overall gas transfer units, dimensionless
p	saturation vapor pressure, F/L ²
$\bar{p}$	partial pressure, F/L ²
p_t	total pressure, F/L ²
Pr	Prandtl number = $C_p \mu/k$, dimensionless
q	flux of heat received at the drying surface, batch drying, $FL/L^2\Theta$
	flux of heat received by the solid/area of drier cross section, continuous drying, $FL/L^2\Theta$
q_c	heat flux for convection, $FL/L^2\Theta$
q_G	flux of heat transferred from the gas per unit drier cross section, $FL/L^2\Theta$
q_k	heat flux for conduction, $FL/L^2\Theta$
q_R	heat flux for radiation, $FL/L^2\Theta$
Q	net flux of heat loss, per unit drier cross section, $FL/L^2\Theta$
Re_c	Reynolds number for duct, $d_e G/\mu$, dimensionless
s	slope of drier, L/L
S	interfacial surface of solid/bed cross section, L^2/L^2
S_S	mass of dry solid in a batch, batch drying, M
	mass velocity of dry solid, continuous drying, $M/L^2\Theta$
Sc	Schmidt number, $\mu/\rho D$, dimensionless
t_G	dry-bulb temperature of a gas, T
t_R	temperature of radiating surface, T
t_s	surface temperature, T
t_S	solid temperature, T
t_0	reference temperature, T
T_D	diameter of drier, L
T_G	absolute temperature of a gas, T
T_R	absolute temperature of radiating surface, T

T_s	absolute temperature of surface, T
U	overall heat-transfer coefficient, $\text{FL/L}^2\text{T}\Theta$
U_k	overall heat-transfer coefficient [Eq. (12.16)], $\text{FL/L}^2\text{T}\Theta$
X	moisture content of a solid, mass moisture/mass dry solid, M/M
X^*	equilibrium moisture content of a solid, mass moisture/mass dry solid, M/M
z_M	thickness of tray material, L
z_S	thickness of drying solid, L
Z	length of drier, L
Δ	difference
ϵ	emissivity of drying surface, dimensionless
θ	time, Θ
λ_s	latent heat of vaporization at t_s , FL/M
μ	viscosity, $\text{M/L}\Theta$
π	3.1416
ρ	density, M/L^3
ρ_S	apparent density of solid, mass dry solid/wet volume, M/L^3
ϕ_D	holdup of solid in a continuous drier, vol solids/vol drier, L^3/L^3
ϕ_{D0}	holdup of solid at no gas flow, L^3/L^3

Subscripts

as	adiabatic saturation
c	at critical moisture content
G	gas
m	average
max	maximum
s	surface
S	dry gas, dry solid
1	at beginning, batch drying; at solids-entering end, continuous drier
2	at end, batch drying; at solids-leaving end, continuous drier
I, II, III	zones I, II, and III in a continuous drier

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PROBLEMS

12.1 A plant wishes to dry a certain type of fiberboard in sheets 1.2 by 2 m by 12 mm (4 by 6 ft by $\frac{1}{2}$ in). To determine the drying characteristics, a 0.3- by 0.3-m (1- by 1-ft) sample of the board, with the edges sealed so that drying took place from the two large faces only, was suspended from a balance in a laboratory cabinet drier and exposed to a current of hot, dry air. The initial moisture content

was 75%. The sheet lost weight at the constant rate of 1×10^{-4} kg/s (0.8 lb/h) until the moisture content fell to 60%, whereupon the drying rate fell. Measurements of the rate of drying were discontinued, but after a long period of exposure to this air it was established that the equilibrium moisture content was 10%. The dry mass of the sample was 0.9 kg (2 lb). All moisture contents are on the wet basis.

Determine the time for drying the large sheets from 75 to 20% moisture under the same drying conditions.

12.2 A sample of a porous, manufactured sheet material of mineral origin was dried from both sides by cross circulation of air in a laboratory drier. The sample was 0.3 m square and 6 mm thick, and the edges were sealed. The air velocity over the surface was 3 m/s, its dry-bulb temperature was 52°C, and its wet-bulb temperature 21°C. There were no radiation effects. The solid lost moisture at a constant rate of 7.5×10^{-5} kg/s until the critical moisture content, 15% (wet basis), was reached. In the falling-rate period, the rate of evaporation fell linearly with moisture content until the sample was dry. The equilibrium moisture was negligible. The dry weight of the sheet was 1.8 kg.

Estimate the time for drying sheets of this material 0.6 by 1.2 m by 12 mm thick from both sides, from 25 to 2% moisture (wet basis), using air of dry-bulb temperature 66°C but of the same absolute humidity at a linear velocity over the sheet of 5 m/s. Assume no change in the critical moisture with the changed drying conditions. **Ans.:** 3.24 h.

12.3 Estimate the rate of drying during the constant-rate period for the conditions existing as the air enters the trays of the drier of Illustration 12.2. The solid being dried is a granular material of thermal conductivity when wet = $1.73 \text{ W/m} \cdot \text{K}$, and it completely fills the trays. The metal of the trays is stainless steel, 16 BWG (1.65 mm thick). Include in the calculations an estimate of the radiation effect from the undersurface of each tray upon the drying surface.

12.4 A laboratory drying test was made on a 0.1-m^2 sample of a fibrous boardlike material. The sample was suspended from a balance, its edges were sealed, and drying took place from the two large faces. The air had a dry-bulb temperature of 65°C, wet-bulb temperature 29°C, and its velocity was 1.5 m/s past the sample. The following are the weights recorded at various times during the test:

Time, h	Mass, kg	Time, h	Mass, kg	Time, h	Mass, kg
0	4.820	3.0	4.269	7.0	3.885
0.1	4.807	3.4	4.206	7.5	3.871
0.2	4.785	3.8	4.150	8.0	3.859
0.4	4.749	4.2	4.130	9.0	3.842
0.8	4.674	4.6	4.057	10.0	3.832
1.0	4.638	5.0	4.015	11	3.825
1.4	4.565	5.4	3.979	12	3.821
1.8	4.491	5.8	3.946	14	3.819
2.2	4.416	6.0	3.933	16	3.819
2.6	4.341	6.5	3.905		

The sample was then dried in an oven at 110°C, and the dry mass was 3.765 kg.

(a) Plot the rate-of-drying curve.

(b) Estimate the time required for drying the same sheets from 20 to 2% moisture (wet basis) using air of the same temperature and humidity but with 50% greater air velocity. Assume that the critical moisture remains unchanged. **Ans.:** 8.55 h.

12.5 A pigment material which has been removed wet from a filter press is to be dried by extruding it into small cylinders and subjecting them to through-circulation drying. The extrusions are 6 mm in diameter, 50 mm long, and are to be placed on screens to a depth of 65 mm. The surface of the particles is estimated to be $295 \text{ m}^2/\text{m}^3$ of bed and the apparent density $1040 \text{ kg dry solid}/\text{m}^3$. Air at a mass velocity $0.95 \text{ kg dry air}/\text{m}^2 \cdot \text{s}$ will flow through the bed, entering at 120°C, humidity 0.05 kg water/kg dry air.

(a) Estimate the constant rate of drying to be expected. *Note:* For long cylinders it is best to take the equivalent diameter as the actual cylinder diameter. **Ans.:** $0.0282 \text{ kg/m}^2 \cdot \text{s}$.

(b) Estimate the constant rate of drying to be expected if the filter cake is dried on trays by cross circulation of the air over the surface at the same mass velocity, temperature, and humidity. Neglect radiation and heat conduction through the solid. **Ans.:** $4.25 \times 10^{-4} \text{ kg/m}^2 \cdot \text{s}$.

12.6 A louver-type continuous rotary drier (Fig. 12.22) was used to dry wood chips from 40 to 15% moisture [Horgan, *Trans. Inst. Chem. Eng.*, 6, 131 (1928)]. The wood was at 33°F (0.56°C), while the dried product was discharged at 100°F (37.8°C) at a rate of 3162 lb/h (0.398 kg/s). The drying medium was the gas resulting from the combustion of fuel, but for the present calculation it may be assumed to have the characteristics of air. It entered the drier at 715°F (380°C), with a humidity 0.038 kg water vapor/kg dry gas, at a rate of 275 lb/min (2.079 kg/s) wet. The gas was discharged at 175°F (77°C). The heat capacity of the dry wood can be taken as $0.42 \text{ Btu/lb} \cdot ^\circ\text{F}$ ($1758 \text{ J/kg} \cdot \text{K}$), and the heat of wetting can be ignored. Estimate the rate of heat loss.

12.7 A direct-heat, cocurrent-flow rotary drier, 8 ft diameter, 60 ft long (2.44 by 18.3 m), was used to dry chopped alfalfa [see Gutzeit and Spraul, *Chem. Eng. Prog.*, 49, 380 (1953)]. Over a 5-h test period, the drier delivered an average of 2200 lb/h (0.28 kg/s) of dried product at 11% moisture and 145°F (63°C) when fed with alfalfa containing 79% moisture at 80°F (26.7°C). The drying medium was the combustion products resulting from the burning of $13\,074 \text{ ft}^3/\text{h}$ (80°F , 4 oz/in² gauge pressure) [$0.1029 \text{ m}^3/\text{s}$ (26.7°C , 862 N/m^2 gauge pressure)] of natural gas (85% methane, 10% ethane, 5% nitrogen by volume) with air at 80°F (26.7°C), 50% humidity. The gas analyzed 2.9% CO_2 , 15.8% O_2 , 81.3% N_2 by volume on a dry basis; it entered the drier at 1500°F (816°C) and left at 195°F (91°C). The heat capacity of dry alfalfa is estimated to be $0.37 \text{ Btu/lb} \cdot ^\circ\text{F}$ ($1549 \text{ J/kg} \cdot \text{K}$), and the heat of wetting can be neglected. Compute the volumetric rate of gas flow through the exhaust fan and the heat losses. **Ans.:** $6.4 \text{ m}^3/\text{s}$ ($13\,550 \text{ ft}^3/\text{min}$), 1500 kW ($5.12 \times 10^6 \text{ Btu/h}$).

12.8 A direct-heat countercurrent rotary hot-air drier is to be chosen for drying an insoluble crystalline organic solid. The solid will enter at 20°C , containing 20% water. It will be dried by air entering at 155°C , $0.01 \text{ kg water/kg dry air}$. The solid is expected to leave at 120°C , with a moisture content 0.3%. Dried product delivered will be 450 kg/h . The heat capacity of the dry solid is $837 \text{ J/kg} \cdot \text{K}$, and its average particle size is 0.5 mm . The superficial air velocity should not exceed 1.6 m/s in any part of the drier. The drier will be insulated, and heat losses can be neglected for present purposes. Choose a drier from the following standard sizes and specify the rate of airflow which should be used: 1 by 3 m, 1 by 9 m, 1.2 by 12 m, 1.4 by 9 m, 1.5 by 12 m. **Ans.:** 1.2 by 12 m.

12.9 A manufactured material in the form of sheets 0.6 by 1.2 m by 12 mm is to be continuously dried in an adiabatic countercurrent hot-air tunnel drier at the rate of 100 sheets per hour. The sheets will be supported on a special conveyor carrying the material in tiers 30 sheets high, and they will be dried from both sides. The dry mass of each sheet is 12 kg , and the moisture content will be reduced from 50 to 5% water by air entering at 120°C , humidity $0.01 \text{ kg water/kg dry air}$; 40 kg dry air will be passed through the drier per kilogram dry solid.

In a small-scale experiment, when dried with air at constant drying conditions, dry-bulb temperature 95°C , wet-bulb temperature 50°C , and at the same velocity to be used in the large drier, the constant-drying rate was $3.4 \times 10^{-4} \text{ (kg water evaporated)/m}^2 \cdot \text{s}$ and the critical moisture content 30%. The equilibrium-moisture content was negligible.

(a) Calculate the value of k_y from the data of the small-scale experiment.

Ans.: $0.01545 \text{ kg/m}^2 \cdot \text{s} \cdot \Delta Y$.

(b) For the large drier, calculate the humidity of the air leaving and at the point where the solid reaches the critical moisture content.

(c) Estimate the time of drying in the large drier. **Ans.:** 9.53 h.

(d) How many sheets of material will be in the drier at all times?

12.10 A continuous countercurrent hot-air tunnel drier is to be designed to dry a filter-press cake of coarse crystals of an inorganic substance, insoluble in water. The filter-press cake will be placed on trays 1.0 m long by 0.9 m wide by 25 mm , with 20 trays to a truck, with 50 mm between trays. The tunnel drier will have a cross section 2 m high by 1 m wide. The trays have a reinforced screen bottom, so that drying takes place from both top and bottom of each tray. Production permits introducing one

truck load per hour. Each tray contains 30 kg dry solid, which will enter the drier at 25°C, 50% moisture, and will be dried to negligible moisture content. The critical moisture content is 15%, and the equilibrium moisture is negligible. The trucks are steel, each weighing about 135 kg. The air is to enter at 150°C, humidity 0.03 kg water/kg dry air, and the discharged solid is expected to leave at 135°C. The air is to be blown over the trays so that the average velocity at the air entrance is to be 4.5 m/s over the trays. The heat capacity of the dry solid is 1.255 kJ/kg · K. The drier is to be well insulated.

(a) Calculate the length of the drier required. Ans.: 8.5 m by Eq. (12.20a), 6.5 m by Eq. (12.20).

(b) The entering air is to be prepared by recycling a portion of the discharged air with atmospheric air (25°C, humidity 0.01 kg water/kg dry air) and heating the mixture to 150°C. Calculate the percentage of discharge air to be recycled and the heat requirement. Calculate the heat also expressed per unit mass of water evaporated. Ans.: 3515 kJ/kg.

LEACHING

Leaching is the preferential solution of one or more constituents of a solid mixture by contact with a liquid solvent. This unit operation, one of the oldest in the chemical industries, has been given many names, depending to some extent upon the technique used for carrying it out. *Leaching* originally referred to percolation of the liquid through a fixed bed of the solid, but is now used to describe the operation generally, by whatever means it may be done. *Lixiviation* is used less frequently as a synonym for leaching, although originally it referred specifically to the leaching of alkali from wood ashes. The term *extraction* is also widely used to describe this operation in particular, although it is applied to all the separation operations as well, whether mass-transfer or mechanical methods are involved. *Decoction* refers specifically to the use of the solvent at its boiling temperature. When the soluble material is largely on the surface of an insoluble solid and is merely washed off by the solvent, the operation is sometimes called *elutriation* or *elution*. This chapter will also consider these washing operations, since they are frequently intimately associated with leaching.

The metallurgical industries are perhaps the largest users of the leaching operation. Most useful minerals occur in mixtures with large proportions of undesirable constituents, and leaching of the valuable material is a separation method which is frequently applied. For example, copper minerals are preferentially dissolved from certain of their ores by leaching with sulfuric acid or ammoniacal solutions, and gold is separated from its ores with the aid of sodium cyanide solutions. Leaching similarly plays an important part in the metallurgical processing of aluminum, cobalt, manganese, nickel, and zinc. Many naturally occurring organic products are separated from their original structure by leaching. For example, sugar is leached from sugar beets with hot water, vegetable oils are recovered from seeds such as soybeans and cottonseed by

leaching with organic solvents, tannin is dissolved out of various tree barks by leaching with water, and many pharmaceutical products are similarly recovered from plant roots and leaves. Tea and coffee are prepared both domestically and industrially by leaching operations. In addition, chemical precipitates are frequently washed of their adhering mother liquors by techniques and in equipment quite similar to those used in true leaching operations, as in the washing of caustic soda liquor from precipitated calcium carbonate following the reaction between soda ash and lime.

Preparation of the Solid

The success of a leaching and the technique to be used will very frequently depend upon any prior treatment which may be given the solid.

In some instances, small particles of the soluble material are completely surrounded by a matrix of insoluble matter. The solvent must then diffuse into the mass, and the resulting solution must diffuse out, before a separation can result. This is the situation with many metallurgical materials. Crushing and grinding of such solids will greatly accelerate the leaching action, since then the soluble portions are made more accessible to the solvent. A certain copper ore, for example, can be leached effectively by sulfuric acid solutions within 4 to 8 h if ground to pass through a 60-mesh screen, in 5 days if crushed to 6-mm granules, and only in 4 to 6 years if 150-mm lumps are used [43]. Since grinding is expensive, the quality of the ore will have much to do with the choice of size to be leached. For certain gold ores, on the other hand, the tiny metallic particles are scattered throughout a matrix of quartzite which is so impervious to the leaching solvent that it is essential to grind the rock to pass through a 100-mesh screen if leaching is to occur at all. When the soluble substance is more or less uniformly distributed throughout the solid or even in solid solution, the leaching action may provide channels for the passage of fresh solvent and fine grinding may not be necessary. Collapse of the insoluble skeleton which remains after solute removal may present problems, however.

Vegetable and animal bodies are cellular in structure, and the natural products to be leached from these materials are usually found inside the cells. If the cell walls remain intact upon exposure to a suitable solvent, the leaching action involves osmotic passage of the solute through the cell walls. This may be slow, but it is impractical and sometimes undesirable to grind the material small enough to release the contents of individual cells. Thus, sugar beets are cut into thin, wedge-shaped slices called *cossettes* before leaching in order to reduce the time required for the solvent water to reach the individual plant cells. The cells are deliberately left intact, however, so that the sugar will pass through the semipermeable cell walls while the undesirable colloidal and albuminous materials largely remain behind. For many pharmaceutical products recovered from plant roots, stems, and leaves, the plant material is frequently dried before treatment, and this does much toward rupturing the cell walls and releasing the solute for direct action by the solvent. Vegetable seeds and beans, such as soybeans, are usually rolled or flaked to give particles in the size range 0.15 to

0.5 mm. The cells are, of course, smaller than this, but they are largely ruptured by the flaking process, and the oils are then more readily contacted by the solvent.

When the solute is adsorbed upon the surface of solid particles or merely dissolved in adhering solution, no grinding or crushing is necessary and the particles can be washed directly.

Temperature of Leaching

It is usually desirable to leach at as high a temperature as possible. Since higher temperatures result in higher solubility of the solute in the solvent, higher ultimate concentrations in the leach liquor are possible. The viscosity of the liquid is lower and the diffusivities larger at higher temperatures, leading to increased rates of leaching. With some natural products such as sugar beets, however, temperatures which are too high may lead to leaching of excessive amounts of undesirable solutes or chemical deterioration of the solid.

Methods of Operation and Equipment

Leaching operations are carried out under batch and semibatch (unsteady-state) as well as under completely continuous (steady-state) conditions. In each category, both stagewise and continuous-contact types of equipment are to be found. Two major handling techniques are used: spraying or trickling the liquid over the solid, and immersing the solid completely in the liquid. The choice of equipment to be used in any case depends greatly upon the physical form of the solids and the difficulties and cost of handling them. This has led in many instances to the use of very specialized types of equipment in certain industries.

UNSTEADY-STATE OPERATION

The unsteady-state operations include those where the solids and liquids are contacted in purely batchwise fashion and also those where a batch of the solid is contacted with a continually flowing stream of the liquid (semibatch method). Coarse solid particles are usually treated in fixed beds by percolation methods, whereas finely divided solids, which can be kept in suspension more readily, can be dispersed throughout the liquid with the help of some sort of agitator.

In-Place (in Situ) Leaching

This operation, also called *solution mining*, refers to the percolation leaching of minerals in place at the mine, by circulation of the solvent over and through the ore body. It is used regularly in the removal of salt from deposits below the earth's surface by solution of the salt in water which is pumped down to the deposit. It has been applied to the leaching of low-grade copper ores containing as little as 0.2 percent copper [34] and to ores as deep as 335 m (1100 ft) below

the surface [3]. In the solution mining of uranium, the ore must be oxidized in place in order to solubilize it in carbonate solutions. The reagent may be injected continuously through one set of pipes drilled down to the ore and the resulting liquor removed through a different set. Alternatively, the reagent may be pumped into the ground intermittently and withdrawn through the same well.

Heap Leaching

Low-grade ores whose mineral values do not warrant the expense of crushing or grinding can be leached in the form of run-of-mine lumps built into huge piles upon impervious ground. The leach liquor is pumped over the ore and collected as it drains from the heap. Copper has been leached from pyritic ores in this manner in heaps containing as much as 2.2×10^7 t of ore, using over $20\,000\text{ m}^3$ (5×10^6 gal) of leach liquor per day. It may require up to 7 or more years to reduce the copper content of such heaps from 2 to 0.3 percent. In a typical case of heap leaching of uranium, after laying a pattern of perforated drain pipe on an impervious clay base, the ore is built into piles on top of the pipes, in heaps some 6 to 8 m tall, trapezoidal in cross section and 120 m wide at the base, as much as 800 m (0.5 mi) long. The leach solution, introduced into ponds in the top of the heap, percolates down to the drain pipes at the base, whence it is led away.

Percolation Tanks

Solids of intermediate size can conveniently be leached by percolation methods in open tanks. The construction of these tanks varies greatly, depending upon the nature of the solid and liquid to be handled and the size of the operation, but they are relatively inexpensive. Small tanks are frequently made of wood if it is not chemically attacked by the leach liquid. The solid particles to be leached rest upon a false bottom, which in the simplest construction consists of a grating of wooden strips arranged parallel to each other and sufficiently close to support the solid. These in turn may rest upon similar strips arranged at right angles, 150 mm or more apart, so that the leach liquor flows to a collection pipe leading from the bottom of the tank. For supporting fairly fine particles, the wood grating may be further covered by a coconut matting and a tightly stretched canvas filter cloth, held in place by caulking a rope into a groove around the periphery of the false bottom. Small tanks may also be made entirely of metal, with perforated false bottoms upon which a filter cloth is placed, as in the leaching of pharmaceutical products from plants. Very large percolation tanks (45 by 34 by 5.5 m deep) for leaching copper ores have been made of reinforced concrete and lined with lead or bituminous mastic. Small tanks may be provided with side doors near the bottom for sluicing away the leached solid, while very large tanks are usually emptied by excavating from the top. Tanks should be filled with solid of as uniform a particle size as practical, since then the percentage of voids will be largest and the pressure drop required for flow of the leaching liquid least. This also leads to uniformity of the extent of leaching

individual solid particles and less difficulty with channeling of the liquid through a limited number of passageways through the solid bed.

The operation of such a tank may follow any of several procedures. After the tank is filled with solid, a batch of solvent sufficient to immerse the solid completely may be pumped into the tank and the entire mass allowed to steep or soak for a prescribed period of time. During this period the batch of liquid may or may not be circulated over the solid by pumping. The liquid is then drained from the solid by withdrawing it through the false bottom of the tank. This entire operation represents a single stage. Repetition of this process will eventually dissolve all the solute. The only solute then retained is that dissolved in the solution wetting the drained solid. This can be washed out by filling the tank with fresh solvent and repeating the operation as many times as necessary. An alternative method is continuously to admit liquid into the tank and continuously to withdraw the resulting solution, with or without recirculation of a portion of the total flow. Such an operation may be equivalent to many stages. Since the solution which results is usually denser than the solvent, convective mixing is reduced by percolation in the downward direction. Upward flow is sometimes used, nevertheless, in order to avoid clogging the bed or the filter with fines, but this may result in excessive entrainment of the fines in the overflow liquid. Still a further modification, less frequently used, is to spray the liquid continuously over the top and allow it to trickle downward through the solid without fully immersing the solid at any time. An excellent review of the processes, techniques, and chemistry of leaching of ores is available [6].

Retention of Liquid after Drainage

Imagine a bed of granular solids whose void space is completely filled with liquid. When the liquid is allowed to drain under the influence of gravity, with admission of air to the voids from the top of the bed, the rate of liquid flow is at first very rapid. The rate gradually falls, and after a relatively long period of time no additional drainage occurs. The bed still contains liquid, however. The fraction of the void volume still occupied by liquid is termed the *residual saturation* s . Figure 13.1 shows the variation of s with height of the bed [11]. In the upper part of the bed the value of s is constant at s_0 , and this represents the liquid which remains in the crevasses and small angles between the particles as fillets, held in place by surface tension. In the lower part of the bed, the liquid is held up in the voids, filling them completely ($s = 1.0$) by capillary action. The drain height Z_D is defined as the height where the value of s is the average in the range s_0 to unity, as shown in the figure. The average value of s for the entire bed will be the area between the ordinate axis and the curve of the figure, divided by the bed height Z ,

$$s_{av} = \frac{(Z - Z_D)s_0}{Z} + \frac{Z_D}{Z} \quad (13.1)$$

A large number of measurements of Z_D under a wide variety of conditions showed that, approximately [11],

$$Z_D = \frac{0.275(g_c/g)}{(K/g)^{0.5}(\rho_L/\sigma)} \quad (13.2)$$

where K = "permeability" of bed

ρ_L = liquid density

σ = surface tension of liquid

The value of s_0 was found to depend upon the group $(K\rho_L/g\sigma)(g/g_c)$, called the *capillary number*, as

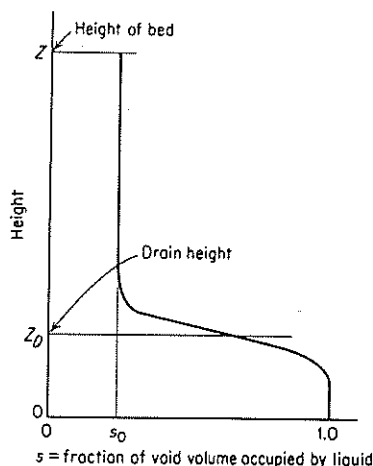


Figure 13.1 Drainage of packed beds [11].

follows:

$$S_0 = \begin{cases} 0.075 & \frac{K\rho_L}{g\sigma} \frac{g}{g_c} < 0.02 \\ \frac{0.0018}{(K\rho_L/g\sigma)(g/g_c)} & \frac{K\rho_L}{g\sigma} \frac{g}{g_c} > 0.02 \end{cases} \quad (13.3)$$

(13.4)

In these expressions it is assumed that drainage has occurred under the action of the force of gravity only and that the contact angle between liquid and solid surfaces is 180° .

The permeability K is the proportionality constant in the flow equation for laminar flow through the bed,

$$G = \frac{K\rho_L \Delta p}{\mu_L Z} \frac{g}{g_c} \quad (13.5)$$

where Δp is the drop in pressure across the bed and G is the mass velocity of flow based on the entire cross section of the bed. Equation (6.66) describes the flow through beds of granular solids, and for laminar flow only the first term of the right-hand side of this expression is used. If $\Delta p/Z$ from this equation is substituted in Eq. (13.5), with Re replaced by $d_p G/\mu_L$, simplification leads to

$$K = \frac{d_p^2 \epsilon^3 g}{150(1 - \epsilon)^2} \quad (13.6)$$

where d_p is the diameter of a sphere of the same surface/volume ratio as the particles of the bed and ϵ is the fraction-void volume. For fibrous material and others whose value of d_p may be difficult to estimate, K can be obtained from Eq. (13.5) after experimental measurement of the pressure drop for laminar flow through the bed.

Illustration 13.1 The sugar remaining in a bed of bone char used for decolorization is leached by flooding the bed with water, following which the bed is drained of the resulting sugar solution. The bed diameter is 1 m, the depth 3.0 m, the temperature is 65°C . The sugar solution which drains has a density 1137 kg/m^3 (71 lb/ft^3) and a surface tension 0.066 N/m . The bulk density of the char is 960 kg/m^3 (60 lb/ft^3) and the individual particle density 1762 kg/m^3 (110 lb/ft^3). The particles have a specific external surface $16.4 \text{ m}^2/\text{kg}$ ($80 \text{ ft}^2/\text{lb}$).

Estimate the mass of solution still retained by the bed after dripping of the solution has stopped. Express this also as mass solution/mass dry bone char.

SOLUTION The fractional void volume $= \epsilon = 1 - (\text{bulk density}/\text{particle density}) = 1 - 960/1762 = 0.455 \text{ m}^3 \text{ voids/m}^3 \text{ bed}$. The particle surface $= a_p = (16.4 \text{ m}^2/\text{kg})(960 \text{ kg/m}^3) = 15\,744 \text{ m}^2/\text{m}^3 \text{ bed}$. Eq. (6.67):

$$d_p = \frac{6(1 - \epsilon)}{a_p} = \frac{6(1 - 0.455)}{15\,744} = 2.077 \times 10^{-4} \text{ m}$$

Eq. (13.6):

$$K = \frac{d_p^2 \epsilon^3 g}{150(1 - \epsilon)^2} = \frac{(2.077 \times 10^{-4})^2 (0.455)^3 (9.807)}{150(1 - 0.455)^2} = 8.94 \times 10^{-10} \text{ m}^3/\text{s}$$

$$\frac{K \rho_s}{g \sigma} \frac{g}{g_c} = \frac{(8.94 \times 10^{-10})(1137)}{9.81(0.066)} \frac{9.81}{1} = 1.54 \times 10^{-5}$$

Eq. (13.3):

$$s_0 = 0.075$$

Eq. (13.2):

$$Z_D = \frac{0.275(1/9.81)}{[(8.94 \times 10^{-10})/9.81]^{0.5} (1137/0.066)} = 0.1705 \text{ m}$$

$$Z = 3.0 \text{ m}$$

Eq. (13.1):

$$s_{av} = \frac{(3 - 0.1705)(0.075)}{3} + \frac{0.1705}{3} = 0.1276$$

$$\frac{\text{Vol liquid retained}}{\text{Vol bed}} = 0.1276 \epsilon = 0.1276(0.455) = 0.0581 \text{ m}^3/\text{m}^3$$

$$\text{Mass liquid in bed} = 0.0581 \frac{\pi(1)^2 3}{4} (1137) = 155.6 \text{ kg}$$

$$\frac{\text{Mass liquid}}{\text{Mass dry solid}} = \frac{0.0581(1137)}{1(960)} = 0.069 \text{ kg/kg}$$

Countercurrent Multiple Contact; the Shanks System

Leaching and washing of the leached solute from the percolation tanks by the crosscurrent methods described above will inevitably result in weak solutions of the solute. The strongest solution will result if a countercurrent scheme is used, wherein the final withdrawn solution is taken from contact with the freshest solid and the fresh solvent is added to solid from which most of the solute has already been leached or washed. In order to avoid moving the solids physically from tank to tank in such a process, the arrangement of Fig. 13.2, shown schematically for a system of six tanks, is used. This *Shanks system*,† as it is called, is operated in the following manner:

1. Assume at the time of inspecting the system at Fig. 13.2a that it has been in operation for some time. Tank 6 is empty, tanks 1 to 5 are filled with solid, tank 5 most recently and tank 1 for the longest time. Tanks 1 to 5 are also

† Named after James Shanks, who first introduced the system in 1841 into England for the leaching of soda ash from the "black ash" of the Le Blanc process. It was, however, apparently a German development.

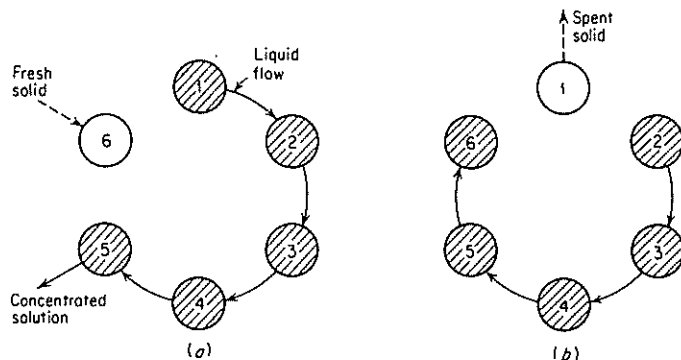


Figure 13.2 Countercurrent multiple contact, Shanks system.

- filled with leach liquid, and the most concentrated is in tank 5 since it is in contact with the freshest solid. Fresh solvent has just been added to tank 1.
- Withdraw the concentrated solution from tank 5, transfer the liquid from tank 4 to tank 5, from 3 to 4, from 2 to 3, and from 1 to 2. Add fresh solid to tank 6.
- Refer to Fig. 13.2*b*. Discard the spent solid from tank 1. Transfer the liquid from tank 5 to tank 6, from 4 to 5, from 3 to 4, and from 2 to 3. Add fresh solvent to tank 2. The circumstances are now the same as they were at the start in Fig. 13.2*a*, except that the tank numbers are each advanced by one.
- Continue the operation in the same manner as before.

The scheme is identical with the batch simulation of a multistage countercurrent operation shown in Fig. 10.38. After several cycles have been run through in this manner, the concentrations of solution and in the solid in each tank approach very closely the values obtaining in a truly continuous countercurrent multistage leaching. The system can, of course, be operated with any number of tanks, and anywhere from 6 to 16 are common. They need not be arranged in a circle but are better placed in a row, called an *extraction battery*, so that additional tanks can conveniently be added to the system if desired. The tanks may be placed at progressively decreasing levels, so that liquid can flow from one to the other by gravity with a minimum of pumping.

Such leaching tanks and arrangements are used extensively in the metallurgical industries, for recovery of tannins from tree barks and woods, for leaching sodium nitrate from Chilean nitrate-bearing rock (*caliche*), and in many other processes.

Percolation in Closed Vessels

When the pressure drop for flow of liquid is too high for gravity flow, closed vessels must be used and the liquid is pumped through the bed of solid. Such

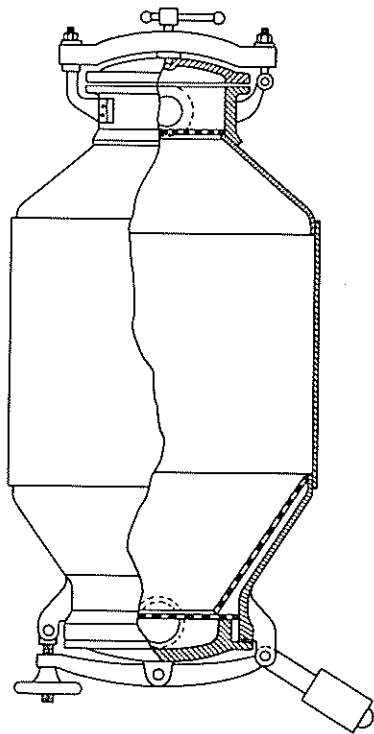


Figure 13.3 Sugar-beet diffuser [12]. (Courtesy of the Institution of Chemical Engineers.)

vessels are sometimes called *diffusers*. Closed tanks are also necessary to prevent evaporation losses when the solvent is very volatile or when temperatures above the normal boiling point of the solvent are desired. For example, some tannins are leached with water at 120°C , 345 kN/m^2 ($50 \text{ lb}_f/\text{in}^2$) pressure, in closed percolation tanks.

Designs vary considerably, depending upon the application. In leaching sugar from sugar-beet slices, or cossettes, a diffuser of the type shown in Fig. 13.3 is used. They are arranged in a battery containing up to 16 vessels, and the beets are leached with hot water in the countercurrent fashion of the Shanks system. Heaters are placed between the diffusers to maintain a solution temperature of 70 to 78°C . In this manner 95 to 98 percent of the sugar, in beets containing initially about 18 percent, can be leached to form a solution of 12 percent concentration. Countercurrent, continuous equipment is also used in the sugar-beet industry [4, 20].

Filter-Press Leaching

Finely divided solids, too fine for treatment by percolation in relatively deep percolation tanks, can be filtered and leached in the filter press by pumping the

solvent through the press cake. This is, of course, common practice in washing mother liquor from precipitates which have been filtered.

Agitated Vessels

Channeling of the solvent in percolation or filter-press leaching of fixed beds, with its consequent slow and incomplete leaching, can be avoided by stirring the liquid and solid in leaching vessels. For coarse solids, many types of special stirred or agitated vessels have been devised [36]. In such cases, closed cylindrical vessels are arranged vertically (Fig. 13.4*a*) and are fitted with power-driven paddles or stirrers on vertical shafts, as well as false bottoms for drainage of the leach solution at the end of the operation. In others, the vessels are horizontal, as in Fig. 13.4*b*, with the stirrer arranged on a horizontal shaft. In some cases, a horizontal drum is the extraction vessel, and the solid and liquid are tumbled about inside by rotation of the drum on rollers, as in Fig. 13.4*c*. These devices are operated in batchwise fashion and provide a single leaching stage. They can be used singly but frequently are also used in batteries arranged for countercurrent leaching. They have been used extensively in the older European and South American installations for leaching vegetable oils from seeds but relatively little in the United States.

Finely divided solids can be suspended in leaching solvents by agitation, and for batch operation a variety of agitated vessels are used (see Chaps. 6 and 11). The simplest is the Pachuca tank (Fig. 13.5), which is employed extensively

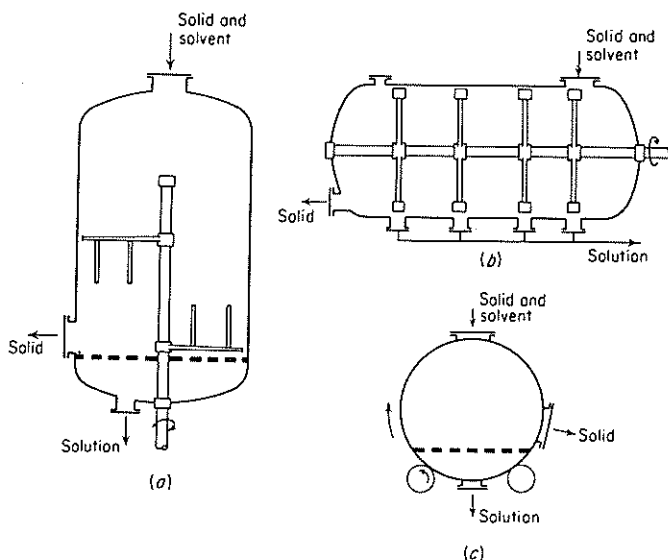


Figure 13.4 Agitated batch leaching vessels (see also Chap. 11).

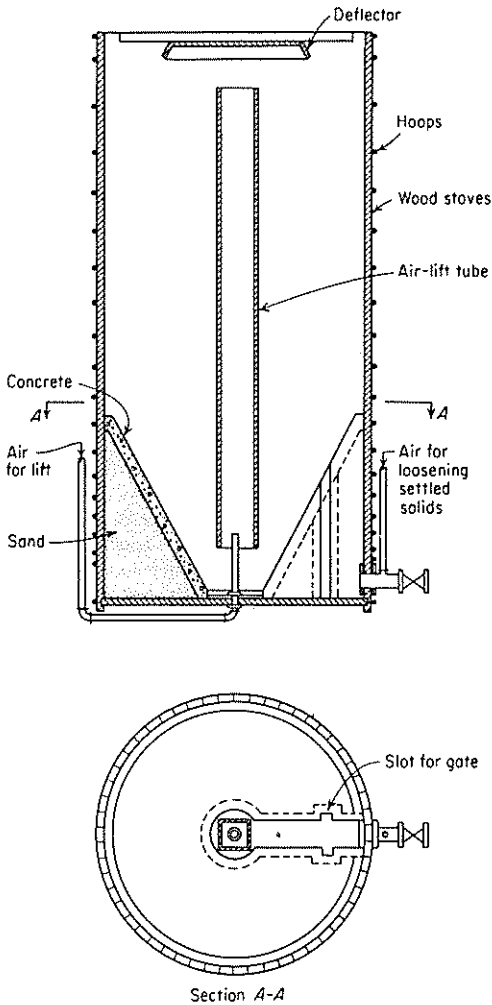


Figure 13.5 Pachuca tank. (Fran Liddell, "Handbook of Non-ferrous Metallurgy," 2d ed., McGraw-Hill Book Company, New York, 1945. Used by permission of the publisher.)

in the metallurgical industries. These tanks are constructed of wood, metal, or concrete and may be lined with inert metal such as lead, depending upon the nature of the leaching liquid. Agitation is accomplished by an air lift: the bubbles of air rising through the central tube cause the upward flow of liquid and suspended solid in the tube and consequently vertical circulation of the tank contents [26]. The standard mechanical agitators, with turbine-type impellers, for example, can also be used to keep the finely divided solids suspended in the liquid. After the leaching has been accomplished, the agitation is stopped, the solid is allowed to settle in the same or a separate vessel, and the clear,

supernatant liquid is decanted by siphoning over the top of the tank or by withdrawal through discharge pipes placed at an appropriate level in the side of the tank. If the solids are finely divided and settle to a compressible sludge, the amount of solution retained in the settled solids will be considerable. Agitation and settling with several batches of wash solvent may then be necessary to recover the last traces of solute, and this may be done in a countercurrent fashion. Rates of leaching, provided diffusivities within the solid are known, can be computed in the manner described for adsorption in such vessels, Chap. 11. Alternatively, the solid may be filtered and washed in the filter.

Batch Settling

The settling characteristics of a slurry consisting of a finely divided solid, of uniform density and reasonably uniform particle size, which is dispersed in a liquid are easily followed by observing a sample of the slurry allowed to stand undisturbed in a vertical cylinder of transparent glass. If the slurry is initially very dilute, the particles will be observed to settle down through the liquid individually, each at a rate dependent upon the particle size, the relative density of solid and liquid, and the viscosity of the liquid, eventually to collect in a pile at the bottom. Ultimately the liquid becomes clear, but at no time until the end is there a sharp line of demarcation between clear liquid and the settling slurry. For more concentrated slurries, of the sort usually encountered in leaching and washing operations, the behavior is different, however. It will usually be observed that the particles settle more slowly owing to mutual interference (hindered settling). Furthermore, except for a few particles of relatively large size which may be present, there is little classification according to size, and the particles largely settle together. As a result there is usually a reasonably sharp line of demarcation between the clear, supernatant liquor in the upper part of the cylinder and the settling mass of solids in the lower part.

Consider the cylinder of Fig. 13.6, initially filled to a height Z_0 with a slurry of uniform concentration w_0 weight fraction solids, in which some settling has already taken place. At the time

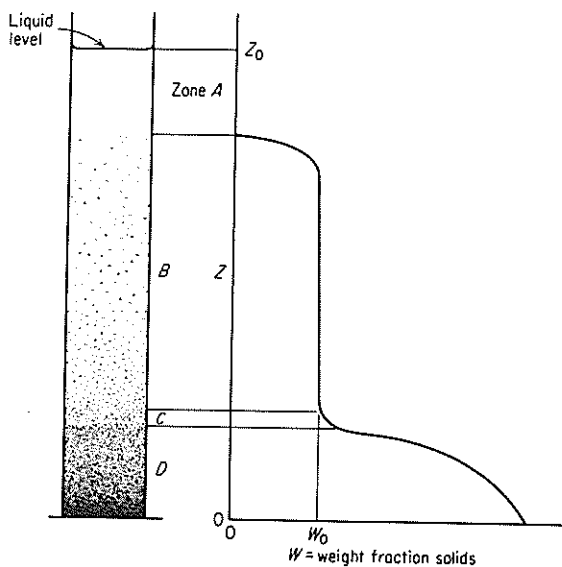


Figure 13.6 Batch settling.

of observation, there is a zone *A* of clear liquid at the top. Directly beneath this is zone *B*, throughout which the solids concentration is reasonably uniform at the initial value w_0 , as shown by the accompanying graph [10]. In zone *D* at the bottom, usually called the *compression zone*, the particles accumulating from above have come to rest upon each other, and, owing to their weight, liquid is squeezed out from between the particles. For compressible sludges, this results in increasing solids concentration with depth in this zone, as shown by the curve. Zone *C* is a transition zone between *B* and *D* which may not always be clearly defined. As settling continues beyond the time corresponding to that in the figure, the line of demarcation between zones *A* and *B* falls and the height of zone *D* rises, until eventually zone *B* disappears and only a compression zone containing all the solids remains. This then slowly subsides to some ultimate height.

The rate of settling is usually followed by plotting the height of the line of demarcation between zones *A* and *B* against time, as shown by the solid curve of Fig. 13.7. The broken curve represents the position of the upper level of zone *D*. The top of zone *B* settles at constant rate (curve of Z vs. time straight) from the beginning until zone *B* has nearly disappeared and all the solids are in the compression zone. The rate of settling of the compression zone to its ultimate height Z_∞ is then relatively slow and is not constant. In a few cases, two constant-rate settling periods can be observed, with substantially no compression period. The appearance of the curves depends not only upon the type of slurry (nature and particle size of the solid, and nature of the liquid) but also upon the initial height and concentration of the slurry, as well as the extent of flocculation and whether or not any stirring is done during settling.

Flocculation If the finely divided solid particles are all similarly electrically charged, they repel each other and remain dispersed. If the charge is neutralized by addition, for example, of an electrolyte (flocculating agent) to the mixture, the particles may form aggregates, or flocs. Since the flocs are of larger size, they settle more rapidly. Slurries and suspensions encountered in chemical operations are usually flocculated.

Stirring Very slow stirring, so slow that eddy currents are not formed within the liquid, changes the character of the settling profoundly. The floc structure is altered so that the solids concentration in zone *B* is no longer uniform at the initial value and zone *D* may not be clearly defined. The ultimate height of the settled slurry may be only a fraction of that obtained without stirring [21], owing to breakdown of bridged floc structures in the compression zone, and the ultimate concentration of

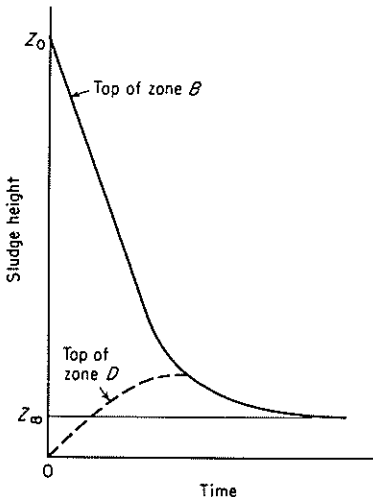


Figure 13.7 Rate of settling.

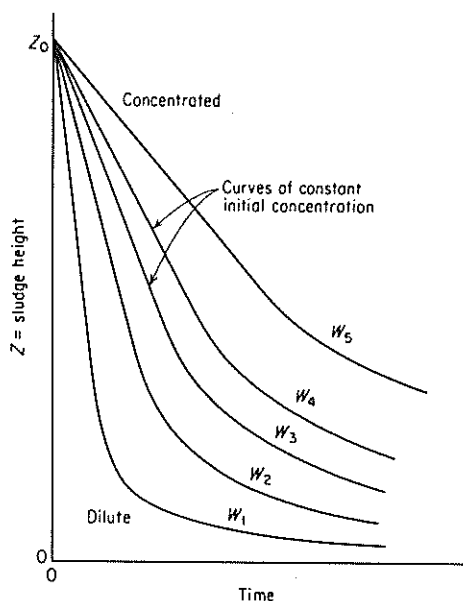


Figure 13.8 Batch settling of slurries. Effect of slurry concentration.

solids in the settled mass is correspondingly greater. Generally, however, the zones of constant- and falling-rate settling are still observed, although the rates will be different from those obtained without stirring [44].

Concentration The rate of settling decreases with increased initial concentration of the solids owing to the increase of the effective density and viscosity of the medium through which the particles settle. Figure 13.8 illustrates the effect usually to be expected when slurries of increasing concentration of the same substance are settled in columns of the same height. Various attempts have been made to predict the effect of concentration on the settling rate, from knowledge of the curves at one or more concentrations. This has been successful only for slurries which are not at one or more concentrations. This has been successful only for slurries which are not compressible [39].

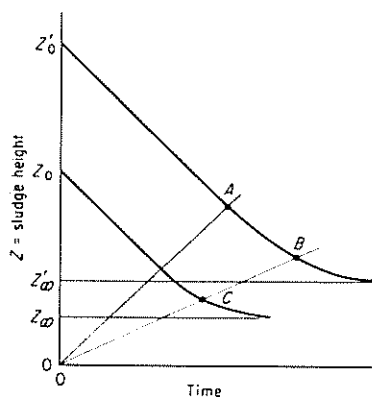


Figure 13.9 Batch settling of slurries at different initial heights.

Height Refer to Fig. 13.9, which shows settling curves for the same slurry begun at different initial heights. The initial constant settling rate is independent of height, and provided some critical minimum value of Z_0 is exceeded, the ultimate value of Z_∞/Z_0 will apparently also be constant. The constant-settling-rate lines both terminate on a line OA radiating from the origin, and in general any line [44] radiating from the origin such as OB will be cut so that line OC /line $OB = Z_0/Z'_0$. It follows that the time for a slurry to settle to a fixed fractional height Z/Z_0 is proportional to the initial height Z_0 . In this way it is possible reasonably well to predict the settling curves for deep tanks from results obtained in small laboratory cylinders. In making such laboratory tests, however, it is important [21] to use cylinders at least 1 m tall and at least 50 mm in diameter and to maintain all other conditions in the laboratory test identical with those expected to prevail on the large scale.

Percolation vs. Agitation

If a solid in the form of large lumps is to be leached, a decision must frequently be made whether to crush it to coarse lumps and leach by percolation or whether to grind it fine and leach by agitation and settling. No general answer can be given to this problem because of the diverse leaching characteristics of the various solids and the values of the solute, but among the considerations are the following. Fine grinding is more costly but provides more rapid and possibly more thorough leaching. It suffers the disadvantages that the weight of liquid associated with the settled solid may be as great as the weight of the solid, or more, so that considerable solvent is used in washing the leached solute free of solute and the resulting solution is dilute. Coarsely ground particles, on the other hand, leach more slowly and possibly less thoroughly but on draining may retain relatively little solution, require less washing, and thus provide a more concentrated final solution.

For more fibrous solids, such as sugar cane, which is leached with water to remove the sugar, it has been shown [35] that leaching is generally more efficient in a thoroughly agitated vessel than by percolation, probably because the large amount of static liquid holdup (see Chap. 6) makes important amounts of solute unavailable.

STEADY-STATE (CONTINUOUS) OPERATION

Equipment for continuous steady-state operations can be broadly classified into two major categories, according to whether it operates in stagewise or in continuous-contact fashion. Stagewise equipment is sometimes assembled in multiple units to produce multistage effects, whereas continuous-contact equipment may provide the equivalent of many stages in a single device.

Leaching during Grinding

As pointed out earlier, many solids require grinding in order to make the soluble portions accessible to the leaching solvents, and if continuous wet grinding is practiced, some of the leaching may be accomplished at this time. As much as 50 to 75 percent of the soluble gold may be dissolved by grinding the ore in the presence of cyanide solution, for example. Similarly, castor seeds are ground in

an attrition mill with solvent for the castor oil. The liquid and solid flow through a grinding mill in parallel and consequently tend to come to a concentration equilibrium. Such operations are therefore single-stage leachings and are usually supplemented by additional agitation or washing operations, as described later.

Agitated Vessels

Finely ground solids which can be readily suspended in liquids by agitation may be continuously leached in any of the many types of agitated tanks or vessels. These must be arranged for continuous flow of liquid and solid into and out of the tank and must be carefully designed so that no accumulation of solid occurs. Thanks to the thorough mixing ordinarily obtained, these devices are single-stage in their action, the liquid and solid tending to come to equilibrium within the vessel.

Mechanically agitated vessels may be used, for which the turbine-type agitator is probably most generally suitable (see Chaps. 6 and 11). Pachuca tanks are frequently used in the metallurgical industries. The Dorr agitator (Fig. 13.10) utilizes both the air-lift principle and mechanical raking of the solids and is extensively used in both the metallurgical and the chemical industry for continuous leaching and washing of finely divided solids. The central hollow shaft of the agitator acts as an air lift and at the same time revolves slowly. The arms attached to the bottom of the shaft rake the settled solids toward the center of the tank bottom, where they are lifted by the air lift through the shaft to the revolving launders attached to the top. The launders then distribute the elevated mixture of liquid and solid over the entire cross section of the tank. The rake arms can be lifted to free them of solids which may settle during a shutdown, and they are also provided with auxiliary air lines to assist in freeing them from settled solid. For unevenly sized solids, operation of the agitator can be adjusted so that coarse particles, which may require longer leaching time, remain in the tank longer than the finer. These agitators are regularly built in sizes ranging from 1.5 to 12 m diameter.

The average holding time in an agitated vessel can be calculated by dividing the vessel contents by the rate of flow into the vessel. This can be done separately for solid and liquid, and the holding time for each will be different if the ratio of the amounts of one to the other in the vessel is different from that in the feed. The average holding time of the solid must be sufficient to provide the leaching action required. Individual solid particles, of course, may short-circuit the tank (axial mixing), i.e., pass through in times much shorter than the calculated average, and this will lead to low stage efficiency. Short circuiting can be eliminated by passing the solid-liquid mixture through a series of smaller agitated vessels, one after the other, the sum of whose average holding time is the necessary leach time. This can readily be accomplished with gravity flow of the slurry by placing the individual tanks in the series at progressively lower levels. Three vessels in series are usually sufficient to reduce short circuiting to a negligible amount. It should be noted that since liquid and solid pass through these vessels in parallel flow, the entire series is still equivalent to only a single stage.

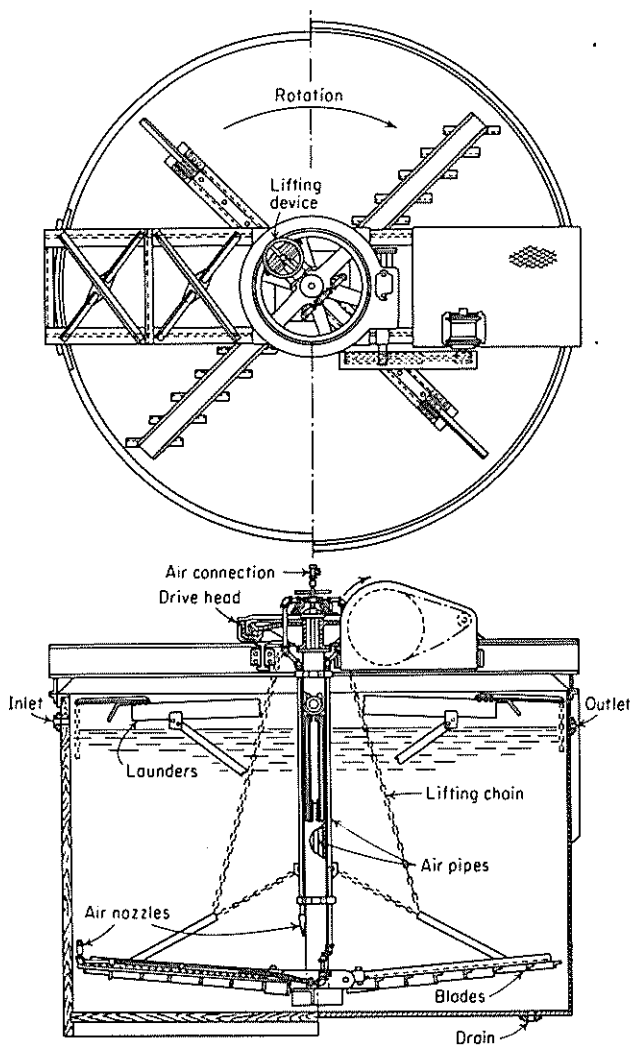


Figure 13.10 Dorr agitator. (Dorr-Oliver, Inc.)

The effluent from continuous agitators may be sent to a filter for separating liquid from solid, upon which the solid may be washed free of dissolved solids, or to a series of thickeners for countercurrent washing.

Thickeners

Thickeners are mechanical devices designed especially for continuously increasing the ratio of solid to liquid in a dilute suspension of finely sized particles by

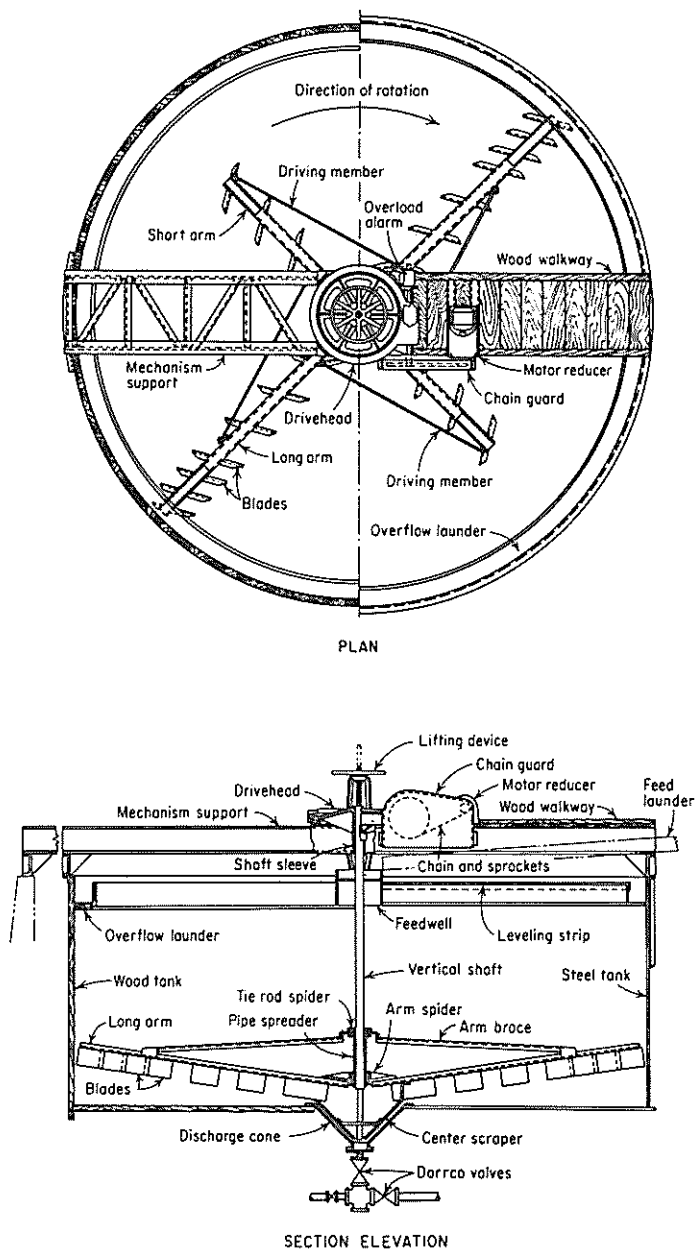


Figure 13.11 Dorr thickener. (Dorr-Oliver, Inc.)

settling and decanting, producing a clear liquid and a thickened sludge as two separate products. Thickeners may be used before any ordinary filter in order to reduce filtering costs, but since both effluents are pumpable and consequently readily transported, thickeners are frequently used to wash leached solids and chemical precipitates free of adhering solution in a continuous multistage countercurrent arrangement, and it is in this application that they are of interest here.

A typical single-compartment thickener of the Dorr-Oliver Company's design is shown in Fig. 13.11. The thin slurry of liquid and suspended solids enters a large settling tank through a feed well at the top center, in such a manner as to avoid mixing the slurry with the clear liquid at the top of the tank. The solids settle from the liquid which fills the tank, and the settled sludge is gently directed toward the discharge cone at the bottom by four sets of plow blades or rakes, revolving slowly to avoid disturbing the settled solid unduly. The sludge is pumped from the discharge cone by means of a diaphragm pump. The clear, supernatant liquid overflows into a launder built about the upper periphery of the tank. Thickeners are built in sizes ranging from 2 to 180 m (6 to 600 ft) in diameter, for handling granular as well as flocculent solids, and of varying detail design depending upon the size and service. In order to reduce the ground-area requirements, several thickeners operating in parallel and superimposed as in Fig. 13.12 may be used. Such a device delivers a single sludge product.

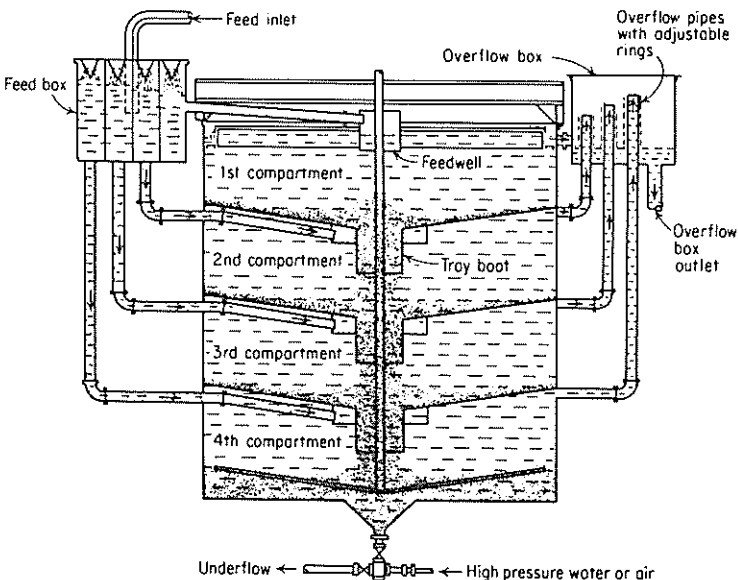


Figure 13.12 Dorr balanced-tray thickener. (Dorr-Oliver, Inc.)

The liquid content of the sludge is greatly dependent upon the nature of the solids and liquid and upon the time allowed for settling but in typical cases might be in the range 15 to 75 percent liquid. The less liquid retained, the more efficient the leaching or washing process being carried on.

Continuous Countercurrent Decantation (CCD)

Leaching equipment such as agitators or grinding mills may discharge their effluent into a cascade of thickeners for continuous countercurrent washing of the finely divided solids free of adhering solute. The same type of cascade can also be used to wash the solids formed during chemical reactions, as in the manufacture of phosphoric acid, by treatment of phosphate rock with sulfuric acid, or of blanc fixe, by reaction of sulfuric acid and barium sulfide, or of lithopone.

A simple arrangement is shown in Fig. 13.13a. The solids to be leached (or the reagents for a reaction), together with solution from the second thickener, are introduced into the leaching agitators at the left, and the strong solution thus produced is decanted from the solids by the first thickener. The agitators together with the first thickener then constitute a single stage. The sludge is passed through the cascade to be washed by the solvent in true countercurrent fashion, and the washed solids are discharged at the right. There may, of course,

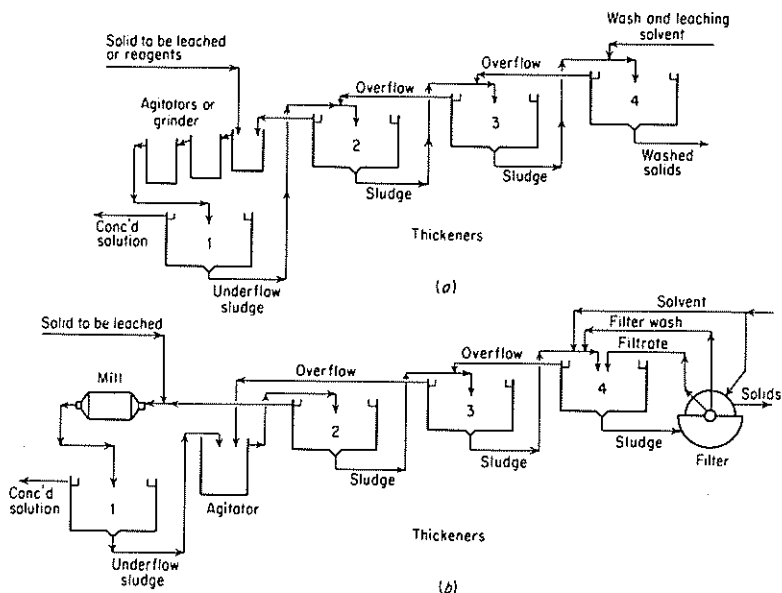


Figure 13.13 Continuous countercurrent decantation (CCD): (a) simple flowsheet; (b) flowsheet with intermediate agitation and filtration of washed solids.

be more or fewer than the four stages shown, and the agitators may be replaced by any continuous-leaching device, such as a grinding mill. Many variations in the flowsheet are regularly made. For example, the sludge from each stage can be *repulped*, or vigorously beaten with the solvent, between stages in order to improve the washing efficiency. Figure 13.13*b* shows an arrangement whereby the underflow from the first thickener is agitated with overflow from the third, for the purpose of bringing about the additional leaching possible with dilute solution. The sludge from the final stage can be filtered, as shown, when the solid is valuable and is to be delivered reasonably dry or when the solute is valuable and solution adhering to the washed solids must be reduced to a minimum. For successful operation of these plants, very carefully controlled rates of flow of both sludges and solution are necessary to avoid disturbing the steady-state conditions prevailing.

For small decantation plants, where ground area may be limited, it is possible to obtain a countercurrent cascade of thickeners built in superimposed fashion into a single shell.

Continuous Settling

The concentrations existing at the various levels of a continuous thickener under steady-state operation differ considerably from those found in batch settling. The solid curve of Fig. 13.14 shows typical concentrations during normal operation [10*a*], and four clearly defined zones are found in the thickener corresponding to the various sections of the curve. The feed slurry is diluted as it issues from the feed well of the thickener, and the bulk of the liquid passes upward to overflow into the launder about the thickener periphery. The solid concentration in the top zone is negligible if the overflow is clear. The solids and the remainder of the feed liquid move downward through the lower three zones and leave in the thickened underflow. The solids concentration in the settling zone is much lower than that in the feed, because of the dilution, but rises rapidly in the compression zone immediately below. In the bottom zone, the action of the rake disturbs arched structures which the settling solids may form, the weight of the solids presses out the liquid, and the concentration rises to the value in the underflow. If the feed rate to the thickener is increased, the concentration of solids in the settling zone rises and reaches a constant maximum value not related to the feed concentration when the settling capacity of this zone is exceeded. The excess solids, which cannot settle,

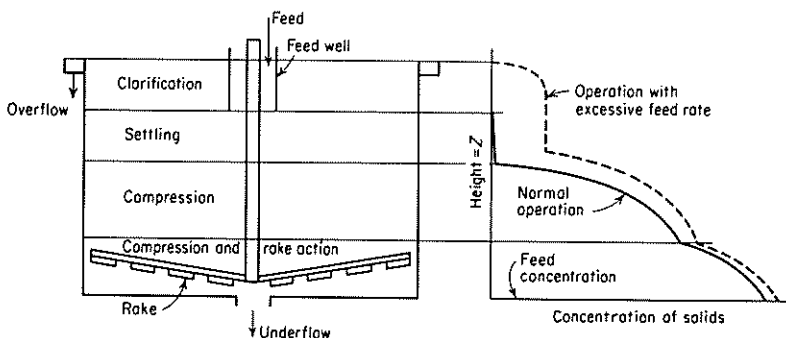


Figure 13.14 Continuous thickener characteristics [10]. (Courtesy of Industrial and Engineering Chemistry.)

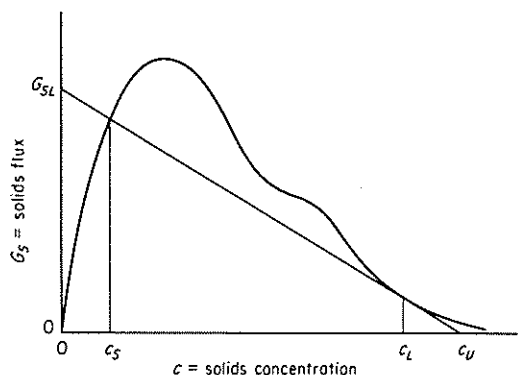


Figure 13.15 Determination of thickener area from flux curve.

overflow with the liquid, as indicated by the broken curve of concentrations in Fig. 13.14 for this condition.

The concentration of solids in the underflow sludge for a given rate and concentration of feed can be increased by reducing the rate of withdrawal of sludge. This increases the depth of the compression zone and increases the detention time of the solids within the thickener, although it is important not to raise the level of the compression zone to such an extent that solids appear in the overflow liquid.

The capacity of continuous thickeners, or their cross-sectional area required for a given solids throughput, can be roughly estimated from batch-settling tests [13, 39]. Initial settling velocities V for slurries of the solid at various initial uniform concentrations c are determined from the slopes of curves like those of Fig. 13.8, covering the entire range of solids concentration to be dealt with (it is best to determine these curves for slurries made by suspending a given weight of solids in varying amounts of liquid by adding and subtracting liquid). The flux of solids during settling, $G_s = cV$, is then plotted against c to produce a curve like that of Fig. 13.15. The tangent of smallest negative slope is then drawn to the curve from point ($G_s = 0$, $c = c_U$), where c_U is the desired under-flow concentration, to intersect the ordinate at G_{SL} , the limiting solids flux. The minimum required thickener cross section for handling W mass/time of solids is then

$$A = \frac{W}{G_{SL}} \quad (13.7)$$

Concentration c_L is that at the point of compression and c_S that in the settling zone.

Hydrocyclones (Hydroclones)

Hydrocyclones, similar to those used for size classification of solids (Fig. 13.16) can also be used as liquid-solid separators in place of thickeners in countercurrent washing of solids in a slurry.

Continuous Leaching of Coarse Solids

Many ingenious devices have been used for moving the solids continuously through a leaching device so that countercurrent action can be obtained. With the exception of the classifiers, which are used mainly in the metallurgical industries, these machines were principally developed for the special solids-handling problems arising in the leaching of sugar beets and of vegetable seeds such as cottonseed, soybeans, and the like. Donald [12] has described many of the

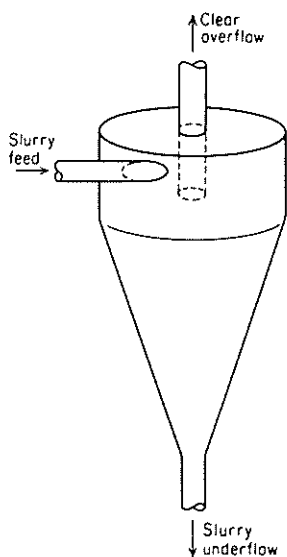


Figure 13.16 Hydrocyclone.

early devices used for sugar beets. Only the more important of the currently used machines can be described here.

Classifiers

Coarse solids can be leached, or more usually washed free of adhering solution or solute, in some types of machinery ordinarily used in the metallurgical industries for classification according to particle size. One such device is shown in Fig. 13.17. The solids are introduced into a tank, made with a sloping bottom and partly filled with the solvent. The rakes, which are given a reciprocating and circular lifting motion by the driving mechanism, rake the solids upward along the bottom of the tank and out of the liquid. In the upper part of the tank the solids are drained and discharged. The liquid overflows at the deep end of the tank. The solute concentration in the liquid is reasonably uniform throughout the tank as a result of the agitation by the rakes, so that the apparatus produces a single-stage action. Several classifiers can be placed in a cascade for continuous multistage countercurrent action, however, in which case they are operated by a single drive mechanism.

Fitch has presented an excellent review of the techniques of separating liquids and solids [14].

Leaching of Vegetable Seeds

Cottonseeds, soybeans, linseeds (flaxseeds), peanuts, rice bran, castor beans, and many other similar products are regularly leached, or *extracted*, with organic

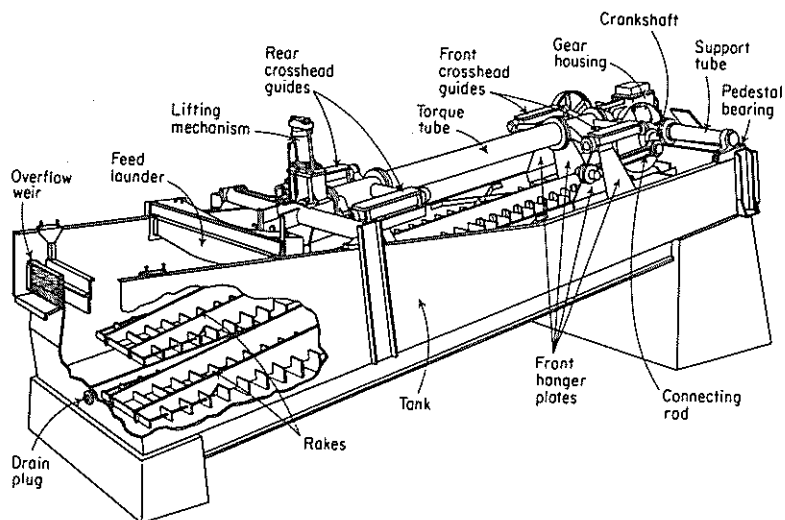


Figure 13.17 Single Dorr classifier for washing coarse solids. (Dorr-Oliver, Inc.)

solvents for removing the vegetable oils they contain. The seeds must usually be specially prepared for most advantageous leaching, and this may involve dehulling, precooking, adjustment of the moisture (water) content, and rolling or flaking. Sometimes a portion of the oil is first removed mechanically by expelling or expression. Leaching solvents are usually petroleum naphthas, for most oils a fraction corresponding closely to hexane; chlorinated hydrocarbons leave too toxic a residue for the leached meal to be used as an animal feed. The oil-solvent solution, which usually contains a small amount of finely divided, suspended solids, is called *miscella* and the leached solids *marc*. The various leaching devices are usually called *extractors* in this industry.

The *Rotocel* [2, 27] is essentially a modification of the Shanks system wherein the leaching tanks are continuously moved, permitting continuous introduction and discharge of the solids. Figure 13.18 is a schematic representation of the device, simplified to show the working principle. A circular rotor, containing 18 cells, each fitted with a hinged screen bottom for supporting the solids, slowly revolves above a stationary compartmented tank. As the rotor revolves, each cell passes in turn under a special device for feeding the prepared seeds and then under a series of sprays by which each is periodically drenched with solvent for leaching. After nearly one revolution, the leached contents of each cell are automatically dumped into one of the lower stationary compartments, from which they are continuously conveyed away. The solvent from each spray percolates downward through the solid and the supporting screen into the appropriate compartment of the lower tank, from which it is continuously pumped to the next spray. The leaching is countercurrent, and the strongest

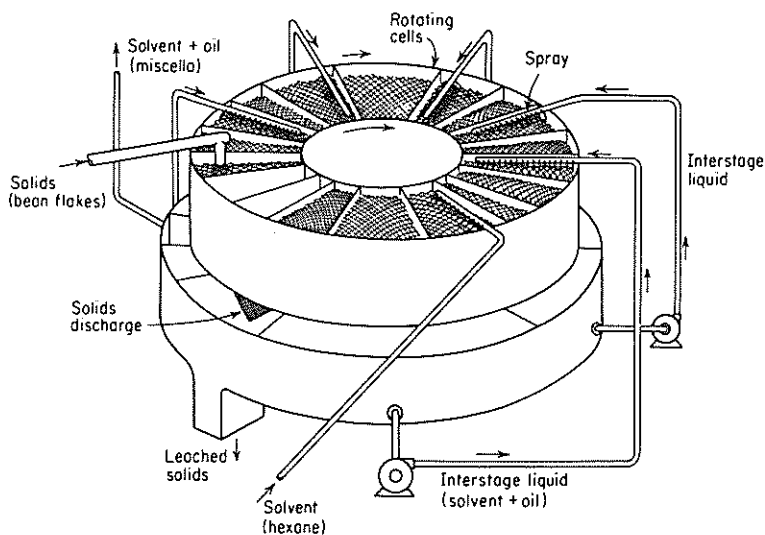


Figure 13.18 Schematic arrangement of the Rotocel.

solution is taken from the freshest seeds. A number of ingenious mechanical devices are necessary for maintaining smooth operation, and the entire machine is enclosed in a vaportight housing to prevent escape of solvent vapors.

The *French stationary-basket extractor* is a variant of this. The flakes are contained in stationary, compartmented beds filled by a rotating spout to feed the solids and are leached with solvent and miscella countercurrently [15, 33].

The *Kennedy extractor* [38], a modern arrangement of which is indicated schematically in Fig. 13.19, is another stagewise device which has been in use since 1927, originally for leaching tannins from tanbark. It is now used for oilseed and other chemical leaching operations. The solids are leached in a series

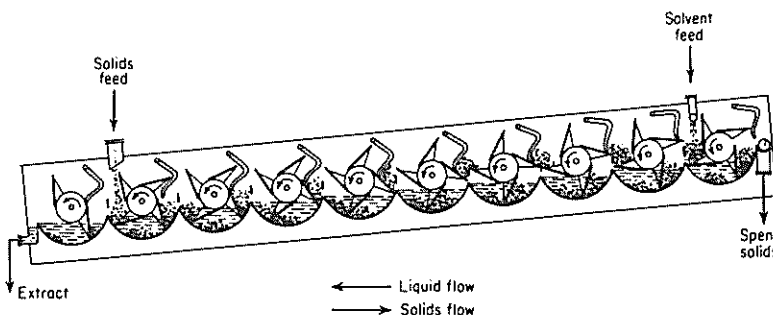


Figure 13.19 Kennedy extractor. (The Vulcan Copper and Supply Co.)

of tubs and are pushed from one to the next in the cascade by paddles, while the solvent flows in countercurrent. Perforations in the paddles permit drainage of the solids between stages, and the solids are scraped from each paddle as shown. As many tubs may be placed in a cascade as are required.

The *Bollman* extractor [38] (Fig. 13.20) is one of several basket-type machines. Solids are conveyed in perforated baskets attached to a chain conveyor, down on the right and up on the left in the figure. As they descend, they are leached in parallel flow by a dilute solvent-oil solution (*half miscella*) pumped from the bottom of the vessel and sprayed over the baskets at the top. The liquid percolates through the solids from basket to basket, collects at the bottom as the final strong solution of the oil (*full miscella*), and is removed. On the ascent, the solids are leached countercurrently by a spray of fresh solvent to provide the *half miscella*. A short drainage time is provided before the baskets are dumped at the top. There are many variants of this device, e.g., the horizontal arrangement of Fig. 13.21.

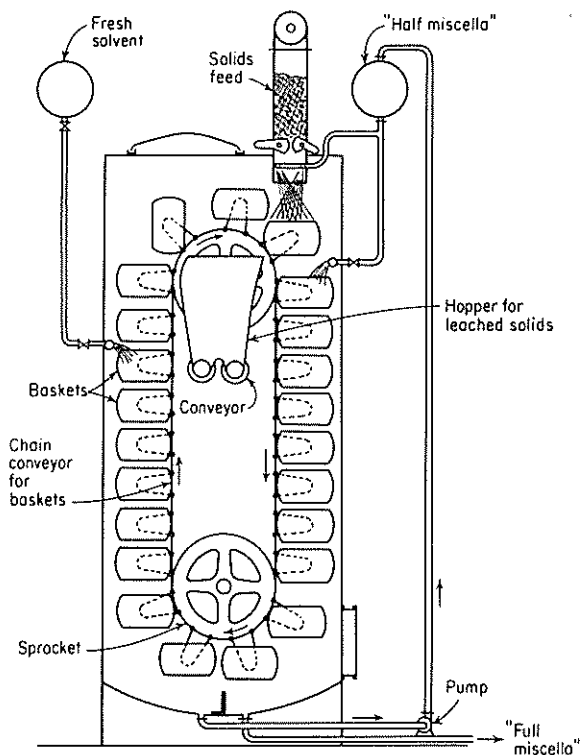


Figure 13.20 Bollman extractor.

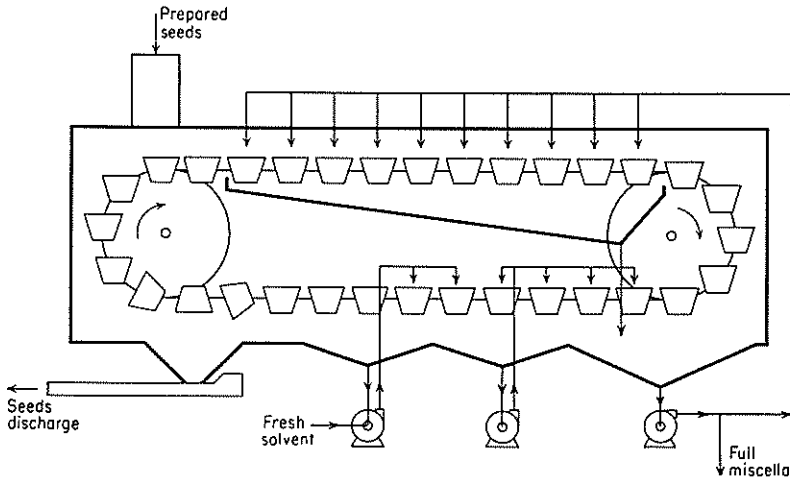


Figure 13.21 Continuous horizontal extractor (schematic).

Continuous tilting-pan filters and horizontal filters [16] are also commonly used. Figure 13.22 shows a typical flowsheet arrangement for a horizontal filter. The filter, in the form of a circular wheel, is divided into a number of sectors and revolves in the horizontal plane. Prepared seeds are slurried with solvent which has already been used for leaching, and the slurry is sent to the filter. The first filtrate is passed again through the filter cake to remove finely divided solids (polishing) before being discharged as miscella. The principle is much the same as that of the Rotocel. Horizontal moving screen-type belts [40] are also used for conveying the solids during leaching.

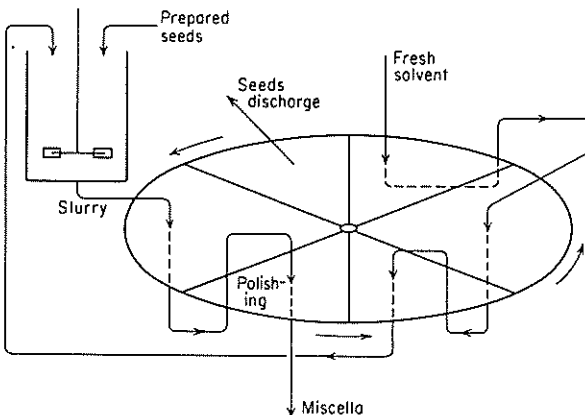


Figure 13.22 Flowsheet for horizontal-filter leaching.

The recovery of solvent from both the miscella and the leached seeds or beans is an essential part of the vegetable-oil leaching process. In a typical arrangement, the filtered miscella is passed to an evaporator for removal of solvent, sometimes followed by final stripping in a tray column, to produce the solvent-free oil. The wet seeds are steamed to remove residual solvent and air-cooled. Vented gas from condensers may be sent to an absorber to be scrubbed with petroleum white oil, and the resulting solvent-white-oil solution stripped to recover any solvent.

METHODS OF CALCULATION

It is important to be able to make an estimate of the extent of leaching which can be obtained for a given procedure, i.e., to calculate the amount of soluble substance leached from a solid, knowing the initial solute content of the solid, the number and amount of washings with leaching solvent, the concentration of solute in the leaching solvent, if any, and the method, whether batch or continuous countercurrent. Alternatively, it may be necessary to compute the number of washings, or number of stages, required to reduce the solute content of the solid to some specified value, knowing the amount and solute concentration of the leaching solvent. The methods of calculation are very similar to those used for liquid extraction.

Stage Efficiency

Consider a simple batch leaching operation, where the solid is leached with more than enough solvent to dissolve all the soluble solute and there is no preferential adsorption of either solvent or solute by the solid. If adequate time of contact of solid and solvent is permitted, all the solute will be dissolved, and the mixture is then a slurry of insoluble solid immersed in a solution of solute in the solvent. The insoluble phases are then separated physically by settling, filtration, or drainage, and the entire operation constitutes one stage. If the mechanical separation of liquid and solid were perfect, there would be no solute associated with the solid leaving the operation and complete separation of solute and insoluble solid would be accomplished with a single stage. This would be an equilibrium stage, of 100 percent stage efficiency. In practice, stage efficiencies are usually much less than this: (1) the solute may be incompletely dissolved because of inadequate contact time; (2) most certainly it will be impractical to make the liquid-solid mechanical separation perfect, and the solids leaving the stage will always retain some liquid and its associated dissolved solute. When solute is adsorbed by the solid, even though equilibrium between the liquid and solid phases is obtained, imperfect settling or draining will result in lowered stage efficiency.

Practical Equilibrium

In the general case, it will be easiest to make calculations graphically, as in other mass-transfer operations, and this will require graphical representation of

equilibrium conditions. It is simplest to use practical equilibrium conditions which take stage efficiencies into account directly, either entirely or in part, much as was done in the case of gas absorption and distillation. In the simplest cases, we must deal with three-component systems containing pure solvent (A), insoluble carrier solid (B), and soluble solute (C). Computations and graphical representation can be made on triangular coordinates for any ternary system of this sort, and the details of this have been worked out [17, 37]. Because of frequent crowding of the construction into one corner of such a diagram, it is preferable to use a rectangular coordinate system patterned after that used for fractional adsorption.

The concentration of insoluble solid B in any mixture or slurry will be expressed as N mass B/mass (A + C), whether the solid is wet with liquid solution or not. Solute C compositions will be expressed as weight fractions on a B-free basis: x = wt fraction C in the effluent solution from a stage (B-free basis), and y = wt fraction C in the solid or slurry (B-free basis). The value of y must include all solute C associated with the mixture, including that dissolved in adhering solution as well as undissolved or adsorbed solute. If the solid is dry, as it may be before leaching operations begin, N is the ratio of weights of insoluble to soluble substance, and $y = 1.0$. For pure solvent A, $N = 0$, $x = 0$.

The coordinate system then appears as in Fig. 13.23. Consider first a simple case of a mixture of insoluble solid from which all the solute has been leached, suspended in a solution of the solute in a solvent, as represented by point M_1 on the figure. The concentration of the clear solution is x , and the insoluble solid/solution ratio is N_{M_1} . Let the insoluble solid be nonadsorbent. If this mixture is allowed to settle, as in a batch-settling tank, the clear liquid which can be drawn off will be represented by point R_1 and the remaining sludge will consist of the insoluble solid suspended in a small amount of the solution. The composition of the solution in the sludge will be the same as that of the clear liquid withdrawn, so that $y^* = x$. The concentration of solid B in the sludge N_{E_1} will depend upon the length of time θ_1 allowed for settling, so that point E_1 then represents the slurry. Line E_1R_1 is a vertical tie line joining the points representing the two effluent streams, clear liquid and slurry. If the circumstances

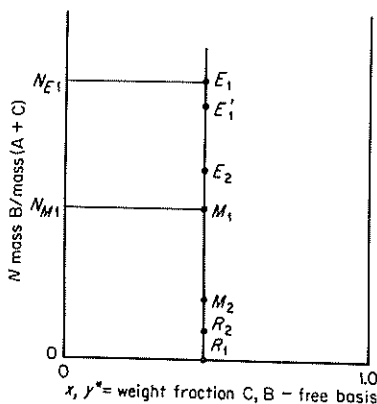


Figure 13.23 Concentrations in leaching and washing.

described are maintained in an actual leaching, points E_1 and R_1 can be taken as the practical conditions of equilibrium for that leaching. Clearly if less time is allowed for settling, say θ'_1 , the sludge will be less concentrated in insoluble solids and may be represented by point E'_1 . There will be some maximum value of N for the sludge, corresponding to its ultimate settled height, in accordance with the description of batch settling given earlier, but usually in practice insufficient time is allowed for this to be attained. Since the concentration of insoluble solid in a sludge settled for a fixed time depends upon the initial concentration in the slurry, a mixture M_2 settled for time θ_1 might result in a sludge corresponding to point E_2 . If the solid does not settle to give an absolutely clear solution, if too much solution is withdrawn from the settled sludge so that a small amount of solid is carried with it, or if solid B dissolves to a small extent in the solution, the withdrawn solution will be represented by some point such as R_2 , somewhat above the lower axis of the graph. Similar interpretations can be made for compositions obtained when wet solids are filtered or drained of solution rather than settled, or when continuously thickened.

The settling or thickening characteristics of a slurry depend, as shown earlier, upon the viscosity and relative density of the liquid in which the solid is suspended. Since these in turn depend upon the solution composition, it is possible to obtain experimental data showing the variation of compositions of thickened solids with composition of solution and to plot them on the diagram as practical equilibrium conditions. It is evident, however, that in every case they must be obtained under conditions of time, temperature, and concentrations identical with those pertaining in the plant or process for which the calculations are being made. For drained beds of impervious solids, the equilibrium corresponding to the residual saturation after long-time drainage can be estimated by the methods of Illustration 13.1. Data for short-time drainage must be obtained experimentally.

In washing operations where the solute is already dissolved, uniform concentration throughout all the solution is rapidly attained, and reduced stage efficiency is most likely to be entirely the result of incomplete drainage or settling. In leaching an undissolved solute interspersed throughout the solid, on the other hand, lowered stage efficiency may be the result of inadequate time of contact as well as incomplete mechanical separation of liquid and solid. In this case it is possible (but not necessary) to distinguish experimentally between the two effects by making measurements of the amount and composition of liquid retained on the solid after short and after long contact time and to use the latter to establish the equilibrium conditions.

Let us now examine a few of the types of equilibrium curves which may occur. Figure 13.24*a* represents data which might be obtained for cases where solute C is infinitely soluble in solvent A, so that x and y may have values over the entire range from 0 to 1.0. This would occur in the case of the system soybean oil (C)-soybean meal (B)-hexane (A), where the oil and hexane are infinitely soluble. The curve DFE represents the separated solid under conditions actually to be expected in practice, as discussed above. Curve GHJ , the composition of the withdrawn solution, lies above the $N = 0$ axis, and in this

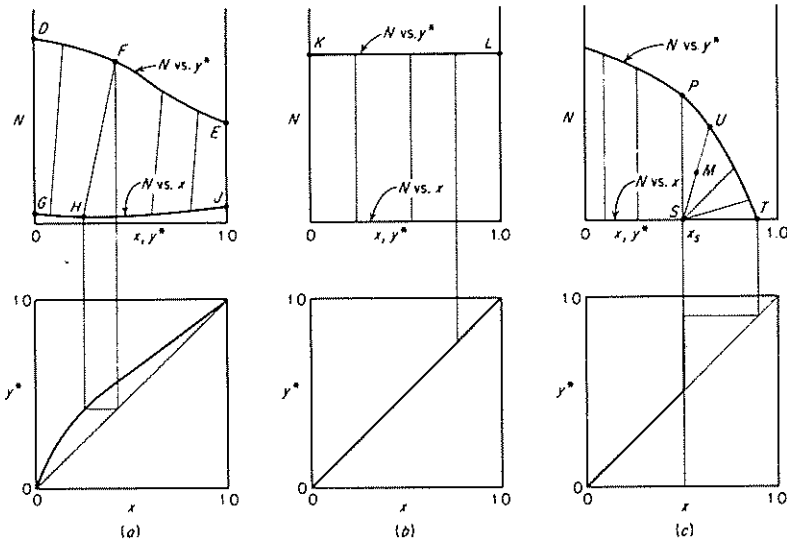


Figure 13.24 Typical equilibrium diagrams.

case, therefore, either solid B is partly soluble in the solvent or an incompletely settled liquid has been withdrawn. The tie lines such as line FH are not vertical, and this will result (1) if insufficient time of contact with leaching solvent to dissolve all solute is permitted, (2) if preferential adsorption of the solute occurs, or (3) if the solute is soluble in the solid B and distributes unequally between liquid and solid phases at equilibrium. The data may be projected upon a plot of x vs. y , as in the manner of adsorption or liquid-extraction equilibria.

Figure 13.24b represents a case where no adsorption of solute occurs, so that withdrawn solution and solution associated with the solid have the same composition and the tie lines are vertical. This results in an xy curve in the lower figure identical with the 45° line, and a distribution coefficient m , defined as y^*/x , equals unity. Line KL is horizontal, indicating that the solids are settled or drained to the same extent at all solute concentrations. It is possible to regulate the operation of continuous thickeners so that this will occur, and the conditions are known as *constant underflow*. The solution in this case contains no substance B, either dissolved or suspended. Figure 13.24c represents a case where solute C has a limited solubility x_s in solvent A. No clear solution stronger than x_s can be obtained, so that the tie lines joining slurry and saturated solution must converge, as shown. In this case any mixture M to the right of line PS will settle to give a clear saturated solution S and a slurry U whose composition depends on the position of M . Point T represents the composition of pure solid solute after drainage or settling of saturated solution. Since the tie lines to the left of PS are shown vertical, no adsorption occurs, and

overflow liquids are clear. It will be appreciated that combinations of these various characteristics may appear in a diagram of an actual case.

Single-Stage Leaching

Consider the single real leaching or washing stage of Fig. 13.25. The circle represents the entire operation, including mixing of solid and leaching solvent and mechanical separation of the resulting insoluble phases by whatever means may be used. Weights of the various streams are expressed as mass for a batch operation or as mass/time [or mass/(area)(time)] for continuous flow. Since for most purposes the solid B is insoluble in the solvent and a clear liquid leach solution is obtained, the B discharged in the leached solids will be taken as the same as that in the solids to be leached. By definition of N ,

$$B = N_F F = E_1 N_1 \quad (13.8)$$

A solute (C) balance gives

$$F y_F + R_0 x_0 = E_1 y_1 + R_1 x_1 \quad (13.9)$$

and a solvent (A) balance gives

$$F(1 - y_F) + R_0(1 - x_0) = E_1(1 - y_1) + R_1(1 - x_1) \quad (13.10)$$

and a "solution" (solute + solvent) balance gives

$$F + R_0 = E_1 + R_1 = M_1 \quad (13.11)$$

Mixing the solids to be leached and leaching solvent produces a mixture of B-free mass M_1 such that

$$N_{M1} = \frac{B}{F + R_0} = \frac{B}{M_1} \quad (13.12)$$

$$y_{M1} = \frac{y_F F + R_0 x_0}{F + R_0} \quad (13.13)$$

These relations can be shown on the coordinate system of Fig. 13.26. Point F represents the solids to be leached and R_0 the leaching solvent. Point M_1 , representing the overall mixture, must fall on the straight line joining R_0 and F , in accordance with the characteristics of these diagrams described in Chap. 9. Points E_1 and R_1 , representing the effluent streams, are located at opposite ends of the tie line through M_1 , and their compositions can be read from the diagram.

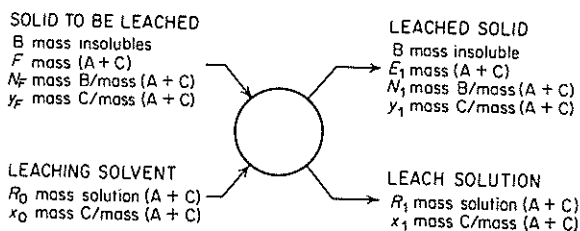


Figure 13.25 Single-stage leaching or washing.

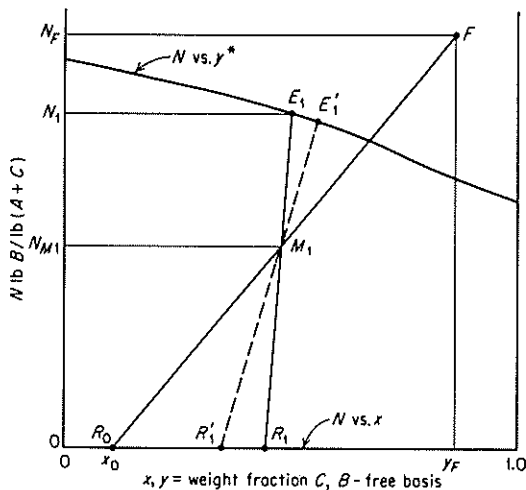


Figure 13.26 Single-stage leaching or washing.

Equation (13.8) permits calculation of the weight of E_1 and Eq. (13.11) that of R_1 . Modification to allow for the presence of B in the liquid withdrawn, necessitating an equilibrium diagram of the type shown in Fig. 13.24a, is readily made by analogy with the corresponding problem in liquid extraction.

If the equilibrium data of Fig. 13.26 were obtained experimentally after long contact time of solid and liquid and therefore represent inefficiency of mechanical separation of liquid and solid only, then in a real stage there may be an additional inefficiency owing to short time of contact. The effluent streams can then be represented by points E'_1 and R'_1 on the figure, and a stage efficiency $(y_F - y'_1)/(y_F - y_1)$ can be ascribed to this cause. If the equilibrium curve was obtained under conditions of contact time corresponding to the actual leaching, the tie line E_1R_1 will give the effluent composition directly.

Multistage Crosscurrent Leaching

By contacting the leached solids with a fresh batch of leaching solvent, additional solute can be dissolved or washed away from the insoluble material. The calculations for additional stages are merely repetitions of the procedure for a single stage, with the leached solids from any stage becoming the feed solids to the next. Equations (13.8) to (13.13) apply, with only obvious changes in the subscripts to indicate the additional stages. When the number of stages for reducing the solute content of a solute to some specified value must be determined, it must be recalled that we are dealing with real stages, owing to the use of "practical" equilibrium data, and that the number found must therefore be integral. This may require adjustment by trial of either the amount of solute to be leached or the amount and apportioning of solvent to the stages.

Illustration 13.2 Caustic soda is being made by treatment of slaked lime, $\text{Ca}(\text{OH})_2$, with a solution of sodium carbonate. The resulting slurry consists of particles of calcium carbonate,

CaCO_3 , suspended in a 10% solution of sodium hydroxide, NaOH , 0.125 kg suspended solid/kg solution. This is settled, the clear sodium hydroxide solution withdrawn and replaced by an equal weight of water, and the mixture thoroughly agitated. After repetition of this procedure (a total of two freshwater washes), what fraction of the original NaOH in the slurry remains unrecovered and therefore lost in the sludge? The settling characteristics of the slurry, determined under conditions representing the practice to be followed in the process [Armstrong and Kammermeyer, *Ind. Eng. Chem.*, 34, 1228 (1942)], show adsorption of the solute on the solid.

x = wt fraction NaOH in clear soln	N = kg CaCO_3 /kg soln in settled sludge	y^* = wt fraction NaOH in soln of the settled sludge
0.0900	0.495	0.0917
0.0700	0.525	0.0762
0.0473	0.568	0.0608
0.0330	0.600	0.0452
0.0208	0.620	0.0295
0.01187	0.650	0.0204
0.00710	0.659	0.01435
0.00450	0.666	0.01015

SOLUTION The equilibrium data are plotted in Fig. 13.27. *Basis:* 1 kg solution in the original mixture, containing 0.1 kg NaOH (C) and 0.9 kg H_2O (A). $B = 0.125$ kg CaCO_3 .

The original mixture corresponds to M_1 with $N_{M1} = 0.125$ kg CaCO_3 /kg soln, $y_{M1} = 0.10$ kg NaOH /kg soln. M_1 is plotted on the figure, and the tie line through this point is drawn. At

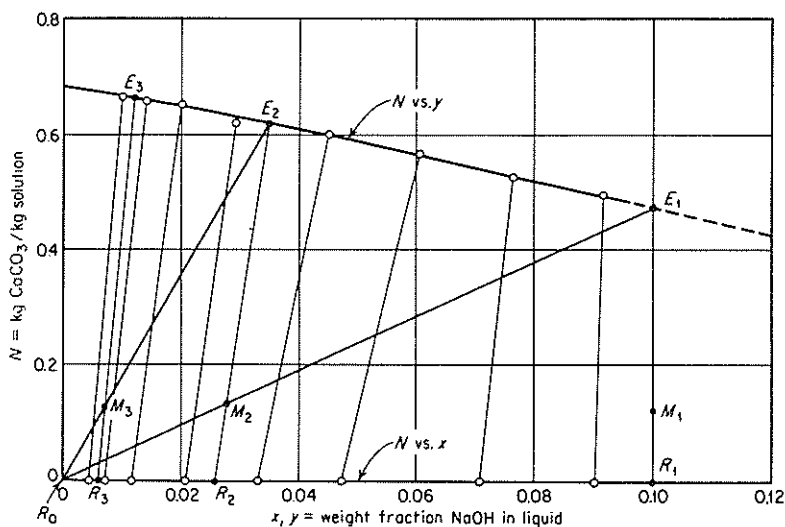


Figure 13.27 Solution to Illustration 13.2.

point E_1 representing the settled sludge, $N_1 = 0.47$, $y_1 = 0.100$. Eq. (13.8):

$$E_1 = \frac{B}{N_1} = \frac{0.125}{0.47} = 0.266 \text{ kg soln in sludge}$$

$$1 - 0.266 = 0.734 \text{ kg clear soln withdrawn}$$

Stage 2 $R_0 = 0.734$ kg water added, $x_0 = 0$ kg NaOH/kg soln. Eq. (13.11) adapted to this stage:

$$M_2 = E_1 + R_0 = E_2 + R_2 = 0.266 + 0.734 = 1.0 \text{ kg liquid}$$

Eq. (13.12):

$$N_{M2} = \frac{B}{E_1 + R_0} = \frac{B}{M_2} = \frac{0.125}{1.0} = 0.125$$

M_2 is located on line R_0E_1 at this value of N , and the tie line through M_2 is drawn. At E_2 , $N_2 = 0.62$, $y_2 = 0.035$. Eq. (13.8):

$$E_2 = \frac{B}{N_2} = \frac{0.125}{0.62} = 0.202 \text{ kg}$$

$$1 - 0.202 = 0.789 \text{ kg soln withdrawn}$$

Stage 3 $R_0 = 0.798$ kg water added, $x_0 = 0$. Eq. (13.11):

$$M_3 = E_2 + R_0 = 0.202 + 0.798 = 1.0$$

$$N_{M3} = \frac{B}{M_3} = \frac{0.125}{1} = 0.125$$

Tie line E_3R_3 is located through M_3 as in the case of stage 2, and, at E_3 , $N_3 = 0.662$, $y_3 = 0.012$. By Eq. (13.8), $E_3 = B/N_3 = 0.125/0.662 = 0.189$ kg soln in final sludge. $E_3y_3 = 0.189(0.012) = 0.00227$ kg NaOH in sludge, or $(0.00227/0.1)100 = 2.27\%$ of original.

The process permits an appreciable loss and produces three solutions, two of which (R_2 and R_3) are quite dilute. It should be compared with the countercurrent washing operation of Illustration 13.3.

Multistage Countercurrent Leaching

A general flowsheet for either leaching or washing is shown in Fig. 13.28. Operation must necessarily be continuous for steady-state conditions to prevail, although leaching according to the Shanks system will approach the steady state after a large number of cycles have been worked through. In the flowsheet shown, it is assumed that solid B is insoluble and is not lost in the clear solution,

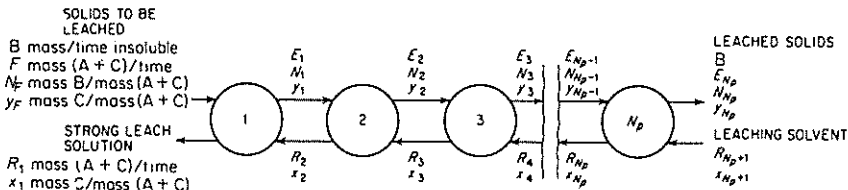


Figure 13.28 Multistage countercurrent leaching or washing.

but the procedure outlined below is readily modified to take care of cases where this may not be true.†

A solvent balance for the entire plant is

$$F + R_{N_p+1} = R_1 + E_{N_p} = M \quad (13.14)$$

and a "solution" (A + C) balance is

$$Fy_F + R_{N_p+1}x_{N_p+1} = R_1x_1 + E_{N_p}y_{N_p} = My_M \quad (13.15)$$

M represents the hypothetical B-free mixture obtained by mixing solids to be leached and leaching solvent. Refer to Fig. 13.29, the operating diagram for the plant. The coordinates of point M are

$$N_M = \frac{B}{F + R_{N_p+1}} \quad (13.16)$$

$$y_M = \frac{Fy_F + R_{N_p+1}x_{N_p+1}}{F + R_{N_p+1}} \quad (13.17)$$

The points E_{N_p} and R_1 , representing the effluents from the cascade, must lie on a line passing through M , and E_{N_p} will be on the "practical" equilibrium curve. Equation (13.14) can be rearranged to read

$$F - R_1 = E_{N_p} - R_{N_p+1} = \Delta_R \quad (13.18)$$

Similarly, a solution balance about any number of stages, such as the first three, can be arranged in this form,

$$F - R_1 = E_3 - R_3 = \Delta_R \quad (13.19)$$

Δ_R represents the constant difference in flow $E - R$ (usually a negative quantity) between each stage. In Fig. 13.29 it can be represented by the intersection of lines FR_1 and $E_{N_p}R_{N_p+1}$ extended, in accordance with the characteristics of these coordinates. Since the effluents from each stage are joined by the practical tie line for the particular conditions which prevail, E_1 is found at the end of the tie line through R_1 . A line from E_1 to Δ_R provides R_2 , and so forth. Alternatively the stage constructions may be made on the x, y coordinates in the lower part of the figure after first locating the operating line. This can be done by drawing random lines from point Δ_R and projecting their intersections with the equilibrium diagram to the lower curve in the usual manner. The usual staircase construction then establishes the number of stages. The stages are real rather than equilibrium, the practical equilibrium data having already taken into account the stage efficiency, and hence there must be an integral number. Especially when the number of stages required is the unknown quantity, some trial-and-error adjustment of the concentrations of the effluents or amount of solvent will be required to obtain an integral number.

† See Illustration 13.4, for example.

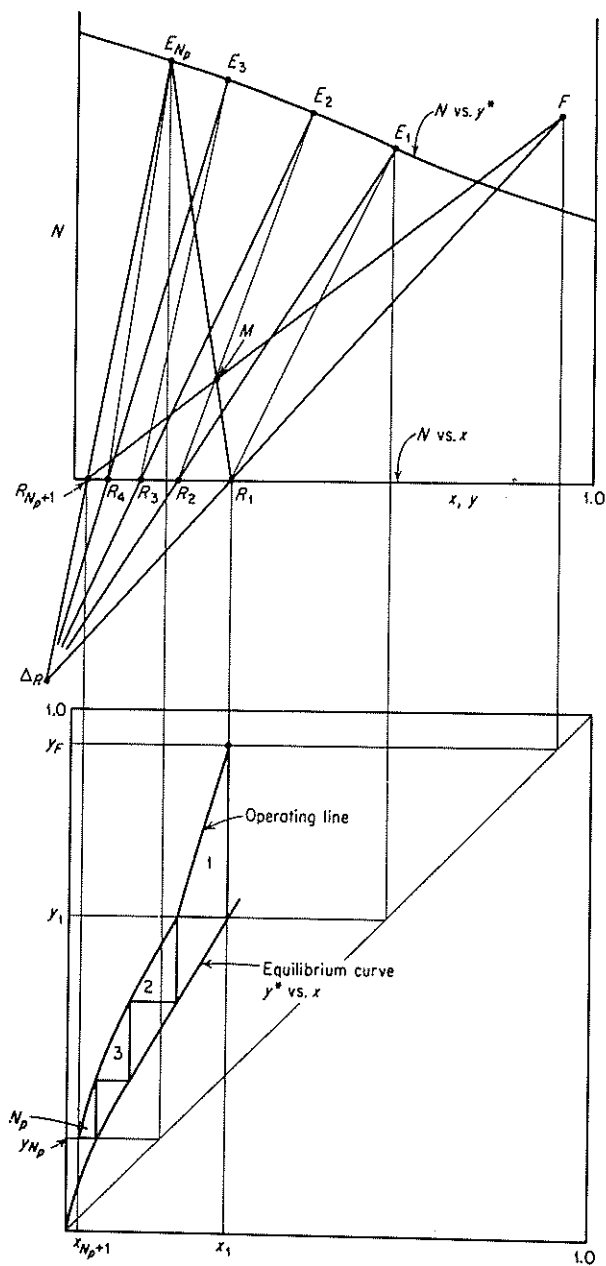


Figure 13.29 Multistage countercurrent leaching or washing.

If the equilibrium curve of Fig. 13.29 represents inefficiency of mechanical separation of liquid and solid only, and not that resulting from short contact time of solvent and solid, the effect of the latter, if known, may be taken care of by drawing a new equilibrium curve on the x, y coordinates. This should be located between the equilibrium curve shown and the operating line, at a fractional distance from the operating line corresponding to the stage efficiency due to the short contact time, in the manner used earlier in gas absorption and distillation.

In the special case where *constant underflow*, or constant value of N for all sludges, pertains, the operating line on the xy diagram is straight and of constant slope R/E . If in addition the practical equilibrium curve on this plot is straight, so that $m = y^*/x = \text{const}$, then Eqs. (5.54) and (5.55) apply. Adapting the former to the present situation gives

$$\frac{y_F - y_{N_p}}{y_F - mx_{N_p+1}} = \frac{(R/mE)^{N_p+1} - R/mE}{(R/mE)^{N_p+1} - 1} \quad (13.20)$$

Figure 5.16 can be used to solve this rapidly, with $(y_{N_p} - mx_{N_p+1})/(y_F - mx_{N_p+1})$ as ordinate, R/mE as parameter. If in addition the tie lines of the equilibrium diagram are vertical, $m = 1.0$. The form of the equation shown is that which is applicable when the value of F for the feed solids is the same as E , so that R/E is constant for all stages, including the first. It frequently may happen, especially when dry solids constitute the feed, that the ratio R_1/E_1 for stage 1 will be different from that pertaining to the rest of the cascade. In this case Eq. (13.20) or Fig. 5.16 should be applied to that part of the cascade excluding the first stage, by substitution of y_1 for y_F and N_p for $N_p + 1$. In general, the equation or chart can be applied to any part of the cascade where operating line and equilibrium line are both straight, and this may be particularly useful for cases where the solute concentration in the leached solution is very small. Just as in liquid extraction and gas absorption, there is an economic optimum combination of treating solvent/solids ratio, number of stages, and extent of leaching [8, 9].

The calculations of these stagewise separation operations have been adapted to the digital computer [24, 41].

Illustration 13.3 Sodium hydroxide, NaOH, is to be made at the rate of 400 kg/h (dry weight) by reaction of soda ash, Na_2CO_3 , with slaked lime, $\text{Ca}(\text{OH})_2$, using a flowsheet of the type shown in Fig. 13.13a. The reagents will be used in stoichiometric proportions, and for simplicity it will be assumed that reaction is complete. Pure water is to be used to wash the calcium carbonate, CaCO_3 , precipitate, and it is desired to produce as overflow from the first thickener a solution containing 10% NaOH. It will be assumed that the settling data of Illustration 13.2 apply.

(a) If three thickeners are used, determine the amount of wash water required and the percentage of the hydroxide lost in the discharged sludge.

(b) How many thickeners would be required to reduce the loss to at least 0.1% of that made?

SOLUTION (a) Mol wt of CaCO_3 (B) = 100, of NaOH (C) = 40. NaOH produced = 400 kg/h or $400/40 = 10 \text{ kmol/h}$. CaCO_3 produced = $10/2 = 5.0 \text{ kmol/h}$ or $5.0(100) = 500 \text{ kg/h} = B$. The water required is that leaving in the strong solution plus that in the final sludge. The amount in the final sludge, according to the settling data, depends upon the NaOH concentration in the final sludge, which is not known. After a trial calculation, it is assumed that the

solution in the final sludge will contain 0.01 wt fraction NaOH ($y_3 = 0.01$), and the settling data indicate $N_3 = 0.666$ kg CaCO_3/kg soln in the final sludge.

$$E = \frac{B}{N_3} = \frac{500}{0.666} = 750 \text{ kg/h soln list} \quad \text{NaOH lost} = E_3 y_3 = 750(0.01) = 7.50 \text{ kg/h}$$

$$\text{Water in sludge} = 750 - 7.5 = 742.5 \text{ kg/h} \quad \text{NaOH in overflow} = 400 - 7.5 = 392.5 \text{ kg/h}$$

$$x_1 = 0.1 \text{ wt fraction NaOH in overflow} \quad R_1 = \frac{392.5}{0.1} = 3925 \text{ kg overflow or strong soln/h}$$

$$\text{Water in } R_1 = 3925 - 392.5 = 3532.5 \text{ kg/h}$$

$$\text{Fresh water required} = R_{N_p+1} = 3532.5 + 742.5 = 4275 \text{ kg/h}$$

For purposes of calculation, it may be imagined that the agitators are not present in the flowsheet and that the first thickener is fed with a dry mixture of the reaction products, CaCO_3 and NaOH, together with overflow from the second thickener.

$$F = 400 \text{ kg NaOH/h} \quad N_F = \frac{B}{F} = \frac{500}{400} = 1.25 \text{ kg } \text{CaCO}_3/\text{kg NaOH}$$

$$y_F = 1.0 \text{ wt fraction NaOH in dry solid, } \text{CaCO}_3\text{-free basis}$$

Plot points R_1 , E_3 , R_{N_p+1} , and F on Fig. 13.30, and locate the difference point Δ_R at the intersection of lines FR_1 and $E_3R_{N_p+1}$ extended. The coordinates of point Δ_R are $N_{\Delta_R} = -0.1419$, $y_{\Delta_R} = -0.00213$. (These can be determined analytically, if desired, by simultaneous solution of the equations representing the intersecting lines.) Further computations must be done on an enlarged section of the equilibrium diagram (Fig. 13.31). Point Δ_R is plotted and the stages stepped off in the usual manner. The construction can be projected onto the xy diagram as shown, if desired. Three stages produce a value $y_3 = 0.01$, so that the assumed value of y_3 is correct. The NaOH lost in the sludge = $(7.5/400)100 = 1.87\%$ of that made.

(b)

$$\text{NaOH lost} = 0.001(400) = 0.4 \text{ kg/h}$$

$$\text{kg } \text{CaCO}_3/\text{kg NaOH in final sludge} = \frac{500}{0.4} = 1250 = \frac{N_{N_p}}{y_{N_p}}$$

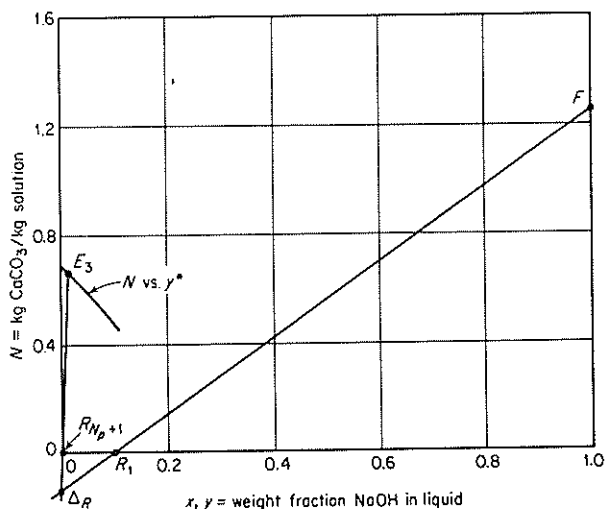


Figure 13.30 Solution to Illustration 13.3.

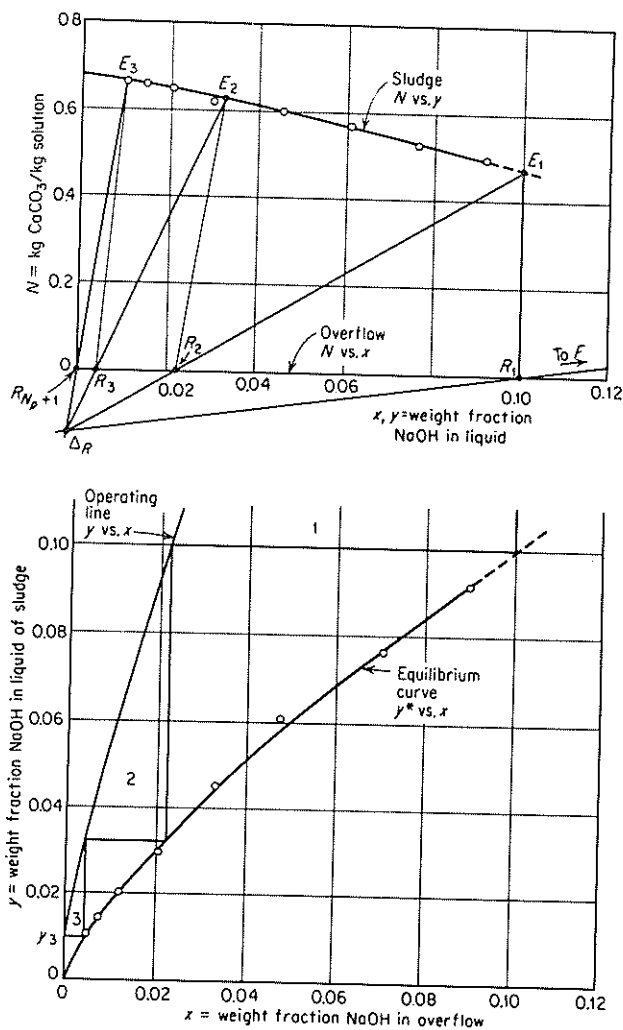


Figure 13.31 Solution to Illustration 13.3.

In order to determine the liquid content of the final sludge, convert the equilibrium data for dilute mixtures into the following form:

N	0.659	0.666	0.677	0.679	0.680
y^*	0.01435	0.01015	0.002†	0.001†	0.0005†
N/y^*	45.6	65.6	338	679	1360

† Estimated values.

By interpolation for $N/y^* = 1250$, $N_{N_p} = 0.680$ kg CaCO_3/kg soln, and $y_{N_p} = 0.680/1250 = 0.000544$ wt fraction NaOH in the liquid of the final sludge.

$$E_{N_p} = \frac{B}{N_{N_p}} = \frac{500}{0.680} = 735 \text{ kg/h} \quad \text{Water in } E_{N_p} = 735 - 0.4 = 734.6 \text{ kg/h}$$

$$\text{NaOH in overflow} = 400 - 0.4 = 399.6 \text{ kg/h} \quad R_1 = \frac{399.6}{0.1} = 3996 \text{ kg/h}$$

$$\text{Water in } R_1 = 3996 - 399.6 = 3596 \text{ kg/h} \quad \text{Fresh water} = R_{N_p+1} = 3596 + 734.6 = 4331 \text{ kg/h}$$

On the operating diagram (Fig. 13.32) point Δ_R is located in the same way as before, and the stages are constructed in the usual fashion. It becomes impractical to continue graphical

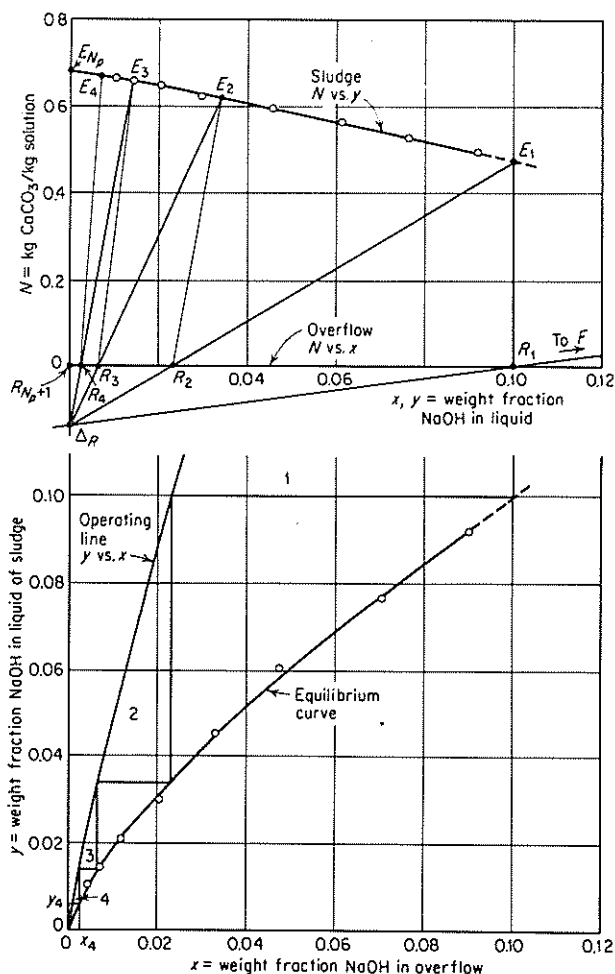


Figure 13.32 Solution to Illustration 13.3.

construction beyond the fourth stage unless considerable magnification of the chart is made, but computations beyond this point can be made with the help of Fig. 5.16. Beyond the fourth stage, the ratio of overflow to liquid in the sludge becomes substantially constant and equal to $R_{N_p+1}/E_{N_p} = 4331/735 = 5.90 = R/E$. This is the initial slope of the operating line on the lower part of Fig. 13.32. The slope of the equilibrium curve at these low concentrations is also substantially constant, $m = y^*/x = 0.01015/0.00450 = 2.26$ and $R/mE = 5.90/2.26 = 2.61$. $x_{N_p+1} = 0$, and $y_4 = 0.007$. Therefore $(y_{N_p} - mx_{N_p+1})/(y_4 - mx_{N_p+1}) = 0.000544/0.007 = 0.0777$. From Fig. 5.16, an additional 2.3 stages beyond the 4 computed graphically are required.

An additional two stages (six thickeners) would make $y_{N_p}/y_4 = 0.099$, or $y_{N_p} = 0.099(0.007) = 0.000693$, corresponding to 0.51 kg NaOH lost/h, while an additional three stages (seven thickeners) would make $y_{N_p} = 0.0365(0.007) = 0.000255$, corresponding to 0.187 kg NaOH lost/h.

It must be emphasized that the cost of these numbers of thickeners probably could not be justified when balanced against the value of the lost NaOH. The very low NaOH loss was specified in order to demonstrate the computation methods.

Illustration 13.4 Flaked soybeans are to be leached with hexane to remove the soybean oil. A 0.3-m-thick layer of the flakes (0.25-mm flake thickness) will be fed onto a slowly moving perforated endless belt which passes under a series of continuously operating sprays [40]. As the solid passes under each spray, it is showered with liquid which percolates through the bed, collects in a trough below the belt, and is recycled by a pump to the spray. The spacing of the sprays is such that the solid is permitted to drain 6 min before it reaches the next spray. The solvent also passes from trough to trough in a direction countercurrent to that of the moving belt, so that a truly continuous countercurrent stagewise operation is maintained, each spraying and draining constituting one stage. Experiments [40] show that the flakes retain solution after 6 min drain time to an extent depending upon the oil content of the solution, as follows:

Wt % oil in soln	0	20	30
kg soln retained/kg insoluble solid	0.58	0.66	0.70

It will be assumed that the retained solution contains the only oil in the drained flakes.

The soybean flakes enter containing 20% oil and are to be leached to 0.5% oil (on a solvent-free basis). The net forward flow of solvent is to be 1.0 kg hexane introduced as fresh solvent per kilogram flakes, and the fresh solvent is free of oil. The solvent draining from the flakes is generally free of solid except in the first stage: the rich miscella contains 10% of the insoluble solid in the feed as a suspended solid, which falls through the perforations of the belt during loading. How many stages are required?

SOLUTION The tie lines are vertical, $x = y^*$. Rearrange the drainage data as follows:

Percent oil in soln = 100y*	kg soln retained kg insoluble solid = $\frac{1}{N}$	N	kg oil kg insoluble solid = $\frac{y^*}{N}$
0	0.58	1.725	0
20	0.66	1.515	0.132
30	0.70	1.429	0.210

Basis: 1 kg flakes introduced.

Soybean feed $B = 0.8$ kg insoluble; $F = 0.2$ kg oil; $N_F = 0.8/0.2 = 4.0$ kg insoluble solid/kg oil; $y_F = 1.0$ mass fraction oil, solid-free basis.

stages determined with the $N = 0$ axis for all stages but the first. Between four and five stages are necessary. Adjustment of the amount of solvent or the amount of unextracted oil, by trial, will provide an integral number.

Rate of Leaching

The many diverse phenomena encountered in the practice of leaching make it impossible to apply a single theory to explain the leaching action. As has been shown, leaching may involve simple washing of the solution from the surface of a solid, or dissolving of a solute from a matrix of insoluble matter, osmosis, and possibly other mechanisms. Our knowledge of them is very limited. Washing a solution from the surface of impervious solid particles may be expected to be very rapid, requiring only the blending of solution and solvent, and stage efficiencies are then quite likely to be governed entirely by the completeness of the mechanical separation of liquid from solid.

Leaching a solute from the internal parts of a solid, on the other hand, will be relatively slow. Solids made up of a skeletal structure of insoluble substance, with the pores impregnated with the solute, can be described in terms of a pore-shape factor, as outlined in Chap. 4. The factor is a function of the solid, independent of the nature of the solute and solvent, and is a measure of the complexity of the path through which the solute diffuses. In natural products such as plant materials, the complexity of the structure may make application of these methods difficult. For sugar-beet cossettes [45], for example, about one-fifth of the cells are ruptured in producing the cossettes, and leaching of sugar from these cells is probably a simple washing process. The remaining cells lose sugar by a diffusion process, and the combination of events produces curves on coordinates such as Fig. 4.2, which deviate considerably from those developed from simple diffusion with constant effective diffusivity or pore-shape factor [25]. Many mechanisms have been considered in an attempt to explain such observations [7]. In another example, wood will show different rates of leaching of an impregnating solute depending upon whether diffusion is in a direction parallel to or across the grain of the wood [30]. If solutes must pass through cell walls by dialysis, it may not be possible to apply the concepts at all. The rates of diffusion of soybean oil from soybean flakes, which do not permit simple interpretation, have been attributed to the presence of several types of structures in the matrix [23] as well as to the presence of a slowly dissolving constituent in the oil [18, 22]. Whole seeds cannot be leached; rolling and flaking evidently crushes cell walls and opens up passageways for penetration of the solvent by capillary action [28, 29]. The fact that the rate of leaching increases with increased surface tension of the solvent-oil solutions and that even for flaked seeds there is a residue of unextractable oil which increases with flake thickness supports this view. That the leached oil is composed of several different substances is evident from the different properties of oil obtained after short and long leaching times. A method of dealing with such differently leached substances has been suggested [19], but these examples serve at least to indicate the complexity of many practical leaching processes. Very little study has been given most of them.

When solids like those described above are immersed in leaching solvents, it is reasonable to suppose that the resistance to mass transfer within the solid itself is likely to be the controlling resistance and that of the liquid surrounding the solid to be quite minor [45]. In such cases increasing the rate of movement of liquid past the solid surface will not appreciably influence the rate of leaching. Applications of the theory of unsteady-state diffusion within solids of various shapes (see Chap. 4) to leaching calculations, including those associated with multistage cascades and those associated with chemical reaction in the solid, have been worked out in considerable detail [1, 31, 32, 42].

NOTATION FOR CHAPTER 13

Any consistent set of units may be used, except as noted.

a	const
a_p	specific surface of particles, area/packed space, L^2/L^3
A	pure leaching solvent
A	thickener cross section, Eq. (13.7), L^2
b	const
B	pure insoluble carrier solid
B	insoluble carrier solid, mass (batch operation), M ; mass/time, M/Θ , or mass/(area)(time), $M/L^2\Theta$ (continuous operation)
c	solids concentration in a slurry, M/L^3
c_u	underflow solids concentration, M/L^3
C	soluble solute
d_p	diameter of a sphere of same surface/volume ratio as a particle, L
E	solvent and solute associated with the leached solids, mass (batch operation), M ; mass/time, M/Θ , or mass/(area)(time), $M/L^2\Theta$ (continuous operation)
F	solute and solvent in solids to be leached, mass (batch operation), M ; mass/time, M/Θ , or mass/(area)(time), $M/L^2\Theta$ (continuous operation)
g	acceleration of gravity, L^2/Θ
g_c	conversion factor, $ML/F\Theta^2$
G	mass velocity, $M/L^2\Theta$
G_S	flux of settling solids, $M/L^2\Theta$
G_{SL}	limiting solids flux, $M/L^2\Theta$
K	permeability, L^3/Θ^2
m	slope of equilibrium curve, dy^*/dx , dimensionless
M	solvent and solute content of a slurry or mixture, mass (batch operation), M ; mass/time, M/Θ , or mass/(area)(time), $M/L^2\Theta$ (continuous operation)
N	mass of insoluble solid B /(mass of solute A and solvent C), M/M
N_p	number of stages, dimensionless
Δp	pressure drop, F/L^2
R	solvent and solute in a leaching solution, mass (batch operation), M ; mass/time, M/Θ , or mass/(area)(time), $M/L^2\Theta$ (continuous operation)
s	residual saturation of a bed of drained solids, fraction of void volume occupied by liquid, dimensionless
s_0	residual saturation in the upper part of a packed bed, dimensionless
s_{av}	average residual saturation, dimensionless
V	initial settling rate, L/Θ
w	concentration of insoluble solids in a slurry, mass fraction
W	rate of solids flow, M/Θ
x	concentration of solute in solution, mass fraction, B -free basis

y	concentration of solute in a mixture, mass fraction, B-free basis
y^*	y at equilibrium
Z	height of a percolation bed or of settled solid, L
Z_D	drain height, L
Z_∞	ultimate height of settled solids, L
ϵ	fractional void volume of a packed bed, dimensionless
θ	time, Θ
μ_L	liquid viscosity, $M/L\Theta$
ρ_L	liquid density, M/L^3
σ	surface tension, F/L

Subscripts

F	feed; solids to be leached
S	saturated
1, 2, etc.	stage 1, stage 2, etc.

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PROBLEMS

13.1 A 1-m-diameter tank fitted with a false bottom and canvas filter is partly filled with 1000 kg (dry weight) of sea sand wet with seawater. The sand is allowed to drain until it stops dripping, whereupon 750 kg fresh water is added and recirculated to reach a uniform salt concentration. The sand is again allowed to drain until dripping stops and is then removed from the tank and dried. Estimate the salt content of the dried sand.

The sand particles have an average size $d_p = 0.4$ mm, a particle density 2660 kg/m^3 , and a bulk density $1490 \text{ kg (dry weight)/m}^3$. Seawater contains 3.5% salt; its density is 1018 kg/m^3 and surface tension 0.0736 N/m . The surface tension of water is 0.728 N/m .

13.2 Derive expressions for the coordinates of point $\Delta_R(y_{\Delta R}, N_{\Delta R})$ (Fig. 13.29), and check the results by determining the numerical values in the case of Illustration 13.3a.

13.3 In order to eliminate the solids in the final miscella of Illustration 13.4, it is decided to pass liquid from stage 3 to stage 1, where the liquid will contact fresh solids. The drained liquid from stage 1, containing the suspended solids, will then be passed to stage 2, where it is filtered by passage of the liquid through the bed of solids in this stage. The final miscella is then withdrawn as a clear solution from stage 2. How many stages will then be required for the same solvent/seeds ratio and the same oil concentration in the discharged solids? Ans.: 6.

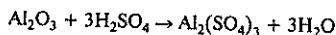
13.4 A mineral containing 20% elemental sulfur is to be leached with hot gas oil, in which the sulfur is soluble to the extent of 10% by weight. The solvent will be repeatedly pumped over the batch of ground mineral, using 1.5 kg fresh solvent/kg mineral. After no further solution of sulfur is obtained, the liquid will be drained and replaced with a fresh batch of 1.5 kg oil/kg original mineral, and the operation repeated. On drainage, the solid materials retain the solution to the extent of one-tenth the weight of undissolved solid (sulfur and gangue). No preferential adsorption takes place.

(a) Calculate the equilibrium data and plot them in the usual manner.

(b) Determine the amount of sulfur unextracted and the sulfur concentration of the composited leach liquors.

(c) Repeat part (b) if a two-stage Shanks system is used, with 3 kg fresh solvent/kg unleached solid. Assume steady state has been reached.

13.5 Aluminum sulfate, $\text{Al}_2(\text{SO}_4)_3$, is to be produced by action of sulfuric acid, H_2SO_4 , on bauxite in a series of agitators, with a cascade of continuous thickeners to wash the insoluble mud free of aluminum sulfate.



The flowsheet is similar to that of Fig. 13.13a. The reaction agitators are fed with (1) 25 t bauxite/day, containing 50% Al_2O_3 and the rest insoluble; (2) the theoretical quantity of aqueous acid containing 60% H_2SO_4 ; and (3) the overflow from the second thickener. Assume the reaction is complete. The strong product solution is to contain 22% $\text{Al}_2(\text{SO}_4)_3$, and no more than 2% of the $\text{Al}_2(\text{SO}_4)_3$ produced is to be lost in the washed mud. The last thickener is to be fed with pure wash water. The underflow from each thickener will contain 4 kg liquid/kg insoluble solid, and the concentration of solubles in the liquid of the underflow for each thickener may be assumed to be the same as that in the overflow. Calculate the number of thickeners required and the amount of wash water required per day. **Ans.:** 3 thickeners.

Note: In solving this problem, be certain to account for the water in the acid as well as that produced by the reaction. Adapt Fig. 5.16 to all but the first thickener in the cascade.

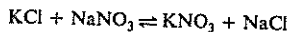
13.6 Barium, occurring naturally as the sulfate, BaSO_4 , is put in water-soluble form by heating with coal, thus reducing the sulfate to the sulfide, BaS . The resulting reaction mixture, barium "black ash" containing 65% soluble BaS , is to be leached with water. Black ash is fed to a tube mill at 100 t/day, together with the overflow from the second of a cascade of thickeners, and the effluent from the mill is fed to the first thickener. All the barium is dissolved in the mill. The strong solution overflowing from the first thickener is to contain 20% BaS by weight. The thickeners will each deliver a sludge containing 1.5 kg liquid/kg insoluble solid. The solution in the overflow and that in the sludge leaving any thickener may be assumed to have the same BaS concentration. It is desired to keep the BaS lost with the final sludge to at most 1 kg/day.

(a) How many thickeners are required? [Adapt Eq. (5.55) to all except the first thickener.]

Ans.: 6.

(b) It is decided to pass the final leached sludge to a continuous filter, as in Fig. 13.13b, where the liquid content of the filtered solids will be reduced to 15% by weight. The filtrate will be returned to the last thickener, but the filter cake will not be washed. How many thickeners will then be required? **Ans.:** 5.

13.7 In the manufacture of potassium nitrate, KNO_3 , potassium chloride, KCl is added to a hot, concentrated aqueous solution of sodium nitrate, NaNO_3 ,



Because of its relatively low solubility, part of the sodium chloride, NaCl , precipitates and is filtered off. A little water is added to the filtrate to prevent further precipitation of NaCl , the mixture is cooled to 20°C , and pure KNO_3 crystallizes. The resulting slurry contains, per 100 kg precipitated KNO_3 , 239 kg of a solution analyzing 21.3% KNO_3 , 21.6% NaCl , and 57.1% water. The slurry is fed to the first of a cascade of four continuous classifiers, where each 100 kg of crystals is countercurrently washed with 75 kg of a saturated solution of KNO_3 , containing 24.0% KNO_3 , in order to free them of NaCl . The wet crystals leaving each classifier retain 25% liquid, and the liquid overflows are clear. The washed crystals discharged from the fourth classifier, containing 25% liquid, are sent to a continuous drier. All liquid except that discharged with the washed crystals leaves in the overflow from the first classifier. Equilibrium between solid and liquid is attained in each classifier, and the clear overflows have the same composition as the liquid retained by the crystals. The solubility of KNO_3 in NaCl solutions (KNO_3 is the equilibrium solid phase) at the prevailing temperature is given

by the following table:

% NaCl	0	6.9	12.6	17.8	21.6
% KNO ₃	24.0	23.3	22.6	22.0	21.3

(a) Plot the equilibrium data [$N = \text{kg KNO}_3/\text{kg (NaCl + H}_2\text{O)}$] for both clear overflow and wet crystals; x and $y = \text{kg NaCl/kg (NaCl + H}_2\text{O)}$.

(b) Calculate the percentage NaCl content which can be expected on the dried KNO₃ product.

Ans.: 0.306%.

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